HYPERBOLIC BILLIARDS AS DYNAMICAL SYSTEMS

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Preface

These lectures grow out of a series of seminars on Billiards I gave at Stony Brook University in September 1992. They have been further developed as a brief graduate course in Dynamical Systems I taught in 1995 at the University of Rome Tor Vergata *and again at Rome and at the University of Padova in 1999. Some parts of it are the offspring of various mini-courses I gave during these years (Jerusalem (1997), Torino ??, Warwick (1998), Bologna (1999), Trieste (2002)). In addition, some of the material is a rielaboration of the article [63] written in collaboration with Maciej Wojtkowski.*

Although the main subject is billiards, I have made an effort to present them as a part and a motivation of general results in the ergodic theory of dynamical systems with some degree of smoothness.[1] *In particular, the book can be used in several different ways, at the end of the preface I will suggest some possible reading itineraries.*

The form in which I present the material is a consequence of two conflicting objectives: on the one hand to be understandable to students with no prior knowledge in the field of dynamical systems; on the other hand to get immediately to topics that are very close to current research in the field.

The goal it is not to present a complete account of the subject–this would take an enormous effort–but to present in detail the main ideas in the simplest possible context. Ideally, the reader should gain the capability of understanding, without further help, the more technical (and complete) accounts of the various parts of the theory or the current research papers. In addition, researchers that do not wish to enter too deeply in the field but want to be familiar, in an operative fashion, with the main techniques should be able to profit form this exposition.

Consequently, I give without proof all the general theorems and, at the same time, I spell out in full details baby theorems that contain all the main ideas–but not all the technical difficulties–of the general results.

[1] Let me emphasize, and I will do it over and over, that I have no pretenses of completeness, for example I almost do not mention "entropy," by all mean a fundamental concept in Ergodic Theory and Dynamical Systems, the same for Markov partitions and many other subjects. Also, very little space is devoted to more advances subjects that have experienced a great attention lately such as rates of mixing. Many times, the choice of the arguments is connected more to my personal interests than to any value judgment, other times it simply reflects my ignorance.

In order to make the book easily accessible to readers of different backgrounds I have tried to make each chapter as self contained as possible–when material from previous chapters is used I give precise indications on where to find it. Thus, more experienced readers can enter in the book at any place with minimum effort. The only prerequisite are elementary facts from topology, algebra, measure theory, differential geometry and functional analisys.

The bibliography has no pretense whatsoever of being complete but it can constitute a starting point for people who want to acquire the main results of this research area.

Each chapter is supplemented by several problems. These are an integral part of the text since they are used to develop many important ideas.

Let me conclude these comments by suggesting some possible ways of using this book–beside reading it from beginning to end, which may be an unrealistic pretense.

Reading Itineraries

Existence of a problem

If one wishes only to be convinced that a problem exists and that, even for relatively short times, the numerical integration of the equation of motion cannot be the all story: read chapter 1.

Ergodic Theory of Dynamical Systems–brief introduction

If one would like to get a very quick idea of what ergodicity is all about, and some feeling for the relevant statistical properties of Dynamical Systems: read chapter 1 (skipping the proofs) only if a total beginner, then chapter 2 and 11.

Ergodic Theory of Dynamical Systems–a short course

If one wishes to get acquainted not just with the ideas of the subject but also to acquire a working knowledge of some relevant techniques and ideas: read chapters 1-2-3-4-7-11.

Existence and properties of invariant foliations

For an introduction to the techniques used in establishing existence and properties of distributions and foliations (but not in the most general setting): read chapter 4 for the distributions (in particular section 4.4) and chapter 7 for the foliations

Ergodicity of Hamiltonian systems

To have a fast overview of the techniques available to establish ergodicity and a panoramic of the available results: read section 1.5 if a total beginner and chapter 8 for a more complete discussion. Chapter 9 con-

tain more advanced results concerning systms with discontinuities. A quick survey of the awailable results can be found in ...

Existence of invariant measures and statistical properties

Just some general facts about invariant measures and some techniques to establish their statistical properties: read section 1.4, see Examples 4.3.1 and look at chapter 11.

Billiards–basic facts

If the goal is simply to gain a descriptive idea of what billiards are and to have an idea of current research in the field: look at sections sections 5.1, 5.2, 5.4, 5.5, 6.1, 6.2, 10.1 and, for a review of the awailable results, ?

Billiards–advanced knowledge

This path reflects my opinion that the best way to understand a theory is to see it in action on a simple but non trivial example. The following concatenation of sections provides a completely self-contained proof of mixing for a two-dimensional Sinai billiard with finite horizon. This is the simplest possible example but it goes really a long way: section 5.1 defines what a Sinai Billiard with finite horizon is; sections 5.3 and 5.4 establish some basic properties. Sections 6.1 (which uses section 4.1) proves hyperbolicity; the necessary results concerning the foliations can be found in chapter 7; chapter 9 contains the general theory used to establish ergodicity; ergodicity is proven in secton 10.1; mixing in section ??

At last, I want to acknowledge my immense debit to all the dynamical systems and, in particular, to the billiard community. It is impossible to thank individually all the people that helped me through the years but I would like to mention some: Giovanni Gallavotti (he started me in the subject), Leonid Bunimovich (who spent quite a lot of time explaining articles that I found completely blaffing and set me straight when I was totally confused), Domocos Szasz (listening to him I felt I was starting to grasp the full picture for the first time), Victor Donnay (he choaced me into thinking in more geometrical terms) and Maciej Wojtkowski (his vision has largely shaped my point of view on the field).

In addition, I learned a lot from countless discussions especially with Nikolai Chernov and Nandor Symani but also V.Baladi, G.Benettin, A.Katok, G.Keller, F.Ledrappier, Ya.Pesin, L.-S.Young.

A special thanks goes to all the students that listened, and reacted, to the presentation of the preliminary version of various parts of this manuscript.

Also, I wish to thank Mark Pollicott, without his encouragement this book would have never started, and ???? for looking at preliminary versions; a *special mention goes to the editor Roger Astley as well, for his encouragement and patience with my endless delays.*

Finally, but most of all, I want to thank Ya. G. Sinai that, almost single handedly, invented the subject. To him I want to say: thanks for all the fun.

Liverani Carlangelo
Rome, 7th February 2005

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COMMENTS

Things to do do, to change, to talk about

W^u **always horizontal!**

Towers (one example)

Markow partitions (for automorphism of the torus)

put in appendix BET

appendix with KAM (just statment and comments)

geodesic flows (hyperbolicity ergodicity)

twist maps–standard map

billiards as twist maps

linked-twist maps

billiards in a wedge with gravity (wojtkowski stuff)

billiards with potential (D-L)

anosov maps, pseudo anosov

d**efine systems with syngularities, so that it is possible to give general statements on the foliations and chapters 5 6 are example of such systems**

Introduction

A *Dynamical Systems* is any system that evolves in time.[2] This encompass quite a few phenomena familiar to us. Of course, at this level of generality, there exists a manifold of very relevant questions that can be asked. In some sense the type of questions investigated through the centuries has shaped what today is meant by Dynamical Systems. Although this is not the proper occasion for an historical excursus, it is worthwhile to stress that the first Dynamical Systems widely investigated have been the planetary motions. Not surprisingly the main emphasis in such investigations was accurate prediction of future positions. Nevertheless, exactly from the effort of predicting accurately future motions stemmed the consciousness of the existence of very serious obstructions to such a program. Specifically, in the work of Poincaré [72] appeared for the first time the phenomena of instability with respect to initial conditions, a central concept in the understanding of modern Dynamical Systems. In fact, we will see briefly that such instability phenomena can be already observed in very simple systems–such as a periodically forced pendulum–that exhibit a so called "homoclinic tangle" [67, 70].

The realization that many relevant systems are very sensitive with respect to the initial conditions dealt a strong blow to the idea that it is always possible to predict the future behavior of a system,[3] yet the work of many physicist (and we must mention at least Boltzmann) and mathematicians (in particular, the so called *Russian School* with people like Kolmogorov, Anosov, Sinai, Arnold,...) led to the understanding that, although precise predictions where not possible, it was possible and, at times, even easy to make statistical predictions. The concept of statistical properties of a Dynamical System is

[2] That is, a set X consisting of all possible configurations of the system and maps $\phi^t : X \to X$ associating to each configuration $x \in X$ the configuration $\phi^t x$ representing the configuration reached by the system at time t when starting from the configuration x at time 0.

[3] Without going to the extreme of some authors of the eighteen century arguing that, given the present state of the universe, a sufficiently powerful mind (maybe God) could predict all the future. Think, more reasonably, of an isolated system and imagine to use some numerical scheme to try to solve the equations of motion for an arbitrarily long time with an arbitrary precision.

exactly at the base of the material treated in this book.

This introduction is dedicated to making precise, in a simple example, the nature of the above mentioned instability.

0.1 A pendulum–The model and a question

We will study a seemingly trivial example: a forced pendulum. To be more concrete, let us imagine a pendulum of length $l = 1$ meter, mass $m = 1$ kilogram and remember that the gravitational constant (on the earth surface) is approximately $g = 9.8$ meters per second squared. The Hamiltonian of the system reads [41]

$$H = \frac{1}{2l^2 m} p^2 - mgl \cos\theta, \tag{0.1.1}$$

where θ is the angle, counted counterclockwise, formed by the pendulum with the vertical direction ($\theta = 0$ corresponds to the configuration in which the pendulum assumes the lowest possible position) and $p = l^2 m\dot{\theta}$ is the associated momentum. Thus (θ, p) are the coordinates of the pendulum. The phase space $\mathcal{M}$ where the motion takes place conists of $\mathbb{T}^1 \times \mathbb{R}$.

The equations of motion associated to the Hamiltonian (0.1.1) represent the motion of an ideal pendulum in the vacuum feeling only the force of gravity. Clearly, this is an highly idealized situation with no counterpart in realty. Every system interacts with the rest of the universe. Thus the only hope for the idea of *isolated systems* to be fruitful is that the interaction with the exterior does not affect significantly the behavior of the system. Let us try to see what this can mean in reality.

The first issue is clearly friction. Let us imagine that we have set up the pendulum in a reasonable vacuum and reduced the friction at the suspension point so that the loss of energy is negligible on the time scale of few minutes. Does such a system behaves as an isolated pendulum within such a time frame? One problem is that the suspension point is still in contact with the rest of the world. If the pendulum is in a lab not so distant from an street (a rather common situation), then the traffic will induce some vibrations. It is then natural to ask: what happens if the suspension point of the pendulum vibrates?

In fact, nothing much happens for small pendulum oscillations (this is a consequence of Komogorv-Arnold-Moser theory, an highly non trivial fact), but if we start close to the vertical configuration it is conceivable that a motion that would be oscillatory for the unperturbed pendulum could gather enough energy from the external force as to change its nature and become rotatory, this would create a substantial difference between the unperturbed (ideal) and the perturbed (more realistic) case.

This is exactly the question we want to address:

Question: *Can we really predict the motion for a reasonable time if the initial condition is close to the vertical ?*

We will assume that the frequency of vibration ω is of one hertz[4] and the amplitude of the oscillations is very very small. Hence, as good mathematicians, we will call such an amplitude ε. In other words, the suspension point moves vertically according to the law $\varepsilon \cos \omega t$.

The Hamiltonian of the vibrating pendulum is then given by (see Problem 0.1)

$$H_\varepsilon = \frac{1}{2l^2 m} p^2 - mgl \cos \theta - \varepsilon m \omega^2 l \cos \omega t \cos \theta. \tag{0.1.2}$$

Accordingly the equation of motion are (see Problem 0.1)[5]

$$\begin{aligned} \dot{\theta} &= \frac{\partial H_\varepsilon}{\partial p} = \frac{p}{l^2 m} \\ \dot{p} &= -\frac{\partial H_\varepsilon}{\partial \theta} = -mgl \sin \theta - \varepsilon m \omega^2 l \cos \omega t \sin \theta. \end{aligned} \tag{0.1.3}$$

It is well known that the function H is an integral of motion for the solutions of (0.1.3) for $\varepsilon = 0$, that is: H computed along the solutions of the associated equations of motion is constant.[6] The physical meaning of H is the energy of the system. Clearly, the energy H_ε is not constant in general since the vibration can add or subtract energy to the pendulum.

0.2 Instability–unperturbed case

Let us first recall few basic facts about the unperturbed pendulum. The equation of motions are given by the (0.1.3) setting $\varepsilon = 0$. It is obvious that there exists two fixed points: $(0, 0)$ which corresponds to the pendulum at rest and is clearly stable , and $(\pi, 0)$ which corresponds to the pendulum in the vertical position and is certainly unstable. Our interest here is to analyze the motions that start close to the unstable equilibrium and to make more precise what it is meant by *instability*.

[4]One hertz corresponds to one oscillation every 2π seconds, a typical vibration transmitted through the ground at a reasonable distance has a frequency of this order of magnitude.

[5]Here we write the Hamilton equations associated to the Hamiltonian, see [5, 41] for the general theory.

[6]See [5, 41] for this general fact or do Problem 0.4 for the simple case at hand.

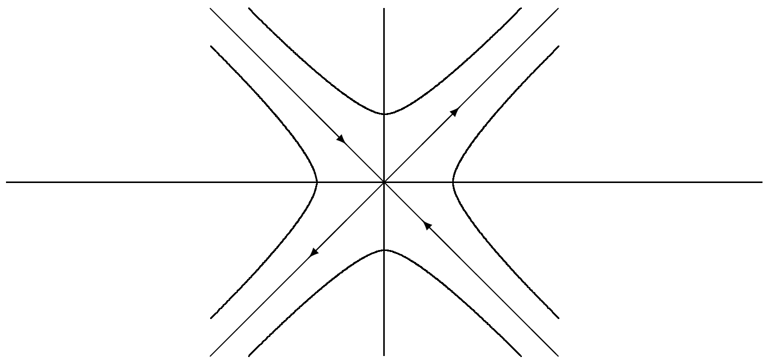

Figure 0.1: Unstable fixed point (phase portrait)

0.2.1 Unstable equilibrium

If we want to have an idea of how the motion looks like near a fixed point the natural first step is to study the linearization of the equation of motion near such a point. In our case, using the coordinates $(\theta_0, p) = (\theta - \pi, p)$, they look like

$$\begin{aligned}\dot{\theta}_0 &= \frac{p}{l^2 m}\\ \dot{p} &= mgl\theta_0.\end{aligned} \tag{0.2.1}$$

Let $\omega_p = \sqrt{\frac{g}{l}}$, it is immediate to verify that the general solution of (0.2.1) is

$$(\theta_0(t), p(t)) = (\alpha e^{\omega_p t} + \beta e^{-\omega_p t}, ml^2\omega_p\{\alpha e^{\omega_p t} - \beta e^{-\omega_p t}\}),$$

where α and β are determined by the initial conditions. Note that if the initial condition has the form $\alpha(1, ml\sqrt{gl})$ it will evolve as $\alpha e^{\omega_p t}(1, ml\sqrt{gl})$. While if the initial condition is of the form $\beta(1, -ml\sqrt{gl})$ it will evolve as $\beta e^{-\omega_p t}(1, -ml\sqrt{gl})$. In other words the directions $(1, ml\sqrt{gl})$ and $(1, -ml\sqrt{gl})$ are invariant for the linear dynamics. The first direction is expanded (and because of this is called *unstable direction*) while the second is contracted (*stable direction*).

Let us imagine to start the motion from an initial condition of the type $(\pi + \theta_0, 0)$, $\theta_0 \in [-\delta, \delta]$, where $\delta \le 10^{-4}$ represents the precision with which we are able to set the initial condition (one tenth of a millimeter); what will happen under the linear dynamics?

Our initial condition correspond to choosing, at time zero, $\alpha = \beta \le \frac{\delta}{2}$. As time goes on the coefficient of β becomes exponentially small while the coefficient of α increases exponentially, thus a good approximation of the position of the pendulum after some time is given by

$$\theta_0(t) \approx \alpha e^{\omega_p t}. \tag{0.2.2}$$

Since $\omega_p \approx 3.13$ seconds^{-1}, it follows that after about 2.5 seconds the position of the pendulum can be anywhere up to a distance of about 10 centimeters from the unstable position.

This means that the unstable position is really unstable and if we tray, as best as we can, to put the pendulum in the unstable equilibrium (always imagining that the friction has been properly reduced) it will typically fall after few seconds and it will fall in a direction that we are not able to predict (since it depend on the sign of δ, our unknown mistake). Nevertheless, after the ideal pendulum starts falling in one direction the subsequent motion is completely predictable, as we will see shortly.

An obvious objection to the above analysis is that I did not show that the linearized equation describes a motion really close to the one of the original equations. The answer to this question is particularly simple in this setting and is addressed in the next subsection.

0.2.2 The unstable trajectories (separatrices)

Given the already noted fact that, for $\varepsilon = 0$, H is a constant of motion, the phase space $\mathcal{M}$ is naturally foliated in the level curves of H, on which the motion must take place. This allows us to obtain a fairly accurate picture of the motions of the unperturbed pendulum. In fact, the level curves are given by the equations

$$\frac{p^2}{2l^2m} - mgl\cos\theta = E$$

where E is the energy of the motion. It is easy to see that $E = -mgl$ corresponds to the stable fixed point $(\theta, p) = (0, 0)$; $-mgl < E < mgl$ corresponds to oscillations of amplitude $\arccos\left[\frac{E}{mgl}\right]$; $E > mgl$ corresponds to rotatory motions of the pendulum. The last case $E = mgl$ is of particular interest to us: obviously it corresponds to the unstable fixed point $(\pi, 0)$, yet there are other two solution that travel on the two curves

$$p = \pm ml\sqrt{2lg(1 + \cos\theta)}.$$

This two curves are the ones that separate the oscillatory motions from the rotatory ones and, for this reasons, are called *separatrices*. It is very important to understand the motion along such trajectories, luckily the two differential equations

$$\dot{\theta} = \pm\sqrt{2\frac{g}{l}(1 + \cos\theta)}. \tag{0.2.3}$$

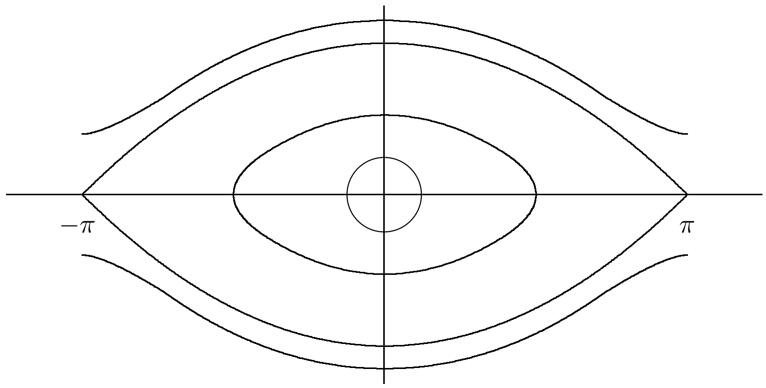

Figure 0.2: Unperturbed pendulum (phase portrait)

can be integrated explicitly (see Problem 0.5) yielding, for $\theta(0) = 0$,

$$\theta(t) = 4 \arctan e^{\pm\omega_p t} - \pi. \tag{0.2.4}$$

This orbits are asymptotic to the unstable fixed point both at $t \to +\infty$ and at $-\infty$ and, for $|t|$ large, agree with the linear behaviour of section 0.2.1. This situation is somewhat atypical as we will see briefly.

0.3 The perturbed case

0.3.1 Reduction to a map

The motion of the above system takes place on the cylinder $\mathcal{M} = S^1 \times \mathbb{R}$. By the theorem of existence and uniqueness for the solutions of differential equations [31, 6] follows immediately the possibility to define the maps $\phi_\varepsilon^t : \mathcal{M} \to \mathcal{M}$ associating to the point (θ, p) the point reached by the solution of (0.1.3) at time t, when starting at time 0 from the initial condition (θ, p). In such a way we define the *flow* ϕ_ε^t associated to the (0.1.3).

Clearly $\phi_\varepsilon^0(\theta, p) = (\theta, p)$, that is the map corresponding at time zero is the identity. Moreover, if $\varepsilon = 0$ the system is autonomous (the vector field does not depend on the time) hence the flow defines a group: for each $t, s \in \mathbb{R}$

$$\phi_0^{t+s}(\theta, p) = \phi_0^t(\phi_0^s(\theta, p)).$$

This corresponds to the obvious fact that the motion for a time $t + s$ can be obtained first as the motion from time 0 to time s, and then pretending that the time s is the initial time and following the motion for time t.

Of course, the above fact does not hold anymore when $\varepsilon \neq 0$. In this case, the maps ϕ_ε^t depend from our choice of the initial time (if we define them by starting from time 1 instead then time 0, in general we obtain different maps). Nevertheless, due to the fact that the external force is periodic something can be saved of the above nice property.

Let us define the map $T_\varepsilon : \mathcal{M} \to \mathcal{M}$ by

$$T_\varepsilon = \phi_\varepsilon^{\frac{2\pi}{\omega}},$$

then (see Problem 0.3), for each $n \in \mathbb{Z}$,

$$T_\varepsilon^n = \phi_\varepsilon^{\frac{2n\pi}{\omega}}. \tag{0.3.1}$$

The interest of (0.3.1) is that, for many purposes, we can study the map T_ε instead than the more complex object ϕ_ε^t. Morally, it means that if we look at the system *stroboscopically*, that is only at the times $\frac{2\pi}{\omega}n$ with $n \in \mathbb{Z}$, then it behaves like an autonomous (time independent) system.[7] Another interesting fact is that the flow ϕ_ε^t (and hence also the map T_ε) is area preserving (see Problem 0.7).[8]

0.3.2 Perturbed pendulum, $\varepsilon \neq 0$

The situation for the case $\varepsilon \neq 0$ is more complex and no easy way exists to study these motions.

As a general strategy, to study the behavior of a system (in our case the map T_ε) it is a good idea to start by investigating simple cases and then move on from there. In our systems the simplest motion consists of the equilibrium solutions. These are the time independent solutions.[9] Because of the special type of perturbation chosen the fixed points of the system for the case $\varepsilon = 0$ remain unchanged when $\varepsilon \neq 0$ (see Problem 0.8 for a brief discussion of a more general case).

Next, we can study the infinitesimal nature of the fixed points. It is natural to expect that the nature of the two fixed points does not change if ε is small, yet to verify this requires some checking. We will discuss explicitly only the fixed point $(\pi, 0)$.

The first step is to make precise the sense in which the case $\varepsilon \neq 0$ is a perturbation of the case $\varepsilon = 0$. This can be achieved by obtaining an explicit

[7] This is a very simple case of a very fruitful an general strategy: to look at the system only when some special event happens–in our case at each time in which the suspension point has its maximum height. See 1.2 if you want to know more.

[8] This also is a special instance of a more general fact: the Hamiltonian nature of the system, see [41], [5] if you want to know more.

[9] That is, equilibrium solutions for the map T_ε. These are *periodic* solutions for the flows of period $\frac{2\pi}{\omega}$. In fact, $T_\varepsilon x = x$ means $\phi^{\frac{2\pi}{\omega}} x = x$.

estimate on the size of

$$R_\varepsilon = \varepsilon^{-1}(T_0 - T_\varepsilon).$$

Let $z(t) = (z_1(t), z_2(t)) = \phi_0^t(x) - \phi_\varepsilon^t(x)$, then substituting in (0.1.3) and subtracting the general case to the case $\varepsilon = 0$ it yields

$$\begin{aligned}|\dot z_1| &\le \frac{|z_2|}{ml^2}\\ |\dot z_2| &\le mgl|z_1| + \varepsilon m\omega^2 l.\end{aligned}$$

In order to get better estimates it is convenient to define the new variables $\zeta_1 = ml^2\omega_p z_1$ and $\zeta_2 = z_2$. In these new variables the preceding equations read

$$\begin{aligned}|\dot\zeta_1| &\le \omega_p|\zeta_2|\\ |\dot\zeta_2| &\le \omega_p|\zeta_1| + \varepsilon m\omega^2 l.\end{aligned}$$

Which implies $\|\dot\zeta\| \le \omega_p\|\zeta\| + \varepsilon m\omega^2 l$. Taking into account that, in our situation, $ml^2\omega_p > 1$, it follows (see Problem 0.9)

$$\|R\|_{\mathcal{C}^0} \le \frac{ml\omega^2}{\omega_p}(e^{2\pi\frac{\omega_p}{\omega}} - 1).$$

Unfortunately, the above norm does not suffices for our future needs. We will see quite soon that it is necessary to estimate also the first and second derivatives of R, that is the $\mathcal{C}^2$ norm.

To do so the easiest way is to use the differentiability with respect to the initial conditions of the solutions of our differential equation (see [6, 31]). Fixing any point $x \in \mathcal{M}$ and calling $\xi^\varepsilon(t) = d_x\phi_\varepsilon^t\xi(0)$ we readily obtain:[10]

$$\begin{aligned}\dot\xi_1^\varepsilon &= \frac{\xi_2^\varepsilon}{l^2 m}\\ \dot\xi_2^\varepsilon &= -mgl\cos\theta\,\xi_1^\varepsilon - \varepsilon m\omega^2 l\cos\omega t\cos\theta\,\xi_1^\varepsilon\end{aligned} \tag{0.3.2}$$

One can then estimate the $\mathcal{C}^1$ norm of R by estimating $\|\xi^\varepsilon(\frac{2\pi}{\omega}) - \xi^0(\frac{2\pi}{\omega})\|$, since $\xi^\varepsilon(\frac{2\pi}{\omega}) = D_{(\theta,p)}T_\varepsilon\xi^\varepsilon(0)$. Analogously one can proceed to estimate the second derivatives.

[10]The vector $\xi_\varepsilon(t)$ is nothing else than the derivative $\frac{d\phi_\varepsilon^t(x+s\xi(0))}{ds}|_{s=0}$, the following equation is then obtained by exchanging the derivative with respect to t with the derivative with respect to s.

Doing so one obtains, for ε small enough,[11]

$$
\begin{aligned}
\|R\|_{\mathcal{C}^1} &\leq \frac{2ml\omega^2}{\omega_p} e^{3\pi\frac{\omega_p}{\omega}} := d_1 \\
\|T_0\|_{\mathcal{C}^2} &\leq ?? := C_2 \\
\|R\|_{\mathcal{C}^2} &\leq ?? := d_2
\end{aligned}
\tag{0.3.3}
$$

0.4 Infinitesimal behavior (linearization)

As a first application of the above considerations let us study the linearization of T_ε at $x_f = (\pi, 0)$. From (0.3.2) follows (see Problem 0.11)

$$
\begin{aligned}
D_{x_f} T_0 &= \begin{pmatrix} \cosh\frac{2\pi\omega_p}{\omega} & \frac{\sinh\frac{2\pi\omega_p}{\omega}}{ml^2\omega_p} \\ ml^2\omega_p \sinh\frac{2\pi\omega_p}{\omega} & \cosh\frac{2\pi\omega_p}{\omega} \end{pmatrix} \\
D_{x_f} T_\varepsilon &= D_{x_f} T_0 + \mathcal{O}(\varepsilon)
\end{aligned}
\tag{0.4.1}
$$

The eigenvalues of $D_{x_f} T_\varepsilon$ are then $\lambda_\varepsilon = e^{\frac{2\pi\omega_p}{\omega}} + \mathcal{O}(\varepsilon)$, λ_ε^{-1}.[12]

Clearly, if ε is sufficiently small, then $\lambda_\varepsilon > 1$. This means that the hyperbolic nature of the unstable fixed point remains unchanged under small perturbations (see Problem 0.12 for a case when the perturbation is not so small).[13]

If one does a similar analysis at the fixed point $(0, 0)$ one finds that the eigenvalues have modulus one: that is the infinitesimal motion is a rotation around the fixed point, exactly as in the $\varepsilon = 0$ case.

Hence the comments made at the end of subsection 0.2.1 for the unperturbed pendulum hold for the perturbed pendulum as well. Only now the is no longer an integral of motion (the energy) that control globally the behavior of the system.

Imagining that the map is linear (which is clearly false but, as we will see, qualitatively not so wrong) this would mean that the distance between two trajectories can be expanded by almost a factor 23 in a second. Initial conditions that are δ close at time zero will be about 23δ far apart after 1 second. If such a state of affair could persist (and we will see it may) after one

[11] The following bounds are not sharp, working more one can obtain better estimates but this would not make much of a difference in the sequel.

[12] This follows by the fact that the eigenvalues of $D_{x_f} T_0$ are $e^{\pm\frac{2\pi\omega_p}{\omega}}$, a simple perturbation theory of matrices (see Problem 0.10) and the already mentioned fact that the map T_ε is area preserving, thus the determinant of its derivative must be one.

[13] As we will see later in detail, hyperbolicity means that there is a direction in which the maps expands (the eigenvector v_ε^u associated to the eigenvalue λ_ε) and a direction in which the map contracts (the eigenvector v_ε^s associated to the eigenvalue λ_ε^{-1})

minute the two configurations would differ roughly by a factor $10^{80}\delta$, which means that not even knowing the initial condition plus or minus a quark could we predict the final one. This is certainly a rather worrisome perspective but much more work it is needed to decide if this may be indeed the case.

0.5 Local behavior (Hadamard-Perron Theorem)

The next step is to try to go from the above infinitesimal analysis to a local picture in a small neighborhood of the fixed points.

It is natural to expects that the two fixed points are still stable and unstable respectively, yet this is a far from trivial fact.

The stability of the point $(0,0)$ can be proven by invoking the so called KAM Theorem (this exceeds the scope of the present book and we will not discuss such matters, see [41] for such a discussion).[14]

The study of the local behavior around the point x_f is instead a bit easier and can be performed by applying the Hadamard-Perron Theorem 3.3.2 to conclude that, in a neighborhood of $(\pi,0)$, there exists two curves $x^u_\varepsilon(s) = (\theta^u_\varepsilon(s), p^u_\varepsilon(s))$, $x^s_\varepsilon(s)$ that are invariant with respect to the map T_ε. Namely, $T_\varepsilon x^s_\varepsilon([-\delta,\delta]) \subset x^s_\varepsilon([-\delta,\delta])$ and $T_\varepsilon^{-1} x^u_\varepsilon([-\delta,\delta]) \subset x^u_\varepsilon([-\delta,\delta])$ for all δ smaller than some δ_0 independent on ε; this are called the local stable and unstable manifold of zero, respectively. Essentially δ_0 is determined by the requirement that the non-linear part of T_ε be smaller than the linear part, for example

$$\inf_{\|v\|=1} \|D_{(\pi,0)} T_0 v\| \geq 10 \sup_{\|x-(\pi,0)\|\leq\delta_0} \|D^2_x T_0\|$$

will do.

Clearly, for $\varepsilon = 0$ $x^s = x^u$ and it coincides with the homoclinic orbit of the unperturbed pendulum. In addition, there exists $c_*, a > 0$, independent on ε, such that

$$\|T_\varepsilon^{-n} x^u_\varepsilon(s)\| \leq c_* e^{-an} \|x^u_\varepsilon(s)\| \quad \text{and} \quad \|x^u_\varepsilon - x^u_0\| \leq c_*\varepsilon. \tag{0.5.1}$$

and the analogous for the stable manifold, only forward in time. We have so obtained a local picture of the behavior of the map T_ε, yet this does not suffice

[14] In some sense this implies that we can indeed predict the motion for an extremely long time if we consider only oscillations close to the configuration $(0,0)$, so in that case the assumption that the pendulum is isolated is legitimate. Yet, this depends on the precision we are interested in and tend to degenerate if the amplitude of the oscillations is rather large. A complete analysis would be a very complicated matter but we will have an idea of the type of problems that can arise by considering extremely large oscillations, close to a full rotation of the pendulum.

to answer to our original question. To do so we need to follow the motion for at least a full oscillation: this requires really a global information.

To gain a more global knowledge we can try to construct larger invariant set for the map T_ε. A natural way to do so is to iterate: define $W^u = \cup_{n=0}^{\infty} T_\varepsilon^n x^u([-\delta_0, \delta_0])$. Since $T_\varepsilon x^u([-\delta_0, \delta_0]) \supset x^u([-\delta_0, \delta_0])$, it is clear that each time we iterate we get a longer and longer curve. The set W^u is then clearly a manifold and it is called the global unstable manifold.[15]

Let us see how much we can control such a growth process. Given $x, y \in \mathcal{M}$ we have

$$\|T_\varepsilon x - T_0 y\| \leq \varepsilon \|R_\varepsilon\|_{\mathcal{C}^0} + \|DT_0\| \|x - y\|.$$

Accordingly in a ball of radius δ_0 around $(\pi, 0)$ we have

$$\|T_\varepsilon x - T_0 y\| \leq d_1 \varepsilon + 2\lambda_0 \|x - y\|.$$

On the other hand, in the same neighborhood, if x belongs to the unstable manifold

$$\|T_\varepsilon x\| \geq \frac{\lambda_0}{2} \|x\|.$$

Thus

$$\|T_\varepsilon^n x_\varepsilon^u(s) - T_0^n x_0^u(s)\| \leq \varepsilon d_1 \sum_{i=0}^{n-1} (2\lambda_0)^i \leq 2 d_1 \varepsilon (2\lambda_0)^n.$$

This means that if we want a distance between the perturbed and the unperturbed unstable manifold let us say smaller of $2d_1 L\varepsilon$, for some L, we can iterate at most $(\ln L)/(\ln 2\lambda_0)$. This means that we can grow the manifold for a length of at least $\delta_0 L^a$, $a := \frac{\ln \lambda_0/2}{\ln 2\lambda_0}$, and still keep a strong control on the position of the unstable manifold. It is then enough to choose $L := (2\pi)^{1/a} \delta_0^{-2/a}$ to insure that the perturbed unstable manifold is very close to the unperturbed one until the separatrix does not enter again in the $\delta - 0$ neighborhood of $(\pi, 0)$.

The global manifold, as the name clearly states, it is a global object: it carries information on the dynamics for arbitrarily long times. Yet, the procedure by which it has been defined is far from constructive and the truth is that, besides the sketchy considerations above, at the moment we know very little of it. The next step is to gain some more detailed understanding of a large portion of W^u.

0.6 A more global understanding (the Melnikov method)

From the above considerations follows that the stable and unstable manifolds $x_\varepsilon^s(s)$, $x_\varepsilon^u(s) = (\theta_\varepsilon^u(s), p_\varepsilon^u(s))$ are ε close to the homoclinic orbit of the un-

[15] If one applies the above procedure to the unperturbed problem it gets the full separatrix.

perturbed pendulum. Let us call $x_0(t) = (\theta_0(t),\, p_0(t))$, $\theta_0(0) = 0$ such an orbit.

Clearly, we can assume, without loss of generality $\theta^u(s) = \theta^s(s) = \theta_0(s)$ (this corresponds simply to fixing the parameterization of the curve).

It is quite natural to ask what this does mean for the original flow. Clearly we can consider the orbit associated to any point on the above manifolds. Doing so we can write

$$\begin{aligned} x^u_\varepsilon(s,\,t) &= \phi^t_\varepsilon x^u_\varepsilon(s) = x_0(t+s) + \varepsilon x^u_1(s,\,t) \\ x^s_\varepsilon(s,\,t) &= \phi^t_\varepsilon x^s_\varepsilon(s) = x_0(t+s) + \varepsilon x^s_1(s,\,t). \end{aligned} \tag{0.6.1}$$

The above are the unstable and stable manifolds for the flow.[16]

Once we do this, it turns out to be specially smart to consider the distance between x^u and x^s in the orthogonal direction to x_0, clearly such a direction is given by ∇H, where H is the Hamiltonian of the unperturbed pendulum.[17]

It is then convenient to define

$$\Delta(s,\,t) = \varepsilon^{-1}\langle x^u_\varepsilon(s,t) - x^s_\varepsilon(s,t),\, \nabla_{x_0(t+s)}H\rangle =: \Delta^u(s,\,t) - \Delta^s(s,\,t).$$

Using (0.1.3) we can differentiate the above quantity with respect to time.[18]

$$\begin{aligned} \frac{d\Delta^u}{dt}(s,\,t) =& \varepsilon^{-1}\langle \dot x^u - \dot x_0, \nabla_{x_0}H\rangle + \langle x^u_1,\, D^2_{x_0}H\dot x_0\rangle \\ =& \varepsilon^{-1}\langle J\nabla_{x^u}H_\varepsilon, \nabla_{x_0}H\rangle + \langle x^u_1,\, D^2_{x_0}H\dot x_0\rangle \\ =& \varepsilon^{-1}\langle J\nabla_{x_0}H_\varepsilon + \varepsilon J D^2_{x_0}H_\varepsilon x^u_1,\, \nabla_{x_0}H\rangle \\ &+ \langle x^u_1,\, D^2_{x_0}HJ\nabla_{x_0}H\rangle + \mathcal{O}(\varepsilon\|D^3H_\varepsilon\|\,\|x^u_1\|^2) \\ =& \langle J\nabla_{x_0}H_1,\, \nabla_{x_0}H\rangle + \mathcal{O}(\varepsilon\|D^2H_1\|\,\|x^u_1\|\|\dot x_0\| + \varepsilon\|D^3H_\varepsilon\|\,\|x^u_1\|^2) \\ =& \{H_1,\, H\}_{x_0(t+s)} + \varepsilon d_3\mathcal{O}(\|x^u_1\|\|\dot x_0\| + \|x^u_1\|^2) \end{aligned}$$

where we have used the fact that D^2H is symmetric, J is antisymmetric, $d_3 := 2mgl \geq \sup\|D^3H_\varepsilon\| + \|D^2H_1\|$ and by $\mathcal{O}(h)$ we mean a quantity that is

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[16] Actually, they correspond to two dimensional surfaces (parametrized by s and t). Such surfaces are called *weak* stable and unstable manifolds since they contain the stable and unstable direction proper and the flow direction.

[17] See Problem 0.13

[18] As we have already seen the Hamiltonian is given by $H_\varepsilon = \frac{1}{2l^2m}p^2 - mgl\cos\theta - \varepsilon m\omega^2 l\cos\omega t\cos\theta := H_0 + \varepsilon H_1$, $H = H_0$, and, introducing the matrix

$$J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

(0.1.3) takes the more compact form

$$\dot x = J\nabla_x H_\varepsilon.$$

smaller, in absolute value, than h. The curly brackets stand for the so called *Poisson brackets* ($\{f, g\}_x = \langle J\nabla_x f, \nabla_x g\rangle$). Accordingly,

$$\Delta^u(s,0) = \int_{-\infty}^{0} \{H_1, H\}_{x_0(\tau+s)} d\tau + \lim_{t\to-\infty} \Delta^u(s,t) + \varepsilon d_4 \mathcal{O}(1)$$

which implies[19]

$$\Delta^u(s,0) = \int_{-\infty}^{0} \{H_1, H\}_{x_0(\tau+s)} d\tau + \varepsilon d_4 \mathcal{O}(1).$$

In complete analogy one can obtain

$$\Delta^s(s,0) = -\int_{0}^{\infty} \{H_1, H\}_{x_0(\tau+s)} d\tau + \varepsilon d_4 \mathcal{O}(1)$$

which implies

$$\Delta(s,0) = \int_{-\infty}^{\infty} \{H_1, H\}_{x_0(\tau+s)} d\tau + \varepsilon d_4 \mathcal{O}(1). \tag{0.6.2}$$

The integral in (0.6.2) is called *Melnikov integral* and provides an expression, at first order in ε, of the distance between the stable and the unstable manifold. All we are left with is to compute the integrals in (0.6.2). This turns out to be an exercise in complex analysis and it is left to the reader (see Problem 0.14), the results are:[20]

$$\int_{-\infty}^{\infty} \{H_1(\cdot,\tau), H\}_{x_0(\tau+s)} d\tau = -8\pi ml \frac{\omega^4 e^{-\frac{\pi\omega}{2\omega_p}}}{\omega_p^2(1-e^{\frac{\pi\omega}{\omega_p}})} \sin\omega s + \mathcal{O}(d_4\varepsilon).$$

We have thus gained very sharp control on the shape of the above manifolds.[21] In particular they have a transversal intersection ε close to $\theta = 0$

[19]Since $\lim_{t\to-\infty} \Delta^u(s,t) \le \lim_{t\to-\infty} \varepsilon^{-1}\|x^u(s,t) - x_0(t+s)\| = 0$ by definition of unstable manifold.

[20]A simple computation yields:

$$\{H_1, H\}_{x_0(t+s)} = \frac{\omega^2}{l} p(t+s)\cos\omega t \sin\theta(t+s).$$

Then, by using (0.2.4) and looking at Problem 0.6, one readily obtains:

$$\{H_1, H\}_{x_0(t)} = 4lm\omega^2\omega_p \frac{\cos\omega(t-s)\sinh\omega_p t}{(\cosh\omega_p t)^3}.$$

Finally, use Problem 0.14.

[21]Note that ε must be exponentially small with respect to ω. In many concrete problems (notably the so called *Arnold diffusion* [1]) it happens that this it is not the case. One can try to solve such an obstacle by computing the next terms of the ε expansion of Δ. In fact, it turns out that it is possible to express Δ as a power series in ε with all the terms exponentially small in ω [1]. Yet this is a quite complex task far beyond our scopes.

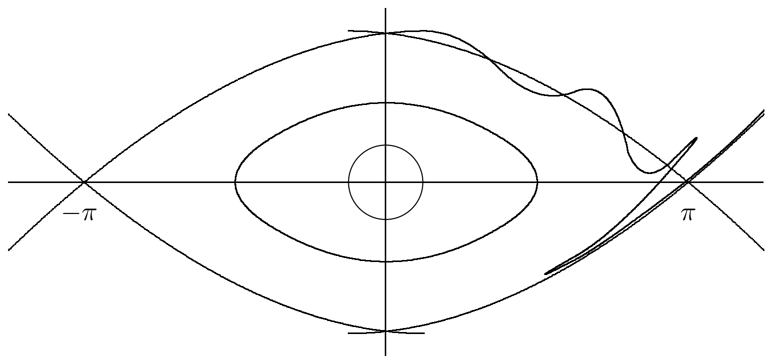

Figure 0.3: Perturbed pendulum

and the angle at which they intersect is $\mathcal{O}(\varepsilon)$. This intersections are called *homoclinic* intersection and their very existence is responsible for extremely interesting phenomena as can be readily seen by trying to draw the stable and unstable manifolds (see Figure 0.3 for an approximate first idea); we will discuss this issue in detail shortly.[22]

We have gained much more global information on the map T_ε, yet it does not suffice to answer to our question. The next section is devoted to obtaining a really global picture. Up to now we have used mainly analytic tools. Next, geometry will play a much more significant rôle.[23]

0.7 Global behavior (an horseshoe)

We want to explicitly construct trajectories with special properties. A standard way to do so is to start by studying the evolution of appropriate regions and to use judiciously the knowledge so gained. Let us see what this does mean in practice.

We start by gaining more control on the evolution of the points on the unstable manifold.

Since points on the unstable manifolds are pulled apart by the dynamics, the estimate must be done with a bit of care. In fact, we will use a way of arguing which it typical when instabilities are present, we will see many other instances of this type of strategy in the sequel.

[22]Note that the intersection corresponds to an homoclinic orbit for the map T_ε (that is, an orbit which approaches the fixed point x_f both in the future and in the past). This is what it is left of the homoclinic orbit of the unperturbed pendulum.

[23]What comes next it is the first example in this book of what it is loosely called *a dynamical argument*.

For each x in the unstable manifold (zero included) let us call $D^u_x T_\varepsilon := D_x T_\varepsilon v^u(x)$, where $v^u(0) = v^u$ and if $x = x^u_\varepsilon(t)$ then $v^u(x) = \|\dot{x}^u_\varepsilon(t)\|^{-1}\dot{x}^u_\varepsilon(t)$, that is the derivative of the map computed along the unstable manifold.

Lemma 0.7.1 (Distortion) *For each x in the unstable manifold and $n \in \mathbb{N}$ such that $\|T^n_\varepsilon x\| \leq \frac{\delta_0}{4}$, holds*[24]

$$e^{-\delta_0 C_*} \leq \left|\frac{D^u_0 T^n_\varepsilon}{D^u_x T^n_\varepsilon}\right| \leq e^{\delta_0 C_*},$$

with $C_* = \frac{C_2}{4(1-\sigma^{-1})}$.

Proof. The proof is a direct application of the chain rule:

$$\begin{aligned}\left|\frac{D^u_0 T^n_\varepsilon}{D^u_x T^n_\varepsilon}\right| &= \prod_{i=1}^{n}\left|\frac{D^u_0 T_\varepsilon}{D^u_{T^i x} T_\varepsilon}\right| \\ &\leq \operatorname{Exp}\left[\sum_{i=1}^{n} |\log(|D^u_0 T_\varepsilon) - \log(|D^u_{T^i x} T_\varepsilon|)|\right] \\ &\leq \operatorname{Exp}\left[\sum_{i=1}^{n} C_2\|T^i x\|\right]\end{aligned}$$

Since we know (by the Hadamard-Perron Theorem) that for points z on the unstable manifold, and in $B_{\frac{\delta_0}{4}}(0)$, $\|T_\varepsilon z\| \geq \sigma^{-1}\|z\|$, it follows that $\|T^i x\| \leq \delta_0 \sigma^{n-i}$. Thus

$$\left|\frac{D^u_0 T^n_\varepsilon}{D^u_x T^n_\varepsilon}\right| \leq \operatorname{Exp}\left[\frac{C_2\delta_0}{4(1-\sigma^{-1})}\right].$$

The other inequality can be proven in exactly the same way. □

A consequence of the above lemma is that

$$\|T^n_\varepsilon x^u_\varepsilon(t_0)\| \leq |D^u_z T^n|\|x^u_\varepsilon(t_0)\| \leq e^{\delta_0 C_*}\lambda^n_\varepsilon\|x^u_\varepsilon(t_0)\|,$$

where $z \in x^u_\varepsilon([0, t_0])$, and

$$\|T^n_\varepsilon x^u_\varepsilon(t_0)\| \geq e^{-\delta_0 C_*}\lambda^n_\varepsilon\|x^u_\varepsilon(t_0)\|.$$

Next we would like to consider the evolution of a small box constructed around the fix point.

[24] This quantity is commonly called *Distortion* because it measures how much the map differs from a linear one (notice that if T is linear then $\frac{D_x T}{D_y T} = 1$). Although apparently an innocent quantity it is hard to overstate its importance in the study of hyperbolic dynamics.

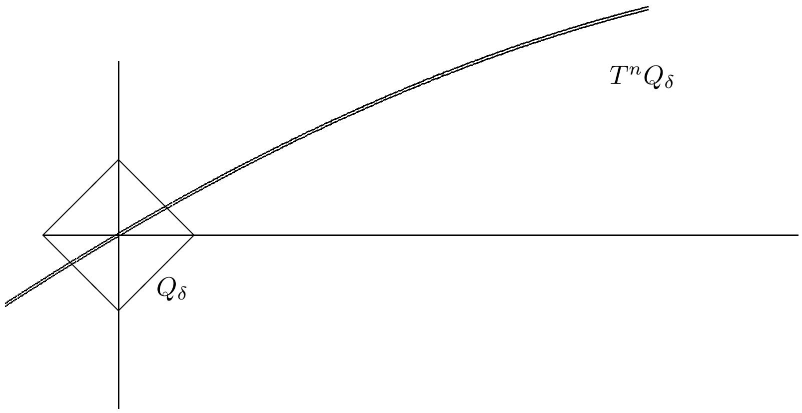

Figure 0.4: The evolution of the small box Q_δ

Consider the following small parallelogram: $Q_\delta := \{\xi \in \mathbb{R}^2 \mid \xi = av^u + bv^s$ for some $a, b \in [-\frac{\delta}{2}, \frac{\delta}{2}]\}$, $\delta \ll \delta_0$. Our first task is to understand the shape of $T_\varepsilon^n Q_\delta$. For each ξ in Q_{δ_0} let ξ^u be the closest point to ξ on the unstable manifold x_ε^u. Then it is easy to see the following.

Lemma 0.7.2 *For each $\xi \in Q_{\delta_0}$ holds*

$$\|T_\varepsilon\xi - (T_\varepsilon\xi)^u\| \leq 2\lambda_0^{-1}\|\xi - \xi^u\|$$
$$\|T_\varepsilon\xi^u - (T_\varepsilon\xi)^u\| \leq d_6\delta_0\|\xi^u - \xi\|$$

PROOF. Consider $D_{\xi^u}T_\varepsilon$, such a matrix must have determinant one and maps the $v^u(\xi^u)$ onto the $v^u(T_\varepsilon\xi^u)$ direction expanding it by $D_{\xi^u}^u T_\varepsilon$, accordingly there must be another direction which is contracted by $(D_{\xi^u}^u T_\varepsilon)^{-1}$. Clearly such a direction must be close to v^s, let us call it w. We can then write $\xi - \xi^u = \alpha v^u(\xi^u) + \beta w(\xi^u)$. Since $w(\xi^u)$ will be closer that $d_7\|\xi\|$ to v^s, which is the perpendicular direction to v^u in turns $d_7\|\xi\|$ close to $v^u(\xi^u)$, we have $\alpha \leq \beta d_7\delta_0$.

$$\begin{aligned}
T_\varepsilon\xi - T_\varepsilon\xi^u &= D_{\xi^u}^u T_\varepsilon(\xi - \xi^u) + \mathcal{O}(C_2\|\xi - \xi^u\|^2) \\
&= \nu_u\alpha v^u(T_\varepsilon\xi^u) + \beta D_{\xi^u}^u T_\varepsilon w(\xi^u) + C_2\mathcal{O}(\alpha^2 + \beta^2) \\
&= \nu_u\alpha v^u(T_\varepsilon\xi^u) + \nu^{-1}\beta w(T_\varepsilon\xi^u) + C_2\mathcal{O}(\alpha^2 + \beta^2) + \mathcal{O}(2\nu^{-1}\beta^2 d_7)
\end{aligned}$$

where $\nu := D_{\xi^u}^u T_\varepsilon$. From the above estimates the Lemma readily follows. □

Iterating the above estimates one can see that T^nQ_δ turns out to be a very thin coating of the unstable manifold, see Figure 0.4.

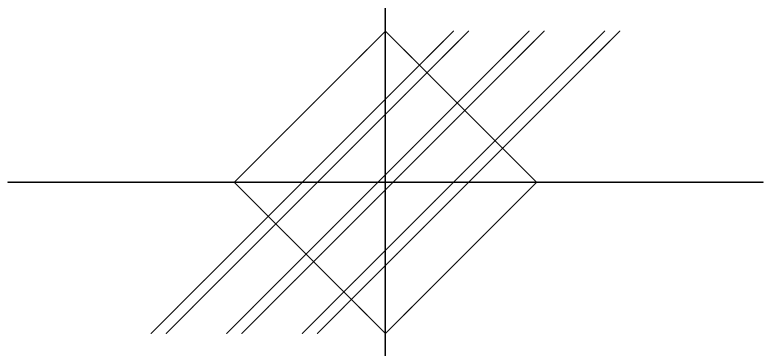

Figure 0.5: Horseshoe construction

In particular by choosing n appropriately ($n = n_0 := \frac{\log \delta^{-1}}{\log \lambda_\varepsilon} + b$) we have that $T^n Q_\delta$ is contained in a ε neighborhood of the unstable manifold at the homoclinic intersection (choose $\delta^2 \lambda^b \leq \varepsilon$). Drawing a picture is is immediately clear that $T^n Q_\delta \cap T^n Q_\delta \neq \emptyset$, moreover they intersect *transversally.*[25]

This is all is needed to construct an horseshoe (see section ???). In particular, in our case it means that $T^{2n_0} Q_\delta \cap Q_\delta \neq \emptyset$, in fact the intersection are transversal and consist of three strips almost parallel to the unstable sides. One contains zero, and it is the lest interesting for us, the other two cross above and below the unstable manifold respectively. The with of such strip is about δ^{-3}. We will discuss in the next chapters all the implications of this situation, here it suffices to notice that if we have two initial conditions in $T^{-2n_0} Q_\delta \cap Q_\delta$ at a distance h, after $2n_0$ iterations the two points will be in Q_δ again but at a distance $h\varepsilon^{-1}$. Since to decide if after that there will be a rotation or an oscillation we need to know the final position with a precision of order δ, we need to know the initial position with a precision $\mathcal{O}(\delta\varepsilon) = \mathcal{O}(\delta^3)$.

Note that in the above construction we have lost almost all the points, only the ones that come back to Q_δ at time $2n_0$ are under control. Nevertheless, we can consider the set $\Lambda := \cup_{k\in\mathbb{Z}} \bigcap T_\varepsilon^{2kn_0} Q_\delta$. This is clearly a measure zero set, yet it is far from empty (it contains uncountably many points) and it is made of points that at times multiple of $2n_0$ are always in Q_δ. When they arrive in Q_δ they will rotate if they are above the separatrices and oscillate otherwise. Let us call this two subset of Q_δ R and O. Given a point $\xi \in Q_\delta$ we can associate to it the doubly infinite sequence $\sigma \in \{0,1\}^{\mathbb{Z}}$ by the rule

[25]The meaning of *transversally* is the following: the square Q_δ has two sides parallel to v^u (the unstable direction), which we will call unstable sides, and two sides parallel to v^s (the stable direction), which we will call stable sides. Then the intersection is transversal if it consists of a region with again four sides: two made of the image of the unstable sides and two made of images of stable sides of Q_δ.

$\sigma_i = 1$ iff $T^{2n_0 i}\xi \in R$. The reader can check that the correspondence is onto.

0.8 Conclusion–an answer

If $\varepsilon = 10^{-6}$ and δ is a millimeter then we need to know the initial condition with a precision of 10^{-9} meters if we want to decide if the point will come back or rotate when it will get almost vertical again (this will happen in about 6 seconds). By the same token if we want to answer the same question, but for the second time the pendulum get close to the unstable position, we need to know the initial condition with a precision of the order 10^{-15} meters, and this just to predict the motion for about 12 seconds.[26]

We can finally answer to our original question:

Answer: *NO!*

Nevertheless, as we mentioned at the beginning, the above answer it is not the end of the story. In fact, there exists many other very relevant questions that can be answered.[27] The rest of the book deals with a particular type of question: can we meaningfully talk about the *average behavior* of a system?

Problems

0.1 Derive the Lagrangian, Hamiltonian and equations of motions for a pendulum attached to a point vibrating with frequency ω and amplitude ε. (Hint: see [59, 41] on how to do such things. Remember that two Lagrangian that differ by a total time derivative give rise to the same equation of motions and are thus equivalent.)

0.2 Consider the systems of differential equations $\dot{x} = f(x)$, $x \in \mathbb{R}^n$ and f smooth and bounded. Prove that the associated flow form a group. (Hint: use the uniqueness of the solutions of the ordinary differential equation)

[26] Remark that it is not just a matter of precision on the initial condition, it is also a matter of how one actually does the prediction. If the method is to integrate numerically the equation of motion, then one has to insure that the precision of the algorithm is of the order of 10^{-15}. This maybe achieved by working in double precision but if one wants to make predictions of the order of one minute it is quite clear that the numerical problem becomes very quickly intractable.

[27] For example: which type of motions are possible? This is a *qualitative* question. Such type of questions give rise to the qualitative theory of Dynamical Systems [70, 51], an extremely important part of the theory of dynamical systems, although not the focus here.

0.3 Consider the systems of differential equations $\dot{x} = f(x,t)$, $x \in \mathbb{R}^n$ and f smooth, bounded and periodic in t of period τ. Let ϕ^t be the associated flow. Define $T = \phi^\tau$, prove that $T^n = \phi^{n\tau}$.

0.4 Show that the hamiltonian is a constant of motion fro the pendulum. (Hint: Compute the time derivative)

0.5 Prove (0.2.4). (Hint: Write (0.2.3) in the integral form

$$t = \int_0^t \frac{\dot{\theta}(s)}{\sqrt{\frac{2g}{l}(1+\cos\theta(s))}} ds.$$

Using some trigonometry and changing variable obtain

$$t = \int_0^{\theta(t)} \frac{1}{2\omega_p \cos\frac{\theta}{2}} d\theta.$$

and compute it.)

0.6 If $\theta(t)$ is the motion obtained in the previous problem, show that

$$\sin\theta(t) = 2\frac{\sinh\omega_p t}{(\cosh\omega_p t)^2}; \quad \cos\theta(t) = \frac{2}{(\cosh\omega_p t)^2} - 1;$$
$$\cos^2\frac{\theta(t)+\pi}{4} = \frac{1}{1+e^{2\omega_p t}}.$$

0.7 Consider the systems of differential equations $\dot{x} = f(x,t)$, $x \in \mathbb{R}^n$ and f smooth. Suppose further that $\mathrm{d}iv f = 0$ (that is $\sum_{i=1}^n \frac{\partial f_i}{\partial x_i} = 0$). Show that the associated flow preserves the volume. (Hint: note that this is equivalent to saying that $|\det d\phi^t| = 1$, moreover by the group property and the chain rule for differentiating it suffices to check the property for small t. See that $d\phi^t = \mathrm{I}d + Dft + \mathcal{O}(t^2) = e^{Dft+\mathcal{O}(t^2)}$. Finally, remember the formula $\det e^A = e^{\mathrm{Tr}A}$.)

0.8 Let $T, T_1 : \mathbb{R}^2 \to \mathbb{R}^2$ be a smooth maps such that $T0 = 0$ and $\det(\mathbb{1} - D_0T) \neq 0$. Consider the map $T_\varepsilon = T + \varepsilon T_1$ and show that, for ε small enough, there exists points $x_\varepsilon \in \mathbb{R}^2$ such that $T_\varepsilon x_\varepsilon = x_\varepsilon$. (Hint: Consider the function $F(x,\varepsilon) = x - T_\varepsilon x$ and apply the Implicit Function Theorem to $F = 0$.)

0.9 Let $x(t) \in \mathbb{R}^n$ be a smooth curve satisfying $\|\dot{x}(t)\| \leq c\|x(t)\| + a$, $x(0) = 0$ prove that

$$\|x(t)\| \leq \frac{a}{c}(e^{ct} - 1).$$

(Hint: Note that $\|x(t)\| \leq \int_0^t \|\dot{x}(s)\|ds$. Transform then the differential inequality into an integral inequality. Show that if $z(t) \leq 0$ and $z(t) \leq \int_0^t z(s)ds$, then $z(t) \leq 0$ for each t. Use the last fact to compare a function satisfying the obtained integral inequality with the solution of the associated integral equation.)

0.10 Given two by two matrices A, B such that A has eigenvalues $\lambda \neq \mu$, show that the matrix $A_\varepsilon = A + \varepsilon B$ has eigenvalues $\lambda_\varepsilon, \mu_\varepsilon$ such that $|\lambda_\varepsilon - \lambda| \leq \varepsilon\|B\|?$ and $|\mu_\varepsilon - \mu| \leq \varepsilon\|B\|?$.[28]

0.11 Compute D_0T. (Hint: solve (0.3.2) for $\varepsilon = 0$, $\theta = \pi$, $p = 0$ and $t = \frac{2\pi}{\omega}$.)

0.12 Compute D_0T_ε and see that, if ω is sufficiently large, the eigenvalues have modulus one (the unstable point becomes stable!).

0.13 Given an Hamiltonian $H : \mathbb{R}^2 \to \mathbb{R}$, for each solution $x(t)$ of the associated equations of motion show that $\langle \nabla_{x(t)}H, \dot{x}(t)\rangle = 0$.

0.14 Compute the following integrals (0.6.2):

$$\int_{\mathbb{R}} e^{iat}(\cosh t)^{-n} \sinh t \, dt,$$

$a \in \mathbb{R}$ and $n \in \mathbb{N}$, $n > 1$.[29] (Hint: By a change of variable one can consider only the case $a > 0$. Consider the integral on the complex plane, show that the integral on the half circle $Re^{i\phi}$, $\phi \in [0, \pi]$, goes to zero as $R \to \infty$, then check that the poles of the integrand, on the complex plane, lie on the imaginary axis, finally use the residue theorem to compute the integrals.)

0.15 Do the same analysis carried out for the pendulum with a vibrating suspension point in the case of a pendulum subject to an external force $\varepsilon \cos \omega t$ and in presence of a small friction $-\varepsilon^2\gamma\dot{\theta}$.

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[28] This is a very simple case of the very general problem of perturbation of point spectrum, see [49] (Kato) if you want to know more.

[29] The result, for $a > 0$, is:

$$\int_{\mathbb{R}} e^{iat}(\cosh t)^{-n} \sinh t = 2\pi i \sum_{k=0}^{\infty} \frac{\phi_{n,k}^{(n-1)}(i\frac{2k+1}{2}\pi)}{(n-1)!},$$

where

$$\phi_{n,k}(z) = e^{iza} \sinh z \left(\frac{z - i\frac{2k+1}{2}\pi}{\cosh z}\right)^n.$$

For $n = 3$ the above formula yields

$$\int_{\mathbb{R}} e^{iat}(\cosh t)^{-3} \sinh t = \pi a^2 e^{-\frac{\pi}{2}a}(1 - e^{-\pi a})^{-1}.$$

Notes

As already mentioned in the text the first to realize that the motions arising from differential equations can be very comnplex was probably Poincaré [72]. At the time the main problem in celestial mechanics (the famouns n-body problem) was to find all the integral of motion. Dirichlet and Weierstrass worked on this problem, but Poincaré was the first to rise serious doubt on the eixstence of such integrals (which would have implied regular motions). For more historical remarks see [67]. In fact, all the content of this chapter is ispired by the more sofisticated analysis in [67].

CHAPTER 1

General facts and definitions

Before entering in the hart of the matter it is necessary the knowledge of some general facts concerning (measurable) Dynamical Systems. This chapter is intended for readers with no previous knowledge of Dynamical Systems. The chapter contains few basic facts, some of which will be used in the following while others are meant to provide a wider context to the material actually discussed. For a much more complete discussion of the relevant concepts the reader is referred to [65], [51].

1.1 Basic Definitions and examples

Definition 1.1.1 *By Dynamical System*[1] *with discrete time we mean a triplet* (X, T, μ) *where* X *is a measurable space,*[2] μ *is a measure and* T *is a measurable map from* X *to itself that preserves the measure (i.e.,* $\mu(T^{-1}A) = \mu(A)$ *for each measurable set* $A \subset X$*).*

An equivalent characterization of invariant measure is $\mu(f \circ T) = \mu(f)$ for each $f \in L^1(X, \mu)$ since, for each measurable set A, $\mu(\chi_A \circ T) = \mu(\chi_{T^{-1}A}) = \mu(T^{-1}A)$, where χ_A is the characteristic function of the set A.

[1] To be really precise this is the definition of "Measurable Dynamical Systems," hopefully the reader will excuse this abuse of language. More generally a Dynamical System can be defined as a set X together with a map $T : X \to X$ or, even more generally, an algebra $\mathcal{A}$ (e.g., the algebra of the continuous functions on X) and an isomorphism $\tau : \mathcal{A} \to \mathcal{A}$ (e.g., $\tau f := f \circ T$). This last definition is so general as to include Stochastic Processes and Quantum Systems. A further generalization consists in realizing that the above setting can be view as the action of the semigroup $\mathbb{N}$ (or the group $\mathbb{Z}$ if T is invertible) on the algebra $\mathcal{A}$. One can then consider other groups (already in the next definition the group is $\mathbb{R}$), for example $\mathbb{Z}^n$ or $\mathbb{R}^n$, this goes in the direction of the Statistical Mechanics and it has receive a lot of attention lately [1]. Of course, such a generality is excessive for the task at hand.

[2] By measurable space we simply mean a set X together with a σ-algebra that defines the measurable sets.

Remark 1.1.2 *In this book we will always assume $\mu(X) < \infty$ (and quite often $\mu(X) = 1$, i.e. μ is a probability measure). Nevertheless, the reader should be aware that there exists a very rich theory pertaining to the case $\mu(X) = \infty$, see [3].*

Definition 1.1.3 *By Dynamical System with continuous time we mean a triplet (X, ϕ^t, μ) where X is a measurable space, μ is a measure and ϕ^t is a measurable group ($\phi^t(x)$ is a measurable function for each t, $\phi^t(x)$ is a measurable function of t for almost all $x \in X$; $\phi^0 =$identity and $\phi^t \circ \phi^s = \phi^{t+s}$ for each $t, s \in \mathbb{R}$) or semigroup ($t \in \mathbb{R}^+$) from X to itself that preserves the measure (i.e., $\mu((\phi^t)^{-1}A) = \mu(A)$ for each measurable set $A \subset X$).*

The above definitions are very general, this reflects the wideness of the field of Dynamical Systems. In the present book we will be interested in much more specialized situations.

In particular, X will always be a topological compact space. The measures will alway belong to the class $\mathcal{M}^1(X)$ of Borel probability measures on X.[3] For future use, given a topological space X and a map T let us define $\mathcal{M}_T$ as the collection of all Borel measures that are T invariant.[4]

Often X will consist of finite unions of smooth manifolds (eventually with boundaries). Analogously, the dynamics (the map or the flow) will be smooth in the interior of X.

Let us see few examples to get a feeling of how a Dynamical System can look like.

1.1.1 Examples

1.1.1.a *Rotations*

Let $\mathbb{T}$ be $\mathbb{R}$ mod 1. By this we mean $\mathbb{R}$ quotiented with respect to the equivalence relations $x \sim y$ if and only if $x - y \in \mathbb{Z}$. $\mathbb{T}$ can be though as the interval $[0, 1]$ with the points 0 and 1 identified. We put on it the topology induced by the topology of $\mathbb{R}$ via the defined equivalence relation. Such a topology is the usual one on $[0, 1]$, apart from the fact that each open set containing 0 must contain 1 as well. Clearly, from the topological point of view, $\mathbb{T}$ is a circle. We choose the Borel σ-algebra. By μ we choose the Lebesgue measure m, while $T : \mathbb{T} \to \mathbb{T}$ is defined by

$$Tx = x + \omega \mod 1,$$

for some $\omega \in \mathbb{R}$. In essence, T translates, or rotates, each point by the same quantity ω. It is easy to see that the measure μ is invariant (Problem 1.4).

[3] Remember that a Borel measure is a measure defined on the Borel σ-algebra, that is the σ-algebra generated by the open sets.

[4] Obviously, for each $\mu \in \mathcal{M}_T$, (X, T, μ) is a Dynamical System.

1.1.1.b *Bernoulli shift*

A Dynamical System needs not live on some differentiable manifold, more abstract possibilities are available.

Let $\mathbb{Z}_n = \{1, 2, ..., n\}$, then define the set of two sided (or one sided) sequences $\Sigma_n = \mathbb{Z}_n^{\mathbb{Z}}$ ($\Sigma_n^+ = \mathbb{Z}_n^{\mathbb{Z}_+}$). This means that the elements of Σ_n are sequences $\sigma = \{..., \sigma_{-1}, \sigma_0, \sigma_1, ...\}$ ($\sigma = \{\sigma_0, \sigma_1, ...\}$ in the one sided case) where $\sigma_i \in \mathbb{Z}_n$. To define the measure and the σ-algebra a bit of care is necessary. To start with, consider the *cylinder sets*, that is the sets of the form

$$A_i^j = \{\sigma \in \Sigma_n \mid \sigma_i = j\}.$$

Such sets will be our basic objects and can be used to generate the algebra $\mathcal{A}$ of the cylinder sets via unions and complements (or, equivalently, intersections and complements). We can then define a topology on Σ_n (the product topology, if $\{1, \dots, n\}$ is endowed by the discrete topology) by declaring the above algebra made of open sets and a basis for the topology. To define the σ-algebra we could take the minimal σ-algebra containing $\mathcal{A}$, yet this it is not a very constructive definition, neither a particular useful one, it is better to invoke the Carathèodory construction.

Let us start by defining a measure on $\mathbb{Z}_n$, that is n numbers $p_i > 0$ such that $\sum_{i=1}^n p_i = 1$. Then, for each $i \in \mathbb{Z}$ and $j \in \mathbb{Z}_n$,

$$\mu(A_i^j) = p_j.$$

Next, for each collection of sets $\{A_{i_l}^{j_l}\}_{l=1}^s$, with $i_l \neq i_k$ for each $l \neq k$, we define

$$\mu(A_{i_1}^{j_1} \cap A_{i_2}^{j_2} \cap ... \cap A_{i_s}^{j_s}) = \prod_{l=1}^{s} p_{j_l}.$$

We now know the measure of all finite intersection of the sets A_i^j. Obviously $\mu(A^c) := 1 - \mu(A)$ and the measure of the union of two sets A, B obviously must satisfy $\mu(A \cup B) = \mu(A) + \mu(B) - \mu(A \cap B)$. We have so defined μ on $\mathcal{A}$. It is easy to check that such a μ is σ-additive on $\mathcal{A}$; namely: if $\{A_i\} \subset \mathcal{A}$ are pairwise disjoint sets and $\cup_{i=1}^\infty A_i \in \mathcal{A}$, then $\mu(\cup_{i=1}^\infty A_i) = \sum_{i=1}^\infty \mu(A_i)$. The next step is to define an outer measure[5]

$$\mu^*(A) := \inf_{\substack{B \in \mathcal{A} \\ B \supset A}} \mu(B) \quad \forall A \subset \Sigma_n.$$

Finally, we can define the σ-algebra as the collection of all the sets that satisfy the *Carathèodory's criterion*, namely A is measurable (that is belongs to the σ-algebra) iff

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c) \quad \forall E \subset \Sigma_n.$$

[5]An outer measure has the following properties: i) $\mu^*(\emptyset) = 0$; ii) $\mu^*(A) \leq \mu^*(B)$ if $A \subset B$; iii)$\mu^*(\cup_{i=1}^\infty A_i) \leq \sum_{i=1}^\infty \mu^*(A_i)$. Note that μ^* need not be additive on all sets.

The reader can check that the sets in $\mathcal{A}$ are indeed measurable.

The Carathèodory Theorem then asserts that the measurable sets form a σ-algebra and that on such a σ-algebra μ^* is numerably additive, thus we have our measure μ (simply the restriction of μ^* to the σ-algebra).[6] The σ-algebra so obtained is nothing else than the completion with respect to μ of the minimal σ-algebra containing $\mathcal{A}$ (all the sets with zero outer measure are measurable).

The map $T : \Sigma_n \to \Sigma_n$ (usually called *shift*) is defined by

$$(T\sigma)_i = \sigma_{i+1}.$$

We leave to the reader the task to show that the measure is invariant (see Problem 1.12).

To understand what's going on, let us consider the function $f : \Sigma \to \mathbb{Z}_n$ defined by $f(\sigma) = \sigma_0$. If we consider T^t, $t \in \mathbb{N}$, as the time evolution and f as an observation, then $f(T^t\sigma) = \sigma_t$. This can be interpreted as the observation of some phenomenon at various times. If we do not know anything concerning the state of the system, then the probability to see the value j at the time t is simply p_j. If $n = 2$ and $p_1 = p_2 = \frac{1}{2}$, it could very well be that we are observing the successive outcomes of tossing a fair coin where 1 means head and 2 tail (or vice versa); if $n = 6$ it could be the outcome of throwing a dice and so on.

1.1.1.c *Dilation*

Again $X = \mathbb{T}$ and the measure is Lebesgue. T is defined by

$$Tx = 2x \mod 1.$$

This map it is not invertible (similarly to the one sided shift). Note that, in general, $\mu(TA) \neq \mu(A)$ (e.g., $A = [0, \frac{1}{2}]$).

1.1.1.d *Toral automorphism (Arnold cat)*

This is an automorphism of the torus and gets its name by a picture draw by Arnold [10]. The space X is the two dimensional torus $\mathbb{T}^2$. The measure is again Lebesgue measure and the map is

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} \mod 1 := L\begin{pmatrix} x \\ y \end{pmatrix} \mod 1.$$

Since the entries of L are integers numbers it is clear that T is well defined on the torus; in fact, it is a linear toral automorphism. The invariance of the measure follows from $\det L = 1$.

[6] See [62] if you want a quick look at the details of the above Theorem or consult [74] if you want a more in depth immersion in measure theory. If you think that the above construction is too cumbersome see Problem 1.14.

1.1.1.e *Hamiltonian Systems*

Up to now we have seen only examples with discrete time. Typical examples of Dynamical Systems with continuous time are the solutions of an ODE or a PDE. Let us consider the case of an Hamiltonian system. The simplest case is when $X = \mathbb{R}^{2n}$, the σ-algebra is the Borel one and the measure μ is the Lebesgue measure m. The dynamics is defined by a smooth function $H : X \to \mathbb{R}$ via the equations

$$\frac{dx}{dt} = J\text{grad}H(x)$$

where $\text{grad}(H)_i = (\nabla H)_i = \frac{\partial H}{\partial x_i}$ and J is the block matrix

$$J = \begin{pmatrix} 0 & \mathbb{1} \\ -\mathbb{1} & 0 \end{pmatrix}.$$

The fact that m is invariant with respect to the Hamiltonian flow is due to the Liouville Theorem (see [5] or Problem 0.7).

Such a dynamical system has a natural decomposition. Since H is an integral of the motion, for each $h \in \mathbb{R}$ we can consider $X_h = \{x \in X \mid H(x) = h\}$. If $X_h \neq \emptyset$, then it will typically consist of a smooth manifold,[7] let us restrict ourselves to this case. Let σ be the surface measure on X_h, then $\mu_h = \frac{\sigma}{\|\text{grad}H\|}$ is an invariant measure on X_h and (X_h, ϕ_t, μ_h) is a Dynamical System (see Problem 1.6).

1.1.1.f *Geodesic flow*

Along the same lines any geodesic flow on a compact Riemannian manifold naturally defines a dynamical system.

1.2 Return maps and Poincaré sections

Normally in Dynamical Systems there is a lot of emphasis on the discrete case. One reason is that there is a general device that allows to reduce the study of many properties of a continuous time Dynamical System to the study of an appropriate discrete time Dynamical System: Poincaré sections (we have already seen an instance of this in the introduction). Here we want to make few comments on this precious tool that we will largely employ in the study of billiards.

Let us consider a smooth Dynamical System (X, ϕ^t, μ) (that is a Dynamical Systems in continuous time where X is a smooth manifold and ϕ^t is a

[7] By the implicit function theorem this is locally the case if $\nabla H \neq 0$.

smooth flow). Then we can define the vector field $V(x) := \frac{d\phi^t(x)}{dt}|_{t=0}$.[8]

Consider a smooth compact submanifold (possibly with boundaries) Σ of codimension one such that $\mathcal{T}_x\Sigma$ (the tangent space of Σ at the point x) is transversal to $V(x)$.[9] We can then define the *return time* $\tau_\Sigma : \Sigma \to \mathbb{R}^+ \cup \{\infty\}$ by

$$\tau_\Sigma = \inf\{t \in \mathbb{R}^+ \backslash \{0\} \mid \phi^t(x) \in \Sigma\},$$

where the inf is taken to be ∞ if the set is empty. Next we define the *return map* $T_\Sigma : D(T) \subset \Sigma \to \Sigma$, where $D(T) = \{x \in \Sigma | \tau_\Sigma(x) < \infty\}$, by

$$T_\Sigma(x) = \phi^{\tau_\Sigma(x)}(x).$$

It is easy to check that there exists $c > 0$ such that $\tau_\Sigma \geq c$ (Problem 1.9).

To define the measure, the natural idea is to project the invariant measure along the flow direction: for all measurable sets $A \subset \Sigma$, define[10]

$$\nu_\Sigma(A) := \lim_{\delta \to 0} \frac{1}{\delta} \mu(\phi^{[0,\,\delta]}(A)). \tag{1.2.1}$$

See Problem 1.8 for the existence of the above limit; see Problem 1.9 for the proof that τ_Σ is finite almost everywhere and Problem 1.10 for the proof that $(\Sigma,\, T_\Sigma,\, \nu_\Sigma)$ is a dynamical system. The reader is invited to meditate on the relation between this Dynamical System and the original one.

1.3 Suspension flows

A natural question is if it is possible to construct a flow with a given Poincaré section, the answer is that there are infinitely many flows with a given section. Let us construct some of them. Given a dynamical system (Σ, T, ν) consider $\tilde{X} := \Sigma \times R^+$. Define the flow $\phi_t((x, s)) = (x, s + t)$. We then define in $\tilde{X}$ the equivalence relation $(x, t) \sim (y, s)$ iff $s = t + n$ and $y = T^n x$ or $t = s + n$ and $x = T^n y$ for some $n \in \mathbb{N}$. A moment of reflection shows that the set X of equivalence classes is nothing else than the set $\Sigma \times [0, 1]$ with the points $(x, 1)$ and $(Tx, 0)$ identified. Clearly the flow is naturally quotiented over the equivalence classes and yields a quotient flow on X, such a flow is called a *suspension flow*.

A more general construction can by obtained by applying a time change to the above example. Alternatively, one can can choose any smooth function $\tau : \Sigma \to \mathbb{R}^+$, that will be called a *ceiling function* and consider the set $X_\tau = \{(x, t) \in \Sigma \times \mathbb{R}^+ \mid t \in [0, \tau(x)]\}$ with the points $(x, \tau(x))$ and $(Tx, 0)$ identified.

[8] Very often it is the other way around: the vector field is given first and then the flow–as we saw in the introduction.

[9] That is $\mathcal{T}_x\Sigma \oplus V(x)$ form the full tangent space at x.

[10] We use the notation: $\phi^I(A) := \cup_{t \in I} \phi^t(A)$ for each $I \subset \mathbb{R}$.

A moment of reflection should show that the topology of X_τ does not depend on τ and is then the same than the suspension defined above. The flow is again defined by $\phi_t(x,s) = (x, s+t)$ for $t \leq \tau(x) - s$. Such flows are called *special flows*.

1.4 Invariant measures

A very natural question is: given a space X and a map T does there always exists an invariant measure μ? A non exhaustive, but quite general, answer exists: Krylov-Bogoluvov Theorem.

First of all we need a useful characterization of invariance.

Lemma 1.4.1 *Given a compact metric space X and map T continuous apart from a compact set K,*[11] *a Borel measure μ, such that $\mu(K) = 0$, is invariant if and only if $\mu(f \circ T) = \mu(f)$ for each $f \in \mathcal{C}^{(0)}(X)$.*

PROOF. To prove that the invariance of the measure implies the invariance for continuous functions is obvious since each such function can be approximate uniformly by simple functions–that is, sum of characteristic functions of measurable sets–for which the invariance it is immediate.[12] The converse implication is not so obvious.

The first thing to remember is that the Borel measures, on a compact metric space, are regular [75]. This means that for each measurable set A the following holds[13]

$$\mu(A) = \inf_{\substack{G \supset A \\ G = \mathring{G}}} \mu(G) = \sup_{\substack{C \subset A \\ C = \overline{C}}} \mu(C). \tag{1.4.1}$$

Next, remember that for each closed set A and open set $G \supset A$, there exists $f \in \mathcal{C}^{(0)}(X)$ such that $f(X) \subset [0,1]$, $f|_{G^c} = 0$ and $f|_A = 1$ (this is Urysohn Lemma for Normal spaces [74]). Hence, setting $B_A := \{f \in \mathcal{C}^{(0)}(X) \mid f \geq \chi_A\}$,

$$\mu(A) \leq \inf_{f \in B_A} \mu(f) \leq \inf_{\substack{G \supset A \\ G = \mathring{G}}} \mu(G) = \mu(A). \tag{1.4.2}$$

Accordingly, for each A closed, we have

$$\mu(T^{-1}A) \leq \inf_{f \in B_A} \mu(f \circ T) = \inf_{f \in B_A} \mu(f) = \mu(A).$$

[11] This means that, if $C \subset X$ is closed, then $T^{-1}C \cup K$ is closed as well.
[12] This is essentially the definition of integral.
[13] This is rather clear if one thinks of the Carathéodory construction starting from the open sets.

In addition, using again the regularity of the measure, for each A Borel holds[14]

$$\mu(T^{-1}A) = \inf_{\substack{U\supset K\\ U=\mathring{U}}} \mu(T^{-1}A\backslash U) \leq \inf_{\substack{U\supset K\\ U=\mathring{U}}} \sup_{\substack{C\subset T^{-1}A\backslash U\\ C=\overline{C}}} \mu(T^{-1}(TC))$$
$$\leq \inf_{\substack{U\supset K\\ U=\mathring{U}}} \sup_{\substack{C\subset A\backslash TU\\ C=\overline{C}}} \mu(T^{-1}C) \leq \sup_{\substack{C\subset A\\ C=\overline{C}}} \mu(T^{-1}C) = \sup_{\substack{C\subset A\\ C=\overline{C}}} \mu(C) = \mu(A).$$

Applying the same argument to the complement A^c of A it follow that it must be $\mu(T^{-1}A) = \mu(A)$ for each Borel set. □

Proposition 1.4.2 (Krylov–Bogoluvov) *If X is a metric compact space and $T : X \to X$ is continuous, then there exists at least one invariant (Borel) measure.*

PROOF. Consider any Borel probability measure ν and define the following sequence of measures $\{\nu_n\}_{n\in\mathbb{N}}$:[15] for each Borel set A

$$\nu_n(A) = \nu(T^{-n}A).$$

The reader can easily see that $\nu_n \in \mathcal{M}^1(X)$, the sets of the probability measures. Indeed, since $T^{-1}X = X$, $\nu_n(X) = 1$ for each $n \in \mathbb{N}$. Next, define

$$\mu_n = \frac{1}{n}\sum_{i=0}^{n-1} \nu_i.$$

Again $\mu_n(X) = 1$, so the sequence $\{\mu_i\}_{i=1}^{\infty}$ is contained in a weakly compact set (the unit ball) and therefore admits a weakly convergent subsequence $\{\mu_{n_i}\}_{i=1}^{\infty}$; let μ be the weak limit.[16] We claim that μ is T invariant. Since μ is a Borel measure it suffices to verify that for each $f \in \mathcal{C}^{(0)}(X)$ holds $\mu(f \circ T) = \mu(f)$ (see Lemma 1.4.1). Let f be a continuous function, then by the weak convergence we have[17]

[14] Note that, by hypothesis, if C is compact and $C \cap K = \emptyset$, then TC is compact.

[15] Intuitively, if we chose a point $x \in X$ at random, according to the measure ν and we ask what is the probability that $T^n x \in A$, this is exactly $\nu(T^{-n}A)$. Hence, our procedure to produce the point $T^n x$ is equivalent to picking a point at random according to the evolved measure ν_n.

[16] This depends on the Riesz-Markov Representation Theorem [75] that states that $\mathcal{M}(X)$ is exactly the dual of the Banach space $\mathcal{C}^{(0)}(X)$. Since the weak convergence of measures in this case correspond exactly to the weak-* topology [75], the result follows from the Banach-Alaoglu theorem stating that the unit ball of the dual of a Banach space is compact in the weak-* topology. But see Problem 1.17 if you want a more elementary proof.

[17] Note that it is essential that we can check invariance only on continuous functions: if we would have to check it with respect to all bounded measurable functions we would need that μ_n converges in a stronger sense (*strong convergence*) and this may not be true. Note as well that this is the only point where the continuity of T is used: to insure that $f \circ T$ is continuous and hence that $\mu_{n_j}(f \circ T) \to \mu(f \circ T)$.

$$\mu(f\circ T)=\lim_{j\to\infty}\frac{1}{n_j}\sum_{i=0}^{n_j-1}\nu_i(f\circ T)=\lim_{j\to\infty}\frac{1}{n_j}\sum_{i=0}^{n_j-1}\nu(f\circ T^{i+1})$$
$$=\lim_{j\to\infty}\frac{1}{n_j}\left\{\sum_{i=0}^{n_j-1}\nu_i(f)+\nu(f\circ T^{n_j})-\nu(f)\right\}=\mu(f).$$

□

The reason why the above theorem is not completely satisfactory is that it is not constructive and, in particular, does not provide any information on the nature of the invariant measure. On the contrary, in many instances the interest is focused not just on any Borel measure but on special classes of measures, for example measures connected to the Lebesgue measure which, in some sense, can be thought as reasonably physical measures (if such measures exists).

In the following examples we will see two main techniques to study such problems: on the one hand it is possible to try to construct explicitly the measure and study its properties in the given situations (expanding maps, strange attractors, solenoid, horseshoe); on the other hand one can try to *conjugate*[18] the given problem with another, better understood, one (logistic map, circle maps). In view of the second possibility the last example is very important (Markov measures). Such an example gives just a hint to the possibility to construct a multitude of invariant measures for the shift which, as we will see briefly, is a standard system to which many other can be conjugated.

1.4.1 Examples

1.4.1.a *Contracting maps*

Let $X\subset\mathbb{R}^n$ be compact and connected, $T:X\to X$ differentiable with $\|DT\|\leq\lambda^{-1}<1$ and $T0=0\in X$. In this case 0 is the unique fixed point and the delta function at zero is the only invariant measure.[19]

1.4.1.b *Expanding maps*

The simplest possible case is $X=\mathbb{T}$, $T\in\mathcal{C}^{(2)}(\mathbb{T})$ with $|DT|\geq\lambda>1$, (see Figure 1.1 for a pictorial example).[20]

[18] See Definition 1.8.2 for a precise definition and Problem 1.40 and 1.41 for some insight.

[19] The reader will hopefully excuse this physicist language, naturally we mean that the invariant measure is defined by $\delta_0(f)=f(0)$. The property that there exists only one invariant measure is called *unique ergodicity*, we will see more of it in the sequel, e.g. see example 1.5.1.a.

[20] Note that this generalizes Examples 1.1.1.c.

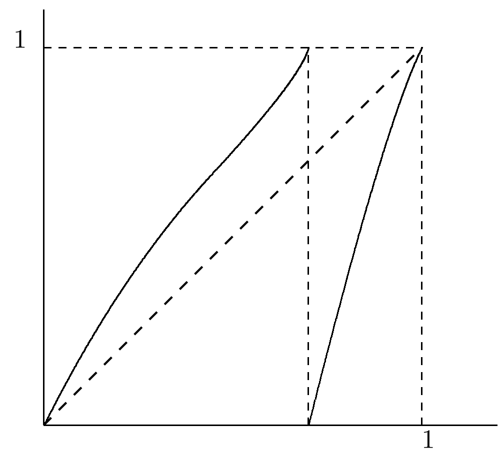


Figure 1.1: Graph of an expanding map on $\mathbb{T}$

We would like to have an invariant measure absolutely continuous with respect to Lebesgue. Any such measure μ has, by definition, the Radon-Nikodym derivative $h = \frac{d\mu}{dm} \in L^1(\mathbb{T}, m)$, [74]. In Proposition 1.4.2 we saw how a measure evolves by defining the operator

$$T_*\mu(f) = \mu(f \circ T) \tag{1.4.3}$$

for each $f \in \mathcal{C}^{(0)}$ and $\mu \in \mathcal{M}(X)$ (see also footnote 16 at page 30). If we want to study a smaller class of measures we must first check that T_* leaves such a class invariant. Indeed, if μ is absolutely continuous with respect to Lebesgue then $T_*\mu$ has the same property. Moreover, if $h = \frac{d\mu}{dm}$ and $h_1 = \frac{dT_*\mu}{dm}$ then (Problem 1.15)

$$h_1(x) = \mathcal{L}h(x) := \sum_{y \in T^{-1}(x)} |D_yT|^{-1}h(y).$$

The operator $\mathcal{L} : L^1(\mathbb{T}, m) \to L^1(\mathbb{T}, m)$ is called *Transfer operator* or Ruelle-Perron-Frobenius operator, and has an extremely important rôle in the study of the statistical properties of the system. Notice that $\|\mathcal{L}h\|_1 \leq \|h\|_1$. The key property of $\mathcal{L}$, in this context, is given by the following inequality (this type of inequality is commonly called of Lasota-York type) (Problem 1.16)

$$\|\frac{d}{dx}\mathcal{L}h\|_1 \leq \lambda^{-1}\|h'\|_1 + C\|h\|_1 \tag{1.4.4}$$

where $C = \frac{\|D^2T\|_\infty}{\|DT\|_\infty^2}$.

The above inequality implies immediately $\|(\mathcal{L}^n h)'\|_1 \leq \frac{C}{1-\lambda^{-1}}\|h\|_1 + \|h'\|_1$, for all $n \in \mathbb{N}$. This, in turn, implies that the $\sup_{n\in\mathbb{N}} \|\mathcal{L}^n h\|_\infty < \infty$. Consequently, the sequence $h_n := \frac{1}{n}\sum_{i=0}^{n-1} \mathcal{L}^i h$ is compact in L^1 (this is a consequence of standard embedding theorems [62] but see Problem 1.17 for an elementary proof). In analogy with Lemma 1.4.2, we have that there exists $h_* \in L^1$ such that $\mathcal{L}h_* = h_*$. Thus $d\mu := h_* dm$ is an invariant measure of the type we are looking for.[21]

1.4.1.c *Logistic maps*

Consider $X = [0, 1]$ and

$$T(x) = 4x(1 - x).$$

This map is not an everywhere expanding map ($D_{\frac{1}{2}}T = 0$), yet it can be conjugate with one, [89].

To see this consider the continuous change of variables $\Psi : [0, 1] \to [0, 1]$ defined by

$$\Psi(x) = \frac{2}{\pi} \arcsin \sqrt{x},$$

thus $\Psi^{-1}(x) = \left(\sin \frac{\pi}{2} x\right)^2$. Accordingly,

$$\begin{aligned}\tilde{T}(x) \quad &:= \Psi \circ T \circ \Psi^{-1}(x) = \Psi(4 \sin^2 \tfrac{\pi}{2} x \cos^2 \tfrac{\pi}{2} x) \\ &= \Psi([\sin \pi x]^2) = \tfrac{2}{\pi} \arcsin[\sin \pi x]\end{aligned}$$

which yields[22]

$$\tilde{T}(x) = \begin{cases} 2x & \text{for } x \in [0, \frac{1}{2}] \\ 2 - 2x & \text{for } x \in [\frac{1}{2}, 1]. \end{cases}$$

The map $\tilde{T}$ is called *tent map* for its characteristic shape, see figure 1.2. What is more interesting is that the Lebesgue measure is invariant for $\tilde{T}$, as the reader can easily check. This means that, if we define $\mu(f) := m(f \circ \Psi^{-1})$, it holds true

$$\mu(f \circ T) = m(f \circ T \circ \Psi^{-1}) = m(f \circ \Psi^{-1} \circ \tilde{T}) = m(f \circ \Psi^{-1}) = \mu(f).$$

Hence, $([0, 1], T, \mu)$ is a Dynamical System. In addition, a trivial computation shows

$$\mu(dx) = \frac{1}{\pi\sqrt{x(1-x)}} dx,$$

thus μ is absolutely continuous with respect to Lebesgue.

[21] In fact, there exists only one such measure, see Examples 4.3.1.c.
[22] Remember that the domain of arcsin is $[-\frac{\pi}{2}, \frac{\pi}{2}]$ and $\sin \pi x = \sin \pi(1 - x)$.

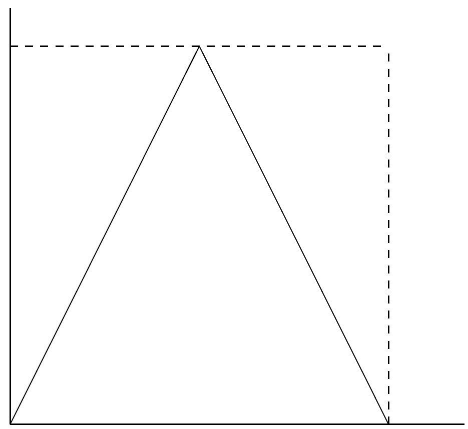


Figure 1.2: Graph of tent map

1.4.1.d *Circle maps*

A circle map is an order preserving continuous map of the circle. A simple way to describe it is to start by considering its lift. Let $\hat{T} : \mathbb{R} \to \mathbb{R}$, such that $\hat{T}(0) \in [0,1]$, $\hat{T}(x+1) = \hat{T}(x) + 1$ ad it is monotone increasing. The circle map is then defined as $T(x) = \hat{T}(x) \mod 1$. Circle maps have a very rich theory that we do not intend to develop here, we confine ourselves to some facts (see [51] for a detailed discussion of the properties below). The first fact is that the *rotation number*

$$\rho(T) = \lim_{n\to\infty} \frac{1}{n} \hat{T}^n(x).$$

is well defined and does not depend on x.

We have already seen a concrete example of circle maps: the rotation R_ω by ω. Clearly $\rho(R_\omega) = \omega$. It is fairly easy to see that if $\rho(T) \in \mathbb{Q}$ then the map has a periodic orbit. We are more interested in the case in which the rotation number is irrational. In this case, with the extra assumption that T is twice differentiable (actually a bit less is needed) the Denjoy theorem holds stating that there exists a continuous invertible function h such that $R_{\rho(T)} \circ h = h \circ T$, that is T is *topologically conjugated* to a rigid rotation. Since we know that the Lebesgue measure is invariant for the rotations, we can obtain an invariant measure for T by pushing the Lebesgue measure by h, namely define

$$\mu(f) = m(f \circ h^{-1}).$$

The natural question if the measure μ is absolutely continuous with respect to Lebesgue is rather subtle and depends, once again, on KAM theory. In essence the answer is positive only if T has more regularity and the rotation number is not very well approximated by rational numbers (in some sense it is 'very irrational') [1].

1.4.1.e *Strange Attractors*

We have seen the case in which all the trajectories are attracted by a point. The reader can probably imagine a case in which the attractor is a curve or some other simple set. Yet, it has been a fairly recent discovery that an attractor may have a very complex (strange) structure. The following is probably the simplest example. Let $X = Q = [0,1]^2$ and

$$T(x,\,y) = \begin{cases} (2x,\,\frac{1}{8}y + \frac{1}{4}) & \text{if } x \in [0,\,1/2] \\ (2x-1,\,\frac{1}{8}y + \frac{3}{4}) & \text{if } x \in]1/2,\,1]. \end{cases}$$

We have a map of the square that stretches in one direction by a factor 2 and contract in the other by a factor 8.

Note that T it is not continuous with respect to the normal topology, so Proposition 1.4.2 cannot be applied directly. This problem can be solved in at least two ways: one is to *code* the system and we will discuss it later (see Examples 1.8.1), the other is to study more precisely what happens iterating a measure in special cases.

In our situation, since T^nQ consists of a multitude of thinner and thinner strips, it is clear that there can be no invariant measure absolutely continuous with respect to Lebesgue.[23] Yet, it is very natural to ask what happens if we iterate the Lebesgue measure by the operator T_*. It is easy to see that T_*m is still absolutely continuous with respect to Lebesgue. In fact, T_* maps absolutely continuous measures into absolutely continuous measures. Once we note this, it is very tempting to define the transfer operator. An easy computation yields

$$\mathcal{L}h(x) = \chi_{TQ}(x) \sum_{y \in T^{-1}(x)} |\det(D_yT)|^{-1} h(y) = 4\chi_{TQ}(x)h(T^{-1}(x)).$$

Since the map expands in the unstable direction, it is quite natural to investigate, in analogy with the expanding case, the *unstable derivative* D^u, that is the derivative in the x direction, of the iterate of the density.

$$\|D^u\mathcal{L}h\|_1 \leq \frac{1}{2}\|D^uh\|_1 \quad \forall h \in \mathcal{C}^{(1)}(Q) \tag{1.4.5}$$

[23] In fact, if μ is an invariant measure, $T_*\mu = \mu$, it follows

$$\mu(\chi_{T^nQ}) = T_*^n\mu(\chi_{T^nQ}) = \mu(\chi_Q) = 1,$$

so μ must be supported on $\Lambda = \cap_{n=0}^{\infty} T^nQ$.

To see the consequences of the above estimate, consider $f \in \mathcal{C}^{(1)}(Q)$ with $f(0,y) = f(1,y) = 0$ for each $y \in [0,1]$, then if μ is a measure obtained by the measure hdm ($h \in \mathcal{C}^{(1)}$) with the procedure of Proposition 1.4.2,[24] we have

$$\begin{aligned}\mu(D^u f) &= \lim_{j\to\infty} \frac{1}{n_j} \sum_{i=0}^{n_j-1} (T_*)^i m(hD^u f) = \lim_{j\to\infty} \frac{1}{n_j} \sum_{i=0}^{n_j-1} m(\mathcal{L}^i h D^u f) \\ &= -\lim_{j\to\infty} \frac{1}{n_j} \sum_{i=0}^{n_j-1} m(f D^u \mathcal{L}^i h)\end{aligned}$$

where we have integrated by part. Remembering (1.4.5) we have

$$\mu(D^u f) = 0,$$

for all $f \in \mathcal{C}^{(1)}_{\text{per}}(Q) = \{f \in \mathcal{C}^{(1)}(Q) \mid f(0,y) = f(1,y)\}$. The enlargement of the class of functions is due to the obvious fact that, if $f \in \mathcal{C}^{(1)}_{\text{per}}(Q)$, then $\tilde{f}(x,y) = f(x,y) - f(0,y)$ is zero on the vertical (stable) boundary and $D^u\tilde{f} = D^u f$.

This means that the measure μ, when restricted to the horizontal direction, is μ-a.e. constant (see Problem 1.32). Such a strong result is clearly a consequence of the fact that the map is essentially linear, one can easily imagine a non linear case (think of dilations and expanding maps) and in that case the same argument would lead to conclude that the measure, when restricted to unstable manifolds, is absolutely continuous with respect to the restriction of Lebesgue (these type of measures are commonly called SRB from Sinai, Ruelle and Bowen).

We can now prove that indeed the measure μ is invariant. The discontinuity line of T is $\{x = \frac{1}{2}\}$. Points close to $\{x = \frac{1}{2}\}$ are mapped close to the boundary of Q, so if $f(0,y) = f(1,y) = 0$, then $f \circ T$ is continuous. Hence, the argument of Proposition 1.4.2 proves that $\mu(f \circ T) = \mu(f)$ for all f that vanish at the stable boundary. Yet, the characterization of μ proves that $\mu(\{(x,y) \in Q \mid x \in \{0,1\}\}) = 0$, thus we can obtain $\mu(f \circ T) = \mu(f)$ for all continuous functions via the Lebesgue dominated convergence theorem and the invariance follows by Lemma 1.4.1.

1.4.1.f *Horseshoe*

This very famous example consists of a map of the square $Q = [0,1]^2$, the map is obtained by stretching the square in the horizontal direction, bending it in the shape of an horseshoe and then superimposing it to the original square in such a

[24] As we noted in the proof of Proposition 1.4.2, the only part that uses the continuity of T is the proof of the invariance. Thus, in general we can construct a measure by the averaging procedure but its invariance is not automatic.

way that the intersection consists of two horizontal strips.[25] Such a description is just topological, to make things clearer let us consider a very special case:

$$T(x,y) = \begin{cases} (5x \mod 1, \frac{1}{4}y) & \text{if } x \in [1/5, 2/5] \\ (5x \mod 1, \frac{1}{4}y + \frac{3}{4}) & \text{if } x \in [3/5, 4/5]. \end{cases}$$

Note that T is not explicitly defined for $x \in [0, 1/5[\cup[\frac{2}{3}, \frac{3}{5}[\cup]4/5, 1]$ since for this values the horseshoe falls outside Q, so its actual shape is irrelevant. Since the map from Q to Q is not defined on the full square, we can have a Dynamical System only with respect to a measure for which the domain of definition of T, and all of its powers, has measure one. We will start by constructing such a measure.

The first step is to notice that the set

$$\Lambda = \cap_{n\in\mathbb{Z}} T^n Q \tag{1.4.6}$$

of the points which trajectories are always in Q is $\neq \emptyset$. Second, note that $\Lambda = T\Lambda = T^{-1}\Lambda$, such an invariant set is called *hyperbolic set* as we will see in ???. We would like to construct an invariant measure on Λ. Since Λ is a compact set and T is continuous on it we know that there exist invariant measures; yet, in analogy with the previous examples, we would like to construct one *coming from Lebesgue*.

As already mentioned we must start by constructing a measure on $\Lambda_- = \cap_{n\in\mathbb{N}\cup\{0\}} T^{-n}Q$ since $T^k\Lambda_- \subset \Lambda_-$. To do so it is quite natural to construct a measure by *subtracting* the mass that leaks out of Q. namely, define the operator $\tilde{T} : \mathcal{M}(X) \to \mathcal{M}(X)$ by

$$\tilde{T}\mu(A) := \mu(TA \cap Q).$$

Again we consider the evolution of measures of the type $d\mu = hdm$. For each continuous f with supp$(f) \subset Q$ holds

$$\tilde{T}\mu(f) = \mu(f \circ T^{-1}\chi_Q) = \int_{T^{-1}Q} fh \circ T|\det DT|dm.$$

We can thus define the operator $\mathcal{L}$ that evolves the densities:

$$\mathcal{L}h(x) = \frac{5}{4}\chi_{T^{-1}Q\cap Q}(x)h(Tx).$$

Clearly $\tilde{T}\mu(f) = m(f\mathcal{L}h)$.

Note that $\tilde{T}m(1) = \frac{1}{2}$, thus $\tilde{T}$ does not map probability measures into probability measures; this is clearly due to the mass leaking out of Q. Calling D^s

[25] We have already seen something very similar in the introduction.

(stable derivative) the derivative in the y direction, follows easily

$$\|D^s\mathcal{L}h\|_1 \le \frac{1}{4}\|D^s h\|_1$$

for each h differentiable in the stable direction.

On the other hand, if $\|D^s h\|_1 \le c$ and $\Delta = [0,1/4] \cup [3/4,1]$,

$$\begin{aligned}|\tilde{T}\mu(1)| &= \int_{Q\cap TQ} h = \int_\Delta dy \int_0^1 dx h(x,y)\\ &= \int_\Delta dy \int_0^1 dx \int_0^1 d\xi h(x,\xi) + \mathcal{O}(\|D^s h\|_1)\\ &= |\Delta|\|h\|_1 + \mathcal{O}(\|D^s h\|_1) = \frac{1}{2}\mu(1) + \mathcal{O}(\|D^s h\|_1).\end{aligned}$$

It is then natural to define $\hat{\mathcal{L}}h := 2\mathcal{L}h$ and $\hat{T} = 2\tilde{T}$. Thus $\|D^s\hat{\mathcal{L}}h\|_1 \le \frac{1}{2}\|D^s h\|_1$. This means that $\{\frac{1}{n}\sum_{i=0}^{n-1}\hat{T}^i\mu\}$ are probability measures. Accordingly, there exists an accumulation point μ_* and $\mu_*(D^s f) = 0$ for each f periodic in the y direction. By the same type of arguments used in the previous examples, this means that μ_* is constant in the y direction, it is supported on Λ_- by construction and $\tilde{T}\mu_* = \frac{1}{2}\mu_*$ (*conformal invariance*) : just the measure we where looking for.

We can now conclude the argument by evolving the measure as usual:

$$T_*\mu_*(f) = \mu_*(f \circ T)$$

for all continuous f with the support in Q. Now the standard argument applies. In such a way we have obtained the invariant measure supported on Λ.

1.4.1.g *Markov Measures*

Let us consider the shift (Σ_n^+, T). We would like to construct other invariant measures bedside Bernoulli. As we have seen it suffices to specify the measure on the algebra of the cylinders. Let us define

$$A(m; k_1, \dots, k_l) = \{\sigma \in \Sigma_n^+ \mid \sigma_{i+m} = k_i \; \forall\, i \in \{1,\dots,l\}\};$$

this are a basis for the algebra of the cylinders.

For each $n \times n$ matrix P, $P_{ij} \ge 0$, $\sum_j P_{ij} = 1$ by the Perron-Frobenius theorem (see Example (4.3.1.a)) there exists $\{p_i\}$ such that $pP = p$. Let us define

$$\mu(A(m; k_1, \dots, k_l)) = p_{k_1}P_{k_1k_2}P_{k_2k_3}\dots P_{k_{l-1}k_l}.$$

The reader can easily verify that μ is invariant over the algebra $\mathcal{A}$ and thus extends to an invariant measure. This is called Markov because it is nothing else than a Markov chain together with its stationary measure [1].[26]

These last examples (strange attractor, solenoid, horseshoe) show only a very dim glimpse of a much more general and extremely rich theory (the study of SRB measures) while the last (Markov measures) points toward another extremely rich theory: Gibbs (or equilibrium) measures. Although this it is not the focus here, we will see a bit more of this in the future.

One of the main objectives in dynamical systems is the study of the long time behavior (that is the study of the trajectories $T^n x$ for large n). There are two main cases in which it is possible to study, in some detail, such a long time behavior. The case in which the motion is rather regular[27] or close to it (the main examples of this possibility are given by the so called KAM [6] theory and by situations in which the motions is attracted by a simple set); and the case in which the motion is very irregular.[28] This last case may seem surprising since the irregularity of the motion should make its study very difficult. The reason why such systems can be studied is, as usual, because we ask the right questions,[29] that is we ask questions not concerning the fine details of the motion but only concerning its statistical or qualitative properties.

The first example of such properties is the study of the invariant sets.

1.5 Ergodicity

Definition 1.5.1 *A measurable set A is invariant for T if $T^{-1}A \subset A$.*

A dynamical system (X, T, μ) is ergodic if each invariant set has measure zero or one.

The definition for continuous dynamical systems being exactly the same.

Note that if A is invariant then $\mu(A \backslash T^{-1}A) = \mu(A) - \mu(T^{-1}A) = 0$, moreover $\Lambda = \cap_{n=0}^{\infty} T^{-n}A \subset A$ is invariant as well. In addition, by definition, $\Lambda = T\Lambda$, which implies $\Lambda = T^{-1}\Lambda$ and $\mu(A\backslash\Lambda) = 0$. This means that, if A is invariant, then it always contains a set Λ invariant in the stronger (maybe more natural) sense that $T\Lambda = T^{-1}\Lambda = \Lambda$. Moreover, Λ is of full measure in A. Our definition of invariance is motivated by its greater flexibility and the

[26] The probabilistic interpretation is that the probability of seeing the state k at time one, given that we saw the state l at time zero, is given by P_{lk}. So the process has a bit of memory: it remembers its state one time step before. Of course it is possible to consider processes that have a longer–possibly infinite–memory. Proceeding in this direction one would define the so called *Gibbs measures*.

[27] Typically, quasi periodic motion, remember the small oscillation in the pendulum.

[28] Remember the example in the introduction.

[29] Of course, the “right questions” are the ones that can be answered.

fact that, from a measure theoretical point of view, zero measure sets can be discarded.

In essence, if a system is ergodic then most trajectories explore all the available space. In fact, for any A of positive measure, define $A_b = \cup_{n\in\mathbb{N}\cup\{0\}} T^{-n}A$ (this are the points that eventually end up in A), since $A_b \supset A$, $\mu(A_b) > 0$. Since $T^{-1}A_b \subset A_b$, by ergodicity follows $\mu(A_b) = 1$. Thus, the points that never enter in A (that is, the points in A_b^c) have zero measure. Actually, if the system has more structure (topology) more is true (see Problem 1.21).

The reader should be aware that there are many equivalent definitions of ergodicity, see Problems 1.25, 1.27, 1.28 and Theorem 1.6.6 for some possibilities.

1.5.1 Examples

1.5.1.a *Rotations*

The ergodicity of a rotation depends on ω. If $\omega \in \mathcal{Q}$ then the system is not ergodic. In fact, let $\omega = \frac{p}{q}$ $(p, q \in \mathbb{N})$, then, for each $x \in \mathbb{T}$ $T^q x = x + p \mod 1 = x$, so T^q is just the identity. An alternative way of saying this is to notice that all the points have a periodic trajectory of period q. It is then easy to exhibit an invariant set with measure strictly larger than 0 but strictly less than 1. Consider $[0, \varepsilon]$, then $A = \cup_{i=1}^{q-1} T^{-i}[0, \varepsilon]$ is an invariant set; clearly $\varepsilon \leq \mu(A) \leq q\varepsilon$, so it suffices to choose $\varepsilon < q^{-1}$.

The case $\omega \notin \mathcal{Q}$ is much more interesting. First of all, for each point $x \in \mathbb{T}$ we have that the closure of the set $\{T^n x\}_{i=0}^{\infty}$ is equal to $\mathbb{T}$, which is to say that the orbits are dense.[30] The proof is based on the fact that there cannot be any periodic orbit. To see this suppose that $x \in \mathbb{T}$ has a periodic orbit, that is there exists $q \in \mathbb{N}$ such that $T^q x = x$. As a consequence there must exist $p \in \mathbb{Z}$ such that $x + p = x + q\omega$ or $\omega \in \mathcal{Q}$ contrary to the hypothesis. Hence, the set $\{T^k 0\}_{k=0}^{\infty}$ must contain infinitely many points and, by compactness, must contain a convergent subsequence k_i. Hence, for each $\varepsilon > 0$, there exists $m > n \in \mathbb{N}$:

$$|T^m 0 - T^n 0| < \varepsilon.$$

Since T preserves the distances, calling $q = m - n$, holds

$$|T^q 0| < \varepsilon.$$

Accordingly, the trajectory of $T^{jq}0$ is a translation by a quantity less than ε, therefore it will get closer than ε to each point in $\mathbb{T}$ (i.e., the orbit is dense). Again by the conservation of the distance, since zero has a dense orbit the same will hold for every other point.

[30] A system with a dense orbit called *Topologically Transitive*.

Intuitively, the fact that the orbits are dense implies that there cannot be a non trivial invariant set, henceforth the system is ergodic. Yet, the proof it is not trivial since it is based on the existence of Lebesgue density points [74] (see Problem 1.43). It is a fact from general measure theory that each measurable set $A \subset \mathbb{R}$ of positive Lebesgue measure contains, at least, one point $\bar{x}$ such that for each $\varepsilon \in (0, 1)$ there exists $\delta > 0$:

$$\frac{m(A \cap [\bar{x} - \delta, \bar{x} + \delta])}{2\delta} > 1 - \varepsilon.$$

Hence, given an invariant set A of positive measure and $\varepsilon > 0$, first choose δ such that the interval $I := [\bar{x} - \delta, \bar{x} + \delta]$ has the property $m(I \cap A) > (1 - \varepsilon)m(I)$. Second, we know already that there exists $q, M \in \mathbb{N}$ such that $\{T^{-kq}x\}_{k=1}^{M}$ divides $[0, 1]$ into intervals of length less that $\frac{\varepsilon}{2}\delta$. Hence, given any point $x \in \mathbb{T}$ choose $k \in \mathbb{N}$ such that $m(T^{-kq}I \cap [x - \delta, x + \delta]) > m(I)(1 - \varepsilon)$ so,

$$\begin{aligned} m(A \cap [x - \delta, x + \delta]) &\geq m(A \cap T^{-kq}I) - m(I)\varepsilon \\ &\geq m(A \cap I) - m(I)\varepsilon \geq (1 - 2\varepsilon)2\delta. \end{aligned}$$

Thus, A has density everywhere larger than $1 - 2\varepsilon$, which implies $\mu(A) = 1$ since ε is arbitrary.

The above proof of ergodicity it is not so trivial but it has a definite dynamical flavor (in the sense that it is obtained by studying the evolution of the system). Its structure allows generalizations to contexts whit a less rich algebraic structure. Nevertheless, we must notice that, by taking advantage of the algebraic structure (or rather the group structure) of $\mathbb{T}$, a much simpler and powerful proof is available.

Let $\nu \in \mathcal{M}_T^1$, then define

$$F_n = \int_{\mathbb{T}} e^{2\pi i n x} \nu(dx), \quad n \in \mathbb{N}.$$

A simple computation, using the invariance of ν, yields

$$F_n = e^{2\pi i n \omega} F_n$$

and, if ω is irrational, this implies $F_n = 0$ for all $n \neq 0$, while $F_0 = 1$. Next, consider $f \in \mathcal{C}^{(2)}(\mathbb{T}^1)$ (so that we are sure that the Fourier series converges uniformly, see Problem 1.31), then

$$\nu(f) = \sum_{n=0}^{\infty} \nu(f_n e^{2\pi i n \cdot}) = \sum_{n=0}^{\infty} f_n F_n = f_0 = \int_{\mathbb{T}} f(x)dx.$$

Hence m is the unique invariant measure (unique ergodicity). This is clearly much stronger than ergodicity (see Problem 1.25)

1.5.1.b *Baker*

This transformation gets its name from the activity of bread making, it bears some resemblance with the horseshoe. The space X is the square $[0, 1]^2$, μ is again Lebesgue, and T is a transformation obtained by squashing down the square into the rectangle $[0, 2] \times [0, \frac{1}{2}]$ and then cutting the piece $[1, 2] \times [0, \frac{1}{2}]$ and putting it on top of the other one. In formulas

$$T(x, y) = \begin{cases} (2x, \frac{1}{2}y) \mod 1 & \text{if } x \in [0, \frac{1}{2}) \\ (2x, \frac{1}{2}(y+1)) \mod 1 & \text{if } x \in [\frac{1}{2}, 1]. \end{cases}$$

This transformation is ergodic as well, in fact much more. We will discuss it later.

1.5.1.c *Translations ($\mathbb{T}^1$)*

Let us consider the flow $(\mathbb{T}^1, \phi_t, m)$ where $\phi_t(x) = x + \omega t \mod 1$, for some $\omega \in \mathbb{R} \setminus \{0\}$. This is just a translation on the unit circle. The proof of ergodicity is trivial and it is left to the reader.

We conclude the chapter with a theorem very helpful to establish the ergodicity of a flow.

Theorem 1.5.2 *Consider a flow (X, ϕ_t, μ) and a Poincarè section Σ such that the set $\{x \in X \mid \cup_{t\in\mathbb{R}} \phi_t(x) \cap \Sigma = \emptyset\}$ has zero measure. Then the ergodicity of the flow (X, ϕ_t, μ) is equivalent to the ergodicity of the section $(\Sigma, T_\Sigma, \mu_\Sigma)$.*

The proof, being straightforward, is left to the reader.

1.5.2 Examples

1.5.2.a *Translations ($\mathbb{T}^2$)*

Let us consider the flow $(\mathbb{T}^2, \phi_t, m)$ where $\phi_t(x) = x + \omega t \mod 1$, for some $\omega \in \mathbb{R}^2 \setminus \{0\}$. This is a translation on the two dimensional torus. To investigate we will use Theorem 1.5.2. Consider the set $\Sigma := \{(x, y) \in \mathbb{T}^2 \mid x = 0\}$, this is clearly a Poincaré section, unless $\omega_1 = 0$ (in which case one can choose the section $y = 0$). Obviously Σ is a circle and the Poincaré map is given by

$$T(y) = y + \frac{\omega_2}{\omega_1} \mod 1.$$

The ergodicity of the flow is then reduced to the ergodicity of a circle rotation, thus the flow is ergodic only if ω_1 and ω_2 have an irrational ratio.

The properties of the invariant sets of a dynamical systems have very important reflections on the statistics of the system, in particular on its time averages. Before making this precise (see Theorem 1.6.6) we state few very general and far reaching results.

1.6 Some basic Theorems

Theorem 1.6.1 *(Birkhoff) Let (X, T, μ) be a dynamical system, then for each $f \in L^1(X, \mu)$*

$$\lim_{n\to\infty} \frac{1}{n} \sum_{j=0}^{n-1} f(T^j x)$$

exists for almost every point $x \in X$. In addition, setting

$$f^+(x) = \lim_{n\to\infty} \frac{1}{n} \sum_{j=0}^{n-1} f(T^j x),$$

holds

$$\int_X f^+ d\mu = \int_X f d\mu.$$

Proof

Since the task at hand is mainly didactic, we will consider explicitly only the case of positive bounded functions, the completion of the proof is left to the reader.

Let $f \in L^\infty(X, d\mu)$, $f \geq 0$, and

$$S_n(x) \equiv \frac{1}{n} \sum_{i=0}^{n-1} f(T^i x).$$

For each $x \in X$, there exists

$$\begin{aligned} \overline{f}^+(x) &= \limsup_{n\to\infty} S_n(x) \\ \underline{f}^+(x) &= \liminf_{n\to\infty} S_n(x). \end{aligned}$$

The first remark is that both $\overline{f}^+$ and $\underline{f}^+$ are invariant functions. In fact,

$$S_n(Tx) = S_n(x) + \frac{1}{n} f(T^n x) - \frac{1}{n} f(x)$$

so, tacking the limit the result follows.[31]

[31] Here we have used the boundedness, this is not necessary. If $f \in L^1(X, d\mu)$ and positive, then $S_n(Tx) \geq S_n(x) - f(x)$, so $\underline{f}^+(Tx) \geq \underline{f}^+(x)$ and it is and easy exercise to check that any such function must be invariant.

Next, for each $n \in \mathbb{N}$ and $k,\, j \in \mathbb{Z}$ we define

$$D_{n,l,j} = \left\{ x \in X \;\middle|\; \overline{f}^{+}(x) \in \left[\frac{l}{n},\, \frac{l+1}{n}\right) ;\; \underline{f}^{+}(x) \in \left[\frac{j}{n},\, \frac{j+1}{n}\right) \right\},$$

by the invariance of the functions follows the invariance of the sets $D_{n,l,j}$. Also, by the boundedness, follows that for each n exists n_0 such as

$$\bigcup_{j,l \in \{-n_0,\, \ldots,\, n_0\}} D_{n,l,j} = X.$$

The key observation is the following.

Lemma 1.6.2 *For each $n \in \mathbb{N}$ and $l,\, j \in \mathbb{Z}$, setting $A = D_{n,l,j}$, holds*

$$\frac{l+1}{n}\mu(A) < \int_A f d\mu + \frac{3}{n}\mu(A)$$
$$\frac{j}{n}\mu(A) > \int_A f d\mu - \frac{3}{n}\mu(A)$$

From the Lemma follows

$$\begin{aligned} 0 \leq \int_X (\overline{f}^{+} - \underline{f}^{+}) d\mu &= \sum_{l,\, j=-n_0}^{n_0} \int_{D_{n,l,j}} (\overline{f}^{+} - \underline{f}^{+}) d\mu \\ &\leq \sum_{l,\, j=-n_0}^{n_0} \left[\frac{l+1}{n} - \frac{j}{n}\right] \mu(D_{n,l,j}) < \frac{6}{n} \sum_{l,\, j=-n_0}^{n_0} \mu(D_{n,l,j}) = \frac{6}{n}. \end{aligned}$$

Since n is arbitrary we have

$$\int_X (\overline{f}^{+} - \underline{f}^{+}) d\mu = 0$$

which implies $\overline{f}^{+} = \underline{f}^{+}$ almost everywhere (since $\overline{f}^{+} \geq \underline{f}^{+}$ by definition) proving that the limit exists. Analogously, we can prove

$$\int_X (f - f^{+}) d\mu = 0.$$

Proof of the Lemma 1.6.2 We will prove only the first inequality, the second being proven in exactly the same way.

For each $x \in A$ we will call $k(x)$ the first $m \in \mathbb{N}$ such that

$$S_m(x) > \frac{l-1}{n},$$

by construction $k(x)$ must be finite for each $x \in A$. Hence, setting $X_k = \{x \in A \mid k(x) = k\}$, $\cup_k X_k = A$, and for each $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$\mu\left(\bigcup_{k=1}^{N} X_k\right) \geq \mu(A)(1-\varepsilon).$$

Let us call

$$Y = A \setminus \bigcup_{k=1}^{N} X_k.$$

Then $\mu(Y) \leq \mu(A)\varepsilon$, also set $L = \sup_{x\in A} |f(x)|$. The basic idea is to follow, for each point $x \in A$, the trajectory $\{T^i x\}_{i=0}^{M}$, where $M > N$ will be chosen sufficiently large. If the point would never visit the set Y, we could group the sum $S_M(x)$ in pieces all, in average, larger than $\frac{l-1}{n}$, so the same would hold for $S_M(x)$. The difficulties come from the visits to the set Y.

For each $n \in \{0, ..., M\}$ define

$$\widetilde{f}_n(x) = \begin{cases} f(T^n x) & \text{if } T^n x \notin Y \\ \dfrac{l}{n} & \text{if } T^n x \in Y \end{cases}$$

and

$$\widetilde{S}_M(x) = \frac{1}{M} \sum_{n=0}^{M-1} \widetilde{f}_n(x).$$

By definition $y \in Y$ implies $y \notin X_1$, i.e. $f(y) \leq \frac{l-1}{n}$. Accordingly, $\widetilde{f}(x) \geq f(T^n x)$ for each $x \in A$. Note that for each n we change the function $f \circ T^n$ only at some points belonging to the set Y and $\frac{l}{n}$ can be taken less or equal than L (otherwise $\mu(A) = 0$), consequently

$$\int_A f d\mu = \int_A S_M d\mu \geq \int_A \widetilde{S}_M d\mu - L\mu(Y) \geq \int_A \widetilde{S}_M d\mu - L\mu(A)\varepsilon.$$

We are left with the problem of computing the sum. As already mentioned the strategy consists in dividing the points according to their trajectory with respect to the sets X_n. To be more precise, let $x \in A$, then by definition it must belong to some X_n or to Y. We set $k_1(x)$ equal to j is $x \in X_j$ and $k_1(x) = 1$ if $x \in Y$. Next, $k_2(x)$ will have value j if $T^{k_1(x)} x \in X_j$ or value 1 if $T^{k_1(x)} \in Y$. If $k_1(x) + k_2(x) < M$, then we go on and define similarly $k_3(x)$. In this way, to each $x \in A$ we can associate a number $m(x) \in \{1, ..., M\}$ and indices $\{k_i(x)\}_{i=1}^{m(x)}$, $k_i(x) \in \{1, ..., N\}$, such that $M - N \leq \sum_{i=1}^{m(x)-1} k_i(x) < M$, $\sum_{i=1}^{m(x)} k_i(x) \geq M$. Let us call $K_p(x) = \sum_{j=1}^{p} k_j(x)$. Using such a division of the orbit in segments of length $k_i(x)$ we can easily estimate

$$\begin{aligned}\widetilde{S}_M(x) =&\frac{1}{M}\left\{\sum_{i=1}^{m(x)-1} k_i(x)\left[\frac{1}{k_i(x)}\sum_{j=K_{i-1}(x)}^{K_i(x)-1}\widetilde{f}_j(x)\right]+\sum_{i=K_{m(x)-1}(x)}^{M-1}\widetilde{f}(T^i x)\right\}\\ \geq&\frac{1}{M}\sum_{i=1}^{m(x)-1} k_i(x)\frac{l-1}{n}\geq\frac{M-N}{M}\frac{l-1}{n}.\end{aligned}$$

Putting together the above inequalities we get

$$\begin{aligned}\int_A f d\mu &\geq \left\{\frac{(M-N)(l-1)}{Mn}-L\varepsilon\right\}\mu(A)\\ &\geq \frac{l+1}{n}\mu(A)-\left\{\frac{2}{n}+\frac{N(l-1)}{Mn}+L\varepsilon\right\}\mu(A).\end{aligned}$$

which, by choosing first ε sufficiently small and, after, M sufficiently large, concludes the proof. □

To prove the result for all function in $L^1(X, \mu)$ it is convenient to deal at first only with positive functions (which suffice since any function is the difference of two positive functions) and then use the usual trick to cut off a function (that is, given f define f_L by $f_L(x) = f(x)$ if $f(x) \leq L$, and $f_L(x) = L$ otherwise) and then remove the cut off. The reader can try it as an exercise. □

Birkhoff theorem has some interesting consequences.

Corollary 1.6.3 *For each $f \in L^1(X, \mu)$ the following holds*

1. *$f_+ \in L^1(X, \mu)$;*
2. *$f_+(Tx) = f_+(x)$ almost surely.*

The proof is left to the reader as an easy exercise (see Problem 1.18).

Another interesting fact, that starts to show some connections between averages and invariant sets, emerges by considering a measurable set A and its characteristic function χ_A. A little thought shows that the ergodic average $\chi_A^+(x)$ is simply the average frequency of visit of the set A by the trajectory $\{T^n x\}$ (Problem 1.28).

Birkhoff theorem implies also convergence in L^1 and L^2 (see also Problem 1.26). Yet, it is interesting to note that convergence in L^2 can be proven in a much more direct way.

Theorem 1.6.4 (Von Neumann) *Let (X, T, μ) be a Dynamical System, then for each $f \in L^2(X, \mu)$ the ergodic average converges in $L^2(X, \mu)$.*

PROOF. We have already seen that it can be useful to lift the dynamics at the level of the algebra of function or at the level of measures. This game assumes different guises according to how one plays it, here is another very interesting version.

Let us define $U : L^2(X, \mu) \to L^2(X, \mu)$ as

$$Uf := f \circ T.$$

Then, by the invariance of the measure, it follows $\|Uf\|_2 = \|f\|_2$, so U is an L^2 contraction (actually, and L^2-isometry). If T is invertible, the same argument applied to the inverse shows that U is indeed unitary, otherwise we must content ourselves with

$$\|U^*f\|_2^2 = \langle UU^*f, f\rangle \leq \|UU^*f\|_2\|f\|_2 = \|U^*f\|_2\|f\|_2,$$

that is $\|U^*\|_2 \leq 1$ (also U^* is and L^2 contraction).

Next, consider $V_1 = \{f \in L^2 \mid Uf = f\}$ and $V_2 = \text{Rank}(\mathbb{1} - U)$. First of all, note that if $f \in V_1$, then

$$\|U^*f - f\|_2^2 = \|U^*f\|_2^2 - \langle f, U^*f\rangle - \langle U^*f, f\rangle + \|f\|_2^2 \leq 0.$$

Thus, $f \in V_1^* := \{f \in L^2 \mid U^*f = f\}$. The same argument applied to $f \in V_1^*$ shows that $V_1 = V_1^*$. To continue, consider $f \in V_1$ and $h \in L^2$, then

$$\langle f, h - Uh\rangle = \langle f - U^*f, h\rangle = 0.$$

This implies that $V_1^\perp = \overline{V_2}$, hence $V_1 \oplus \overline{V_2} = L^2$. Finally, if $g \in V_2$, then there exists $h \in L^2$ such that $g = h - Uh$ and

$$\lim_{n\to\infty} \frac{1}{n}\sum_{i=0}^{\infty} U^i g = \lim_{n\to\infty} \frac{1}{n}(h - U^n h) = 0.$$

On the other hand if $f \in V_1$ then $\lim_{n\to\infty} \frac{1}{n}\sum_{i=0}^{\infty} U^i f = f$. The only function on which we do not still have control are the g belonging to the closure of V_2 but not in V_2. In such a case there exists $\{g_k\} \subset V_2$ with $\lim_{k\to\infty} g_k = g$. Thus,

$$\|\frac{1}{n}\sum_{i=0}^{\infty} U^i g\|_2 \leq \|\frac{1}{n}\sum_{i=0}^{\infty} U^i g_k\|_2 + \|g - g_k\|_2 \leq \|\frac{1}{n}\sum_{i=0}^{\infty} U^i g_k\|_2 + \frac{\varepsilon}{2},$$

provided we choose k large enough. Then, by choosing n sufficiently large we obtain

$$\|\frac{1}{n}\sum_{i=0}^{\infty} U^i g\|_2 \leq \varepsilon.$$

We have just proven that

$$\lim_{n\to\infty} \frac{1}{n} \sum_{i=0}^{n-1} U^i = P$$

where P is the orthogonal projection on V_1. □

Another very general result, of a somewhat disturbing nature, is Poincaré return theorem.

Theorem 1.6.5 (Poincaré) *Given a dynamical systems (X, T, μ) and a measurable set A, with $\mu(A) > 0$, there exists infinitely many $n \in \mathbb{N}$ such that*

$$\mu(T^{-n}A \cap A) \neq 0.$$

The proof is rather simple (by contradiction) and the reader can certainly find it out by herself (see Problem 1.19).[32]

Let us go back to the relation between ergodicity and averages. From an intuitive point of view a function from X to $\mathbb{R}$ can be thought as an "observable," since to each configuration it associates a value that can represent some relevant property of the configuration (the property that we observe). So, if we observe the system for a long time via the function f, what we see should be well represented by the function f^+. Furthermore, notice that there is a simple relations between invariant functions and invariant sets. More precisely, if a measurable set A is invariant, then its characteristic function χ_A is a measurable invariant function; if f is an invariant function then for each measurable set $I \in \mathbb{R}$ the set $f^{-1}(I)$ is a measurable invariant set (if the implications of the above discussions are not clear to you, see Problem 1.27).

As a byproduct of the previous discussion it follows that if a system is ergodic then for each function $f \in L^1(X, \mu)$ the function f_+ is almost everywhere constant and equal to $\int_X f$. We have just proven an interesting characterization of the ergodic systems:

Theorem 1.6.6 *A Dynamical System (X, T, μ) is ergodic if and only if for each $f \in L^1(X, \mu)$ the ergodic average f^+ is constant; in fact, $f^+ = \mu(f)$ a.e..*

[32] An unsettling aspect of the theorem is due to the following possibility. Consider a room full of air, the motion of the molecules can be thought to happen accordingly to Newton equations, i.e. it is an Hamiltonian systems, hence a dynamical system to which Poincaré theorem applies. Let A be the set of configurations in which all the air is in the left side of the room. Since we ignore, in general, the past history of the room, it could very well be that at some point in the past the systems was in a configuration belonging to A–maybe some silly experiment was performed. So there is a positive probability for the system to return in the same state. Therefore the disturbing possibility of sudden death by decompression.

In other words, if we observe the time average of some observable for a sufficiently long time then we obtain a value close to its space average. The previous observation is very important especially because the space average of a function does not depend on the dynamics. This is exactly what we where mentioning previously: the fact that the dynamics is sufficiently ‘complex’ allows us to ignore it completely, provided we are interested only in knowing some average behavior. The relevance of ergodic theory for physical systems is largely connected to this fact.

1.7 Mixing

We have argued the importance of ergodicity, yet from a physical point of view ergodicity may be relevant only if it takes places at a sufficiently fast rate (i.e., if the time average converges to the space average on a physically meaningful time scale). This has prompted the study of stronger statistical properties of which we will give a brief, and by no mean complete, account in the following.

Definition 1.7.1 *A Dynamical System (X, T, μ) is called* mixing *if for every pairs of measurable sets A, B we have*

$$\lim_{n\to\infty} \mu(T^{-n}(A) \cap B) = \mu(A)\mu(B).$$

Obviously, if a system is mixing, then it is ergodic. In fact, if A is an invariant set for T, then $T^{-n}A \subset A$, so, calling A^c the complement of A, we have

$$\mu(A)\mu(A^c) = \lim_{n\to\infty} \mu(T^{-n}A \cap A^c) = 0,$$

and the measure of A is either one or zero.

An equivalent characterization of mixing is the following:

Proposition 1.7.2 *A Dynamical System (X, T, μ) is mixing if and only if*

$$\lim_{n\to\infty} \int_X f \circ T^n g d\mu = \int_X f d\mu \int_X g d\mu$$

for every f, $g \in L^2(X, \mu)$ or for every $f \in L^\infty(X, \mu)$ and $g \in L^1(X, \mu)$.[33]

The proof is rather straightforward and it is left as an exercise to the reader (see Problem 1.29) together with the proof of the next statement.

[33] The quantity $\int_X f \circ T g - \int_X f \int_X g$ is called “correlation,” and its tending to zero–which takes places always in mixing systems–it is called “decay of correlation.”

Proposition 1.7.3 *A Dynamical System (X, T, μ), with X a compact metric space, T continuous and μ Borel, is mixing if and only if for each probability measure λ absolutely continuous with respect to μ*

$$\lim_{n\to\infty} \lambda(f \circ T^n) = \mu(f)$$

for each $f \in \mathcal{C}^{(0)}(\mathbb{T}^2)$.

This last characterization is interesting from a mathematical point of view. Define, as usual, the evolution of a measure via the equation

$$(T_*\lambda)(f) \equiv \lambda(f \circ T)$$

for each continuous function f. If for each measure, absolutely continuous with respect to the invariant one, the evolved measure converges weakly to the invariant measure, then the system is mixing (and thus the evolved measures converge strongly). This has also a very important physical meaning: if the initial configuration is known only in probability, the probability distribution is absolutely continuous with respect to the invariant measure, and the system is mixing, then, after some time, the configurations are distributed according to the invariant measure. Again the details of the evolution are not important to describe relevant properties of the system.

1.7.1 Examples

1.7.1.a *Rotations*

We have seen that the translations by an irrational angle are ergodic. They are not mixing. The reader can easily see why.

1.7.1.b *Bernoulli shift*

The key observation is that, given a measurable set A, for each $\varepsilon > 0$ there exists a set $A_\varepsilon \in \mathcal{A}$, thus depending only on a finite subset of indices,[34] with the property[35]

$$\mu(A_\varepsilon \backslash A) \leq \varepsilon.$$

Then, given A, B measurable, and for each $\varepsilon > 0$, let A_ε, B_ε be such an approximation, and I_A, I_B the defining sets of indices, then

$$\left|\mu(T^{-m}A \cap B) - \mu(A)\mu(B)\right| \leq 4\varepsilon + \left|\mu(T^{-m}A_\varepsilon \cap B_\varepsilon) - \mu(A_\varepsilon)\mu(B_\varepsilon)\right|.$$

[34] Remember, this means that there exists a finite set $I \subset \mathbb{Z}$ such that it is possible to decide if $\sigma \in \Sigma_n$ belongs or not to A_ε only by looking at $\{\sigma_i\}_{i\in I}$.

[35] This follows from our construction of the σ-algebra and by the definition of outer measure, see Examples 1.1.1–Bernoulli shift.

If we choose m so large that $(I_A+m)\cap I_B=\emptyset$, then by the definition of Bernoulli measure we have

$$\mu(T^{-m}A_\varepsilon\cap B_\varepsilon)=\mu(T^{-m}A_\varepsilon)\mu(B_\varepsilon)=\mu(A_\varepsilon)\mu(B_\varepsilon),$$

which proves

$$\lim_{m\to\infty}\mu(T^{-m}A\cap B)=\mu(A)\mu(B).$$

1.7.1.c *Dilation*

This system is mixing. In fact, let $f,\, g\in\mathcal{C}^{(1)}(\mathbb{T})$, then we can represent them via their Fourier series $f(x)=\sum_{k\in\mathbb{Z}}e^{2\pi ikx}f_k$, $f_{-k}=\overline{f}_k$. It is well known that $\sum_{k\in\mathbb{Z}}|f_k|<\infty$ and $|f_k|\leq\frac{c}{|k|}$, for some constant c depending on f. Therefore,

$$f(T^nx)=\sum_{k\in\mathbb{Z}}e^{2\pi i2^nkx}f_k,$$

which implies that the only Fourier coefficients of $f\circ T^n$ different from zero are the $\{2^nk\}_{k\in\mathbb{Z}}$. Hence,

$$\left|\int_{\mathbb{T}}f\circ T^ng-\int_{\mathbb{T}}f\int_{\mathbb{T}}g\right|=\left|\sum_{k\in\mathbb{Z}}f_kg_{2^nk}-f_0g_0\right|\leq c2^{-n}\sum_{k\in\mathbb{Z}}|f_k|.$$

The previous inequalities imply the exponential decay of correlations for each smooth function. The proof is concluded by a standard approximation argument: given $f,\, g\in L^2(X,\,d\mu)$, for each $\varepsilon>0$ exists $f_\varepsilon,\, g_\varepsilon\in\mathcal{C}^{(1)}(X)$: $\|f-f_\varepsilon\|_2<\varepsilon$ and $\|g-g_\varepsilon\|_2<\varepsilon$. Thus,

$$\left|\int_{\mathbb{T}}f\circ T^ng-\int_{\mathbb{T}}f\int_{\mathbb{T}}g\right|\leq\left|\int_{\mathbb{T}}f_\varepsilon\circ T^ng_\varepsilon-\int_{\mathbb{T}}f_\varepsilon\int_{\mathbb{T}}g_\varepsilon\right|+2(\|f\|_2+\|g\|_2)\varepsilon,$$

which yields the result by choosing first ε small and then n sufficiently large.

1.8 Stronger statistical properties

One very fruitful idea in the realm of measurable dynamical systems is the idea of *entropy*. In some sense the entropy measure the complexity of the motions from a measure theoretical point of view.

To define it one starts by considering a partition of the space into measurable sets $\xi:=\{A_1,\dots A_n\}$ and defines[36]

$$H_\mu(\xi)-\sum_i\mu(A_i)\log\mu(A_i).$$

[36]The case of a countable partition, or even an uncountable partition, can be handled and it is very relevant, but outside the aims of this book, see [73] for a complete treatment of the subject.

Given two partitions $\xi = \{A_i\}, \eta = \{B_j\}$ we define $\xi \vee \eta := \{A_i \cap B_j\}$. Let then be

$$\xi_{-n}^T := \xi \vee T^{-1}(\xi) \vee \cdots \vee T^{-n+1}(\xi).$$

It is then possible to prove that the sequence $H_\mu(\xi_{-n}^T$ is sub-additive, hence the limit

$$h_\mu(T, \xi) := \lim_{n\to\infty} \frac{1}{n} H_\mu(\xi_{-n}^T$$

exists.

Definition 1.8.1 *The entropy of T with respect to μ is defined as*

$$h_\mu(T) := \sup\{h_\mu(T, \xi) \mid H(\xi) < \infty\}$$

Clearly if a system has positive metric entropy this means that the motion has a high complexity and it is very far from regular. One of the main property of entropy is that it is a metric invariant, that is if two systems are metrically conjugate (see the following), then they have the same metric entropy.

Even more extreme form statistical behaviors are possible, to present them we need to introduce the idea of equivalent systems. This is done via the concept of conjugation that we have already seen informally in Example 1.4.1 (logistic map, circle map).

Definition 1.8.2 *Two Dynamical Systems (X_1, T_1, μ_1), (X_2, T_2, μ_2) are (measurably) conjugate if there exists a measurable map $\phi : X_1 \to X_2$ almost everywhere invertible*[37] *such that $\mu_1(A) = \mu(\phi(A))$ and $T_2 \circ \phi = \phi \circ T_1$.*

Clearly, the conjugation is an equivalence relation. Its relevance for the present discussion is that conjugate systems have the same ergodic properties (Problem1.41).[38]

We can now introduce the most extreme form of stochasticity.

Definition 1.8.3 *A dynamical system (X, T, μ) is called Bernoulli if there exists a Bernoulli shift (M, ν, σ) and a measurable isomorphism $\phi : X \to M$ (i.e., a measurable map one one and onto apart from a set of zero measure and with measurable inverse) such that, for each $A \in X$,*

$$\nu(\phi(A)) = \mu(A)$$

and

$$T = \phi^{-1} \circ \sigma \circ \phi.$$

[37] This means that there exists a measurable function $\phi^{-1} : X_2 \to X_1$ such that $\phi \circ \phi^{-1} =$ id μ_2-a.e. and $\phi^{-1} \circ \phi =$ id μ_1-a.e..

[38] Of course the reader can easily imagine other forms of conjugacy, e.g. topological or differential conjugation.

That is a system is Bernoulli if it is isomorphic to a Bernoulli shift. Since we have seen that Bernoulli systems are very stochastic (remind that they can be seen as describing a random event like coin tossing) this is certainly a very strong condition on the systems. In particular it is immediate to see that Bernoulli systems are mixing (Problem1.41).

1.8.1 Examples

1.8.1.a *Dilation*

We will show that such a system is indeed Bernoulli. The map ϕ is obtained by dividing $[0, 1)$ in $[0, \frac{1}{2})$ and $[\frac{1}{2}, 1)$. Then, given $x \in \mathbb{T}$, we define $\phi : \mathbb{T} \to \Sigma_2^+$ by

$$\phi(x)_i = \begin{cases} 1 & \text{if } T^i x \in [0, \frac{1}{2}) \\ 2 & \text{if } T^i x \in [\frac{1}{2}, 1) \end{cases}$$

the reader can check that the map is measurable and that it satisfy the required properties. Note that the above shows that the Bernoulli measure with $p_1 = p_2 = \frac{1}{2}$ is nothing else than Lebesgue measure viewed on the numbers written in basis two. This may explain why we had to be so careful in the construction of the Bernoulli measure.

1.8.1.b *Baker*

Let us define ϕ^{-1}; for each $\sigma \in \Sigma_2$

$$x = \sum_{i=0}^{\infty} \frac{\sigma_{-i}}{2^{i+1}},$$
$$y = \sum_{i=1}^{\infty} \frac{\sigma_i}{2^i}.$$

Again the rest is left to the reader.

1.8.1.c *Forced Pendulum*

In the introduction we have seen that there exists a square Q with stable and unstable sides such that, calling T the map introduced by the flow at a proper time, $TQ \cap Q \supset Q_0^u \cup Q_1^u$. Where Q_i^u are rectangles that go from one stable side of Q to the other and, in analogy, $T^{-1}Q \cap Q \supset Q_0^s \cup Q_1^s$.

We can use this fact to code the dynamics similarly to what we have done for the Backer map. Namely, given the set $\Lambda = \bigcap_{n\in\mathbb{Z}} T^n Q$ (this set it is non empty–see Example 1.4.1–Horseshoe) and $\phi : \Lambda \to \Sigma_2$ define by

$$[\phi(x)]_k = \begin{cases} i \in \{0,1\} & \text{if } k \geq 0 \text{ and } T^k x \in Q_i^u \\ i \in \{0,1\} & \text{if } k < 0 \text{ and } T^k x \in Q_i^s. \end{cases}$$

It is easy to verify that ϕ is onto and that it is a.e. invertible. It remains to specify the measure on the Horseshoe, we can just pull back any invariant measure on the shift and we will get an invariant measure on the set Λ.

Let us conclude with a final remark on the physical relevance of the concept just introduced. As we mentioned, if f is an observable, then its ergodic average represents the result of an observation over a very long time (the time scale being determined by the mixing properties of the system). Yet, in reality, it may happen that we look for too short a time or, after studying a certain quantity, we can get a grant to buy the needed apparatus to perform more precise measurements. What would we see in such a case? Clearly, we would not see a constant, even for an ergodic system, and we would interpret the non constant part as fluctuations. In many cases it may happen that this fluctuations have a very special nature: they are Gaussian. In such a case we say that the system satisfies the Central Limit Theorem (CLT). Let us be more precise: define $S_n f := \frac{1}{\sqrt{n}} \sum_{i=0}^{n-1} f \circ T^i$.

Definition 1.8.4 *Given a Dynamical System (X, T, μ) and a class of observables $\mathcal{A} \subset L^2(X, \mu)$ we say that the class $\mathcal{A}$ satisfies the CLT if $\forall f \in \mathcal{A}$, $\mu(f) = 0$,*

$$\lim_{n\to\infty} \mu(\{x \mid S_n f \geq t\}) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{t} e^{-\frac{x^2}{2\sigma^2}} dx,$$

where (the variance) σ is defined by $\sigma^2 = \mu(f) + 2\sum_{i=1}^{\infty} \mu(f \circ T^i f)$.[39]

The relevance of the above theorem is the following: if the system is ergodic and satisfies the CLT, then $\frac{1}{n}\sum_{i=0}^{n-1} f \circ T^i - \mu(f) = \mathcal{O}(\frac{1}{\sqrt{n}})$, we have thus the precise scale on which the fluctuations should appear.

In this book we will be mainly interested in the question of how to establish if a given system is ergodic or not.

Unfortunately, neither ergodicity is a typical property of dynamical systems, nor is regular motion. It is a frustrating fact of life that generically dynamical systems present some kind of mixed behavior. Nevertheless, there are some class of systems that are known to be ergodic and among them the hyperbolic systems are probably the most relevant. We will discuss them in the next chapters.

[39] This definition is a bit stricter than usual because, in general, there may be cases in which the fluctuations are Gaussian but the formula for the variance does not hold as written.

Problems

1.1 Given a measurable Dynamical Systems (X, T, μ) verify that, for each measurable set A, if $T(A)$ is measurable, then $\mu(TA) \geq \mu(A)$.

1.2 Set $\mathcal{M}^1(X) = \{\mu \in \mathcal{M} \mid \mu(X) = 1\}$ and $\mathcal{M}^1_T(X) = \mathcal{M}^1(X) \cap \mathcal{M}_T(X)$. Prove that $\mathcal{M}^1_T(X)$ and $\mathcal{M}^1(X)$ are convex sets in $\mathcal{M}(x)$.

1.3 Call $\mathcal{M}^e(X) \subset \mathcal{M}^1(X)$ the set of ergodic probability measures. Show that $\mathcal{M}^e(X)$ consists of the extremal points of $\mathcal{M}_T(X)$. (Hint: Krein-Milman Theorem [37]).

1.4 Prove that the Lebesgue measure is invariant for the rotations on $\mathbb{T}$.

1.5 Consider a rotation by $\omega \in \mathbb{Q}$, find invariant measures different from Lebesgue.

1.6 Prove that the measure μ_h defined in Examples 1.1.1 (Hamiltonian systems) is invariant for the Hamiltonian flow. (Hint: Use the properties of H to deduce $\langle \nabla_{\phi^t x} H, \, d_x \phi^t \nabla_x H \rangle = \|\nabla_x H\|^2$, and thus $d_x \phi^t \nabla_x H = \frac{\|\nabla_x H\|^2}{\|\nabla_{\phi^t x} H\|^2} \nabla_{\phi^t x} H + v$ where $\langle \nabla_{\phi^t x} H, \, v \rangle = 0$. Then study the evolution of an arbitrarily small parallelepiped with one side parallel to $\nabla_x H$–or look at the volume form if you are more mathematically incline–remembering the invariance of the volume with respect to the flow.)

1.7 Given a Poincaré section prove that there exists $c > 0$ such that $\inf \tau_\Sigma \geq c > 0$.

1.8 Show that ν_Σ, defined in (1.2.1) is well defined.(Hint: use the invariance of μ and the fact that, by Problem 1.7, if $A \subset \Sigma$ then $\mu(\phi^{[0,\delta]}(A) \cap \phi^{[n\delta,\,(n+1)\delta]} A) = 0$ provided $(n+1)\delta \leq c$.)

1.9 Show that the return time τ_Σ is finite ν_Σ-a.e. .(Hint: let $\delta < c$ and $\Sigma_\delta := \phi^{[0,\delta]}\Sigma$, apply Poincaré return theorem to Σ_δ.)

1.10 Show that ν_Σ is T_Σ invariant. Verify that, collecting the results of the last exercises, $(\Sigma, \, T_\Sigma, \, \nu_\Sigma)$ is a Dynamical System.

1.11 something about holomorphic dynamics?

1.12 Prove that the Bernoulli measure is invariant with respect to the shift. (Hint: check it on the algebra $\mathcal{A}$ first.)

1.13 Let Σ_p be the set of periodic configurations of Σ. If μ is the Bernoulli measure prove that $\mu(\Sigma_p) = 0$ (Hint: Σ_p is the countable union of zero measure sets.)

1.14 Consider the Bernoulli shift on $\mathbb{Z}$ and define the following equivalence relation: $\sigma \sim \sigma'$ iff there exists $n \in \mathbb{Z}$ such that $T^n\sigma = \sigma'$ (this means that two sequences are equivalent if they belong to the same orbit). Consider now the equivalence classes (the space of orbits) and choose[40] a representative from each class, call the set so obtained K. Show that K cannot be a measurable set. (Hint: show that $K \cap T^n K \subset \Sigma_p$, then by using Problem 1.13 show that if K is measurable $\sum_{i=-\infty}^{\infty} \mu(T^n K) = 1$ which, by the invariance of μ, is impossible).

1.15 Compute the transfer operator for maps of $\mathbb{T}$. (Hint: Use the equivalent definition $\int g\mathcal{L}f dm = \int fg \circ T dm$.) Prove that $\|\mathcal{L}h\|_1 \leq \|h\|_1$.

1.16 Prove the Lasota-York inequality (1.4.4).

1.17 Prove that for each sequence $\{h_n\} \subset \mathcal{C}^{(1)}(\mathbb{T})$, with the property $\sup_{n\in\mathbb{N}} \|h_n'\|_1 + \|h_n\|_1 < \infty$, it is possible to extract a subsequence converging in L^1. (Hint: Consider partitions $\mathcal{P}_n$ of $\mathbb{T}$ in intervals of size $\frac{1}{n}$. Define the conditional expectation $\mathbb{E}(h|\mathcal{P}_n)(x) = \frac{1}{m(I(x))}\int_{I(x)} hdm$, where $x \in I(x) \in \mathcal{P}_n$. Prove that $\|\mathbb{E}(h|\mathcal{P}_n) - h\|_1 \leq \frac{1}{n}\|h'\|_1$. Notice that the functions $\mathbb{E}(h_n|\mathcal{P}_m)$ have only m distinct values and, by using the standard diagonal trick, construct an subsequence h_{n_j} such that all the $\mathbb{E}(h_{n_j}|\mathcal{P}_m)$ are converging. Prove that h_{n_j} converges in L^1.)

1.18 Prove Corollary 1.6.3.(Hint: ??)

1.19 Prove Theorem 1.6.5 (Hint: Note that $\mu(T^{-n}A \cap T^{-m}A) \neq 0$ then, supposing without loss of generality $n < m$, $\mu(A \cap T^{-m+n}A) \neq 0$. Then prove the theorem by absurd remembering that $\mu(X) < \infty$.)

1.20 Let $U \subset X$ of positive measure, consider

$$f_U(x) = \lim \frac{1}{n}\sum_{i=0}^{n-1} \chi_U(T^i x).$$

Show that the limit exists and that the set $A_0 := \{x \in U \mid f_U(x) = 0\}$ has zero measure. (Hint: The existence follows from Birkhoff theorem, it also follows that A_0 is an invariant set, then

$$0 = \int_{A_0} f_U = \int_{A_0} \chi_U = \mu(A_0).$$

)

[40] Attention !!!: here we are using the *Axiom of choice*.

1.21 A topological Dynamical System (X, T) is called *Topologically transitive*, if it has a dense orbit. Show that if $(\mathbb{T}^d, T, m)$ is ergodic and T is continuous, then the system is topologically transitive. (Hint: For each $n \in \mathbb{N}$, $x \in \mathbb{T}^d$ consider $B_{\frac{1}{m}}(x)$–the ball of radius $\frac{1}{m}$ centered at x. By compactness, there are $\{x_i\}$ such that $\cup_i B_{\frac{1}{m}}(x_i) = \mathbb{T}^d$. Let

$$A_{m,i} = \{y \in \mathbb{T}^d \mid T^k y \cap B_{\frac{1}{M}}(X_I) = \emptyset \ \ \forall k \in \mathbb{N}\},$$

clearly $A_{m,i} = \cap_{k\in\mathbb{N}} T^{-k} B_{\frac{1}{m}}(x_i)^c$ has the property $T^{-1}A_{m,i} \supset A_{m,i}$. It follows that $\tilde{A}_{m,i} = \cup_{n\in\mathbb{N}} T^{-n} A_{m,i} \supset A_{m,i}$ is an invariant set and it holds $\mu(\tilde{A}_{m,i} \backslash A_{m,i}) = 0$. Since $A_{m,i}$ it is not of full measure, $\tilde{A}_{m,i}$, and thus $A_{m,i}$, must have zero measure. Hence, $\bar{A}_m = \cap_i A_{m,i}$ has zero measure. This means that $\cup_{m\in\mathbb{N}} \bar{A}_m$ has zero measure. Prove now that, for each $y \in \mathbb{T}^d$, the trajectories that never get closer than $\frac{2}{m}$ to y are contained in $\bar{A}_m$, and thus have measure zero. Hence, almost every point has a dense orbit.)
Extend the result to the case in which X is a compact metric space and μ charges the open sets (that is: if $U \subset X$ is open, then $\mu(U) > 0$).

1.22 Give an example of a system with a dense orbit which it is not ergodic. (Hint: A system with two periodic orbits, and the measure supported on them. Along such lines more complex examples can be readily constructed)

1.23 Give an example of an ergodic system with no dense orbit. (Hint: A non transitive system with a measure supported on a periodic orbit.)

1.24 Give an example of a Dynamical Systems which does not have any invariant probability measure. (Hint: $X = \mathbb{R}^d$, $Tx = x + v$, $v \neq 0$.)

1.25 Show that a Dynamical Systems (X, T, μ) is ergodic if and only if there does not exists any invariant probability measure absolutely continuous with respect to μ, beside μ itself.

1.26 Prove that Birkhoff theorem implies Von Neumann theorem. (Hint: Note that the ergodic average is a contraction in L^∞, an isometry in L^2 and that $L^1 \subset L^2$ (since the measure is finite). Use Lebesgue dominate convergence theorem to prove convergence in L^2 for bounded functions. Use Fatou to show that if $f \in L^2$ then $f^+ \in L^2$ and a $3 - \varepsilon$ argument to conclude).

1.27 Prove that if (X, T, μ) is ergodic, then all $f \in L^1(X, \mu)$ such that $f \circ T = f$ are a.e. constant. Prove also the converse.

1.28 For each measurable set A, let

$$F_{A,n}(x) = \frac{1}{n}\sum_{i=0}^{n-1}\chi_A(T^i x).$$

be the average number of times x visits A in the time n. Show that there exists $F_A = \lim_{n\to\infty} F_{A,n}$ a.e. and prove that, if the system is ergodic, $F_A = \mu(A)$. (Hint: Birkhoff theorem and Theorem 1.6.6).

1.29 Prove Proposition 1.7.2 and Proposition 1.7.3. (Hint: Note that for each measurable set A and $\varepsilon > 0$ there exists $f \in \mathcal{C}^{(0)}(X)$ such that $\mu(|f - \chi_A|) < \varepsilon$ –by Uryshon Lemma and by the regularity of Borel measures. To prove that $\mu(T^{-n}A \cap B) \to \mu(A)\mu(B)$ choose $d\lambda = \mu(B)^{-1}\chi_B d\mu$ and use the invariance of μ to obtain the uniform estimate $\lambda(|f \circ T^n - \chi_A \circ T^n|) \leq \mu(B)^{-1}\mu(|f - \chi_A|)$.)

1.30 Show that the irrational rotations are not mixing.

1.31 Prove that if $f \in \mathcal{C}^{(2)}(\mathbb{T})$, then its Fourier series converges uniformly.[41] (Hint: Remember that $f_n = \frac{1}{2\pi}\int_{\mathbb{T}} e^{2\pi inx} f(x)dx$. Thus $f_n = \frac{1}{(2\pi in)^2 2\pi}\int_{\mathbb{T}} e^{2\pi inx} f^{(2)}(x)dx$.)

1.32 Let ν be a Borel measure on $Q = [0,1]^2$ such that $\nu(\partial_x f) = 0$ for all $f \in \mathcal{C}^{(1)}_{\text{per}}(Q) = \{f \in \mathcal{C}^{(1)}(Q) \mid f(0,y) = f(1,y)\ \forall\, y \in [0,1]\}$. Prove that there exists a Borel measure ν_1 on $[0,1]$ such that $\nu = m \times \nu_1$. (Hint: The measure ν_1 is nothing else then the marginal with respect to x, that is: for each continuous function $f : [0,1] \to \mathbb{R}$ define $\tilde{f} : Q \to \mathbb{R}$ by $\tilde{f}(x,y) = f(y)$, then $\nu_1(f) = \nu(\tilde{f})$. To prove the statement use Fourier series. If f is smooth enough $f(x,y) = \sum_{k\in\mathbb{Z}} \hat{f}_k(y)e^{2\pi ikx}$ where the Fourier series for f and $\partial_x f$ converge uniformly. Then notice that $0 = \nu(\partial_x e^{2\pi ik\cdot}) = 2\pi ik\nu(e^{2\pi ik\cdot})$ implies $\nu(f) = \nu(\hat{f}_0) = m \times \nu_1(f)$.)

1.33 Prove that is a flow is ergodic (mixing) so is each Poincarè section. Prove that is a map is ergodic so is any suspension on the map. Give an example of a mixing map with a non-mixing suspension (constant ceiling).

1.34 Consider $([0,1], T)$ where

$$T(x) = \frac{1}{x} - \left[\frac{1}{x}\right]$$

($[a]$ is the integer part of a), and

$$\mu(f) = \frac{1}{\ln 2}\int_0^1 f(x)\frac{1}{1+x}dx.$$

[41] This result is far from optimal, see [1] if you want to get deeper in the theory of Fourier series.

Prove that $([0,1], T, \mu)$ is a Dynamical System.[42] (Hint: write $\mu(f \circ T) = \sum_{i=1}^{\infty} \int_{\frac{1}{i+1}}^{\frac{1}{i}} f \circ T(x)\mu(dx)$, change variable and use the identity $\frac{1}{a^2+a} = \frac{1}{a} - \frac{1}{a+1}$ to obtain a series with alternating signs.)

1.35 Prove that for each $x \in \mathbb{Q} \cap [0,1]$ holds $\lim_{n\to\infty} T^n(x) = 0$. (Hint: if $x = \frac{p_0}{q_0}$, $p_0 \leq q_0$, then $q_0 = k_1 p_0 + p_1$, with $p_1 < p_0$, and $T(x) = \frac{p_1}{p_0}$. Let $q_1 = p_0$ and go on noticing that $p_{i+1} < p_i$.)[43]

1.36 In view of the two previous exercises explain why it is problematic to study the statistical properties of the Gauss map on a computer.(Hint: The computer uses only rational numbers. It is quite amazing that these type of pathologies arises rather rarely in the numerical studies carried out by so many theoretical physicist.)

1.37 Prove that any infinite continuous fraction of the form

$$a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{a_3 + \ddots}}}$$

with $a_i \in \mathbb{N}$ defines a real number. (Hint: Note that if you fix the first n $\{a_i\}$, this corresponds to specifying which elements of the partition $\{[\frac{1}{i+1}, \frac{1}{i}]\}$ are visited by the trajectory of $\{T^i x\}$. By the expansivity of the map readily follows that x must belong to an interval of size λ^{-n} for some $\lambda > 1$.)

[42] The above map is often called *Gauss map* since to him is due the discovery of the above invariant measure.

[43] This is nothing else that the *Euclidean algorithm* to find the greatest common divisor of two integers [38] Elements, Book VII, Proposition 1 and 2. The greatest common divisor is clearly the last non-zero p_i. This provides also a remarkable way of writing rational numbers: *continuous fractions*

$$\frac{p_0}{q_0} = \cfrac{1}{k_1 + \cfrac{1}{k_2 + \ddots + \cfrac{1}{k_n}}}.$$

1.38 Prove that, for each $a \in \mathbb{N}$,

$$x = \cfrac{1}{a + \cfrac{1}{a + \cfrac{1}{a + \ddots}}} = \frac{-a + \sqrt{a^2+4}}{2}.$$

(Hint: Note that $T(x) = x$.) Study periodic continuous fractions of period two.

1.39 Choose a number in $[0,1]$ at random according to Lebesgue distribution. Assuming that the Gauss map is mixing (which it is, see ???) compute the average percentage of numbers larger than n in the associated continuous fraction. (Hint: Define $f(x) = [x^{-1}]$, then the entries of the continuous fraction of x are $\{f \circ T^i\}$. The quantity one must compute is then $m(\lim_{k\to\infty} \frac{i}{k} \sum_{i=0}^{k-1} \chi_{[n,\infty)} \circ f \circ T^i) = \mu([n,\infty))$.)

1.40 Let (X_0, T_0, μ_0) be a Dynamical System and $\phi : X_0 \to X_1$ an homeomorphism. Define $T_1 := \phi \circ T_0 \circ \phi^{-1}$ and $\mu_1(f) = \mu_0(f \circ \phi^{-1})$. Prove that (X_1, T_1, μ_1) is a Dynamical System.

1.41 Let (X_0, T_0, μ_0) be measurably conjugate to (X_1, T_1, μ_1), then show that one of the two is ergodic if and only if the other is ergodic. Prove the same for mixing.

1.42 Show that the systems described in Examples ??–strange attractor and horseshoe, are Bernoulli.

1.43 Prove Lebesgue density theorem: for each measurable set A, $m(A) > 0$, there exists $x \in A$ such that for each $\varepsilon > 0$ exists $\delta > 0$ such that $m(A \cap [x-\delta, x+\delta]) > (1-\varepsilon)2\delta$. (Hint: we have seen in Examples 1.8.1-Dilations that Lebesgue measure is equivalent to Bernoulli measure and that the cylinder correspond to intervals. It then suffices to prove the theorem for the latter. Let $A \subset \Sigma^+$ such that $\mu(A) > 0$, then, for each $\varepsilon > 0$,there exists $A_\varepsilon \in \mathcal{A}$ such that $A_\varepsilon \supset A$ and $\mu(A_\varepsilon) - \mu(A) < \varepsilon\mu(A)$. Since $A_\varepsilon \in \mathcal{A}$, it exists $n_\varepsilon \in \mathbb{N}$ such that it is possible to decide if $\sigma \in A_\varepsilon$ only by looking at $\{\sigma_1, \ldots, \sigma_{n_\varepsilon}\}$. Consider all the cylinders $\mathcal{I}\{A(0; k_1, \ldots, k_{n_\varepsilon})\}$, clearly if $I \in \mathcal{I}$ then $I \cap A_\varepsilon$ is either I or $\emptyset$. Let $\mathcal{I}_+ = \{I \in \mathcal{I} \mid I \cap A_\varepsilon = I\}$ and $\mathcal{I}_+ = \{I \in \mathcal{I} \mid I \cap A_\varepsilon = \emptyset\}$. Now suppose that for each $I \in \mathcal{I}_+$ holds $\mu(I \cap A) \le (1-\varepsilon)\mu(I)$ then

$$\mu(A) = \sum_{I \in \mathcal{I}_+} \mu(A \cap I) \le (1-\varepsilon)\mu(A_\varepsilon) < \mu(A),$$

which is absurd. Thus there must exists $I \in \mathcal{I}_+$: $\mu(A \cap I) > (1-\varepsilon)\mu(I)$.)

Notes

Give references for SRB and Gibbs, mention entropy, K-systems. diffeo with holes, strange attractors, history of the field

. . . .

CHAPTER 2

Ergodicity and Mixing (Basic ideas)

The concept of ergodicity is a very important one in dynamical systems, yet it turns out to be surprisingly difficult to establish if a system is or not ergodic and very few examples have been fully analyzed. Nonetheless, in this chapter we will see that a very simple idea introduced by Hopf [47, 48] allows to discuss the ergodicity in some special cases. The relevance of Hopf's idea is that, properly generalized, it allows to prove ergodicity in a vast class of systems. Much in the following chapters will deal with such a generalization.

2.1 A Basic Example

To explain the Hopf approach we will study a very simple case: a slight generalization of Arnold's cat, see Examples 1.1.1. Let $T : \mathbb{T}^2 \to \mathbb{T}^2$ (here by $\mathbb{T}^2$ we mean $\mathbb{R}^2 \mod 1$) be defined by

$$T\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 & a \\ a & 1+a^2 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \mod 1 \tag{2.1.1}$$

It is obvious that if $a \in \mathbb{Z}$, then T is well defined and it is a linear automorphism of $\mathbb{T}^2$. Moreover, for all $x \in \mathbb{T}^2$

$$D_xT = \begin{pmatrix} 1 & a \\ a & 1+a^2 \end{pmatrix} \equiv L.$$

Since $\det L = 1$, Lebesgue measure is preserved. It is immediate to see that there exists $\lambda > 1$; v_+, $v_- \in \mathbb{R}^2$:

$$\begin{aligned} Lv_+ &= \lambda v_+ \\ Lv_- &= \lambda^{-1} v_-. \end{aligned}$$

We will call v_+ the unstable eigenvector (direction) and v_- the stable eigenvector (direction). Remark that, since $L^* = L$, $\langle v_+, v_-\rangle = 0$.

The dynamical system just described is a basic model of hyperbolic systems (see next chapter) and will appear in various disguises in this book.

Proposition 2.1.1 *The Arnold cat is ergodic.*

Sections 2.1.1 and 2.2.1 contain two different proofs of the above proposition.

2.1.1 An algebraic proof

A first idea to studying the ergodic properties of this system is to imitate what we have done for the Rotations (Examples 1.5.1) and the Dilations (Examples 1.7.1): use Fourier series. Let us see how such an approach would work.

Let $f,\ g \in C^{(m)}(\mathbb{T}^2)$, then[1]

$$f \circ T^n(x) = \sum_{k\in\mathbb{Z}^2} e^{2\pi i\langle k,\, L^n x\rangle} f_k = \sum_{k\in\mathbb{Z}^2} e^{2\pi i\langle k,\, x\rangle} f_{L^{-n}k},$$

so

$$\begin{aligned}\int_{\mathbb{T}^2} f\circ T^{2n} g = \int_{\mathbb{T}^2} f\circ T^n g\circ T^{-n} &= \sum_{k\in\mathbb{Z}^2} f_{L^{-n}k} g_{L^n k}\\ &= f_0 g_0 + \sum_{k\in\mathbb{Z}^2\setminus\{0\}} f_{L^{-n}k} g_{L^n k}.\end{aligned}$$

It is well known [75] that $f \in C^{(m)}(\mathbb{T}^2)$ implies[2]

$$|f_k| \le \frac{\|f^{(m)}\|_1}{\|k\|^m} \text{ for } k \neq 0$$

hence

$$\left|\sum_{k\in\mathbb{Z}^2\setminus\{0\}} f_{L^{-n}k} g_{L^n k}\right| \le \sum_{k\in\mathbb{Z}^2\setminus\{0\}} \frac{\|f^{(m)}\|_1 \|g^{(m)}\|_1}{\|L^{-n}k\|^m \|L^n k\|^m}.$$

For each $k \in \mathbb{Z}^2$ holds $\|k\|^2 = \langle k,\, v^+\rangle^2 + \langle k,\, v^-\rangle^2$ hence one of the two terms must be larger than $\|k\|^2/2$.[3] Moreover if $k \neq 0$ $\|L^n k\| \ge 1$ for each $n \in \mathbb{Z}$. Using the above facts it yields

[1] Note that $e^{2\pi i\langle k, T^n x\rangle} = e^{2\pi i\langle k, L^n x\rangle}$.

[2] Here for $\|f^{(m)}\|_1$ we mean $\sup_{\substack{i+j=m\\ i,j\ge 0}} \frac{1}{(2\pi)^m}\int_{\mathbb{T}^2} |\partial^i_{x_1}\partial^j_{x_2} f| dx_1 dx_2$; and $\|k\| = \sqrt{k_1^2 + k_2^2}$.

[3] Here we have normalized the eigenvalues so that $\|v^\pm\| = 1$.

$$\left|\sum_{k\in\mathbb{Z}^2\setminus\{0\}} f_{L^{-n}k}g_{L^nk}\right| \le \sum_{k\in\mathbb{Z}^2\setminus\{0\}} \frac{\|f^{(m)}\|_1\|g^{(m)}\|_1 2^{m/2}}{\lambda^{nm}\|k\|^m}$$
$$\le \text{const.}\|f^{(m)}\|_1\|g^{(m)}\|_1\lambda^{-nm},$$

where the constant does not depend on f or g and we have assumed $m \ge 3$ to insure the convergence of the series.

Accordingly, for each $f,\ g \in C^{(3)}(\mathbb{T}^2)$ we have

$$\left|\int_{\mathbb{T}^2} f\circ T^n g - \int_{\mathbb{T}^2} f\int_{\mathbb{T}^2} g\right| \le \text{const.}\|f^{(3)}\|_1\|g^{(3)}\|_1\lambda^{-3n/2}.$$

To obtain the final result we need an approximation argument. If $f, g \in L^2(\mathbb{T}^2)$ we can choose $f_n, g_n \in C^{(3)}(\mathbb{T}^2)$ such that they converge to f and g, respectively, in L^2.

Then, for each $\varepsilon \ge 0$, choose $m \in \mathbb{N}$ such that

$$\|f - f_m\|_2 + \|g - g_m\|_2 \le \varepsilon.$$

Accordingly,

$$\left|\int_{\mathbb{T}^2} f\circ T^n g - \int_{\mathbb{T}^2} f\int_{\mathbb{T}^2} g\right| \le \left|\int_{\mathbb{T}^2} f_m\circ T^n g_m - \int_{\mathbb{T}^2} f_m\int_{\mathbb{T}^2} g_m\right|$$
$$+ 2\|f - f_m\|_2\|g\|_2 + 2\|f_m\|_2\|g - g_m\|_2$$
$$\le 2(\|g\|_2 + \|f\|_2)\varepsilon + \varepsilon,$$

where we have chosen n large enough depending on m and ε. We have just proven mixing.

The above result is certainly rather satisfactory: non only it proves the mixing–hence the ergodicity–of the map but gives an explicit estimate on the rate of decay and shows how such a rate depends on the regularity of the functions.[4] Therefore, an eventual critique can not concern the type of result but only the method; indeed the method does have a shortcoming.

The use of Fourier series is strictly related to the group structure of $\mathbb{T}^2$ and the linearity of the map. Clearly in more general systems, where both such properties may fail, such a technique has no hope whatsoever of being applied.[5] In some sense, much of the theory of hyperbolic systems may be

[4] In fact, the obtained estimate it is not optimal: using the Diofantine properties of the stable and unstable directions a better estimate can be obtained (see Problem 2.1).

[5] In fact, there are very few cases in which this type of ideas has produced relevant results, noticeably the case of geodesic flows on surfaces of constant negative curvature [1].

view as an attempt to find an alternative proof of the above facts. Such a proof must be *dynamical* meaning that it must use properties of the dynamics and as little as possible of the structure of the space.

The best way to gain a real feeling of what is meant by *dynamical* is to see such type of arguments in action.

2.2 An Idea by Hopf

The following argument, due to Hopf [47], [48] is exactly such a dynamical proof of ergodicity. Let $f : \mathbb{T}^2 \to \mathbb{R}$ be a continuous function. We want to prove that for almost every $x \in \mathbb{T}^2$ the time averages converge as $n \to +\infty$ to the average value of f, i.e., $\int_{\mathbb{T}^2} f d\mu$. Once this is established one can obtain the same property for all integrable functions by an approximation argument, this proves ergodicity due to the characterization provided by Theorem 1.6.6 (see also Problem 1.27). From Birkhoff Ergodic Theorem (BET) we know that the time averages converge almost everywhere to a function $f^+ \in L^1(\mathbb{T}^2, \mu)$ which is invariant on the orbits of T, i.e., $f^+ \circ T = f^+$, and has the same average value as f, i.e., $\int f^+ d\mu = \int f d\mu$. Further, applying BET to f and T^{-1} we obtain that the time averages in the past

$$\frac{f(x) + f(T^{-1}x) + \cdots + f(T^{-n+1}x)}{n}$$

converge almost everywhere as $n \to +\infty$ to $f^- \in L^1(\mathbb{T}^2, \mu)$, $f^- \circ T = f^-$ and $\int f^- d\mu = \int f d\mu$.

The next Lemma is part of the usual magic of the ergodic theory.

Lemma 2.2.1 *The functions f^+ and f^- coincide almost everywhere.*

Proof. Let

$$\mathcal{A}_+ = \{x \in \mathbb{T}^2 \mid f_+(x) > f_-(x)\};$$

by definition $\mathcal{A}_+$ is an invariant set, hence

$$\int_{\mathcal{A}_+} [f_+(x) - f_-(x)]\, d\mu(x) = \int_{\mathcal{A}_+} f(x) d\mu(x) - \int_{\mathcal{A}_+} f(x) d\mu(x) = 0$$

which implies $\mu(\mathcal{A}_+) = 0$ and $f_+ \leq f_-$ μ-almost everywhere. The same argument, this time applied to the set $\mathcal{A}_- = \{x \in \mathbb{T}^2 \mid f_-(x) > f_+(x)\}$, implies the converse inequality. $\square$

2.2.1 A dynamical proof

For $x \in \mathbb{T}^2$ let us denote by $W^u(x)$ ($W^s(x)$) the line in $\mathbb{T}^2$ passing through x and having the direction of the unstable eigenvector (the stable eigenvector), i.e., the eigenvector with eigenvalue λ (λ^{-1}). We call $W^u(x)$ ($W^s(x)$) the unstable (stable) leaf (or manifold) of x. The leaves of x have the following property. If $y \in W^u(x)$ ($y \in W^s(x)$) then the distance (computed along the leaf)

$$d(T^n y, T^n x) = \lambda^{-|n|} d(y, x) \to 0 \quad \text{as} \quad n \to -\infty(+\infty).$$

Hence for $y, z \in W^{u(s)}(x)$

$$|f(T^n y) - f(T^n z)| \to 0 \quad \text{as} \quad n \to -\infty(+\infty).$$

It follows that for $y, z \in W^{u)}(x)$ either $f^-(y)$ and $f^-(z)$ are both defined and equal or they are both undefined; the same can be said for $f^+(y)$ and $f^+(z)$ if $y, z \in W^s(x)$.

It is interesting to notice that $W^u(x)$ is an infinitely long line in the direction v_+ that fills densely the torus (see Problem 2.6). This implies that the collection (foliation) $\{W^u(x)\}_{x\in\mathbb{T}^2}$ of this global manifolds has a quite complex structure (see Problem 2.7). For this reason it turns out to be much more convenient to deal only with *local manifolds*.

A local manifold of size δ is simply a piece of $W^u(x)$ of size δ centered at x. In the following by $W^u(x)$ and $W^s(x)$ we will always mean local manifolds (lines) of some length. The exact length is, most of the times, irrelevant an often will not be specified (in the following it will be frequently chosen so that the lines do not wrap around the torus more than once).

Up to now we have seen that f^+ is constant along a.e. stable lines while f^- is constant along a.e. unstable line, since they are equal a.e. it seems obvious that they must be equal and constant. Yet, in the last sentence there are a lot of almost everywhere and, being measure theory a rather subtle subject, it is better to spell out the reasoning in full detail.[6]

Let us choose any point $x \in \mathbb{T}^2$ and prove that there is a neighborhood of x in which f^+ is a.e. constant. Since x is arbitrary this implies that f^+ is a.e. constant.[7] Chose a square Q_δ of size $2\delta < \frac{1}{4}$ centered at x with sides parallel to v_+ and v_- respectively. Let $\phi : [-\delta, \delta]^2 \to Q_\delta$ be defined by $\phi(\alpha, \beta) = x + \alpha v_+ + \beta v_-$, where we have chosen $\|v_\pm\| = 1$. It is then convenient to transport the problem in $[-\delta, \delta]^2$ by ϕ: doing so the Lebesgue measure is sent in the Lebesgue measure and that $f^+ \circ \phi$ is a.e. constant in the vertical direction (α constant), while $f^- \circ \phi$ is a.e. constant in the horizontal

[6] We have already seen in Examples 1.5.1–Rotations that these type of arguments must employ measure theory in a non trivial way.

[7] Please, note this apparently naïve idea to look at the problem first locally and then globally, we will see much more of it in the following.

direction. This corresponds simply to a change of variables and from now on we will confuse Q_δ and $[-\delta,\delta]^2$ since this does not create any ambiguity.

There are three full measure sets to consider:
$\widetilde{\mathcal{B}}_+ = \{\xi \in Q_\delta \mid f^+(\xi) \text{ is defined}\}$; $\widetilde{\mathcal{B}}_- = \{\xi \in Q_\delta \mid f^-(\xi) \text{ is defined}\}$ and $G = \{\xi \in \widetilde{\mathcal{B}}_+ \cap \widetilde{\mathcal{B}}_- \mid f^+(\xi) = f^-(\xi)\}$.

Let us call $W^s_\alpha := \{(a,b) \in Q_\delta \mid a = \alpha\}$ the segment in Q_δ parallel to the stable direction passing through the point $(\alpha, 0)$, and $W^u_\beta := \{(a,b) \in Q_\delta \mid b = \beta\}$ the segment in Q_δ parallel to the unstable direction passing through the point $(0,\beta)$. The previous discussion proves that there exist $\mathcal{B}_\pm \in [-\delta,\delta]$ such that $\widetilde{\mathcal{B}}_+ = \cup_{\alpha\in\mathcal{B}_+} W^s_\alpha$ and $\widetilde{\mathcal{B}}_- = \cup_{\beta\in\mathcal{B}_-} W^u_\beta$.

Since m is the product of two one dimensional Lebesgue measures[8] Fubini theorem [74] implies that $\mathcal{B}_\pm$ are measurable sets of full measure. Again by Fubini Theorem, it follows

$$4\delta^2 = m(Q_\delta) = m(\widetilde{\mathcal{B}}_+ \cap G) = \int_{\mathcal{B}_+} d\alpha \int_{-\delta}^{\delta} d\beta \chi_{W^s_\alpha \cap G}(\alpha,\beta).$$

This implies immediately that there exists a set $I \subset \mathcal{B}_+$, of full measure, such that, for each $\alpha \in I$ the set $J_\alpha = \{\beta \in \mathcal{B}_- \mid (\alpha,\beta) \in G\}$ is measurable and has full measure as well; the same holds for $E = \cup_{\alpha\in I} W^s_\alpha$.

Finally, let $z, y \in E$, $z = (a,b)$ and $y = (c,d)$. If $a = c$, then $z, y \in W^s_a$ and $f^+(z) = f^+(y)$. On the other hand, if $a \neq c$ then by choosing $\beta \in J_a \cap J_c$ it follows

$$\begin{aligned} f^+(z) &= f^+(W^s_a) = f^+(a,\beta) = f^-(a,\beta) \\ &= f^-(W^u_\beta) = f^-(c,\beta) = f^+(c,\beta) = f^+(y). \end{aligned}$$

That is, f^+ is constant on E, hence f^+ (and f^-) is a.e. constant on Q_δ. By the arbitrariness of Q_δ follows that $f^+ = f^-$ =constant a.e..

Up to now we have proved that f^+ is a.e. constant only if $f \in \mathcal{C}^{(0)}(\mathbb{T}^2)$, to prove ergodicity we need the same result for each $f \in L^1(\mathbb{T}^2)$. This can be easily obtained by an approximation argument; yet, it is probably more interesting to prove directly that all invariant sets have measure zero or one.

Let us consider a T-invariant measurable subset A. Let

$$f_n \to \chi_A \ \text{ in } \ L^1(\mathbb{T}^2, \mu)$$

be a sequence of uniformly bounded continuous approximations to the indicator function.[9] We will use the fact that the time average is continuous with

[8] Here, to have an unambiguous notation, we should use m_n for the Lebesgue measure in $\mathbb{R}^n$, then we just said $m_2 = m_1 \times m_1$. For simplicity, I have suppressed all the subscript hoping not to confuse the reader too much.

[9] If the existence of such a sequence $\{f_n\}$ it is not obvious, consider the following: for

respect to the L^1 norm to establish that the time average of χ_A must be constant on $\mathbb{T}^2$. Indeed, if we denote by $\|\cdot\|_1$ the $L^1(\mathbb{T}^2, m)$ norm, then

$$\begin{aligned}\|f_n^+ - \chi_A^+\|_1 &= \left\| \lim_{N\to\infty} \frac{1}{N} \sum_{i=1}^{N} \left(f_n \circ T^i - \chi_A \circ T^i\right)\right\|_1 \\ &= \lim_{N\to\infty} \frac{1}{N} \left\| \sum_{i=1}^{N} \left(f_n \circ T^i - \chi_A \circ T^i\right)\right\|_1\end{aligned}$$

by the Lebesgue Dominated Convergence Theorem.

Using the invariance of the measure we obtain

$$\|f_n^+ - \chi_A^+\|_1 \le \lim_{N\to\infty} \frac{1}{N} \sum_{i=1}^{N} \|f_n - \chi_A\|_1 = \|f_n - \chi_A\|_1.$$

Since the time averages $f_n^+ = m(f_n)$ a.e. in $\mathbb{T}^2$ and $\lim_{n\to\infty} m(f_n) = m(A)$, the Lebesgue dominated convergence theorem implies $\|m(A) - \chi_A^+\|_1 = 0$, that is $\chi_A^+ = m(A)$ a.e.. In addition, the invariance of A forces $\chi_A^+ = \chi_A$ so that either A or A^c has measure zero. In view of the arbitrariness of the invariant set A it follows that T must be ergodic.

2.2.2 What have we done?

The question remains of how and if such an argument can be extended to more general systems. The answer must lie in the possibility to generalize the main ingredients of the previous proof. Such ingredients are essentially two: a) the *existence* of two foliations on which f^+ (f^- respectively) are constant; b) some *regularity* property of such foliations.

In general the foliations will be provided by the stable and unstable manifolds (the existence of which is the content of the next chapter). A careful look at the previous proof should convince the reader that the needed regularity is a property of the type: consider two manifolds W_1^s, W_2^s and define a map $\phi : W_1^s \to W_2^s$ by $\phi(x) = W^u(x) \cap W_2^s$ (this is often called *holonomy map* or *Poincaré transformation*[10], we will use the first name), then ϕ is measurable and *absolutely continuous* that is : if $A \subset W_2^s$ has positive measure so has $\phi^{-1}A$. The absolutely continuity property of stable and unstable foliations will be the topic of chapter 7.

each $\varepsilon > 0$, by the regularity of the Lebesgue measure, there exists $C_\varepsilon \subset A \subset G_\varepsilon$ (C_ε closed and G_ε open) such that $m(G_\varepsilon) - m(C_\varepsilon) \le \varepsilon$. Then Uryshon lemma implies that there exists $f_\varepsilon \in \mathcal{C}^{(0)}(\mathbb{T}^2)$ such that $f_\varepsilon(\mathbb{T}^2) \subset [0,1]$, $f_\varepsilon|_{C_\varepsilon} = 1$ and $f_\varepsilon|_{G_\varepsilon^c} = 0$. Thus $\|f_\varepsilon - \chi_A\|_1 \le m(G_\varepsilon \backslash C_\varepsilon) \le \varepsilon$.

[10] Note that if one could define a flow along the unstable direction–and in our case it is possible–then the above map would indeed be a Poincaré map with respect to such a flow.

Of course, the above comments are very imprecise, their only aim is to give an idea of what is coming next. In the mean time, to start building some feeling for the foliations and their properties, see Problems 2.12, 2.13 and 2.14.

2.3 About mixing

We continue our investigations with a discussion of an other dynamical proofs in which we will see the role of hyperbolicity and some basic ideas associated to it at work. The final goal will be to obtain a dynamical proof of the following.

Proposition 2.3.1 *The Arnold cat is mixing.*

We will start by proving Topological Mixing.

Definition 2.3.2 *A smooth Dynamical System is topologically mixing if for each two open sets U and V there exists an integer $n \in \mathbb{N}$ such that*

$$T^{-m}U \cap V \neq \emptyset \quad \forall m \geq n.$$

Note that the all point in the above definition is that it holds for <u>all</u> n large enough (see Problem 2.3).

Remark that it suffices to have the above property for any class of sets that can be used as a basis for the topology. The most convenient choice is given by the so called "rectangles." Such sets are an extremely important tool in hyperbolic theory and we have already met them several times–although I will not insist on them in the present book–here they appear in the simplest possible form.

Definition 2.3.3 *By rectangle we mean a quadrilater (i.e. a region with boundaries consisting of four segments) with sides parallel to the stable or unstable directions.*

Proposition 2.3.4 *The Arnold cat is topologically mixing.*

PROOF. Let us consider two rectangles A and B. A first key observation is that, for each $m \in \mathbb{N}$, $T^m A$ and $T^m B$ are rectangles as well. The second key observation is that they have a very special shape: in the stable direction their size has contracted by a factor λ^m while in the unstable direction the size has expanded by the same factor. Hence, provided m is chosen large enough, $T^m A$ and $T^m B$ are very thin in the stable direction and very elongated in the unstable direction. This property of stretching and squeezing, that we are witnessing here, is the cornerstone of almost all arguments in hyperbolic theory. Of course, similar, but symmetrical, arguments hold for $T^{-m}A$ and $T^{-m}B$. We can then choose $m \in \mathbb{N}$ so large that the length of the unstable sides of $T^m B$ is larger than 2 and, at the same time, the same is true for the stable side of $T^{-m}A$. It is then a trivial geometric observation, best seen on the covering of $\mathbb{T}^2$, that $T^n A \cap T^{-n} B \neq \emptyset$, for each $n \geq m$, thus $T^{-2n} A \cap B \neq \emptyset$, which suffices to prove the topological mixing. □

The reader who starts to appreciate the spirit of the game may be unhappy about the previous proof. The problem is that we have used a bit too heavily the structure of the foliation (straight lines) and of $\mathbb{T}^2$ (the covering).

It is then quite natural to wonder if a more flexible and dynamical proof is available. Here it is.

Another proof of Proposition 2.3.4. Let us start by a preliminary result.

Given any rectangle A let us call A_c a rectangle of half the size and situated at its center.[11]

Lemma 2.3.5 *If $T^{-n} A_c \cap A_c \neq \emptyset$ for some $n \in \mathbb{N}$ such that $\lambda^n > 4$, then $T^{-mn} A \cap A \neq \emptyset$ for all $m \in \mathbb{N}$.*

Proof. By construction $T^{-n}A$ intersects A completely from one unstable side to the other (see figure 2.1)

This means that $T^{-2n}A \supset T^{-n}(T^{-n}A \cap A)$, which is a very thin rectangle contained in $T^{-n}A$ and that crosses it from one unstable side to the other. Accordingly $T^{-2n}A$ will intersect A completely (from one unstable side to the other). By induction the result follows. □

Note that the $n \in \mathbb{N}$ required by the above statement always exists (see Problem 2.3).

Next, let $A, B \subset \mathbb{T}^2$ be two rectangles and let $n_B \in \mathbb{N}$ such that Lemma 2.3.5 applies to B. We then consider the Dynamical Systems $(\mathbb{T}^2, T^{n_B}, m)$, this is ergodic as well (see Problem 2.2).[12] Consequently, for each integer $i \in \{1, \dots, n_B - 1\}$ there exists $k_i \in \mathbb{N}$ such that

$$T^{-k_i n_B}(T^{-i} A_c) \cap B_c \neq \emptyset,$$

and the unstable size of A times $\lambda^{-k_i n_B}$ is smaller than one quarter of the unstable size of B (see Problem 2.4). This implies immediately that

$$T^{-k n_B}(T^{-i} A) \cap B \neq \emptyset \quad \forall k \geq k_i. \tag{2.3.1}$$

[11] This may seem a silly construction but it is a rather general trick used to exploit topological mixing and we will see it again under the name of *core* of a rectangle in chapter 8.

[12] This is the crucial property always needed to obtain mixing in hyperbolic systems: ergodicity of all the powers of the map.

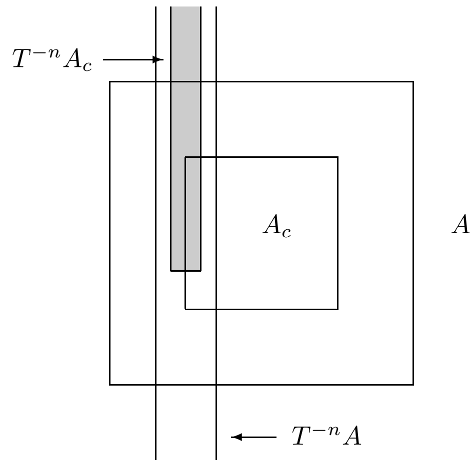


Figure 2.1: Intersection between A and $T^{-n}A$

In fact, $T^{-k_i n_B}(T^{-i}A)$ crosses B from one unstable side to the other and touches B_c, thus (2.3.1) can be proved by the same type of arguments used in Lemma 2.3.5.

Finally, set $k_m := \max\{k_i \mid i \in \{1, \ldots, n_B\}\}$. For each $n > k_m n_B$ we can write $n = kn_B + i$ where $0 < i < n_B$, thus

$$T^{-n}A \cap B = T^{kn_B}(T^{-i}A) \cap B \neq \emptyset,$$

by (2.3.1). □

By the same arguments one can prove the following (see Problem 2.5).

Lemma 2.3.6 *Given any stable segment W^s of length δ, and any unstable segment W^u of length $L > \lambda\delta^{-1}$, then it holds $W^s \cap W^u \neq \emptyset$.*

To start discussing the problem of mixing we need to adopt a point of view among the many possible. We will take the one that looks at the measures (see Proposition 1.7.3 and Problem 1.29) which, by now, should be rather familiar to the reader. Calling μ_0 a measure absolutely continuous with respect to Lebesgue we would like to study the asymptotic behavior of $\mu_n := {T_*}^n\mu_0$. Thanks to Proposition 1.7.3 we need to study only the weak convergence. The first observation is that such a set of measures is compact hence we can study the set of its limit points Γ (of course with the goal of showing that it consists of only one point).[13] Such a set is simply the set of limits of

[13]Note that such accumulation points are not necessarily invariant measures, this is why we considered accumulation points of averages in section 1.4.

convergent subsequences. Since the measure μ_0 is absolutely continuous with respect to m there exists a function $h \in L^1(\mathbb{T}^2)$, $h \geq 0$, such that

$$\mu_0(f) = m(hf).$$

A lesson that we have learned from the computation in Fourier transform and from the Hopf argument is that the regularity of the functions do matter considerably and that it may be useful to consider, at first, regular functions and then obtain the wanted result by an approximation argument. Accordingly, we will restrict ourself to the case $h \in C^{(1)}(\mathbb{T}^2)$ and establish two fundamental facts.[14]

Lemma 2.3.7 *If $\bar{\mu} \in \Gamma$ then $\bar{\mu}$ is absolutely continuous with respect to Lebesgue. In addition, $\bar{h} = \frac{d\bar{\mu}}{dm} \in L^\infty(\mathbb{T}^2, m)$.*

PROOF. We notice that the sequence μ_n is uniformly absolutely continuous with respect to Lebesgue, that is $\forall f \in \mathcal{C}^{(0)}(\mathbb{T}^2)$ such that $f \geq 0$

$$\mu_n(f) = \int_{\mathbb{T}^2} h \circ T^{-n} f \leq \|h\|_\infty \|f\|_1.$$

This implies $\bar{\mu}(f) \leq \|h\|_\infty m(f)$ and

$$\bar{\mu}(A) = \sup_{\substack{C \subset A \\ C = \overline{C}}} \bar{\mu}(C) = \sup_{\substack{C \subset A \\ C = \overline{C}}} \inf_{\{f \in \mathcal{C}^{(0)} \mid f > \chi_C\}} \bar{\mu}(f) \leq \|h\|_\infty m(A), \tag{2.3.2}$$

where we have used (1.4.1) and (1.4.2). Clearly (2.3.2) implies the absolute continuity. Hence, by the Radon-Nikodym theorem [74], there exists $\bar{h} \in L^1(\mathbb{T}^2, m)$ such that $d\bar{\mu} = \bar{h} dm$.

Next, let $A = \{x \in \mathbb{T}^2 \mid \bar{h}(x) > \|h\|_\infty\}$. If $m(A) \neq 0$, then

$$\|h\|_\infty m(A) < \int_A \bar{h} dm = \bar{\mu}(A) \leq \|h\|_\infty m(A)$$

which is a contradiction, thus $\bar{h} \leq \|h\|_\infty$ a.e.. □

The next argument is very similar to what we have already seen in Examples 1.4.1–Strange Attractors. Let us call D^u the derivative along the unstable direction (if v^+ is the normal vector in the unstable direction then $D^u f := \langle \nabla f, v^+ \rangle$).

Lemma 2.3.8 *There exists $c > 0$: for each $f \in \mathcal{C}^{(1)}(\mathbb{T}^2)$*

$$|\mu_n(D^u f)| \leq \lambda^{-n} c \|f\|_\infty.$$

[14] Actually, this regularity condition on h will be needed only in Lemma 2.3.8.

PROOF.

$$\begin{aligned}\mu_n(D^u f) &= \int_{\mathbb{T}^2} h(D^u f)\circ T^n = \int_{\mathbb{T}^2} h\langle(\nabla f)\circ T^n,\, v^+\rangle\\ &= \int_{\mathbb{T}^2} h\langle L^{-n}\nabla(f\circ T^n),\, v^+\rangle = \lambda^{-n}\sum_{i=1}^{2}\int_{\mathbb{T}^2} h\partial_{x_i}(f\circ T^n)v_i^+\\ &= -\lambda^{-n}\int_{\mathbb{T}^2} D^u h f\circ T^n,\end{aligned}$$

where the last equality is obtained by integrating by parts with respect to both coordinates. Accordingly,

$$|\mu_n(D^u f)| \le \lambda^{-n}\|\nabla h\|_1\|f\|_\infty.$$

□

From the above results it follows that if $\bar\mu \in \Gamma$ then there exists $\bar h \in L^\infty(\mathbb{T}^2)$ such that, for each $f \in L^1(\mathbb{T}^2, m)$,

$$\bar\mu(f) = \int f\bar h dm$$

and for each $f \in \mathcal{C}^{(1)}(\mathbb{T}^2)$, $\bar\mu(D^u f) = 0$. This two facts together imply that $\bar h$ is constant almost everywhere.

To see this we start by a **local** argument showing that $\bar h$ is constant along the unstable direction. We have already done a similar argument, in Examples 1.4.1–Strange Attractors, by using Fourier series, let us see here a more measure theoretical argument to convince the reader that the global structure of $\mathbb{T}^2$ has nothing to do with the result.

Let us consider an arbitrary rectangle R of size smaller than 1/4. Consider an arbitrary $f \in \mathcal{C}^{(1)}(\mathbb{T}^2)$ with support contained in $\mathring{R}$. Then consider coordinates in R parallel to its sides (since this is achieved by rotations and rigid translations it leaves invariant the Lebesgue measure). As before, the unstable sides are horizontal. Let us call x the coordinate along the stable direction and y the one along the unstable direction. In such coordinates $R = [0, a] \times [0, b]$ (we have translates the origin at the bottom left corner of R). Given $f \in \mathcal{C}^{(1)}$, we define

$$\begin{aligned}\tilde f(x, y) &= f(x, y) - \frac{1}{a}\int_0^a f(\xi, y)d\xi,\\ F(x, y) &= \int_0^x \tilde f(\xi, y)d\xi.\end{aligned}$$

Then $F|_{\partial R} = 0$ so F can be extended to a function on $\mathbb{T}^2$ by setting $F = 0$ outside R. Note, that F is continuous and differentiable everywhere apart from the boundary ∂R where the derivative can be discontinuous. In the new coordinates D^u becomes simply the derivative with respect to x.

$$\int_{\mathbb{T}^2} \bar{h} f = \int_R \bar{h} f = \int_0^a dx \int_0^b dy \bar{h} \tilde{f} + \frac{1}{a} \int_0^b dy \int_0^a dx \bar{h}(x, y) \int_0^a d\xi f(\xi, y),$$

and, setting $\tilde{h}(y) = \frac{1}{a} \int_0^a d\xi \bar{h}(\xi, y)$, $\bar{f}(y) = \int_0^a d\xi f(\xi, y)$,

$$\int_{\mathbb{T}^2} \bar{h} f = \int_0^a dy \int_0^b dx \bar{h} \partial_x F + \int_0^b dy \tilde{h}(y) \bar{f}(y) = \int_{\mathbb{T}^2} \bar{h} D^u F + \int_0^b dy \tilde{h}(y) \bar{f}(y).$$

At this point a small obstacle appears, due to the fact that F is not $\mathcal{C}^{(1)}$. The problem is easily solved by approximating F by $\mathcal{C}^{(1)}$ functions F_ε such that $\|D^u F - D^u F_\varepsilon\|_1 \leq \varepsilon$. Then

$$\left| \int_{\mathbb{T}^2} \bar{h} D^u F \right| = \left| \int_{\mathbb{T}^2} \bar{h} D^u F - \int_{\mathbb{T}^2} \bar{h} D^u F_\varepsilon \right| \leq \|\bar{h}\|_\infty \varepsilon.$$

Hence, $\int_{\mathbb{T}^2} \bar{h} D^u F = 0$ also if the derivative is not continuous, consequently

$$\int_{\mathbb{T}^2} \bar{h} f = \int_{\mathbb{T}^2} \tilde{h} f. \tag{2.3.3}$$

By the arbitrariness of f (2.3.3) implies that $\bar{h} = \tilde{h}$ almost everywhere in $\mathring{R}$. Since R is arbitrary it follows that $\bar{h}$ is constant a.e. along the unstable direction.

A **global** argument is now needed to show that $\bar{h}$ must be constant.[15]

Proof of Proposition 2.3.1–a shortcut. Consider a line $\ell_a = \{x = a\}$. Clearly for each point $p = (a, y) \in \ell_a$ W^u_p intersects again ℓ_a at the point $(a, y + \omega_+ \mod 1)$ where $(1, \omega_+)$ is the unstable direction. Then we can consider the Dynamical Systems $(\ell_a, R_{\omega_+}, m)$, and the function $h_a = \bar{h}(a, y)$. By the previous discussion (and Fubini Theorem) it follows that, for almost every a, the function h_a is an $L^1(\ell_a, m)$ invariant function for the rotation R_{ω_+}; but we know that the irrational rotations are ergodic (see Examples 1.5.1), thus h_a =constant which implies immediately $\bar{h}$ constant. □

The above proof is simple but uses quite heavily the global properties of the foliation and of $\mathbb{T}^2$ to reduce the problem to one already studied (the irrational rotations). Clearly it is not clear how such a trick could work in more general situations. Again we would like a more flexible and dynamical argument.

[15] The fact that the argument is global, i.e. uses some properties of $\mathbb{T}^2$, reflects the fact that it is not as general has the Hopf argument which, instead, is of a completely local nature, as we will see better later (section 8.3).

PROOF OF PROPOSITION 2.3.1–DYNAMICAL. We will use a strategy already employed to prove the ergodicity of irrational rotations based on the existence of density points. Morally, this allows us to consider only rectangles. By topological mixing we can ensure that any two rectangle are crossed by the same unstable line (although it is more convenient to take preimmages of the rectangle and show that they must intersect a given unstable segment), so it is not possible that $\bar{h}$ has values different in the two rectangles. This very naïve argument can be made precise as follows.

If $\bar{h}$ it is not a.e. constant then there exists two sets A and B of positive measure such that $\bar{h}|_A > \bar{h}|_B$ a.e.. Let x_A and x_B be density points, of A and B respectively, and choose two rectangle R_A and R_B of the same size, smaller than $\frac{1}{4}$, and such that

$$\begin{aligned} m(A \cap R_A) &\geq \alpha m(R_A) \\ m(B \cap R_B) &\geq \alpha m(R_B) \end{aligned} \tag{2.3.4}$$

where $\alpha \in [0, 1)$ will be chosen later.

Let us consider $h \circ T^n$, clearly $h \circ T^n|_{T^{-n}A} > h \circ T^n|_{T^{-n}B}$ and the relations (2.3.4) hold for $T^{-n}A$, $T^{-n}R_A$ and $T^{-n}B$, $T^{-n}R_B$.

Next, let $\hat{R}_A \subset R_A$ and $\hat{R}_B \subset R_B$ be two shorter rectangles obtained by the original ones by chopping off a quarter of the length in the stable direction from each side. Let n_0 be so large that the stable length of the rectangles time λ^{n_0} is larger than one. Now chose another rectangle R, of size $\rho \leq \frac{1}{4}$, as you please. By topological mixing it follows that there exists $n > n_0$ such that $T^{-n}\hat{R}_A \cap R \neq \emptyset$ and $T^{-n}\hat{R}_B \cap R \neq \emptyset$. In addition, by the construction of $\hat{R}_A$ and $\hat{R}_B$ and the choice of n_0, it follows that $T^{-n}R_A$ and $T^{-n}R_B$ cross $\tilde{R}$ completely from one unstable side to the other, where $\tilde{R}$ is a rectangle containing R at its center and of double size. Moreover, the same quantitative argument of Lemma 2.3.6 shows that it is possible to choose n such that the stable length of $T^{-n}R_A$, $T^{-n}R_B$ is shorter than $8\lambda^2$.

Let L_A, L_B the two rectangles contained in $T^{-n}R_A \cap \tilde{R}$ and $T^{-n}R_B \cap \tilde{R}$, respectively, that cross $\tilde{R}$ from an unstable side to the other. Chose

$$\alpha = 1 - \frac{m(L_B)}{4m(R_B)} = 1 - \frac{m(L_A)}{4m(R_A)}.$$

The all point is that, on almost all the unstable lines in $\tilde{R}$, $\bar{h} \circ T^n$ is constant, so if one of this unstable lines intersects both $T^{-n}A$ and $T^{-n}B$ we have a contradiction. Thus, it must be

$$m\left(\left[\bigcup_{x \in T^{-n}A} W^u_x \cap L_B\right] \bigcap \left[\bigcup_{x \in T^{-n}B} W^u_x \cap L_B\right]\right) = 0.$$

Fubini theorem implies

$$m\left(\bigcup_{x\in T^{-n}A} W_x^u \cap L_B\right) = m\left(\bigcup_{x\in T^{-n}A} W_x^u \cap L_A\right) \geq m(T^{-n}A\cap L_A),$$

and

$$m\left(\bigcup_{x\in T^{-n}B} W_x^u \cap L_B\right) \geq m(T^{-n}B\cap L_B),$$

This yields:

$$\begin{aligned}
m(L_B) &\geq m(T^{-n}A\cap L_A) + m(T^{-n}B\cap L_B)\\
&\geq m(T^{-n}A\cap T^{-n}R_A) - m(T^{-n}R_A\backslash L_A)\\
&\quad + m(T^{-n}B\cap T^{-n}R_B) - m(T^{-n}R_B\backslash L_B)\\
&\geq 2\{\alpha m(T^{-n}R_B) - m(T^{-n}R_B) + m(L_B)\}\\
&\geq \frac{3}{2}m(L_B)
\end{aligned}$$

which is a contradiction. This shows that is not possible that the unstable manifolds starting at $T^{-n}A$ systematically avoid $T^{-n}B$.

Hence, $\bar{h}$ is constant, but then $\bar{h} = \int_{\mathbb{T}^2}\bar{h} = \bar{\mu}(1) = \mu_0(1)$. We have just proved that Γ consists of only one measure: the Lebesgue measure. Thus

$$\lim_{n\to\infty}\int_{\mathbb{T}^2} hf\circ T^n dm = \int_{\mathbb{T}^2} hdm\int_{\mathbb{T}^2} fdm,$$

for each $g,\, f \in C^{(1)}(\mathbb{T}^2)$. The mixing follows by the same approximation argument used in the Fourier series analyses. $\square$

2.4 Shadowing

In this section we explore the topological complexity of the dynamics of our model systems. I have already remarked that when such a strong instability with respect to the initial condition is present it is impossible to follow exactly an orbit of the system. In fact if we compute (e.g. with a computer) the orbit of the initial point $x\in\mathbb{T}^2$, due to round off errors we do not get an orbit but rather a *pseudo-orbit*.

Definition 2.4.1 *Give an systems (X,T), X Riemannian manifold, an infinite sequence $\{x_i\}_{i\in\mathbb{Z}}\subset\mathbb{T}^2$ is called an ε-pseudo orbit if, for all $i\in\mathbb{Z}$,*

$$d(x_{i+1}, Tx_i)\leq\varepsilon.$$

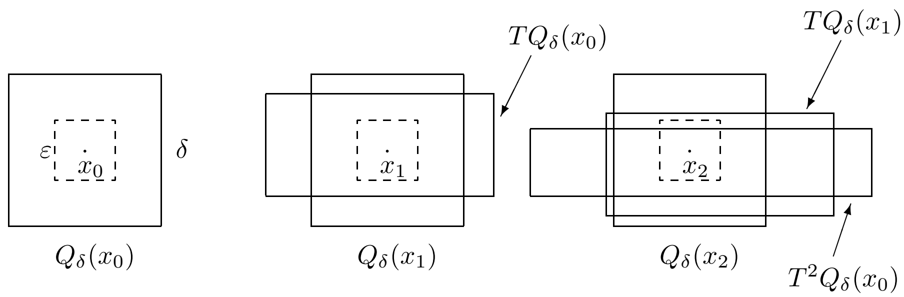


Figure 2.2: Intersection between $T^nQ_\delta(x_0)$ and $Q_\delta(x_n)$

Which means exactly that at each step an error of order ε is allowed.

The following result, beside being very useful, is a partial replay to the argument that it is not possible to follow orbits on a computer. Although the result is quite general, we state, and prove, it in our special context.

Proposition 2.4.2 *For each $\delta > 0$ there exists and $\varepsilon > 0$ such that, if $\{x_i\}$ is a ε-pseudo-orbit for the Arnold cat, then there exists $\xi \in \mathbb{T}^2$ such that*

$$d(x_i, T^i\xi) \leq \delta \quad \forall i \in \mathbb{Z}.$$

That is, there exists an orbit that δ-shadows the pseudo-orbit, moreover such an orbit is unique.

Proof. As usual we consider rectangular (better yet: square) neighborhood of points. So, let $Q_\varepsilon(x)$ be a square of size ε centered at x with sides parallel to the stable and unstable direction, respectively.

Next, let us consider $TQ_\delta(x_0)$, since $d(Tx_0, x) \leq \varepsilon$, if $\frac{\delta}{2\lambda} + \varepsilon < \frac{\delta}{2}$ and $\frac{\lambda\delta}{2} - \varepsilon > \frac{\delta}{2}$, then $TQ_\delta(x_0)$ crosses $Q_\delta(x_1)$ completely from the stable side to the other stable side. Thus, provided we choose $\delta \geq \frac{2\lambda}{\lambda-1}\varepsilon$, we have the picture of the intersection between rectangle that we have already learned to like.

Of course the same transversal intersection takes place for each $TQ_\delta(x_i)$ and $Q_\delta(x_{i+1})$. This immediately implies that $T^nQ_\delta(x_0)$ crosses $Q_\delta(x_n)$ from one stable side to the other (see figure 2.2)

Thus $K_n = T^{-n}(T^nQ_\delta(x_0) \cap Q_\delta(x_n))$ is a sequence of nested ($K_{n+1} \subset K_n$) vertical rectangles. The unstable side of K_n is of size $\lambda^{-n}\delta$ while the stable side is of size δ.

Clearly, if $\xi \in K_n$, then

$$d(T^i\xi, x_i) < \delta \quad \forall i \in \{0, \dots, n\}.$$

We can then consider the vertical line $K_\infty = \cap_{n\in\mathbb{N}} K_n$, by construction K_∞ consists of points whose orbit δ shadows $\{x_i\}_{i\in\mathbb{N}}$. By doing the same exact construction in the past we obtain an horizontal line $\tilde{K}_\infty$ of points that δ shadows $\{x_{-i}\}_{i\in\mathbb{N}}$. The theorem is then proven by choosing $\{\xi\} = \tilde{K}_\infty \cap K_\infty$.

the uniqueness should be obvious from the construction. In alternative the reader can prove it by contradiction. □

The above theorem is not so helpful from the measure theoretical point of view, since it could happen that the set of trajectories that shadow pseudo-orbits are of measure zero. (*say more*)

Nevertheless, it is very useful from the topological point of view (see Problem 2.15 for a dim glimpse to such possibilities).

2.5 Markov partitions

In all the above constructions the concept of rectangle has played a key rôle. In this section we present a construction that is the glorification of such a point of view.

Consider the stable and unstable manifolds of zero and prolong them until they meet (of course when they meet we meet an old friend: an homoclinic intersection) few times.

Clearly in such a way we have obtained a partition of $\mathbb{T}^2$. Such a partition consists of rectangles with sides that are either stable or unstable manifolds. We call them respectively the stable and the unstable sides of the rectangles. A partition is Markov if the preimage of each unstable side of a rectangle is contained in the unstable side of a rectangle and the image of every stable side is contained in the stable side of a rectangle. The reader can check that it is possible to use the above construction to have a Markov partition with (for example) three rectangles (see Figure 2.3 where the case $a = 1$ is drown).

Problems

2.1 Use the Diofantine properties of the stable and unstable direction to obtain better estimates of the decay of correlations. The Diofantine property refers to the following fact: if we normalize the eigenvectors in such a way that $v_\pm = (1, \omega_\pm)$, then $\omega_\pm$ are irrational numbers that are badly approximated by rationals: there exists $c \geq 0$ such that $|\omega_\pm - \frac{p}{q}| \geq \frac{c}{q^2}$ for each $p, q \in \mathbb{N}$, [1]. (Hint: ???)

2.2 Prove that the dynamical System $(\mathbb{T}^2, T^n, m)$ (where T is the Arnold cat map) is ergodic for each $n \in \mathbb{N}$. (Hint: the same proof as for $n = 1$.)

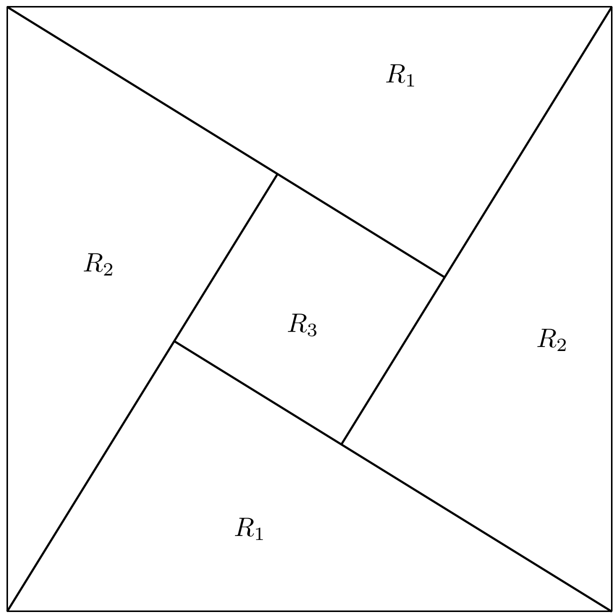


Figure 2.3: Markov partition

2.3 Let (X, T, μ) be a Dynamical Systems where X is a compact metric space, T is continuous, and μ charges the open sets (i.e. if $U \subset X$ is open, then $\mu(U) > 0$). Prove that for each $U \subset X$ open, there exist infinitely many $n \in \mathbb{N}$ such that $T^{-n}U \cap U \neq \emptyset$. (Hint: Poincaré Theorem.)

2.4 Let (X, T, μ) be an ergodic Dynamical Systems where X is a compact metric space, T is continuous, and μ charges the open sets. Prove that for each $U, V \subset X$ open, there exist infinitely many $n \in N$ such that $T^{-n}U \cap V \neq \emptyset$. (Hint:For each $k \in \mathbb{N}$, $A = \cup_{n \leq k} T^{-n}U$ is an invariant open set, if it does not intersect V, then $m(A) < 1$, thus, by ergodicity, $m(A) = 0$ which implies $U = \emptyset$.)

2.5 Prove Lemma 2.3.6. (Hint: As in the proof of Topologically mixing consider $T^{-n}W^s$, T^nW^u and chose n so large that $\lambda\delta > 2$ while the length L of W^u must satisfy $\lambda^{-n}L > 2$.)

2.6 Show that for each $x \in \mathbb{T}^2$ the global unstable manifold $W^u(x)$ is dense in $\mathbb{T}^2$. (Hint: *An algebraic proof*–Let us normalize $v_+ = (1, \omega)$, then ω is irrational. Clearly $W^u(x) = \{x + tv_+ \text{ mod } 1\}_{t \in \mathbb{R}}$. Consider a point $y = (y_1, y_2)$ and chose $t_0 = y_1 - x_1$, then, for each $n \in \mathbb{Z}$, $x + (t_0 +$

$n)v_1^+ \bmod 1 = (y_1, R_\omega^n \xi \bmod 1)$ where $\xi = x_2 + (y_1 - x_1)\omega \bmod 1$. Now, we know that R_ω has dense orbits (see Examples 1.5.1–Rotations), thus the result.
A dynamical proof–It follows Lemma 2.3.6 plus the fact that $T^{-n}W^u$ is shorter than W^u.)

2.7 Consider the global unstable foliation $\{W^u(x)\}$ and choose an interval of length (in the horizontal direction) one from each fiber.[16] Let K be the set obtained by the union of all such segments. Prove that K it is not measurable. (Hint: Define $R : \mathbb{T}^2 \to \mathbb{T}^2$ by $R(x, y) = (x, R_\omega y)$. Then, remember Problem 1.14.)

2.8 Let $W^u(x)$, $W^s(x) \subset U \subset \mathbb{R}^2$, $U = \overset{\circ}{U}$ and $\bar{U}$ compact, smooth manifolds ($\mathcal{C}^{(1)}$ curves) such that, the $\{W^{u(s)}(x)\}_{x\in U}$ are pairwise disjoint, $\partial W^{u(s)}(x) \subset \partial U$, if $z \in W^u(x) \cap W^s(y)$, then the angle between $W^u(x)$ and $W^s(y)$ at z is larger than some $\theta > 0$. In addition, assume that, calling $v^{u(s)}(x)$ the unit tangent vector to $W^{u(s)}(x)$ at x, $v^{u(s)} \in \mathcal{C}^{(1)}$. We will call such two foliation " $\mathcal{C}^{(1)}$ uniformly transversal foliations." Show that to each such a foliation it is associated a change of variable (a diffeomorphism $\Psi : U \to U$) and that to each change of variables is associated such a foliation. (Hint: ...)

2.9 Consider two $\mathcal{C}^{(1)}$ uniformly transversal foliations (as in Problem 2.8). Prove that if $f \in L^\infty$ is constant along almost every fiber of the two foliations, then it is constant almost everywhere. (Hint: Do the argument locally and change variables so that the foliations becomes straight.)

2.10 Consider the Bernoulli measures μ_p^B defined on Σ_2^+ (the one sided sequences with two symbols) by choosing $p_0 = p$ and $p_1 = 1 - p$ (see Examples 1.1.1–Bernoulli shift). Show that, if $p \neq p'$ then μ_p^B and $\mu_{p'}^B$ are mutually singular. (Hint: All the dynamical systems $(\Sigma_2^+, \tau, \mu_p^B)$ are ergodic–See Examples ?? and ??.)

2.11 Let μ_p be the measure on $[0, 1]$ obtained from μ_p^B by the binary representation of the real numbers (see Examples), let

$$F_p(x) := \mu_p([0, x]).$$

Show that, for each $p \in (0, 1)$, $F_p : [0, 1] \to [0, 1]$ is one one, onto, continuous. In addition, show that there exists $c \in \mathbb{R}^+$ such that, for each $p, q \in [\frac{1}{4}, \frac{3}{4}]$, holds

$$|F_p(x) - F_q(x)| \leq c|p - q|.$$

[16] The Axiom of choice again.

(Hint: Note that the cylinder correspond to intervals with end points made of binary rationals. It is then immediately clear that all the measures μ_p give positive measures to the open sets. To prove the last inequality prove the representation

$$F_p(x) = \sum_{n=0}^{\infty} \sigma_n \prod_{i=0}^{n} p^{\sigma_i}(1-p)^{1-\sigma_i}$$

where σ is the binary representation of x.)

2.12 Construct $\phi : [0,1] \to [0,1]$, invertible and continuous, such that there exists $A \subset [0,1]$ with $m(A) = 0$ while $m(\phi(A)) = 1$. (Hint: Any of the above F_p will do.)

2.13 Construct a continuous foliation Ψ in $[0,1]^2$made of $\mathcal{C}^\infty$ leaves (that is Ψ is a isomomorphism of $[0,1]^2$ and $\Psi(\cdot, y) \in \mathcal{C}^\infty$). In addition, the foliation must be made of straight lines in $\{(x,y) \in [0,1]^2 \mid x \in [0, \frac{1}{4}] \cup [\frac{3}{4}, 1]\}$ but is should not be absolutely continuous in the region $\{(x,y) \in [0,1]^2 \mid x \in [\frac{1}{4}, \frac{3}{4}]\}$. (Hint: Let $\varphi \in \mathcal{C}^{(\infty)}(\mathbb{R})$, $\varphi(\mathbb{R}) = [0,1]$, $\varphi(x) = 0$ for $x < 0$ and $\varphi(x) = 1$ for $x > \frac{1}{2}$. Then, using ϕ from Problem 2.12, define

$$\Psi(x,y) = \begin{cases} (x,y) & \text{if} \quad x \in [0, \frac{1}{4}] \\ (x, [1-\varphi(x-\frac{1}{4})]y + \varphi(x-\frac{1}{4})\phi(y)) & \text{if} \quad x \in [\frac{1}{4}, \frac{3}{4}] \\ (x, \phi(y)) & \text{if} \quad x \in [\frac{3}{4}, 1]. \end{cases}$$

Clearly the leaves $\Psi(\cdot, y)$ are $\mathcal{C}^{(\infty)}$, yet the foliation it is not absolutely continuous.)

2.14 Find two $\mathcal{C}^{(0)}$ uniformly transversal foliations in $[0,1]^2$, with $\mathcal{C}^{(\infty)}$ leaves, such that the Hopf argument does not apply. (Hint: Call Ψ_p, $p \in [\frac{1}{4}, \frac{3}{4}]$ the foliation constructed in the Problem 13 starting from the function F_p defined in the Problem 11. Choose a sequence p_n converging to one quarter, e.g. $p_n = \frac{1}{4} + \frac{1}{4^n}$, then let $x_n = \frac{1}{2} - \frac{1}{2n}$. Finally define the foliation

$$\Psi(x,y) = \begin{cases} \Psi_{p_n}(x_n + (x_{n+1} - x_n)x, y) & \text{for } x \in [x_n, x_{n+1}] \\ (x, F_{\frac{1}{4}}(y)) & \text{for } x \in [\frac{1}{2}, 1] \end{cases}$$

Further define the function $g : [0,1] \to [0,1]$ to be one on a set of full measure for $\mu_{\frac{1}{4}}$ and of zero measure for μ_{p_n} and zero otherwise. The functions f^+, f^- defined by

$$f^-(x,y) = \begin{cases} 0 & \text{for } x \in [0, \frac{1}{2}) \\ 1 & \text{for } x \in [\frac{1}{2}, 1] \end{cases}$$

and

$$f^+(x, \Psi(x, y)) = g(x),$$

are then constant on the vertical and the Ψ foliation respectively. Moreover they clearly are equal Lebesgue almost everywhere, nevertheless they are certainly not constant.)

2.15 Show (first without using Markov Partitions and then by using Markov partitions) that the Arnold cat has at least e^{cn} periodic orbits of period n, for some $c > 0$. (Hint: If we have a rectangle R of size ε, then $T^{-n}R \cap R \neq \emptyset$ for some $n \leq c \ln \varepsilon^{-1}$. Then, if $x \in T^{-n}R \cap R$ we consider the pseudo orbit $x_k = T^i x$ where $i = k \mod n$. Then Proposition 2.4.2 implies the existence of a periodic orbit in an ε-neighborhood R_ε of R. On the other hand the boxed $T^{-k}R_\varepsilon$, $k \in \{0, \dots, n\}$ invade a part of $\mathbb{T}^2$ of measure $c\varepsilon^2 \ln \varepsilon^{-1}$. The argument is then concluded taking boxes in the remaining space and continuing until all the available space is exhausted. On the other hand, if one takes in account Markov partions, then the number of periodic orbits is given–appart from the non-invertibility of the coding–by the number of periodic simbolic sequences of period n.)

Notes

Hopf history and ref

Mention Young-Robinson example

. . . .

CHAPTER 3

Hyperbolic Systems–general facts

This chapter is design to give an idea of the general results of hyperbolic theory. Since such a theory covers a rather vast landscape and it contains very technical results our exposition is bound to be quite sketchy. Nevertheless, in the following chapters we will prove all the results we need in this book for the special setting of area preserving two dimensional maps (see section 4.4 and Problem3.7 for Oseledets and chapter 7 for the results on foliations).

3.1 Hyperbolicity

Our goal in this section is to introduce and discuss a class of systems for which we can hope to extend the results of the previous section. As we have seen, the key property the we used in the study of the Arnold cat were the expanding and contracting properties of the map. These are generalized in the following definition.

Definition 3.1.1 *By Hyperbolic System (with discrete time) we mean a Dynamical System* (X, T, μ) *such that* X *is a smooth compact Riemannian manifold (possibly with boundary),* T *is almost everywhere differentiable and there exists two measurable families of invariant*[1] *subspaces* $E^u(x)$, $E^s(x) \in \mathcal{T}_x X$ *transversal at almost each point,*[2] *and measurable functions* $\nu(x) > 1$, $c(x) > 0$ *such that for almost all* $x \in X$

$$\begin{aligned} \|D_x T^n v\| &\geq c(x)^{-1}\nu(x)^n \|v\| \quad \forall v \in E^u(x) \\ \|D_x T^n v\| &\leq c(x)\nu(x)^{-n} \|v\| \quad \forall v \in E^s(x). \end{aligned}$$

If the functions c, ν *can be chosen constant, then the system is called Uniformly Hyperbolic. If, in addition,* T *is a diffeomorphism, then the system is called Anosov (or sometimes C or U system).*

[1] That is $D_x T E^{s(u)}(x) = E^{s(u)}(Tx)$.

[2] That is, $E^u(x) \cap E^s(x) = \{0\}$ and $E^u(x) \oplus E^s(x) = \mathcal{T}_x X$ a.e.

The condition in Definition 3.1.1 is essentially equivalent to saying that two very close initial conditions almost certainly will grow apart at an exponential rate. This corresponds to a strong instability with respect to the initial conditions and characterizes the sense in which the dynamics of hyperbolic systems is a very complex one. Such complex behavior has recently captured the popular fantasy under the ambiguous name of *chaos*.

3.1.1 Examples

3.1.1.a *Rotations*

Clearly the rotations are not hyperbolic since $DT = 1$.

3.1.1.b *Dilation*

One can easily see that such a system is expanding, hence $E^u = \mathbb{R}$ and $E^s = \emptyset$.

3.1.1.c *Arnold cat*

We have studied it in detail in the previous chapter.

3.1.1.d *Baker*

In this case one direction is expanding and one is contracting, dimE^u =dimE^s = 1

A more general notion of hyperbolicity is the one of *hyperbolic set*.

Definition 3.1.2 *Given a diffeomorphism T of a manifold X, we say that $\Lambda \subset X$ is hyperbolic if Λ is compact, $T(\Lambda) = \Lambda$ and there exists two measurable families of invariant subspaces $E^u(x)$, $E^s(x) \in \mathcal{T}_x X$ transversal at each point and measurable functions $\nu(x) > 1$, $c(x) > 0$ such that for all $x \in \Lambda$*

$$\begin{aligned} \|D_x T^n v\| &\geq c(x)^{-1}\nu(x)^n \|v\| \quad \forall v \in E^u(x) \\ \|D_x T^n v\| &\leq c(x)\nu(x)^{-n} \|v\| \quad \forall v \in E^s(x). \end{aligned}$$

If the constants c, ν can be chosen independently of $x \in \Lambda$ then Λ is called Uniformly Hyperbolic.

3.1.2 Examples

3.1.2.a *Solenoid*

...

3.1.2.b *Smale Horseshoe*

In this case the set Λ is the one constructed in Examples 1.4.1 and $\dim E^s = \dim E^u = 1$.

3.1.2.c *Hyperbolic Julia sets*

if possible

3.1.2.d *Forced pendulum*

Same situations as for the horseshoe, see Examples 1.8.1.

A first interesting result concerning hyperbolic structures is the following.

Theorem 3.1.3 *For an Anosov map the distributions $E^u(x)$, $E^s(x)$ are everywhere defined, everywhere transversal and continuous.*

PROOF. For each $x \in X$ consider a neighborhood $U(x)$ contianed in a chart. Use the Cartesian structure of the chart to identify tangent vectros at different points.[3] Consider $\{y_i\}$, $\lim_{i\to\infty} y_i = x$. The there must exists a subsequence such that $E^u(y_i)$ converge since the set (Grassmannian) of subspaces (with the obvious topology) is compact. Let us call $E_+(x)$ one such limit. For each $v \in E_+(x)$ holds

$$\|D_x T^n v\| \geq \|D_{y_{i_k}} T^n v_k\|(1-\varepsilon) \geq (1-2\varepsilon)c^{-1}\nu^n\|v\|$$

for some n_k large enough and some $v_k \in E^u(y_{i_k})$. Thus $E_+(x)$ is contained in the unstable subspace at x. In the same manner it is easy to show that if v is a vectro expanding exponentially in the future, then it must belong to $E_+(x)$. The same holds for the constacting space $E_-(x)$, in addition the two subspaces are obviously transversal (see Problem 3.1). Accordingly we have a splitting defined everywhere and continuous. This in particular imples that their dimension is constant and that there must be a minimal "angle" between them. □

Definition 3.1.1 it is not particularly helpful in concrete cases since, in general, it is not clear how to verify if a systems is hyperbolic or not. A first step toward a better understanding is contained in the next section, then in chapter 4 we will study this issue further.

[3] Or, in a Riemannian manifold, one can use the Rienmanian connection. Anyhow it is irrelevant: any connecion will do.

3.2 Lyapunov exponents and invariant distributions

We start by a different and very helpful characterization of hyperbolicity obtained by introducing the so called Lyapunov Exponents (LE).

Definition 3.2.1 *For each $x \in X$, $v \in \mathcal{T}_x X$ we define*

$$\lambda(x, v) = \lim_{n\to\infty} \frac{1}{n} \log \|D_x T^n v\|.$$

If $\lambda(x, v)$ exists it is called "Lyapunov exponent" (LE).

It is interesting to notice that $\lambda(Tx, D_x Tv) = \lambda(x, v)$ (see Problem 3.2). Moreover it should be clear that, if the system is ergodic and the map invertible, then $\lambda(x, v)$, if it exists, can assume only finitely many values (see Problem 3.4).

The existence and properties of the LE have been studied in detail. Here we satisfy ourselves with a basic fact.

Theorem 3.2.2 (Oseledets [69], [76]) *For each (X, T, μ), X union of compact manifolds, T almost everywhere differentiable, if*

$$\int_X \|\log D_x T\| d\mu < \infty, \tag{3.2.1}$$

then, for almost all $x \in X$, there exists numbers $\{\lambda_1, \dots \lambda_{s(x)}\}$ and a flag of subspaces

$$\{0\} = V_0 \subset V_1(x) \subset \dots \subset V_{s(x)-1}(x) \subset V_{s(x)}(x) = \mathcal{T}_x X,$$

such that, for all $v \in V_k(x) \backslash V_{k-1}(x)$, the LE $\lambda(x, v)$ exists and equal λ_k.

Note that if T is invertible, $\{\lambda_i(x)\}$ is equal a.e. to $\{-\lambda_i^-(x)\}$ where $\{\lambda_i^-(x)\}$ are the LE of (X, T^{-1}, μ) (see Problem 3.5).

We will not prove Theorem 3.2.2 in the above generality (see ??? for a proof), but see section 4.4 for a very constructive proof in an important, but special, case.

To appreciate the relevance of the LE, in the present context, consider the following.

Theorem 3.2.3 *A system $(X, T, \mu,)$, where X is a Riemannian manifold and T is a diffeomorphism, is hyperbolic iff for almost all $x \in X$*

$$\lambda(x, v) \neq 0 \quad \forall v \in \mathcal{T}_x X, \;\; v \neq 0.$$

PROOF. Keep in mind that the L.E. are defined a.e. by Theorem 3.2.2 if the system is hyperbolic, then all the LE are different from zero. The other implication is almost as trivial. Define $E^s(x) = \{v \in \mathcal{T}_x \mid \lambda(x, v) < 0\}$; then consider the Dynamical system (X, T^{-1}, μ) and its LE $\lambda^-(x, v)$ and define $E^u(x) = \{v \in \mathcal{T}_x \mid \lambda^-(x, v) < 0\}$. Next, let

$$\rho(x) = \sup\{\lambda(x, v),\ \lambda_-(x, w) \mid v \in E^s(x),\ w \in E^u(x)\}$$

clearly $\rho(x) < 0$ a.e.. Then setting $\nu(x) = e^{-\rho(x)/2}$ and

$$c(x) = \sup\{\nu(x)^n \|D_x T^n v\|,\ \nu(x)^n \|D_x T^{-n} w\| \mid v \in E^s(x);\ w \in E^u(x)\}_{n\in\mathbb{N}},$$

the theorem is proven. □

3.2.1 Examples

3.2.1.a *Logistic map*

do by conjugation and by showing via a dyn. proof that $\lambda(x) > 0$ for each $x \notin \cup_{n\in\mathbb{N}} T^{-n}(\{0\})$.

The above facts show that the hyperbolic systems have an infinitesimal behavior similar to our toy model. Yet, in the arguments of the previous chapter it was essential the existence of local manifolds (lines). In the following sections we will see that hyperbolic systems do have local manifolds as well.

3.3 Invariant manifold of a fixed point

In order to gain a feeling for the type of results we will present in the next section let us consider the simplest possible case in which the existence of invariant manifolds arises: the Hadamard-Perron theorem.

Definition 3.3.1 *Given a smooth map $T : X \to X$, X being a Riemannian manifold, and a fixed point $p \in X$ (i.e. $Tp = p$) we call (local) stable manifold (of size δ) a manifold $W^s(p)$ such that*[4]

$$W^s(p) = \{x \in B_\delta(x) \subset X \mid \lim_{n\to\infty} d(T^n x, p) = 0\}.$$

Analogously, we will call (local) unstable manifold (of size δ) a manifold $W^u(p)$ such that

$$W^u(p) = \{x \in B_\delta(x) \subset X \mid \lim_{n\to\infty} d(T^{-n} x, p) = 0\}.$$

[4] Sometime we will write $W^s_\delta(p)$ when the size really matters. By $B_\delta(x)$ we will always mean the open ball of radius δ centered at x.

It is quite clear that $TW^s(p) \subset W^s(p)$ and $TW^u(p) \supset W^u(p)$ (Problem 3.8). Less clear is that these sets deserve the name "manifold." Yet, if one thinks of the Arnold cat at the point zero (which is a fixed point) it is obvious that the stable and unstable manifolds at zero are just segments in the stable and unstable direction, the next Theorem shows that this is a quite general situation.

Theorem 3.3.2 (Hadamard-Perron) *Consider an invertible map $T : U \subset \mathbb{R}^2 \to \mathbb{R}^2$, $T \in \mathcal{C}^{(1)}(U, \mathbb{R}^2)$, such that $T0 = 0$ and*

$$D_0T = \begin{pmatrix} \lambda & 0 \\ 0 & \mu \end{pmatrix} \tag{3.3.1}$$

where $0 < \mu < 1 < \lambda$.[5] *That is, the map T is hyperbolic at the fixed point 0. Then there exists stable and unstable manifolds at 0. Moreover, $\mathcal{T}_0W^{s(u)}(0) = E^{s(u)}(0)$ where $E^{s(u)}(0)$ are the expanding and contracting subspaces of D_0T.*

Proof. We will deal explicitly only with the unstable manifold since the stable one can be treated exactly in the same way by considering T^{-1} instead of T.

Since the map is continuously differentiable for each $\varepsilon > 0$ we can choose $\delta > 0$ so that, in a 2δ-neighborhood of zero, we can write

$$T(x) = D_0Tx + R(x) \tag{3.3.2}$$

where $\|R(x)\| \leq \varepsilon\|x\|$, $\|D_xR\| \leq \varepsilon$.

3.3.0.b ***Existence–a fixed point argument***

The first step is to decide how to represent manifolds. In the present case, since we deal only with curves, it seems very reasonable to consider the set of curves $\Gamma_{\delta,c}$ passing through zero and "close" to being horizontal, that is the differentiable functions $\gamma : [-\delta, \delta] \to \mathbb{R}^2$ of the form

$$\gamma(t) = \begin{pmatrix} t \\ u(t) \end{pmatrix}$$

and such that $\gamma(0) = 0$; $\|(1,0) - \gamma'\|_\infty \leq c$. It is immediately clear that any smooth curve passing through zero and with tangent vector, at each point, in the cone $\mathcal{C} := \{(a, b) \in \mathbb{R}^2 \mid |\frac{b}{a}| \leq c\}$, can be associated to a unique element of $\Gamma_{\delta,c}$, just consider the part of the curve contained in the strip $\{(x, y) \in \mathbb{R}^2 \mid |x| \leq \delta\}$. Moreover, if $\gamma \in \Gamma_{\delta,c}$ then $\gamma \subset B_{2\delta}(0)$, provided $c \leq 1/2$.

[5] Notice that if D_0T has eigenvalues $0 < \mu < 1 < \lambda$ then one can always perform a change of variables such that (3.3.1) holds.

Notice that it suffice to specify the function u in order to identify uniquely an element in $\Gamma_{\delta,c}$. It is then natural to study the evolution of a curve through the change in the associated function.

To this end let us investigate how the image of a curve in $\Gamma_{\delta,c}$ under T looks like.

$$T\gamma(t) = \begin{pmatrix} \lambda t + R_1(t, u(t)) \\ \mu u(t) + R_2(t, u(t)) \end{pmatrix} := \begin{pmatrix} \alpha_u(t) \\ \beta_u(t) \end{pmatrix}.$$

At this point the problem is clearly that the image it is not expressed in the way we have chosen to represent curves, yet this is easily fixed. First of all, $\alpha_u(0) = \beta_u(0) = 0$. Second, by choosing $\varepsilon < \lambda$, we have $\alpha'_u(t) > 0$, that is, α_u is invertible. In addition, $\alpha_u([-\delta, \delta]) \supset [-\lambda\delta + \varepsilon\delta, \lambda\delta - \varepsilon\delta] \supset [-\delta, \delta]$, provided $\varepsilon \le \lambda^{-1}$. Hence, α_u^{-1} is a well defined function from $[-\delta, \delta]$ to itself. Finally,

$$|\frac{d}{dt}\beta_u \circ \alpha_u^{-1}(t)| = \left|\frac{\beta'_u(\alpha_u^{-1}(t))}{\alpha'_u(\alpha_u^{-1}(t))}\right| \le \frac{\mu c + \varepsilon}{\lambda - \varepsilon} \le c$$

where, again, we have chosen $\varepsilon \le \frac{c(\lambda-\mu)}{1+c}$.

We can then consider the map $\tilde{T} : \Gamma_{\delta,c} \to \Gamma_{\delta,c}$ defined by

$$\tilde{T}\gamma(t) := \begin{pmatrix} t \\ \beta_u \circ \alpha_u^{-1}(t) \end{pmatrix} \tag{3.3.3}$$

which associates to a curve in $\Gamma_{\delta,c}$ its image under T written in the chosen representation. It is now natural to consider the set of functions $B_{\delta,c} = \{u \in \mathcal{C}^{(1)}([-\delta, \delta]) \mid u(0) = 0, \ |u'|_\infty \le c\}$ in the vector space $\mathrm{L}ip([-\delta, \delta])$.[6] As we already noticed $B_{\delta,c}$ is in one-one correspondence with $\Gamma_{\delta,c}$, we can thus consider the operator $\hat{T} : \mathrm{L}ip([-\delta, \delta]) \to \mathrm{L}ip([-\delta, \delta])$ defined by

$$\hat{T}u = \beta_u \circ \alpha_u^{-1} \tag{3.3.4}$$

From the above analysis follows that $\hat{T}(B_{\delta,c}) \subset B_{\delta,c}$ and that $\hat{T}u$ determines uniquely the image curve.

The problem is then reduced to studying the map $\hat{T}$. The easiest, although probably not the most productive, point of view is to show that $\hat{T}$ is a contraction in the sup norm. Note that this creates a little problem since $\mathcal{C}^{(1)}$ it is not closed in the sup norm (and not even $\mathrm{L}ip([-\delta, \delta])$ is closed). Yet, the set $B^*_{\delta,c} = \{u \in \mathrm{L}ip([-\delta, \delta]) \mid u(0) = 0, \sup_{t,s\in[-\delta,\delta]} \frac{|u(s)-u(t)|}{|t-s|} < c\}$ is closed (see Problem 3.9). Thus $\overline{B_{\delta,c}} \subset B^*_{\delta,c}$. This means that, if we can prove that the sup norm is contracting, then the fixed point will belong to $B^*_{\delta,c}$ and we will obtain only a Lipschitz curve. We will need a separate argument to prove that the curve is indeed smooth.

[6] This are the Lipschitz functions on $[-\delta, \delta]$, that is the functions such that $\sup_{t,s\in[-\delta,\delta]} \frac{|u(s)-u(t)|}{|t-s|} < \infty$.

Let us start to verify the contraction property. Notice that

$$\alpha_u^{-1}(t) = \lambda^{-1}t + \lambda^{-1}R_1(\alpha_u^{-1}(t), u(\alpha_u^{-1}(t))),$$

thus, given $u_1, u_2 \in B_{\delta,c}$, by Lagrange Theorem

$$\begin{aligned}|\alpha_{u_1}^{-1}(t) - \alpha_{u_2}^{-1}(t)| &\leq \lambda^{-1}|\langle\nabla_\zeta R_1,\, (\alpha_{u_1}^{-1}(t) - \alpha_{u_2}^{-1}(t), u_1(\alpha_{u_1}^{-1}(t)) - u_2(\alpha_{u_2}^{-1}(t)))\rangle| \\ &\leq \frac{\varepsilon}{\lambda}\left\{|\alpha_{u_1}^{-1}(t) - \alpha_{u_2}^{-1}(t)| + |u_1(\alpha_{u_2}^{-1}(t)) - u_2(\alpha_{u_2}^{-1}(t))|\right\}.\end{aligned}$$

This implies immediately

$$|\alpha_{u_1}^{-1}(t) - \alpha_{u_2}^{-1}(t)| \leq \frac{\lambda^{-1}\varepsilon}{1-\lambda^{-1}\varepsilon}\|u_1 - u_2\|_\infty. \tag{3.3.5}$$

On the other hand

$$\begin{aligned}|\beta_{u_1}(t) - \beta_{u_2}(t)| &\leq \mu|u_1(t) - u_2(t)| + |\langle\nabla_\zeta R_2,\, (0, u_1(t) - u_2(t))\rangle| \\ &\leq (\mu+\varepsilon)\|u_1 - u_2\|_\infty. \end{aligned} \tag{3.3.6}$$

Moreover,

$$|\beta'_u(t)| \leq \mu + \varepsilon. \tag{3.3.7}$$

Collecting the estimates (3.3.5, 3.3.6, 3.3.7) readily yields

$$\begin{aligned}\|\hat{T}u_1 - \hat{T}u_2\|_\infty &\leq \|\beta_{u_1}\circ\alpha_{u_1}^{-1} - \beta_{u_1}\circ\alpha_{u_2}^{-1}\|_\infty + \|\beta_{u_1}\circ\alpha_{u_2}^{-1} - \beta_{u_2}\circ\alpha_{u_2}^{-1}\|_\infty \\ &\leq \left\{[\mu+\varepsilon]\frac{\lambda^{-1}\varepsilon}{1-\lambda^{-1}\varepsilon} + (\mu+\varepsilon)\right\}\|u_1 - u_2\|_\infty \\ &\leq \sigma\|u_1 - u_2\|_\infty,\end{aligned}$$

for some $\sigma \in (0,1)$, provided ε is chosen small enough.

Clearly, the above inequality immediately implies that there exists a unique element $\gamma_* \in \Gamma_{\gamma,c}$ such that $\tilde{T}\gamma_* = \gamma_*$, this is the *local* unstable manifold of 0.

3.3.0.c *Regularity–a cone field*

As already mentioned, a separate argument it is needed to prove that γ_* is indeed a $\mathcal{C}^{(1)}$ curve.

To prove this, one possibility would be to redo the previous fixed point argument trying to prove contraction in $\mathcal{C}^{(1)}_{Lip}$ (the $\mathcal{C}^{(1)}$ functions with Lipschitz derivative); yet this would require to increase the regularity requirements on T. A more geometrical, more instructive and more inspiring approach is the following.

Define the cone field $\mathcal{C}_{\theta,h}(x,u) := \{\xi \in B_h(x) \mid (a,b) = \xi - x;\ a \neq 0;\ |\frac{b}{a} - u| \leq \theta\}$, with $|u| \leq c\delta$, $\theta \leq c\delta$ and $h \leq \delta$. By construction $B_h(x) \cap \gamma_* \subset \mathcal{C}_{c\delta,h}$ for each $x \in \gamma_*$. We will study the evolution of such a cone field on γ_*.

For all $\xi \in \mathcal{C}_{\theta,h}(x,u)$, if $(a,b) = \xi - x$ and $(\alpha,\beta) = T\xi - Tx$, it holds

$$(\alpha,\beta) = D_xT(a,b) + \mathcal{O}(C\|(a,b)\|^2).$$

Thus, setting $(\alpha',\beta') = D_xT(a,b)$ and $u' = \frac{\beta'}{\alpha'}$,

$$\left|\frac{\beta}{\alpha} - u'\right| \leq \mu\lambda^{-1}[c_1h + \theta],$$

for some constant c_1 depending only on T and δ. Accordingly, if h is small enough, there exists $\sigma \in (0,1)$ such that

$$T\mathcal{C}_{\theta,h}(x,u) \subset \mathcal{C}_{\sigma\theta,h}(Tx,u').$$

Hence, if $x \in \gamma_*$, $\gamma_* \cap B_{\sigma^{-n}h}(T^{-n}x) \subset \mathcal{C}_{c\delta,h}(T^{-nx},0)$ and, since $T^{-n}\gamma_* \subset \gamma_*$,

$$\gamma_* \cap B_h(x) \subset \mathcal{C}_{\sigma^n c,h}(x,v_n) \tag{3.3.8}$$

where $(a,av_n) = D_{T^{-n}x}T^n(1,0)$ and $h < \sigma^n c\delta$.

The estimate (3.3.8) clearly implies

$$\gamma_*'(x) = (1, \lim_{n\to\infty} v_n) \tag{3.3.9}$$

which indeed exists (see Problem 3.10). □

There is an issue not completely addresses in our formulation of Hadamard-Perron theorem: the uniqueness of the manifolds.[7] It is not hard to prove that $W^{s(u)}(p)$ are indeed unique (see Problem 3,11).

The point of view employed in the previous theorem is brought to its extreme consequences in [46].

There is another point of view that can be adopted in the study of stable and unstable manifolds: to "grow" the manifolds. This is done by starting with a very short curve in $\Gamma_{\delta,c}$, e.g. $\gamma_0(t) = (t,0)$ for $t \in [\lambda^{-n}\delta,\ \lambda^n\delta]$, and showing that the sequence $\gamma_n := T^n\gamma_0$ converges to a curve in the strip $[-\delta,\ \delta]$, independent of γ_0. From a mathematical point of view, in the present case, it corresponds to spell out explicitly the proof of the fixed point theorem. Nevertheless, as we will see in ???, it is a more suggestive point of view and it is more convenient when the hyperbolicity is non uniform. For example consider the map[8]

$$T\begin{pmatrix} x \\ y \end{pmatrix} := \begin{pmatrix} 2x - \sin x + y \\ x - \sin x + y \end{pmatrix} \tag{3.3.10}$$

[7] Namely the doubt may remain that a less regular set satisfying Definition 3.3.1 exists.

[8] Some times this is called *Lewowicz map*, see [1].

then 0 is a fixed point of the map but

$$D_0T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

is not hyperbolic, yet, due to the higher order terms, there exist stable and unstable manifolds (see Problems 3.12, 3.13, 3.14).

3.4 Invariant manifolds and foliations (existence)

The concept of stable and unstable manifold of a fixed point can be obviously generalized to periodic orbits and, less obviously but more interestingly, to any orbit:

Definition 3.4.1 *For each point $x \in X$ we define its stable (unstable) sets as*

$$W^{s(u)}(x) = \{z \in X \mid \lim_{n\to\infty(-\infty)} d(T^n x, T^n z) = 0\}.$$

Remark 3.4.2 *It will not surprise the reader that very often such sets are indeed manifolds, in this case they are called stable (unstable) manifolds. In the following we will discuss only this case.*

Very often the structure of the stable manifolds is quite complex (as we have already seen in the introduction) and for this reason it may be more convenient to talk about *local* stable (unstable) manifolds as we did already (see Definition 3.3.1).

Definition 3.4.3 *By local stable (unstable) manifolds at $x \in X$ we mean the connected component containing x of $B_\varepsilon(x) \cup W^{s(u)}(x)$, for some $\varepsilon > 0$.*

Clearly the local manifolds are never unique, since their size may vary, but they may be unique once the size is assigned. Sometimes local stable manifolds will be denoted by $W^s_{\text{loc}}(x)$ as well as $W^s_\delta(x)$, we will use the former only if some confusion can arise and the latter only if the size plays an explicit rôle in the argument.

Definition 3.4.4 *By $W^s_\delta(x)$ we mean a local stable manifolds such that $\partial W^s_\delta(x) \subset \partial B_\delta(x)$.*

For simplicity we will restrict ourself to stable manifolds until the end of the section, note that if the map is invertible, then the same considerations hold for the unstable manifold as well.

We will be mostly interested in the case in which almost every point has a stable manifold. Note that is $W^s(x)$ is a local stable manifold at x, then $TW^s(x)$ is a local stable manifold at Tx. This means that we have an invariant foliation

Definition 3.4.5 ....

The first interesting fact is the following.

Proposition 3.4.6 *If we have a stable foliation, then for almost all points $x \in X$ the associated stable manifold contracts exponentially. That is there exists $K \in \mathbb{R}^+$, $\nu \in (0,1)$:*

$$d(T^n\xi, T^n\eta) \leq K\nu^n \ell(x) \quad \xi, \eta \in W^s_{\mathrm{loc}}(x),$$

where $\ell(x)$ is the size of the local stable manifold at x.

PROOF. Let $\Lambda_\delta = \{x \in X \mid x$ has stable manifold of size $\delta\}$, by hypothesis

$$m(\cup_{k\in\mathbb{N}}\Lambda_{\frac{1}{k}}) = m(X).$$

Moreover, by definition, for each $\delta > 0$ and $x \in \Lambda_\delta$ there exists $n_\delta(x) \in \mathbb{N}$ such that

$$d(T^n\xi,\, T^n\eta) < \frac{1}{2}\ell(x) \quad \forall \xi, \eta \in W^s_\delta(x),\ n > n_\delta(x).$$

Let us then define $\Lambda_{\delta,L} := \{x \in \Lambda_\delta \mid n_\delta(x) \leq L\}$. Clearly $m(\cup_{L\in\mathbb{N}}\Lambda_{\delta,L}) = m(\Lambda_\delta)$ for each $\delta > 0$. By the above remarks we can choose δ, L such that $m(\Lambda_{\delta,L}) > 0$. We start by considering the *separated* return map $T_{\delta,L}(x) =: T^{n(\delta,L,x)}$ to the set $\Lambda_{\delta,L}$ (see 1.2) where $n(\delta, L, x) := \min\{n \geq L \ :\ T^n x \in \Lambda_{\delta,L}\}$. It is then natural to define the return times $\{n_j(x)\}$, $x \in \Lambda_{\delta,L}$, such that $T^{n_j(x)}x = T^j_{\delta,L}x$. Thus, for each $n \in \{n_j(x), \dots, n_{j+1}(x) - 1\}$ holds

$$d(T^n\xi,\, T^n\eta) \leq 2^{-j}\ell(x). \tag{3.4.1}$$

Lemma 3.4.7 *Given $U \subset X$ such that (3.4.1) holds, almost every $x \in U$ has a manifold $W_\delta(x)$ that contracts exponentially.*

PROOF. Let $j_n(x) := \frac{1}{n}\sum_{i=1}^{n}\chi_U(T^i x)$, then (3.4.1) reads

$$d(T^n\xi,\, T^n\eta) \leq 2^{-nj_n(x)}\ell(x)$$

but Birkhoff Theorem implies that $j_n \to j^+$ almost everywhere and that $A_0 := \{\in U \mid j^+(x) = 0\}$ has measure zero (see Problem .) thus for almost all $x \in U$ there exists $\bar{n}$ such that $j_n(x) \geq \frac{1}{2}j^+(x)$ for all $n \geq \bar{n}$. From this it follows immediately that there exists $K > 0$:

$$d(T^n\xi,\, T^n\eta) \leq K2^{-\frac{n}{2}j^+(x)}\ell(x)$$

□

□

The use of non-invariant sets on which the systems has some useful property–on which the previous proof is based–reaches its glorification in the notion of *Pesin sets* which are the key ingredoend in the proof of Theorem 3.4.8. In fact, the importance of the notion of hyperbolicity lies almost entirely in the following result.[9]

Theorem 3.4.8 (Pesin [71]) *Let (X, μ, T) be a Smooth Hyperbolic System, almost each point $x \in X$ has a local stable manifolds $W^s_\varepsilon(x)$, for some $\varepsilon > 0$, that is a manifold with the following properties:*

1. $x \in W^s_\varepsilon(x)$;
2. $\mathcal{T}_x W^s_\varepsilon(y) = E^s(y) \quad \forall y \in W^s(x)$;
3. *there exists $K > 0$, and $\nu \in (0, 1)$ such that, for each $y \in W^s_\varepsilon(x)$ $d(T^n x, T^n y) \leq K\nu^n d(x, y)$.*

and such manifolds are unique.

Note that, if T is invertible, the previous theorem applied to T^{-1} yield a local unstable manifold. Moreover, if $W^s(x)$ is a local stable manifold at x, then there must exist local stable manifolds at Tx as well (since $TW^s(x)$ is one of them). From now on we will assume that W^s are the local stable manifolds provided by Pesin Theorem, without entering in a real discussion concerning their size, which can be proven to enjoy some extremely mild form of uniformity.

We will not prove 3.4.8 in its full generality, but see 7 for a proof in a less general setting.

Definition 3.4.9 *At each point x for which the local stable manifold exists we can define the global stable manifold has*

$$\overline{W}^s(x) = \bigcup_{n=0}^{\infty} T^{-n} W^s(T^n x).$$

Remark 3.4.10 *One can then see that the general situation is not too dissimilar from the two dimensional automorphism of the torus, in particular, if the system is Anosov, then $\overline{W^s}$ turns out to be of infinite size. Yet, if the*

[9]To appreciate better the next theorem consider that the Oseledets Theorem provides invariant distributions made of stable and unstable subspaces, then Pesin theorem essentially says that such distributions are integrable. In general, to integrate a distribution are necessary both regularities and geometrical properties (crf. Frobenius Theorem [14]) that, typically, are not verified (regularity) and very hard to verify (geometry) in the present setting. Only the fact that the distribution is of a dynamical origin saves the day.

system is non-uniformly hyperbolic, then $\overline{W^s}$ *can be arbitrarily short or it may be long in its own metric but wiggle so much as to be contained in a very small ball.*

3.5 Invariant manifolds and foliations (regularity)

The only other property that was really used in chapter 2 was the possibility to use the stable and unstable foliation as the basis for new coordinates. The relevant property, since we are dealing with measures, is that the change of coordinates does not substantially alter the measure: that is the new measure (obtained by the invariant one via the change of coordinates) is absolutely continuous with respect to the invariant one. If we want to extend the applicability of such an argument we need some analogous property. The right generalization turns out to be the so called *absolute continuity* . To describe it let us consider $x \in X$ with $W^s(x) \neq \emptyset$. Then, for a.e. such x, there exists a sufficiently small neighborhood $U(x)$ such that, for any two manifolds W_1, W_2 transversal to $W^s(x)$, the set $A = \{z \in W_1 \cap U(x) \mid W^s(z) \cap W_2 \neq \emptyset\}$ is not empty. In addition, if by μ_{W_i} we mean the measure induced on W_i by the Riemannian metric, $\mu_{W_1}(A) > 0$. Let us define $\phi : A \to W_2$ by $\phi(z) = W^s(z) \cap W_2$. We state the following in the simplest case.

Theorem 3.5.1 (Pesin [71]) *If the system* (X, μ, T) *is smooth, hyperbolic and* μ *is the Riemannian volume, then, for almost all* $x \in X$ *the measure* $\phi_* \mu_{W_1}$ *is absolutely continuous with respect to the measure* μ_{W_2}*.*

In our example we had the extreme case $\phi_* \mu_{W_1} = \mu_{W_2}$, yet it turns out that the above weaker property suffices to push the Hopf argument through, as we will see in chapters 8, 9. For the time being let us only quote the following general fact:

Theorem 3.5.2 ([71],[60]) *If the system* (X, μ, T) *is smooth, hyperbolic and* μ *is the Riemannian volume, then it has, at most, countably many ergodic components.*[10]

3.6 Comments on the non-smooth case

The results of the last section, although quite general, have a shortcoming: the smoothness requirements. As we will see in the following, systems that are

[10] An ergodic component is a set which is the support of an invariant ergodic measure. The ergodic components of a measure are the measures associated to its ergodic decomposition. In other words μ can be written as the convex combination of at most countably many ergodic measures.

quite natural both from the mathematical point of view and from the physical one are not smooth–typically they have discontinuities. In this section we will discuss a class of systems called *smooth systems with singularities*. Although the theory of such systems has been done in great generality, here we will give a restrictive definition, just sufficient for our later purposes. See the notes at the end of the chapter for information on more general settings.

Definition 3.6.1 *By Smooth Dynamical System with singularities we mean a Dynamical Systems (X, T, μ), where*

- *X is the union of finitely many compact pieces X_i of $\mathbb{R}^n$, ∂X_i is the union of finitely many $n-1$ dimensional smooth manifolds.*
- *T is smooth outside a compact set $\mathcal{S}$. The singularity set $\mathcal{S}$ is the finite union of smooth $n-1$ dimensional manifolds with boundary $\mathcal{S}_i$, $\mathcal{S}_i \cup \mathcal{S}_j \not\subset \partial\mathcal{S}_i \cap \partial\mathcal{S}_j$ implies $i = j$. In addition, the boundary $\partial\mathcal{S}_i$ is the finite union of smooth $n-2$ dimensional manifolds.*
- *There exists $c_1, c_2 > 0$ such that*
$$\|D_x T\| + \|D_x^2 T\| + \|D_x T^{-1}\| + \|D_x^2 T^{-1}\| \leq c_1 \, dist\,(x, \mathcal{S})^{-c_2}.$$
- *The measure μ is absolutely continuous with respect to Lebesgue.*

Remark 3.6.2 *Note that the fact that (X, T, μ) is a Smooth Dynamical System with singularities does not implies immediately that the same holds for (X, T^k, μ). The problem is that the map T can be very wild near the set $\mathcal{S}$, so it is not clear that the singularity set of T^k will satisfy our requirements. Nevertheless, in the examples we will consider, all the Dynamical System (X, T^k, μ) will always be Smooth Dynamical System with singularities.*

Remark 3.6.3 *We will call a smooth Dynamical System with singularities invertible if T^{-1} is densely defined and (X, T^{-1}, μ) is itself a smooth Dynamical System with singularities.*

Note that the above conditions imply the applicability of Oseledec Theorem (Problem ???).

3.6.1 Examples

3.6.1.a *Backer map*

It is easy to check that the Backer map is a Smooth Dynamical System with singularities.

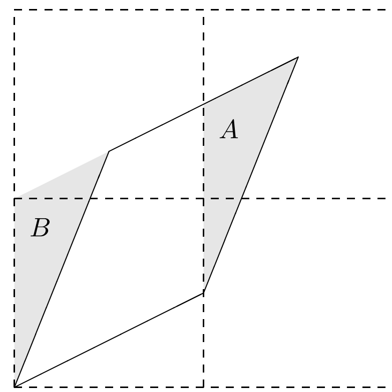


Figure 3.1: Definition of the map T

3.6.1.b *Discontinuous Arnold cat*

If we consider $(\mathbb{R}^2, T_0, m)$ defined by

$$T_0 \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 & a \\ a & 1+a^2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} =: A \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \tag{3.6.1}$$

with $a \notin \mathbb{Z}$, then it is not possible to quotient the system down to a torus preserving the continuity of the map. Yet, it is possible to construct a discontinuous version of the Arnold cat. To be more concrete, let us consider the case $a = 1/2$,

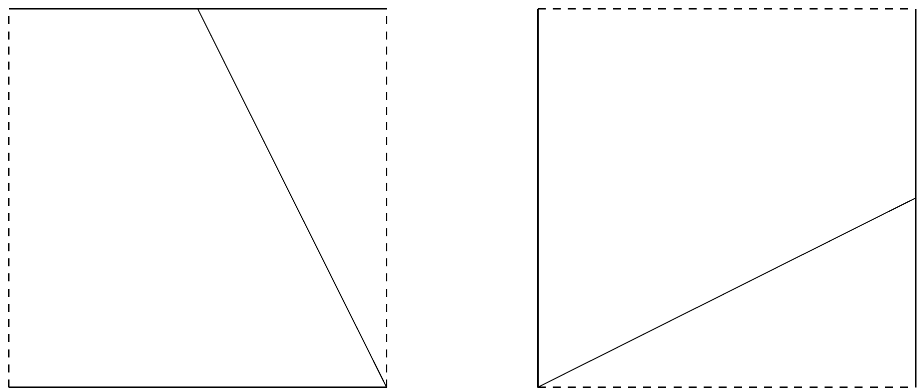
$$A = \begin{pmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & \frac{5}{4} \end{pmatrix}. \tag{3.6.2}$$

To define a map on the torus we must specify how to restric T_0 to the torus. To do so we consider the fundamental domain $D = \{(x, y) \in \mathbb{R}^2 \,|; 0 \leq x \leq 1;\ 0 \leq y \leq 1\}$. Figure 3.1 shows the image $T_0 D \subset \mathbb{R}^2$.

To define a map on the two dimensional torus $\mathbb{T}^2$ it suffices to cut the shaded region A from the image $T_0 D$ and translate it over the sadowed region B. After that one identifies the points $\mod 1$ as usual. We will call $T : \mathbb{T}^2 \to \mathbb{T}^2$ the map just defined. The discontinuities of T are the two lines $\{y = 0\}$ and $\{y = 2(1 - x)\}$. The explicit formula for the map $(x', y') = T(x, y)$ reads

$$\begin{aligned} x' &= x + \frac{y \mod 1}{2} \mod 1 \\ y' &= \frac{1}{2}(x + \frac{y}{2} \mod 1) + y \mod 1. \end{aligned} \tag{3.6.3}$$

We leave to the reader to check that the above formulae yields precisely the map described in Fiure 3.1.



$\mathcal{S}^+$ $\mathcal{S}^-$

Figure 3.2: Discontinuity lines of the maps T and T^{-1}

The stable and unstable directions can still be found by diagonalizing the matrix A. The two eigenvalues are $\lambda = (9 + \sqrt{17})/8$ and $\lambda^{-1} = (9 - \sqrt{17})/8$ while the unstable direction is given by $v^u = (1, (1 + \sqrt{17})/2)$ and the stable by $v^s = (1, (1 - \sqrt{17})/2)$. This means that the angle between the unstable direction and the vertical part of $\mathcal{S}^-$ is about 38 degrees, while the angle between v^u and the diagonal part of $\mathcal{S}^-$ is about 25 degrees (see figure 3.2).

The status of the stable and unstable manifolds is more complex. A line parallel to, say, the unstable direction is cut by $\mathcal{S}^-$ into pieces and if y and z belong to two different pieces the distance $d(T^n y, T^n z)$ does not decrease to zero as $n \to -\infty$. Since this last property is the definition of unstable manifold (and is of crucial importance in the Hopf method), the unstable (and stable) leaves are going to be much shorter than in the smooth case. Here is how we construct them. We will formulate everything for the unstable leaves alone, the construction of the stable leaves being completely similar.

We proceed inductively. Thus, for $x \in int(\mathcal{M}^-)$, we define $W_1^u(x)$ as the open segment of the line through x with the direction of the unstable eigenvector which contains x and has both endpoints on $\mathcal{S}^-$. The preimage $T^{-1}W_1^u(x)$ is by a factor of λ shorter than $W_1^u(x)$ and, in general, is cut into two, three or four pieces by $\mathcal{S}^-$.[11] We pick the piece which contains $T^{-1}x$ and take its image under T; this is our second approximate unstable leaf $W_2^u(x)$, i.e.,

$$W_2^u(x) = T\left(T^{-1}W_1^u(x) \cap W_1^u(T^{-1}(x))\right).$$

[11] It can be cut in three pieces only if it is sufficiently close to $(0,0)$ or $(1,1/2)$, while it can be cut in four only if it is long enough to be close to both points simultaneuosly. This last possibility can take place only for $T^{-1}W_1^u(x)$, teh connected pieces of $T^{-2}W_1^u(x)$ being already too short (see figure 3.2).

Unless $T^{-1}x \in \mathcal{S}^-$ the second approximate unstable leaf $W_2^u(x)$ is again an open segment containing x with endpoints on $\mathcal{S}^- \cup T\mathcal{S}^-$ and naturally $W_2^u(x) \subset W_1^u(x)$. By construction T^{-2} has no discontinuities on $W_2^u(x)$. Given $W_n^u(x)$, $n = 1, 2, \ldots,$ we define the $n+1$ approximate unstable leaf of x, $W_{n+1}^u(x)$, by

$$W_{n+1}^u(x) = T^n \left(T^{-n} W_n^u(x) \cap W_1^u(T^{-n}(x))\right).$$

If $x \notin \bigcup_{i=0}^{+\infty} T^i\mathcal{S}^-$ then this inductive procedure will yield a nested sequence of open segments containing x

$$W_1^u(x) \supset W_2^u(x) \supset \ldots$$

with endpoints on

$$\bigcup_{i=0}^{+\infty} T^i\mathcal{S}^- .$$

We can also describe this construction in the following way. First we consider a fairly long segment $W_1^u(x)$. Then we look at $T\mathcal{S}^-$, if it does not intersect $W_1^u(x)$ then we do not change it, if it splits $W_1^u(x)$ into several segments, then we keep the segment which contains x. We repeat it with $T^2\mathcal{S}^-$ and further images of $\mathcal{S}^-$, so that the segment may be cut shorter infinitely many times. The property $x \notin \bigcup_{i=0}^{+\infty} T^i\mathcal{S}^-$ ensures that x stays always strictly inside the segment. It is quite remarkable that, for almost every x, this inductive process shortens the segment only finitely many times. More precisely we have

Proposition 3.6.4 *For almost all $x \in \mathbb{T}^2 \setminus \bigcup_{i=0}^{+\infty} T^i\mathcal{S}^-$ the sequence of approximate unstable leaves of x stabilizes, i.e., there is a natural $N = N(x)$ such that*

$$\bigcap_{i=1}^{+\infty} W_i^u(x) = \bigcap_{i=1}^{N} W_i^u(x).$$

Proof. For $t > 0$, let

$$X_t = \{x \in \mathbb{T}^2 \mid d(x, \mathcal{S}^-) \leq t\}$$

where $d(\cdot, \cdot)$ is the distance of a point form a set. Because $\mathcal{S}^-$ is the finite union of three segments we have

$$\mu(X_t) \leq 3\ t.$$

Choosing $t_n = \frac{1}{n^2}$ we get

$$\sum_{n=1}^{+\infty} \mu(X_{t_n}) < +\infty,$$

hence also

$$\sum_{n=1}^{+\infty} \mu\left(T^n X_{t_n}\right) < +\infty.$$

It follows by the Borel-Cantelli Lemma that almost every x belongs to only finitely many of the sets

$$T X_{t_1}, T^2 X_{t_2}, \ldots,$$

which means that except for finitely many values of n

$$d(T^{-n}x,\, \mathcal{S}^-) > \frac{1}{n^2}.$$

Choosing $c(x) > 0$ sufficiently small we can take care of the finite number of exceptional values of n so that

$$d(T^{-n}x,\, \mathcal{S}^-) > \frac{c(x)}{n^2}$$

for each $n = 1, 2, \ldots$. Each time $W^u_{n+1}(x)$ is shorter than $W^u_n(x)$ we must have

$$d(T^{-n}x,\, \mathcal{S}^-) < \frac{\text{length}\,(W^u_n(x))}{\lambda^n}.$$

But then

$$\frac{c(x)}{n^2} < \frac{\text{length}\,(\mathcal{W}^u_n(x))}{\lambda^n} \leq \frac{\text{length}\,(W^u_1(x))}{\lambda^n} \leq 2\lambda^{-n},$$

which can hold for at most finitely many values of n. □

We define the unstable leaf only for points x in the set of full measure described in Proposition 4.1, by taking the intersection

$$W^u(x) = \bigcap_{i=1}^{+\infty} W^u_i(x).$$

In view of Proposition 3.6.4, for each $W^u(x)$, there are natural numbers $n_l(x)$ and $n_r(x)$ such that $T^{n_l(x)}W^u(x)$ has the left endpoint on $\mathcal{S}^-$ and $T^{n_r(x)}W^u(x)$ has the right endpoint on $\mathcal{S}^-$. Most importantly we have the exponential contraction of $W^u(x)$, i.e., for $y \in W^u(x)$ the distance

$$d(T^{-n}y,\, T^{-n}x) = \frac{d(y,\, x)}{\lambda^n} \to 0 \quad \text{as} \quad n \to +\infty.$$

Everything that we have done to construct the unstable leaves can be repeated for the stable leaves and they have analogous properties.

This it is encouraging, yet it is clearly not sufficient to perform the Hopf argument. We will discuss this further and with much more care in chapter 9. For the time being it suffices to notice that what we have seen so far implies that the discontinuous Arnold cat has, at most, countably many ergodic components (see Problem counterg).

The interest of the Smooth Dynamical System with singularities is that most of the theory of the previous sections holds for this more general systems. Namely Theorems 3.4.8, 3.5.1, [52].

3.7 Flows

All what we have described so far has a rather straightforward generalization in the case of flows, yet some natural changes are called for.

To appreciate the problem let us consider a flow, on a compact Riemannian manifold, generated by a smooth non-zero vector field V. By definition $\frac{d}{dt}\phi^t|_{t=0} = V(x)$ and $d\phi^t V(x) = V(\phi^t x)$, thus $\lambda(x, V(x)) = 0$. This is a rather general fact: the Lyapunov exponents in the flow direction is zero. The only relevant exception is constituted by hyperbolic fixed points (think of the unstable equilibrium point of the pendulum) that, in the previous example, was ruled out by the assumption that the vector field be non zero. We will consider only such case.

Consequently a flow is hyperbolic if the tangent space is split in three transversal subspaces E^s, E^u, E^0, where E^0 is the flow direction and corresponds to a zero Lyapunov exponents.

Oseledets Theorem (Theorem 3.2.2) holds unchanged with (3.2.1) obviously replaced by

$$\int_X \|\log d\phi^t\| d\mu < \infty.$$

For a smooth flow coming from a non vanishing vector field Theorem 3.2.3 holds unchanged as well, the same for Theorems 3.4.8, 3.5.1 and 3.5.2.

The extension to non-smooth cases require a bit of care. we will do it only for the symplectic case (Hamiltonian flows with collisions), which is the focus of the following chapters.

3.7.1 Examples

3.7.1.a *Smooth flows with collisions*

Let M be a smooth manifold with piecewise smooth boundary ∂M. We assume that the manifold M is equipped with a symplectic structure ω.[12] Given a smooth function H on M with *non vanishing* differential we obtain the non vanishing

[12] That is a non–degenerate closed antisymmetric two form.

Hamiltonian vector field $F = \nabla_\omega H$ on M by $\omega(\nabla_\omega H, v) = dH(v)$. The vector field F is tangent to the level sets of the Hamiltonian $M^c = \{z \in M | H(z) = c\}$.

We distinguish in the boundary ∂M the regular part, ∂M_r, consisting of the points which do not belong to more than one smooth piece of the boundary and where the vector field F is transversal to the boundary. The regular part of the boundary is further split into "outgoing" part, ∂M_-, where the vector field F points outside the manifold M and the "incoming" part, ∂M_+, where the vector field is directed inside the manifold. Suppose that additionally we have a piecewise smooth mapping $\Gamma : \partial M_- \to \partial M_+$, called the collision map. We assume that the mapping Γ preserves the Hamiltonian, $H \circ \Gamma = H$, and so it can be restricted to each level set of the Hamiltonian.

We assume that all the integral curves of the vector field F that end (or begin) in the singular part of the boundary lie in a codimension 1 submanifold of M.

We can now define a flow $\Psi^t : M \to M$, called a flow with collisions, which is a concatenation of the continuous time dynamics Φ^t given by the vector field F, and the collision map Γ. More precisely a trajectory of the flow with collisions, $\Psi^t(x),\ x \in M$, coincides with the trajectory of the flow Φ^t until it gets to the boundary of M at time $t_c(x)$, the collision time. If the point on the boundary lies in the singular part then the flow is not defined for times $t > t_c(x)$ (the trajectory "dies" there). Otherwise the trajectory is continued at the point $\Gamma(\Psi^{t_c} x)$ until the next collision time, i.e., for $0 \le t \le t_c\left(\Gamma(\Psi^{t_c(x)} x)\right)$

$$\Psi^{t_c + t} x = \Phi^t \Gamma \Psi^{t_c} x.$$

We define a flow with collisions to be symplectic, if for the collision map Γ restricted to any level set M^c of the Hamiltonian we have

$$\Gamma^* \omega = \omega,$$

for some non vanishing function β defined on the boundary. More explicitly we assume that for every vectors ξ and η from the tangent space $T_z \partial M^c$ to the boundary of the level set M^c we have

$$\omega(D_z \Gamma \xi, D_z \Gamma \eta) = \omega(\xi, \eta).$$

We restrict the flow with collisions to one level set M^c of the Hamiltonian and we denote the resulting flow by Ψ_c^t. This flow is very likely to be badly discontinuous but we can expect that for a fixed time t the mapping Ψ_c^t is piecewise smooth, so that the derivative $D\Psi_c^t$ is well defined except for a finite union of codimension one submanifolds of M^c. We will consider only such cases.

The symplectic volume $\wedge^d \omega$ is clearly invariant for the flow, so will be the measure μ_c obtained by restricting the symplectic volume to the manifold M^c. Clearly for such an invariant measure all the trajectories that begin (or end) in the singular part of the boundary have measure zero. With respect to the measure μ_c

the flow Ψ_c^t is a measurable flow in the sense of Definition 1.1.3 and we obtain a measurable derivative cocycle $D\Psi_c^t : T_xM^c \to T_{\Psi_c^t x}M^c$. We can define Lyapunov exponents of the flow Ψ_c^t with respect to the measure μ_c, if we assume that

$$\begin{aligned}&\int_{M^c} \log_+ \|D_x\Psi_c^t\| d\mu_c(x) < +\infty \\ &\int_{\partial M^c_-} \log_+ \|D_y\Gamma\| d\mu_{cb}(y) < +\infty \end{aligned} \tag{3.7.1}$$

(cf.[69],[76]).

Problems

3.1 Show that if a system is uniformly hyperbolic, then the distributions are uniformly transversal. (Hint: Let $v \in E^u$ and $w \in E^s$, $\|v\| = \|w\| = 1$, and let $\cos\theta := \langle v, w\rangle$ be the angle betwen them.

$$\|D_xT^n\|\theta^2 \geq \|D_xT^n\|2(1 - cos\theta) \geq \|D_xT^n(v - w)\| \geq c^{-1}\nu^n - c\nu^n.$$

the result follows by choosing n large enough and by the continuity of D_xT.)

3.2 Prove that $\lambda(Tx, D_xTv) = \lambda(x, v)$.

3.3 Prove that $\lambda(x, v + w) \leq \max\{\lambda(x, v), \lambda(x, w)\}$ and $\lambda(x, \alpha v) = \lambda(x, v)$ for each $\alpha \in \mathbb{R}$, if they all exist. (Hint: Just apply the definition of LE and note that

$$\lambda(x, v + w) \leq \lim_{n\to\infty} \max\{\frac{1}{n}\log\|D_xT^nv\|, \frac{1}{n}\log\|D_xT^nw\|\}.$$

)

3.4 Assuming only that the LE are well defined a.e., prove that, if (X, T, μ) is ergodic, X is a d dimensional manifold and T a diffeomorphism, then there exists d numbers $\{\lambda_i\}$ such that the Lyapunov exponents $\lambda(x, v) \in \{\lambda_i\}$ a.e.. (Hint: For each $\alpha \in \mathbb{R}$ define $V_\alpha(x) := \{v \in \mathcal{T}_xX \mid \lambda(x, v) \leq \alpha\}$. By Problem 3.3 $V_\alpha(x)$ is a linear vector space and, by Problem 2 the distribution V_α is invariant. Then $d_\alpha(x) := \dim V_\alpha(x)$ is an invariant function, thus a.e. constant for each α. In addition, d_α is an increasing function of α and can assume only the values $\{0, \ldots, d\}$. Thus there are at most $s \leq d$ $\{\alpha_j\}$ where d_α jumps. But this means that the LE are discrete. In fact, let $v \in V_\alpha(x)\backslash V_\beta(x)$, $\alpha > \beta$, then for each $w \in \text{span}\{v, V_\beta(x)\}$ it is easy to compute that $\lambda(x, w) = \lambda(x, v) > \beta$, which means: the LE is constant over $V_\alpha(x)$ apart for lower dimensional subspaces. In addition, we have a flag of subspaces $\{V_i\}_{i=0}^s$, $s \le d$, such that $V_\alpha \in \{V_i\}_{i=0}^s$ for each $\alpha \in \mathbb{R}$. Hence, if $V_\alpha \supset V_i$ but $V_\alpha \not\supset V_{i+1}$ it must be $V_\alpha = V_i$, thus if $v \in V_\alpha$ but $v \notin V_{i-1}$ $\lambda(x,v) = \alpha_i$ where $\alpha_i = \inf\{\alpha \in \mathbb{R} \mid V_\alpha \supset V_i\}$.)

3.5 Show that, if T is invertible, $\{\lambda_i(x)\}$ is equal a.e. to $\{-\lambda_i^-(x)\}$ where $\{\lambda_i^-(x)\}$ are the LE of (X, T^{-1}, μ).

3.6 Show that

$$\lim_{n\to\infty} \frac{1}{n} \log|\det(D_xT^n)|$$

exists almost everywhere. (Hint: Apply BET.)

3.7 Let (X, T, μ) be a Dynamical Systems, X a compact Riemannian manifold and T a.e. differentiable. Suppose that there exists a one-dimensional distribution $E(x)$ such that $D_xTE(x) = E(Tx)$. Prove, without using Oseledets theorem, that for each $v \in E(x)$ the LE $\lambda(x,v)$ is well defined. (Hint: Let $v(x) \in E(x)$, $\|v(x)\| = 1$, then $D_xTv(x) = \alpha(x)v(Tx)$ and thus $D_xT^nv(x) = \prod_{i=1}^n \alpha(T^ix)v(T^ix)$. Then the result follows by the BET.)

3.8 Show that, if p is a fixed point, then $TW^s(p) \subset W^s(p)$ and $TW^u(p) \supset W^u(p)$.

3.9 Prove that the set $B^*_{\delta,c}$ in section 3.3 is closed with respect to the sup norm $\|u\|_\infty = \sup_{t\in[-\delta,\delta]} |u(t)|$.

3.10 Prove that the limit in (3.3.9) is well defined.

3.11 Prove that, in the setting of Theorem 3.3.2, the unstable manifold is unique. (Hint: This amounts to show that the set of points that are attracted to zero are exactly the manifolds constructed in Theorem 3.3.2. Use the local hyperbolicity to show that.)

3.12 Consider the Levowich map (3.3.10), show that, given the set of curves $\Gamma_{\delta,c} := \{\gamma : [-\delta, \delta] \to \mathbb{R}^2 \mid \gamma(t) = (t, u(t));\ \gamma(0) = 0;\ |u'(t)| \in [c^{-1}t, ct]\}$, it is possible to construct the map $\tilde{T} : \Gamma_{\delta,c} \to \Gamma_{\delta(1+c^{-1}\delta),\, c}$ in analogy with (3.3.3).

3.13 In the case of the previous problem show that for each $\gamma_i \in \Gamma_{\delta,c}$ holds $d(\tilde{T}\gamma_1, \tilde{T}\gamma_2) \le (1 - c\delta)d(\gamma_1, \gamma_2)$.

3.14 Show that for the Levowich map zero has a unique unstable manifold. (Hint: grow the manifolds, that is, for each $n > 1$ define $\delta_n := \frac{\rho}{n}$. Show that one can choose ρ such that $\delta_{n-1} \geq \delta_n(1 + c^{-1}\delta_n)$. according to Problem 12 it follows that $\tilde{T} : \Gamma_{\delta_n,c} \to \Gamma_{\delta_{n-1},c}$. Moreover,

$$d(\tilde{T}^{n-1}\gamma_1, \tilde{T}^{n-1}\gamma_2) \leq \prod_{i=1} n(1 + c^{-1}\delta_i) d(\gamma_1, \gamma_2).$$

Finally, show that, setting $\gamma_n(t) = (0, t) \in \Gamma_{\delta_n,c}$, the sequence $\tilde{T}^{n-1}\gamma_n$ is a Cauchy sequence that converges in $\mathcal{C}^{(0)}$ to a curve in $\Gamma_{1,c}$ invariant under $\tilde{T}$.)

3.15 Prove that Oseledets Theorem can be applied to smooth Dynamical Systems with singularities. (Hint: Just check the (3.2.1) is satisfied.)

3.16 Prove that the discontinuous Arnold cat (Examples ??) has, at most, countably many ergodic components. (Hint: Imitate the Hopf argument perform in chapter 2. First of all notice that, by Fubini theorem, almost all segment has a full measure of point on which f^+ and f^- are well defined and equal. Thus almost all points have a stable manifold of some length with almost all point on it with unstable manifold of some length. This allows immediately to construct that almost all points belong to an invariant set of positive measure on which the map is ergodic.Clearly there can be at most countably many such sets.)

3.17 define cocycles (Hint:The derivative of the flow with collisions can be also naturally factored onto the quotient of the tangent bundle TM^c of M^c by the vector field F, which we denote by $\widehat{T}M^c$. Note that for a point $z \in \partial M^c$ the tangent to the boundary at z can be naturally identified with the quotient space. We will again denote the factor of the derivative cocycle by

$$A^t(x) : \widehat{T}_x M^c \to \widehat{T}_{\Psi_c^t x} M^c.$$

We will call it the transversal derivative cocycle. If the derivative cocycle has well defined Lyapunov exponents then the transversal derivative cocycle has also well defined Lyapunov exponents which coincide with the former ones except that one zero Lyapunov exponent is skipped.)

Notes

This point of view (fixed point for manifolds) is brought to in extreme consequences in the pugh-shub.

Mentions conjugation with the linear part (Hartman-Grobman theorem)

Mentions Katok-Strelchyn

CHAPTER 4

Establishing hyperbolicity–
the magic of cones

In chapter 2 we have taken advantage of the hyperbolic structure of the map in a very explicit manner, yet non linear maps may be hyperbolic but this cannot be seen by naively analyzing their derivative. Hence the necessity to have a tool to establish the hyperbolicity of a given system.

Our next task is to develope a general approach which allows to establish when the Lyapunov exponents are different from zero almost everywhere.

Although the results presented below look quite general, the reader should be cautioned that their concrete application has been possible only in fairly simpe situations, when the distributions are roughly constant on the phase space. The really general case, when the distributions are only measurable and therefore drasically dependent from the geometry of the orbits, is wide open.

4.1 The two dimensional case

We start by dealing with the area preserving two dimensional case in order to explain the basic idea.

Theorem 4.1.1 (Wojtkowski [92]) *Let* (X, μ, T) *be a dynamical system where* X *is a compact two dimensional Riemannian manifold,* μ *is the Riemannian volume,*[1] T *a diffeomorphism of* X *and*

$$\int_X \log \|DT\| d\mu < \infty.$$

[1] In fact, it is the symplectic not the Riemannian structure that matters; yet, in two dimension, there is not much difference. On the contrary, in section 4.2 this difference will be made explicit.

If there exists a measurable family of convex two sided cones $\mathcal{C}(x) \subset \mathcal{T}_x X$ such that, for almost all $x \in X$, there exists $n \in \mathbb{N}$ with the property[2]

$$D_x T^n \mathcal{C}(x) \subset int(\mathcal{C}(T^n x)) \cup \{0\},$$

then the Lyapunov exponents are different from zero almost surely.

A system with a family of cones satisfying the hypotheses of the Theorem 4.1.1 is called *eventually strictly monotone.*[3]

Proof. Let $x \in X$ and $n \in \mathbb{N}$ such that

$$D_x T^n \mathcal{C}(x) \subset \operatorname{int}(\mathcal{C}(T^n x)) \cup \{0\}.$$

The first thing to notice is that it is possible to make an orientation preserving change of coordinates (i.e., a change of coordinates via a matrix with positive determinant) both in $\mathcal{T}_x X$ and in $\mathcal{T}_{T^n x} X$ such that, in the new coordinates, $\mathcal{C}(x)$ and $\mathcal{C}(T^n x)$ become the standard cone $\mathcal{C}_+ = \{v \in \mathbb{R}^2 \mid v_1 v_2 \geq 0\}$ and the Riemannian volume is the standard ones (see Problem 4.2). Viewed in this coordinates $D_x T^n$ becomes a two by two matrix with determinant equal to one, that maps $\mathcal{C}_+$ strictly into itself. Note that, since the cone family is measurable, the change of coordinates depends measurably by x.

To continue it is necessary to study a bit the general properties of the matrices enjoying the above mentioned properties. Notice that if we define a quadratic form $\mathcal{Q} : \mathbb{R}^2 \to \mathbb{R}$ by $\mathcal{Q}(v) = v_1 v_2$, then $\mathcal{C}_+ = \{v \in \mathbb{R}^2 \mid \mathcal{Q}(v) \geq 0\}$, so our task is to study the two by two matrices L with $\det(L) = 1$ and such that $\mathcal{Q}(v) \geq 0$, $v \neq 0$, implies $\mathcal{Q}(Lv) > 0$.[4]

Algebraic considerations

Let $v = (1, u)$ with $u \in \mathbb{R}^+$, which implies $v \in \mathcal{C}_+$, and

$$L = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

[2]A "(two sided) cone," in a linear space, is a set such that if ξ belongs to the set then $\lambda\xi$ belongs to the set for each $\lambda \in \mathbb{R}$. In addition, we require that the set is closed, has open interior and its complement is non void. Actually, this last conditions could be relaxed for x in a zero measure set without changing the following proof (see footnote 8 at page 110).By "two sided convex cone" we mean that each half cone is a convex set. Notice that, in $\mathbb{R}^2$, such a cone is defined uniquely by the two edges. By measurable we mean that the functions from X to the unit vectors, in the direction of the edges, are measurable.

[3]In fact, in the case of billiards a similar condition is called *sufficiency* but I find the above terminology more appropriate.

[4]We will call such matrices *Strictly monotone*. Note that if A, B are strictly monotone so are AB and BA.

with $\det(L) = 1$ (note that this is equivalent to L being symplectic, see section 4.2). Then, for each $u \geq 0$, we must have

$$0 < \mathcal{Q}(Lv) = ac + (ad + bc)u + bdu^2, \tag{4.1.1}$$

Setting $u = 0$, it must be $ac > 0$. On the other hand, since $(0, 1) \in \mathcal{C}_+$ and $L(0, 1) = (b, d)$, it must be $bd > 0$. Finally, if we compute the quadratic polynomial in its minimum $u_0 = -\frac{ad+bc}{2bd}$ we get, calling $v_0 = (1,\, u_0)$,

$$\mathcal{Q}(Lv_0) = -\frac{1}{4bd} < 0.$$

The above relation is possible only if $v_0 \notin \mathcal{C}_+$, which implies $u_0 < 0$ or $ad + bc > 0$, that is $ad > \frac{1}{2}$. Collecting the above results it follows that all the elements of the matrix L must be different from zero and, in addition, they must have the same sign. Since $\mathcal{Q}(v) = \mathcal{Q}(-v)$, without loss of generality we can assume them to be all positive.[5]

The next step is to define some measure of expansion for a strictly monotone matrix. A natural quantity to consider is:

$$\sigma(L) = \inf_{v \in \mathrm{int}(\mathcal{C}_+)} \sqrt{\frac{\mathcal{Q}(Lv)}{\mathcal{Q}(v)}}.$$

Choosing again $v = (1,\, u)$, it follows

$$\frac{\mathcal{Q}(Lv)}{\mathcal{Q}(v)} = \frac{(a + bu)(c + du)}{u} := f(u),$$

a simple computation shows that f attains its minimum value, for $u \geq 0$, at $u = \sqrt{\frac{ac}{bd}}$. Accordingly,

$$\sigma(L) = \sqrt{bc} + \sqrt{ad} = \sqrt{bc} + \sqrt{1 + bc} > 1. \tag{4.1.2}$$

Moreover, given two monotone matrices L_1, L_2, we have

$$\sigma(L_1 L_2) = \inf_{v \in \mathrm{int}(\mathcal{C}_+)} \sqrt{\frac{\mathcal{Q}(L_1 L_2 v)}{\mathcal{Q}(L_2 v)}} \sqrt{\frac{\mathcal{Q}(L_2 v)}{\mathcal{Q}(v)}} \geq \sigma(L_1)\sigma(L_2). \tag{4.1.3}$$

An interesting fact, that follows immediately from (4.1.2), is that $\sigma(L) > 1$ if and only if L is strictly monotone.

[5] Just consider the change of variable $x = y$, $y = x$.

Measure theoretical considerations

The point of measuring the expansion via the $\mathcal{Q}$-form is due to the following Lemma.[6]

Lemma 4.1.2 *If the Dynamical System (X, T, μ), X a two dimensional Riemannian manifold, T differentiable a.e. and μ the Riemannian volume, is eventually strictly monotone, then*

$$\liminf_{n\to\infty} \frac{1}{n} \ln \sigma(D_x T^n) > 0 \quad \mu\text{-a.e.}.$$

PROOF. Let $\nu : X \to \mathbb{R}^+ \cup \{\infty\}$ be defined by

$$\nu(x) := \liminf_{n\to\infty} \frac{1}{n} \ln \sigma(D_x T^n).$$

Then

$$\begin{aligned}
\nu(T^{-1}x) &= \liminf_{n\to\infty} \frac{1}{n} \ln \sigma(D_{T^{-1}x} T^n) \\
&\geq \liminf_{n\to\infty} \frac{1}{n} \{\ln \sigma(D_x T^{n-1}) + \ln \sigma(D_{T^{-1}x} T)\} \\
&\geq \liminf_{n\to\infty} \frac{1}{n} \ln \sigma(D_x T^n) = \nu(x), \qquad (4.1.4)
\end{aligned}$$

where we have used (4.1.3) and the fact that $\sigma(D_\xi T) \geq 1$ by monotonicity.

Let $A_0 = \{x \in X \mid \nu(x) = 0\}$, to prove the Lemma it suffices to show that $\mu(A_0) = 0$. To this end note that, since (4.1.4) implies $\nu(Tx) \leq \nu(x)$, ν and hence A_0 must be invariant.[7]

Consequently, suppose that $\mu(A_0) > 0$, then, for each $m \in \mathbb{N}$,

$$0 = \int_{A_0} \nu(x)\mu(dx) \geq \int_{A_0} \liminf_{n\to\infty} \frac{1}{n} \sum_{i=0}^{[\frac{n}{m}]-1} \ln \sigma(D_{T^{im}x} T^m),$$

where we have used (4.1.3) again. At this point we would like to use BET, yet we face a technical problem: we do not know if $\ln \sigma(D_x T^m)$ is integrable. Nevertheless, we are not interested in large values of $\ln \sigma(D_x T^m)$. It is then natural to define

$$\varphi_m(x) = \min\{\ln \sigma(D_x T^m),\, 1\}.$$

[6] It is interesting to remark that the next Lemma, together with the results of section 4.3, imply the existence of the stable and unstable distribution (see Examples 4.3.1-Cones and $\mathcal{Q}$-forms). Thus, since X is two dimensional, the existence a.e. of the L.E. follows as in Problem 3.7 without invoking Oseledec Theorem. The use of Oseledec Theorem is instead necessary in higher dimensions.

[7] Indeed, $\mu(\nu - \nu \circ T) = 0$ be the invariance of μ, which implies $\nu \circ T = \nu$ μ-almost everywhere.

Now $\varphi_m \in L^\infty(X,\mu)$, thus the ergodic average $\varphi_m^+ \in L^\infty(X,\mu)$; hence,

$$\begin{aligned}0 \geq \int_{A_0} \liminf_{n\to\infty} \frac{1}{n} \sum_{i=0}^{[\frac{n}{m}]-1} \varphi_m(T^{im}x) = \frac{1}{m}\int_{A_0} \lim_{n\to\infty} \frac{1}{n}\sum_{i=0}^{n-1} \varphi_m(T^{im}x) \\ = \frac{1}{m}\int_{A_0} \varphi_m^+(x) = \frac{1}{m}\int_\Lambda \varphi_m^+ = \frac{1}{m}\int_\Lambda \varphi_m. \qquad (4.1.5)\end{aligned}$$

That is $\varphi_m = 0$ a.e. in Λ. But this is a contradiction since, calling $B_m = \{x \in X \mid \sigma(D_xT^m) > 1\}$, the definition of eventually strictly monotone is equivalent to $\mu(\cup_{m\in\mathbb{N}} B_m) = \mu(X)$. Therefore there must exists an $m \in \mathbb{N}$ such that $\mu(\Lambda \cap B_m) > 0$, which implies

$$\int_\Lambda \varphi_m \geq \int_{\Lambda\cap B_m} \varphi_m > 0,$$

whereby contradicting (4.1.5). □

The relevance of what we have seen up to now for the estimation of the Lyapunov exponents depends on the trivial inequality

$$\|v\|^2 \geq 2\mathcal{Q}(v). \qquad (4.1.6)$$

The only real problem left is that, due to our change of variable to put the cones into their standard form, the euclidean norm $\|\cdot\|$ in the new variables no longer correspond to the original norm in X (let us call such original norm, at the point x, $\|\cdot\|_x$). Nevertheless, the two norms must be equivalent by construction,[8] hence there must exists an everywhere strictly positive measurable function $a(x) \leq 1$ such that, for each $v \in \mathcal{T}_xX$

$$a(x)^{-1}\|v\| \geq \|v\|_x \geq a(x)\|v\|.$$

Let us introduce the set $A(\varepsilon) = \{x \in X \mid a(x) > \varepsilon\}$, clearly $\cup_{\varepsilon>0} A(\varepsilon)$ has full measure. Now Poincaré theorem implies, if $\mu(A(\varepsilon)) \neq 0$, that almost all points in $A(\varepsilon)$ return to $A(\varepsilon)$ infinitely often. Let $x \in A(\varepsilon)$ be one of such points, then there exists a sequence n_k such that $T^{n_k}x \in A(\varepsilon)$ for each $k \in \mathbb{N}$.

Accordingly, for each $x \in X$ for which the L.E. exists,[9] $v \in \text{int}(\mathcal{C}(x))$, $\|v\|_x = 1$, and $m \in \mathbb{N}$ holds

[8]If we admit that $\mathcal{C}(x)$ can have an empty interior on a set of zero measure in X–see footnote 2 at page 107–then the two norms would be equivalent only almost everywhere. Nevertheless, this does not change the proof: call X_1 the incriminated set, then $X_2 := \cup_{n\in\mathbb{Z}} T^n X_1$ has also zero measure and it is an invariant set. We can then discard such a set and work on its complement without any other change in the following.

[9]This is a full measure set by Oseledets Theorem [69].

$$
\begin{aligned}
\lambda(x,\,v) &= \lim_{n\to\infty}\frac{1}{n}\log\|D_xT^nv\|_{T^nx} \geq \liminf_{k\to\infty}\frac{1}{n_k}\log\|D_xT^{n_k}v\|_{T^{n_k}x}\\
&\geq \liminf_{k\to\infty}\frac{1}{n_k}\log\|D_xT^{n_k}v\| + \liminf_{k\to\infty}\frac{1}{n_k}\log a(T^{n_k}x)\\
&\geq \liminf_{n\to\infty}\frac{1}{n}\log\|D_xT^nv\| - \lim_{k\to\infty}\frac{1}{n_k}\log\varepsilon^{-1}
\end{aligned}
$$

and, since $\mathcal{Q}(v)\neq 0$ and by (4.1.6),

$$
\begin{aligned}
\lambda(x,\,v) &\geq \liminf_{n\to\infty}\frac{1}{n}\log\sqrt{\frac{\mathcal{Q}(D_xT^nv)}{\mathcal{Q}(v)}}\\
&\geq \liminf_{n\to\infty}\frac{1}{n}\log\sigma\left(D_xT^n\right) > 0,
\end{aligned}
$$

due to Lemma 4.1.2.

We have then seen that, for almost every $x\in X$ and $v\in Int\,\mathcal{C}(x)$, $\lambda(x,\,v)\neq 0$. Next, note that the Lyapunov exponents of T are given by minus the Lyapunov exponents of T^{-1} (seeProblem 3.5 and vicinity). Moreover T^{-1} is strictly monotone with respect to $\mathcal{C}(x)^c$. Thus, by Oseledets Theorem 3.2.2 (or see section 4.4), almost every point must have a vector $v_-(x)$ such that $\lambda(x,v_-(x))<0$.[10] Obviously, given any vector $v\in\mathcal{T}X$ it follows $\lambda(x,\alpha v+\beta v_-(x))=\lambda(x,v)$, provided $\alpha\neq 0$, and this concludes the story. □

Remark 4.1.3 *The measurability assumption is a very weak hypothesis but cannot be eliminated. Indeed, if one constructs a cone family along the trajectories it can easily be made strictly monotone. Hence, if the system has zero Lyapunov exponents such a cone family cannot be measurable (see Problem 4.4).*

The above theorem provides us with a very powerful instrument to establish hyperbolicity for a given dynamical system.

To see how it works let us consider some simple examples.

4.1.1 Examples

4.1.1.a *Linear automorphysms of the Torus*

Consider the matrix

$$
L=\begin{pmatrix}1 & a\\ a & 1+a^2\end{pmatrix}
$$

[10] See Prob45 for an alternative argument.

with $a \in \mathbb{N}$ and the standard cone $\mathcal{C}_+ = \{(v, v) \in \mathbb{R}^2 \mid uv \geq 0\}$. Then

$$L \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} u + av \\ au + (1 + a^2)v \end{pmatrix}$$

shows that $\mathcal{C}_+$ is strictly monotone for L. Of course, this is a rather silly example since it is completely obvious that the map is hyperbolic, the next example is a little less trivial.

4.1.1.b *Perturbations of linear automorphysms of the Torus*

Consider a diffeomorphism $\Phi : \mathbb{T}^2 \to \mathbb{T}^2$ such that

$$\|(D\Phi - \mathbb{1})L\| < 1; \tag{4.1.7}$$

where L is defined as in the previous example, then the map T defined by $Tx := \Phi(Lx)$ is hyperbolic. To see this write

$$DT \begin{pmatrix} u \\ v \end{pmatrix} = L \begin{pmatrix} u \\ v \end{pmatrix} + (D\Phi - \mathbb{1})L \begin{pmatrix} u \\ v \end{pmatrix} := \begin{pmatrix} u + av \\ au + (1 + a^2)v \end{pmatrix} + \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

but $\max\{|\alpha|, |\beta|\} \leq \|(D\phi - \mathbb{1})L\| \|(u, v)\| \leq u + v$. Thus $\mathcal{C}_+$ is strictly monotone for DT.

It is interesting to notice that, already for this simple example, it would be not immediately clear how to establish hyperbolicity without using a cone language. In addition, remark that the full strength of Wojtkowski theorem it is not used here–since the cone family is strictly monotone.

4.1.1.c *Levowich map*

Let us consider the map $T : \mathbb{T}^2 \to \mathbb{T}^2$ defined by[11]

$$T \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x - \sin x + y \\ x - \sin x + y \end{pmatrix},$$

It is immediate to verify that

$$D_{(x,y)}T = \begin{pmatrix} 2 - \cos x & 1 \\ 1 - \cos x & 1 \end{pmatrix}$$

Since $\det(DT) = 1$, it follows that $(\mathbb{T}^2, T, m)$ is a Dynamical Systems. In addition, $D_x T\mathcal{C}_+ \subset \mathcal{C}_+$ strictly, apart from the zero measure set $\{x = 0\}$, so Theorem 4.1.1 applies.

[11] Note that here our torus has the periodicity of 2π instead than one as in the previous examples, this is just to have simpler formulae; the reader can easily reformulate the problem on the torus $\mathbb{R}^2 \mod 1$.

4.2 Higher dimension–an overview

The difficulties in extending to higher dimensions the previous results stem mainly from the large variety of possible cone shapes in higher dimension. It is far from obvious how to relate monotone properties of a given cone to the behavior of the Lyapunov exponents. One possible way is to generalize the approach based on quadratic forms. We will comment on this possibility in section 4.6. Yet, in the special case of *Symplectic Systems*, it is possible to develop a very rich theory which is astonishingly similar to the two dimensional one. Here we give a brief insight into this theory, but see [63], [64] and [92] for a much more detailed account.

Definition 4.2.1 *By symplectic Systems we mean a Dynamical System* (X, T, μ) *where* X *is a symplectic manifold,*[12] μ *is the symplectic volume*[13] *and* T *is a symplectic map.*[14]

Clearly one can also define a Symplectic Systems in continuous time, we have already seem the typical example: Hamiltonian systems (see Examples 1.1.1).

For the convenience of the reader we will present here some of the material from [94] and [63].

Let $\mathcal{W}$ be a linear symplectic space of dimension $2d$ with the symplectic form ω. For instance we call $\mathcal{W} = \mathbb{R}^d \times \mathbb{R}^d$ the standard linear symplectic space if

$$\omega(w_1, w_2) = \langle \xi^1, \eta^2 \rangle - \langle \xi^2, \eta^1 \rangle, \tag{4.2.1}$$

where $w_i = (\xi^i, \eta^i),\ i = 1, 2$, and $\langle \xi, \eta \rangle = \xi_1\eta_1 + \cdots + \xi_d\eta_d$.

The symplectic group $Sp\,(d, \mathbb{R})$ is the group of linear maps of $\mathcal{W}$ ($2d \times 2d$ matrices if $\mathcal{W} = \mathbb{R}^d \times \mathbb{R}^d$) preserving the symplectic form i.e., $L \in Sp\,(d, \mathbb{R})$ if

$$\omega(Lw_1, Lw_2) = \omega(w_1, w_2) \tag{4.2.2}$$

for every $w_1, w_2 \in \mathcal{W}$.

By definition a Lagrangian subspace of a linear $2d$ dimensional symplectic space $\mathcal{W}$ is a maximal subspace on which ω vanishes (Lagrangian subspaces are d dimensional, seeProblem 4.13).

[12]A symplectic manifold is a smooth manifold of even dimensions together with a symplectic form. By *symplectic form* we mean an antisymmetric differential two form ω which is closed and non degenerate. By "closed" we meand that $d\omega = 0$, by "non degenerate" that for all $v \in \mathcal{T}_xX$ if $\omega(v, w) = 0$ fora each $w \in \mathcal{T}_xX$, then $w = 0$, see [5] for more details.

[13]Given a symplectic form ω on a manifold of dimension $2d$ the $2d$ form $\wedge^d\omega$ is a volume form: the symplectic volume.

[14]A map is symplectic if is conserves the symplectic two form ω, that is, for each $x \in X$ and vectors $v, w \in \mathcal{T}_xX$, holds $\omega(D_xTv, D_xTw) = \omega(v, w)$.

Definition 4.2.2 *Given two transversal Lagrangian subspaces V_1 and V_2 we define the sector between V_1 and V_2 by*

$$\mathcal{C} = \mathcal{C}\,(V_1, V_2) = \{w \in \mathcal{W} \mid \omega(v_1, v_2) \geq 0 \text{ for } w = v_1 + v_2, v_i \in V_i, i = 1,\, 2\}$$

Equivalently, if we define the quadratic form associated with an ordered pair of transversal Lagrangian subspaces,

$$\mathcal{Q}(w) = \omega(v_1, v_2)$$

where $w = v_1 + v_2$, is the unique decomposition of w with the property $v_i \in V_i, i = 1, 2$, then we have

$$\mathcal{C} = \{w \in \mathcal{W} \mid \mathcal{Q}(w) \geq 0\}.$$

In the case of the standard symplectic space, $V_1 = \mathbb{R}^d \times \{0\}$ and $V_2 = \{0\} \times \mathbb{R}^d$ we get

$$\mathcal{Q}\left((\xi, \eta)\right) = \langle \xi, \eta \rangle$$

and

$$\mathcal{C}_+ = \{(\xi, \eta) \in \mathbb{R}^d \times \mathbb{R}^d \mid \langle \xi, \eta \rangle \geq 0\}.$$

We will refer to this $\mathcal{C}_+$ as the standard sector. Since any two pairs of transversal Lagrangian subspaces are symplectically equivalent (see Problem 4.15) we may consider only this case without any loss of generality.

It is natural to ask if a sector determines uniquely its sides. It is not a vacuous question since, for $d > 1$, there are many Lagrangian subspaces in the boundary of a sector. The answer is positive.

Proposition 4.2.3 *For two pairs of transversal Lagrangian subspaces V_1, V_2 and V_1', V_2' if*

$$\mathcal{C}\,(V_1, V_2) = \mathcal{C}\,(V_1', V_2')$$

then

$$V_1 = V_1' \quad \textit{and} \quad V_2 = V_2'.$$

Moreover V_1 and V_2 are the only isolated Lagrangian subspaces contained in the boundary of the sector $\mathcal{C}\,(V_1, V_2)$.

Based on the notion of the sector between two transversal Lagrangian subspaces (or the quadratic form $\mathcal{Q}$) we define two monotonicity properties of a linear symplectic map. By $int\,\mathcal{C}$ we denote the interior of the sector, i.e.,

$$int\,\mathcal{C} = \{w \in \mathcal{W} | \mathcal{Q}(w) > 0\}.$$

Definition 4.2.4 *Given the sector $\mathcal{C}$ between two transversal Lagrangian subspaces we call a linear symplectic map L monotone if*

$$L\mathcal{C} \subset \mathcal{C}$$

and strictly monotone if

$$L\mathcal{C} \subset int\, \mathcal{C} \cup \{0\}.$$

A very useful characterization of monotonicity is given in the following

Proposition 4.2.5 *L is (strictly) monotone if and only if $\mathcal{Q}(Lw) \geq \mathcal{Q}(w)$ for every $w \in \mathcal{W}$ ($\mathcal{Q}(Lw) > \mathcal{Q}(w)$ for every $w \in \mathcal{W},\ w \neq 0$). In particular, $Q(Lw) = Q(w)$, that is, L is a Q-isometry iff $L\mathcal{C} = \mathcal{C}$.*

The fact that monotonicity implies the increase of the quadratic form defining the cone is a manifestation of a very special geometric structure of a sector and does not hold for cones defined by general quadratic forms.

Proposition 4.2.6 *A monotone map L is strictly monotone if and only if*

$$LV_i \subset int\, \mathcal{C} \cup \{0\},\ \ i = 1, 2.$$

For the proofs of all the above facts see [94] and [64] or look at Problem 4.20, Problem 4.22 and Problem 4.24.

Remark 4.2.7 *Proposition 4.2.3 and 4.2.6 are trivial in the two dimensional case. As already notice, proposition 4.2.5 follows, in the two dimensional case, by* (4.1.2).

The relevance of the above discussion is the possibility to extend Theorem 4.1.1 to the present setting.

Theorem 4.2.8 (Wojtkowski [92]) *Let (X, μ, T) be a dynamical system where X is the finite union of Symplectic Manifolds, μ is the symplectic volume, T an invertible almost everywhere differentiable symplectic map of X and*

$$\int_X \log \|DT\| d\mu < \infty.$$

If there exists a measurable, a.e. non degenerate, eventually strictly invariant family of sectors then the Lyapunov exponents are different from zero almost surely.

PROOF. The proof follows the one of Theorem 4.1.1 where the algebraic considerations are replaced by Propositions 4.2.3, 4.2.5, 4.2.6, while the measure theoretical part is exactly the same. □

4.2.1 Examples

4.2.1.a *Linear symplectic maps*

We will consider the following generalization of the Arnold cat. Let us consider the Dynamical Systems $(\mathbb{T}^{2d}, T, m)$, where m is the Lebesgue measure and $Tx = Lx \mod 1$, with the following matrix L

$$L = \begin{pmatrix} \mathbb{1} & \mathbb{1} \\ M & \mathbb{1} + M \end{pmatrix}$$

where $M > 0$ and $M_{ij} \in \mathbb{Z}$ (see Problem 4.9 for a more concrete examples and Problem 4.21 to realize how general the example is). Then the system is symplectic and strictly monotone with respect to the standard sector, thus this Dynamical Systems is hyperbolic.

4.3 Metrics and cones

This section is devoted to a little digression on semi–metrics that can be associated to one sided convex cones. This is a rather vast field [90], here we will consider only few basic facts.

There exists a very geometric approach: consider the projectivization of the cone (that is the set of equivalence classes with respect to the equivalence relation $\sim$ defined by $v \sim w$ iff there exists $\lambda \in \mathbb{R}^+$ such that $v = \lambda w$) whereby obtaining a convex set in the projective space and then use the associated projective metric [28]. We will use a more direct, yet equivalent, approach (see Problem 4.28 and Problem 4.30 for further informations on the above point of view and the connection with the following). Hopefully, the reader will excuse the setting which is quite abstract in order to be applicable to some unexpected situations.

We start by illustrating some results in lattice theory originally due to Garrett Birkhoff. For more details see [12, 13], and [29], [68] for an overview of the field. Consider a topological vector space $\mathbb{V}$ with a partial ordering "$\preceq$," that is a vector lattice.[15] We require the partial order to be *continuous*, i.e. given $\{v_n\} \in \mathbb{V}$ $\lim_{n\to\infty} v_n = v$, if $v_n \succeq w$ for each n, then $v \succeq w$. We call such vector lattices *integrally closed*.[16]

[15] We are assuming the partial order to be well behaved with respect to the algebraic structure: for each $v, w \in \mathbb{V}$ $v \succeq w \iff v - w \succeq 0$; for each $v \in \mathbb{V}$, $\lambda \in \mathbb{R}^+ \setminus \{0\}$ $v \succeq 0 \Longrightarrow \lambda v \succeq 0$; for each $v \in \mathbb{V}$ $v \succeq 0$ and $v \preceq 0$ imply $v = 0$ (antisymmetry of the order relation).

[16] To be precise, in the literature "integrally closed" is used in a weaker sense. First, $\mathbb{V}$ does not need a topology. Second, it suffices that for $\{\alpha_n\} \in \mathbb{R}$, $\alpha_n \to \alpha$; $v, w \in \mathbb{V}$, if $\alpha_n v \succeq w$, then $\alpha v \succeq w$. Here we will ignore these and other subtleties: our task is limited to a brief account of the results relevant to the present context.

We define the closed convex cone[17] $\mathcal{C} = \{v \in \mathbb{V} \mid v \neq 0,\, v \succeq 0\}$ (hereafter, the term "closed cone" $\mathcal{C}$ will mean that $\mathcal{C} \cup \{0\}$ is closed). Conversely, given a closed convex cone $\mathcal{C} \subset \mathbb{V}$, enjoying the property $\mathcal{C} \cap -\mathcal{C} = \emptyset$, we can define an order relation by (see Problem 4.27)

$$v \preceq w \iff w - v \in \mathcal{C} \cup \{0\}.$$

Henceforth, each time that we specify a convex cone we will assume the corresponding order relation and vice versa. The reader must therefore be advised that "$\preceq$" will mean different things in different contexts.

It is then possible to define a projective metric Θ (Hilbert metric),[18] in $\mathcal{C}$, by the construction:

$$\begin{aligned} \alpha(v,\, w) &= \sup\{\lambda \in \mathbb{R}^+ \mid \lambda v \preceq w\} \\ \beta(v,\, w) &= \inf\{\mu \in \mathbb{R}^+ \mid w \preceq \mu v\} \\ \Theta(v,\, w) &= \log\left[\frac{\beta(v,\, w)}{\alpha(v,\, w)}\right] \end{aligned} \tag{4.3.1}$$

where we take $\alpha = 0$ and $\beta = \infty$ if the corresponding sets are empty.

Lemma 4.3.1 *The function Θ is a semi-metric in $\mathcal{C}$.*

PROOF. Clearly $\Theta(v, w) = 0$ is equivalent to $v = \lambda w$ for some $\lambda \in \mathbb{R}^+$, also $\Theta(v, w) = \Theta(w, v)$ and the tringle inequlity can be easily checked (see Problem 4.27). $\square$

The importance of the previous constructions is due, in our context, to the following theorem.

Theorem 4.3.2 ([12]) *Let $\mathbb{V}_1$, and $\mathbb{V}_2$ be two integrally closed vector lattices; $L : \mathbb{V}_1 \to \mathbb{V}_2$ a linear map such that $L(\mathcal{C}_1) \subset \mathcal{C}_2$, for two closed convex cones $\mathcal{C}_1 \subset \mathbb{V}_1$ and $\mathcal{C}_2 \subset \mathbb{V}_2$ with $\mathcal{C}_i \cap -\mathcal{C}_i = \emptyset$. Let Θ_i be the Hilbert metric corresponding to the cone $\mathcal{C}_i$. Setting $\Delta = \sup_{v,\, w \in L(\mathcal{C}_1)} \Theta_2(v,\, w)$ we have*

$$\Theta_2(Lv,\, Lw) \leq \tanh\left(\frac{\Delta}{4}\right)\Theta_1(v,\, w) \qquad \forall v,\, w \in \mathcal{C}_1$$

[17] **Attention**: here, by "cone," we mean any set such that, if v belongs to the set, then λv belongs to it as well, for each $\lambda > 0$. The reason for this change in the definition of cone is that two sided cones, viewed as sets, are never convex, while convexity plays a central rôle in the following. As we will see in Examples 4.3.1–Cones and Q-forms this change in definition does not limit the applicability of the present theory to the cones introduced in the previous section.

[18] In fact, we define a semi–metric, since $v \sim w \Rightarrow \Theta(v,\, w) = 0$. The metric that we describe corresponds to the conventional Hilbert metric on $\widetilde{\mathcal{C}}$, the quotient of $\mathcal{C}$ with respect to the relation "$\sim$".

($\tanh(\infty) \equiv 1$*).*

PROOF. Let $v,\, w \in \mathcal{C}_1$. On the one hand if $\alpha := \alpha(v,w) = 0$ or $\beta := \beta(v,w) = \infty$, then the inequality is obviously satisfied. On the other hand, if $\alpha \neq 0$ and $\beta \neq \infty$, then

$$\Theta_1(v,\, w) = \ln \frac{\beta}{\alpha}$$

where $\alpha v \preceq w$ and $\beta v \succeq w$, since $\mathbb{V}_1$ is integrally closed. Notice that $\alpha \geq 0$, and $\beta \geq 0$ since $v \succeq 0$, $w \succeq 0$. If $\Delta = \infty$, then the result follows from $\alpha L v \preceq L w$ and $\beta L v \succeq L w$. If $\Delta < \infty$, then, by hypothesis,

$$\Theta_2\left(L(w - \alpha v),\, L(\beta v - w)\right) \leq \Delta$$

which means that there exist $\lambda,\, \mu \geq 0$ such that

$$\begin{aligned} \lambda L(w - \alpha v) &\preceq L(\beta v - w) \\ \mu L(w - \alpha v) &\succeq L(\beta v - w) \end{aligned}$$

with $\ln \frac{\mu}{\lambda} \leq \Delta$. The previous inequalities imply

$$\begin{aligned} \frac{\beta + \lambda\alpha}{1+\lambda} L v &\succeq L w \\ \frac{\mu\alpha + \beta}{1+\mu} L v &\preceq L w. \end{aligned}$$

Accordingly,

$$\begin{aligned} \Theta_2(Lv,\, Lw) \leq \ln \frac{(\beta+\lambda\alpha)(1+\mu)}{(1+\lambda)(\mu\alpha+\beta)} &= \ln \frac{e^{\Theta_1(v,\,w)} + \lambda}{e^{\Theta_1(v,\,w)} + \mu} - \ln \frac{1+\lambda}{1+\mu} \\ &= \int_0^{\Theta_1(v,\,w)} \frac{(\mu-\lambda)e^{\xi}}{(e^{\xi}+\lambda)(e^{\xi}+\mu)} d\xi \leq \Theta_1(v,\, w) \frac{1 - \frac{\lambda}{\mu}}{\left(1 + \sqrt{\frac{\lambda}{\mu}}\right)^2} \\ &\leq \tanh\left(\frac{\Delta}{4}\right) \Theta_1(v,\, w). \end{aligned}$$

$\square$

Remark 4.3.3 *In general, it suffices to know* $L(\mathcal{C}_1) \subset \mathcal{C}_2$ *in order to conclude* $\Theta_2(Lv,\, Lw) \leq \Theta_1(v,\, w)$. *However, a strict contraction depends on the diameter of the image being finite.*[19]

[19] In the theory of Markov processes this corresponds to the so called *positivity improving* (see also Example 4.3.1.a).

In particular, if an operator maps a convex cone strictly inside itself (in the sense that the diameter of the image is finite), then it is a contraction in the Hilbert metric. This implies the existence of a "positive" eigenfunction (provided the cone is complete with respect to the Hilbert metric), and, with some additional work, the existence of a gap in the spectrum of L (see [12] for details or solve Problem 4.35).

Usually the space $\mathbb{V}$ comes endowed with its own metric, in such a case it is natural to wonder about the strength of the Hilbert metric compared to other metrics. While, in general, the answer depends on the cone, it is nevertheless possible to state an interesting result.

Definition 4.3.4 *A function $\rho : \mathbb{V} \to \mathbb{R}^+$ is called homogeneous of degree one if for all $\lambda \in \mathbb{R}$ and $v \in \mathbb{V}$*

$$\rho(\lambda v) = |\lambda|\rho(v).$$

Remark 4.3.5 *Note that a norm or a linear functional are both homogeneous function of degree one.*

Definition 4.3.6 *A homogeneous function of degree one is called* adapted to a cone $\mathcal{C}$ *if, for each $v, w \in \mathbb{V}$,*

$$-v \preceq w \preceq v \Longrightarrow \rho(v) \geq \rho(w),$$

and $v \in int\, \mathcal{C}$ implies $\rho(v) > 0$.

Lemma 4.3.7 *Let ρ_i be two homogeneous functions of degree one adapted to the cone $\mathcal{C} \subset \mathbb{V}$. Then, given $v,\, w \in int\, \mathcal{C} \subset \mathbb{V}$ for which $\rho_1(v) = \rho_1(w)$,*

$$\rho_2(v - w) \leq \left(e^{\Theta(v,\, w)} - 1\right) \min\{\rho_2(v), \rho_2(w)\}.$$

Proof. We know that $\Theta(v,\, w) = \ln \frac{\beta}{\alpha}$, where $\alpha v \preceq w \preceq \beta v$. This implies that $-w \preceq 0 \preceq \alpha v \preceq w$, i.e. $\rho_1(w) \geq \alpha \rho_1(v)$, or $\alpha \leq 1$. In the same manner it follows that $\beta \geq 1$. Hence,

$$\begin{aligned}
w - v &\preceq (\beta - 1)v \preceq (\beta - \alpha)v \\
w - v &\succeq (\alpha - 1)v \succeq -(\beta - \alpha)v \\
w - v &\preceq (1 - \beta^{-1})w \preceq (\alpha^{-1} - \beta^{-1})w \\
w - v &\succeq (1 - \alpha^{-1})w \succeq -(\alpha^{-1} - \beta^{-1})w
\end{aligned}$$

which implies

$$\begin{aligned}
&\|w - v\| \leq (\beta - \alpha)\|v\| \leq \frac{\beta - \alpha}{\alpha}\|v\| = \left(e^{\Theta(v,\, w)} - 1\right)\|v\| \\
&\|w - v\| \leq (\alpha^{-1} - \beta^{-1})\|w\| \leq (\frac{\beta}{\alpha} - 1)\beta^{-1}\|w\| \leq \left(e^{\Theta(v,\, w)} - 1\right)\|w\|.
\end{aligned}$$

$\square$

Many normed vector lattices satisfy the hypothesis of Lemma 1.3 (e.g. Banach lattices[20]).

4.3.1 Examples

4.3.1.a *Perron-Frobenius Theorem*

Consider a matrix $L : \mathbb{R}^n \to \mathbb{R}^n$ of all strictly positive elements: $L_{ij} \geq \gamma > 0$. The Perron-Frobenius theorem states that there exists a unique eigenvector v^+ such that $v_i^+ > 0$, in addition the corresponding eigenvalue λ is simple, maximal and positive. There quite a few proofs of this theorem a possible one is based on Birkhoff theorem. Consider the cone $\mathcal{C}^+ = \{v \in \mathbb{R}^2 \mid v_i \geq 0\}$, then obviously $L\mathcal{C}^+ \subset \mathcal{C}^+$. Moreover an explicit computation (see Problem 4.) shows that

$$\Theta(v, w) = \ln \sup_{ij} \frac{v_i w_j}{v_j w_i}. \tag{4.3.2}$$

Then, setting $M = \max_{ij} L_{ij}$, it follows that

$$\Theta(Lv, Lw) \leq 2 \ln \frac{M}{\gamma} := \Delta < \infty.$$

We have then a contraction in the Hilbert metric and the result follows from usual fixed points theorems. Note that, since $\Theta(v, \lambda v) = 0$, for all $\lambda \in \mathbb{R}^+$, the fixed point $v_+ \in \mathbb{R}^n$ is only projective, that is $Lv_+ = \lambda v_+$ for some $\lambda \in \mathbb{R}$; in other words, we have an eigenvalue.

Remark that L^* satisfies the same conditions as L, thus there exists $w^+ \in \mathcal{C}^+$, $\mu \in \mathbb{R}^+$, such that $L^* w^+ = \mu w^+$. Next, define $\rho_1(v) = |\langle w^+, v\rangle|$ and $\rho_2(v) = \|v\|$. It is easy to check that they are two homogeneous forms of degree one adapted to the cone.

In addition, if $\rho_1(v) = \rho_2(v)$, then $\rho_1(L^n v) = \rho_1(L^n w)$. Hence, by Lemma 4.3.7

$$\begin{aligned} \|L^n v - L^n w\| &\leq \left(e^{\Theta(L^n v, L^n w)} - 1\right) \min\{\|L^n v\|, \|L^n w\|\} \\ &\leq K\Lambda^n \min\{\|L^n v\|, \|L^n w\|\}, \end{aligned} \tag{4.3.3}$$

for some constant K depending only on v, w. The estimate 4.3.3 means that all the vectors in the cone grow at the same rate. In fact, for all $v \in \text{int}\mathcal{C}$,

$$\|\lambda^{-n} L^n v - \lambda^{-n} L^n w\| \leq K\Lambda^n.$$

[20] A Banach lattice $\mathbb{V}$ is a vector lattice equipped with a norm satisfying the property $\| \, |v| \, \| = \|v\|$ for each $f \in \mathbb{V}$, where $|v|$ is the least upper bound of v and $-v$. For this definition to make sense it is necessary to require that $\mathbb{V}$ is "directed," i.e. any two elements have an upper bound.

Hence, $\lim_{n\to\infty} \lambda^{-n} L^n v = v_+$.

Finally, consider $\mathbb{V}_1 = \{v \in \mathbb{V} \mid \langle w^+, v\rangle = 0\}$. Clearly $L\mathbb{V}_1 \subset \mathbb{V}_1$ and $\mathbb{V}_1 \oplus \text{span}\{v_+\} = \mathbb{V}$. Let $w \in \mathbb{V}_1$, clearly there exists $\alpha \in \mathbb{R}^+$ such that $\alpha v_+ + w \in \mathcal{C}$,[21] thus

$$\|L^n w\| \leq \|L^n(\alpha v_+ + w) - \alpha L^n v_+\| \leq L\Lambda^n \lambda^n.$$

This immediately implies that L restricted to the subspace $\mathbb{V}_1$ has spectral radius less that $\lambda\Lambda$. In other words, λ is the maximal eigenvalue, it is simple and any other eigenvalue must be smaller than $\lambda\Lambda$. We have thus obtained an estimate of the spectral gap between the first and the second eigenvalue.

4.3.1.b *Cones and Q-forms*

Here we would like to consider only half of the cone defined by the Q-form in order to apply the present theory. If $L_{ij} > 0$, then we can choose the first quadrant; on the other hand , if $L_{ij} < 0$, then L maps the first into the third quadrant. In both cases a monotone matrix L maps a one sided cone into a one sided cone cone. Here we will consider only the first case and leave the other–essentially identical–to the reader.[22] Consequently, L is a monotone matrix with respect to the standard sector $\mathcal{C}_+$ and $L_{ij} > 0$, then $L\mathcal{C}^+ \subset \mathcal{C}^+$ where, as in the previous example, $\mathcal{C}^+ = \{v \in \mathbb{R}^2 \mid v_i \geq 0\}$. Thus all the results of the previous example apply.

In particular, remembering (4.3.2), if $v = (1, \alpha)$, $w = (1, \beta) \in \mathcal{C}_+$, then

$$\Theta(v, w) = \left|\ln \frac{\alpha}{\beta}\right|. \tag{4.3.4}$$

Another interesting formula is (see Problem 4.34)

$$2\sinh(\frac{1}{2}\Theta(v, w)) = \frac{|\omega(v, w)|}{\sqrt{Q(v)Q(w)}}. \tag{4.3.5}$$

This means that here exists a relation between the Hilbert metric and the Q-form.

To understand this relation better, let us compute

$$\text{diam}(L\mathcal{C}^+) = \sup_{\alpha,\beta>0} \left|\ln \frac{(a + \alpha b)(c + \beta d)}{(a + \beta b)(c + \alpha d)}\right|.$$

Since

$$\frac{a}{c} = \frac{ad}{cd} = \frac{1 + bc}{cd} = \frac{b}{d} + \frac{1}{cd} > \frac{b}{d}.$$

[21] this is a special case of the general fat that any vector can be written as the linear combination of two vectors belonging to the cone.

[22] An easy way out is to consider L^2 instead of L.

Thus

$$\mathrm{diam}(L\mathcal{C}_+) = \left|\ln \frac{ad}{cb}\right| = \ln \frac{1+bc}{bc}, \tag{4.3.6}$$

which implies that, if L is strictly monotone, then $\mathrm{diam}(L\mathcal{C}^+) < \infty$. Accordingly, the rate of contraction of the Hilbert metric is given by

$$\Lambda = \frac{1-\sqrt{\frac{bc}{1+bc}}}{1+\sqrt{\frac{bc}{1+bc}}} = \frac{1}{(\sqrt{1+bc}+\sqrt{bc})^2} = \frac{1}{\sigma(L)^2}, \tag{4.3.7}$$

where the last equality follows by $(4.1.2)$.

Remark 4.3.8 *It is not immediately clear how to extend the above considerations to the higher dimensional setting discussed in section 4.2. In fact, to do so it is necessary to introduce a different metric [63] of Caratheodory type [90]. We will not do it here but the reader should be aware that such a generalization is possible.*

4.3.1.c *Expanding maps–uniqueness of the a.c. measure*

A remarkable fact of Birkhoff theorem is that it applies to infinite dimensional vector spaces. In Example 1.4.1 we have studied the properties of $\mathcal{L}$. A computation similar to the one done there shows that, given a twice differentiable expanding map of the torus, the cone

$$\mathcal{C}_\alpha = \{h \in \mathcal{C}^{(0)}(\mathbb{T}) \mid h \geq 0; \frac{h(x)}{h(y)} \leq e^{\alpha d(x,y)}\} \tag{4.3.8}$$

is invariant. In fact, if $h \in \mathcal{C}_\alpha$

$$\begin{aligned}\mathcal{L}h(x) &= \sum_{z\in T^{-1}x} |D_zT|^{-1}h(z) \leq \sum_{w\in T^{-1}y} \frac{|D_wT|}{|D_zT|}|D_wT|^{-1}h(w)e^{\alpha d(z,w)} \\ &\leq \sum_{w\in T^{-1}y} |D_wT|^{-1}h(w)e^{(\lambda^{-1}\alpha+C)d(x,y)} = e^{(\lambda^{-1}\alpha+C)d(x,y)}\mathcal{L}h(y).\end{aligned}$$

By choosing α large enough, there exists $\sigma \in (\lambda^{-1}, 1)$ such that $\mathcal{L}\mathcal{C}_\alpha \subset \mathcal{C}_{\sigma\alpha}$.

A direct computation shows that the diameter is finite (see Problem 4.33). Accordingly, we have a contraction in the Hilbert metric. This implies that there exists only one invariant measure μ which is absolutely continuous with respect to Lebesgue ($d\mu = h_* dm$). Moreover, if $\rho_1(f) = |\int f|$ and $\rho_2(f) = \|f\|_\infty$, we have that Lemma 4.3.7 applies whereby showing that $\mathcal{L}h \to h_*$ in the sup norm for all $h \in \mathcal{C}_\alpha$, $\rho_1(h) = \rho_2(h_*)$. By arguments similar to the one employed in 4.3.1.a it is possible to see that $\mathcal{L}$, viewed as an operator in $\mathcal{C}^{(1)}(\mathbb{T})$, has a maximal eigenvalue one while all the rest of the spectrum is separate by a gap (see Problem 4.35), clearly this implies not only the mixing but provides as well an estimate on the mixing rate for $\mathcal{C}^{(1)}(\mathbb{T})$ functions (see Problem 4.36).

4.4 Cones and invariant distributions

Here we use the machinery developed in the previous section to obtain a constructive proof of the existence of the unstable distribution in a special, but very interesting, case.

Lemma 4.4.1 *Given a smooth Symplectic Dynamical Systems with syngularities (X, T, μ), X a symplectic two dimensional manifold, μ the symplectic volume, if the systems is eventually strictly monotone, then $\{E^u(x)\}$ is almost everywhere well defined. Moreover, if the system is smooth and $\mathcal{C}(x)$ is continuous, then $\{E^u(x)\}$ is continuous (where it is defined). In addition, if the cone family is strictly monotone, then $\{E^u(x)\}$ is everywhere defined.*

The same holds for $\{E^s(x)\}$ and, in the last case, the two distributions are uniformly transversal.

Proof. Let $\mathcal{C}_n(x) := D_{T^{-n}x}T^n\mathcal{C}(T^{-n}x)$ and $\Delta_n(x) := \operatorname{diam}(\mathcal{C}_n(x))$, then Δ_n is decreasing, thus we can define

$$\Delta_\infty(x) := \lim_{n\to\infty} \Delta_n(x).$$

The key consequence of the results of section 4.3 (in particular Examples 4.3.1–Cones and Q-forms) is

$$\begin{aligned}\Delta_\infty(T^m x) &= \lim_{n\to\infty} \operatorname{diam}(D_{T^{m-n}x}T^n\mathcal{C}(T^{-n+m}x)) \\ &= \lim_{n\to\infty} \operatorname{diam}(D_xT^m D_{T^{-n}x}T^n\mathcal{C}(T^{-n}x)) \\ &\le \frac{1}{\sigma(D_xT^m)^2}\Delta_\infty(x).\end{aligned}$$

Next, let $\Omega = \{x \in X \mid \Delta_\infty(x) = \infty\}$, we claim that $\mu(\Omega) = 0$. In fact, let $B_m = \{x \in X \mid \sigma(D_xT^m) \ge 2\}$, by eventual strict monotonicity of the cone field and Lemma 4.1.2 follows $\mu(\cup_{m\in\mathbb{N}}B_m) = \mu(X)$. In addition, $B_m \supset B_{m_0}$ for all $m > m_0$. Moreover, if $x \in B_m$, then $\Delta_\infty(T^m x) < \infty$ (see 4.3.7). Thus $T^{-n}\Omega \cap B_m = \emptyset$ for all $n \ge m$, and

$$\mu(\Omega) = \lim_{n\to\infty}\mu(T^{-n}\Omega) \le \lim_{n\to\infty}\mu(X\setminus \cup_{m\le n} B_m) = 0.$$

Finally, let $\Omega_L = \{x \in X \mid \frac{L}{2} \leq \Delta_\infty(x) \leq L\}$ and suppose $\mu(\Omega_L) > 0$. Then, there exists $n \in \mathbb{N}$ such that $\mu(\Omega_L \cap B_m) > 0$. Consequently, for almost all $x \in \Omega_L \cap B_m$ there exists a return time $\bar{n}m \in \mathbb{N}$ in the past (that is $T^{-m\bar{n}}x \in \Omega_L \cap B_m$). Accordingly,

$$\frac{L}{2} \leq \Delta_\infty(x) \leq \frac{1}{\sigma(D_xT^m)^2}\Delta_\infty(T^{-m\bar{n}}x) \leq \frac{L}{4},$$

which is a contradiction unless $L = 0$. We have so proven that $\mu(\Omega_0) = \mu(X)$. In other words the cones $\mathcal{C}_\infty = \cap_{n\geq 0}\mathcal{C}_n(x)$ is almost everywhere degenerate since, having zero diameter, it consists of a single direction, such a direction is precisely the unstable direction.

To prove the continuity of the above distribution note that the cone family $\mathcal{C}_n(x)$ is continuous. Let x be such that $\Delta_\infty(x) = 0$, for each $\varepsilon > 0$ there exists $m \in \mathbb{N}$ such that $\Delta_m(x) < \frac{\varepsilon}{2}$. Then one can chose δ such that the edges of $\mathcal{C}_m(y)$ vary, in the Hilbert metric, by an amount less than $\frac{\varepsilon}{2}$ if $d(x,y) < \delta$. This means that the distance between $E^u(x)$ and $E^u(y)$ is less than ε in the Hilbert metric of $\mathcal{C}(x)$, provided $d(x,y) < \delta$. In addition, all the unstable manifolds in such a neighborhood are unifrom;ly in the interior of the cone $\mathcal{C}(x)$. The result follows then taking into account that the Hilbert metric is equivalent to teh normal distance in each compact contained in the interior of the cone (see (4.3.4)).

The proof of the last fact is obvious: just a simplification of the above arguments. □

With similar techniques it is also possible to construct the stable and unstable foliations, as we will see in chapter 7.

Let us conclude with an interesting simple fact.

Lemma 4.4.2 *A smooth two-dimensional Symplectic Dynamical System* (X, T, μ) *is Anosov iff it admits a strictly monotone continuous cone family.*

PROOF. By Lemma 4.4.1 it follows that the stable and unstable distribution are continuous. But then, by continuity, there exists $\alpha > 0$ and $\sigma > 1$ such that

$$\begin{array}{l} \alpha\sqrt{Q(v)} \leq \|v\| \leq \alpha^{-1}\sqrt{Q(v)} \quad \forall x \in X \text{ and } v \in E^u(x) \\ \sigma(D_xT) \geq \sigma \quad \forall x \in X. \end{array}$$

Thus,

$$\|D_xT^n v\| \geq \alpha\sqrt{Q(D_xT^n v)} \geq \alpha\sigma^n\sqrt{Q(v)} \geq \alpha^2\sigma^n\|v\|.$$

Analogously one can obtain the statement for the stable direction by using the cone family given by the complementary cones (see Problem 4.6).

The proof that an Anosov systems admit a continuous strictly invariant cone family is obvious and it is left to the reader.[23] □

[23] See Problem 7.5.

Theorem 4.4.3 *The set of Anosov two dimensional symplectic Dynamical Systems is open in the $\mathcal{C}^{(1)}$ topology.*

PROOF. Consider an Anosov map, let $\mathcal{C}$ be an associated continuos cone family. Clearly this will be a strictly invariant cone family also for any sufficiently close map. For more detail see Problem 4.7 and Problem 4.8. □

In fact the above theorem can be easily generalized, see Problem 4.38.

4.5 The case of Hamiltonian flows

talk about cocycles and derivative cocycles, forms in ambient space

....

4.5.1 Examples

4.5.1.a *Geodesic flows*

use Jacobi fields

4.5.1.b *Smooth flows with collisions*

4.5.1.c *Billiards with potential*

maybe in next chapter?

....

4.6 General quadratic forms–the non-symplectic case

Potapov stuff, kobayashy and caratheodory metric

. . . .

Before continuing in the development of the theory it can be helpful to develop and study some more interesting and totally non-trivial examples. To this end are dedicated the next two chapters.

Problems

4.1 Consider $\mathbb{R}^2$ endowed with the scalar product $\langle v, w\rangle_G := \langle v, Gw\rangle$, where $\langle\cdot,\cdot\rangle$ is the standard scalar product and $G > 0$. Show that there exists a change of coordinates $M : \mathbb{R}^2 \to \mathbb{R}^2$ such that, in the new coordinates $\langle\cdot,\cdot\rangle_G$ becomes the standard scalar product.

4.2 Consider the cone $\mathcal{C}$ defined by the two transversal vectors $v_1, v_2 \in \mathbb{R}^2$. This means that $v \in \mathbb{R}^2$ belongs to the cone iff $v = \alpha v_1 + \beta v_2$ with $\alpha\beta \geq 0$. Show that there is a linear change of coordinates $M : \mathbb{R}^2 \to \mathbb{R}^2$ such that $M\mathcal{C} = \mathcal{C}_+$ and $\det M = 1$.

4.3 Show that the hypothesis of Theorem 4.1.1 can be relaxed, in particular it holds for smooth systems with singularities (see section 3.6). (Hint: Just follow the proof step by step and notice that nothing substantial need to be changed.)

4.4 Construct a strictly invariant cone family for the irrational translation on $\mathbb{T}^2$ (see Examples 1.1.1) and show that it is not measurable. (Hint: For each trajectory choose a point x. At such a point choose the standard cone $\mathcal{C}_+$, let $\mathcal{C}_n^- = \{(v_1, v_2) \in \mathbb{R}^2 \mid 1 + \frac{1}{n} \leq \frac{v_2}{v_1} \leq 2 + \frac{1}{n}\}$ and $\mathcal{C}_n^+ = \{(v_1, v_2) \in \mathbb{R}^2 \mid -2 - \frac{1}{n} \leq \frac{v_2}{v_1} \leq -1 - \frac{1}{n}\}^c$. Then set $\mathcal{C}(T^n x) = \mathcal{C}_n^+$ and $\mathcal{C}(T^{-n}x) = \mathcal{C}_n^-$. Such a cone family is strictly monotone by construction (since $D_xT = 1$), yet the system has obviously zero Lyapunov exponents. Since all the other hypothesis of Theorem 4.1.1 are satisfied, it follows that the above cone family cannot be measurable.)

4.5 Show that for two dimensional symplectic maps the sum of the Lyapunov exponent is zero (*pairing of the Lyapunov exponents*). (Hint: If $\omega(v, w) = 1$ then it holds that $1 = \omega(DT^n v, DT^n w) \sim \|DT^n v\|\|DT^n w\|$.)

4.6 Check that $\inf_{v\in\mathcal{C}_+} \sqrt{\frac{Q(Lv)}{Q(v)}} = \inf_{v\in\mathcal{C}_-} \sqrt{\frac{Q(L^{-1}v)}{Q(v)}}$, remember that $\mathcal{C}_- = \overline{(\mathcal{C}_+)^c}$. (Hint: see [63])

4.7 Show that, in a two dimensional area preserving systems, if the LE are different from zero then there exists and eventually strictly invariant cone family.(Hint: By Oseledec and Problem ?? there exists the unstable distributions, then construct the cones around it.)

4.8 Show that, Anosov two imensiona area preserving systems form an open set in the $\mathcal{C}^{(1)}$ topology in the set of area preserving diffeo.

4.9 Prove that if M is the two by two matrix

$$M = \begin{pmatrix} a & b \\ b & c \end{pmatrix},$$

with $a, b, c \in \mathbb{Z}$, then $M > 0$ iff $a, c > 0$ and $c > \frac{b^2}{a}$.

4.10 Show that a $2d \times 2d$ matrix L of the form

$$L = \begin{pmatrix} A & B \\ C & D \end{pmatrix},$$

where A, B, C, D are $d \times d$ blocs, is symplectic, iff $C^*A = A^*C$, $D^*B = B^*D$ and $A^*D - C^*B = \mathbb{1}$. (Hint: Note that, by introducing the matrix

$$J = \begin{pmatrix} 0 & \mathbb{1} \\ -\mathbb{1} & 0 \end{pmatrix}$$

the standard symplectic form 4.2.1 can be written in terms of the usual scalar product as

$$\omega((\xi^1, \xi^2), (\eta^1, \eta^2)) = \langle (\xi^1, \xi^2), J(\eta^1, \eta^2) \rangle.$$

From this point of view the definition of symplectic matrix 4.2.2 can be written as

$$L^* J L = J.$$

A trivial algebraic computation yields now the result.)

4.11 Prove that is L is symplect then $\det L = 1$. (Hint: The determinant of a matrix is nothing else than the volume of the parallelepiped of sides $(Le_1, \dots, Le_{2d})$ (where $e_1, \dots, e_{2d}$ is the standard orthonormal basis of $\mathbb{R}^{2d}$). On the other hand the volume form can be written has $\wedge^d \omega$ (since that is a $2d$ form with the right normalization and the space of $2d$ forms is one dimensional). Thus $\det L = \wedge^d \omega(Le_1, \dots, Le_{2d}) = \wedge^d \omega(e_1, \dots, e_{2d}) = 1$ where we have used the fact that $\omega(Lv, Lu) = \omega(v, u)$. The reader that wants to appreciate the power of the above geometrical interpretation of the determinant and of the external forms can try to prove the statement by purely algebraic means.)

4.12 Given a subspace $E \in \mathbb{R}^{2d}$ we define the skew orthogonal subspace $E^{\angle} = \{v \in \mathbb{R}^{2d} \mid \omega(w, v) = 0 \ \forall \ w \in E\}$. Show that $E^{\angle}$ is a vector spae and that $\dim(E) + \dim(E^{\angle}) = 2d$. (Hint: Given a basis $\{w_i\}$ of E, $\ell = \dim(E)$, consider the map $F : \mathbb{R}^{2d} \to \mathbb{R}^{\ell}$ defined by

$$F(v) := (\omega(w_1, v), \dots, \omega(w_\ell, v)).$$

Then $\operatorname{Ker}(F) = E^{\angle}$. Hence $\dim E^{\angle} = 2d - \dim R(F)$. Clearly $\dim R(F) \leq \ell$, in addition if $R(F) < \ell$ then there must exiat a vector $(\alpha_1, \dots, \alpha_\ell) \in \mathbb{R}^{\ell}$ such that for each $v \in \mathbb{R}^{2d}$

$$0 = \sum_i \alpha_i \omega(w_i, v) = \omega(\sum_i \alpha_i w_i, v)$$

which, be the non degenearacy of ω, would imply $\sum_i \alpha_i w_i = 0$, a contradiction.)

4.13 Show that Lagrangian subspaces are d dimensional. (Hint: Le E be a Lagrangian subsapce, then $E^{\angle} \supset E$. Suppose there exists $v \in E^{\angle} \setminus E$, then for each $w_i = \alpha_i v + \bar{v}_i$, $i \in \{1, 2\}$, $\bar{v}_i \in E$

$$\omega(w_1, w_2) = \omega(\alpha_1 v + \bar{v}_1, \alpha_2 v + \bar{v}_2) = 0.$$

Thus E it is not maximal. Accordingly $E^{\angle} = E$ and the result follows by Problem 4.12.)

4.14 Show that all the Lagrangian subspaces transversal to $V = \{(0, \eta) \in \mathbb{R}^{2d} \mid \eta \in \mathbb{R}^d\}$ can be represented as $\{(\xi, U\xi \in \mathbb{R}^{2d} \mid \xi \in \mathbb{R}^d\}$ for some symmetric matrix U. (Hint: Let $V_U := \{(\xi, U\xi \in \mathbb{R}^{2d} \mid \xi \in \mathbb{R}^d\}$, then $\omega((\xi, U\xi), (\zeta, U\zeta)) = 0$, thus V_U is Lagrangian. On the other hand, if $\tilde{V}$ is Lagrangian, then it is a d dimensional space (Problem 4.13). Let $\{(\xi_i, \eta_i)\}_{i=1}^d$ be a base for $\tilde{V}$, then $\xi_i \neq 0$ by the transversality assumption and we can define the matrix U via $U\xi := \eta_i$. It is immediate that $\tilde{V}$ Lagrangian implies $U = U^*$.)

4.15 Show that any two transversal Lagrangian subspaces can be mapped into $\{(u, 0)\}$ and $\{(0, u)\}$, respectively, by a symplectic transformation.

4.16 Show that all symplectic Q-isometrys L (that is $Q(Lv) = Q(v)$) have the form

$$L = \begin{pmatrix} A & 0 \\ 0 & A^{*-1} \end{pmatrix}.$$

(Hint: Start by considering the vector $(0, u)$, $U \in \mathbb{R}^d$, clearly $Q((0, u)) = 0$ thus $Q(L(0, u)) = 0$ if L is a Q-isometry. But if

$$L = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

it follows $\langle Bu, Du\rangle = 0$ for each $u \in \mathbb{R}^d$, that is $B^*D = 0$. The same argument applied to the vector $(u, 0)$ yields $A^*C = 0$. Accordingly, by symplecticity (see Problem 4.10),

$$\begin{aligned} Q(L(v,u)) =& \langle Au + Bv,\, Cu + Dv\rangle = \langle u, (A^*D + C^*B)v\rangle \\ =& \langle u, (\mathbb{1} + 2C^*B)v\rangle \end{aligned}$$

thus $Q(L(v,u)) = Q(v,u)$ iff $C^*B = 0$ which implies $A^*D = \mathbb{1}$.)

4.17 Show that if the matrix

$$L = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

is symplectic then

$$L^{-1} = \begin{pmatrix} D^* & -B^* \\ -C^* & A^* \end{pmatrix}$$

(Hint: multiply and use Problem 4.10.)

4.18 Show that the symplectic matrices form a multiplicative group. (Hint: Use the definition and the above problems.)

4.19 A symplectic map L is a Q-isometry iff $L\mathcal{C} = \mathcal{C}$. (Hint: One direction is trivial. On the other hand, if $L\mathcal{C} = \mathcal{C}$ it follows that L maps the boundary, of $\mathcal{C}$, to the boundary. Accordingly, if $\langle v, u\rangle = 0$ it must be

$$0 = \langle Av + bu,\, Cv + Du\rangle. \tag{4.6.1}$$

Choosing in 4.6.1 $u = 0$ yields $A^*C = 0$, choosing $v = 0$ shows that it must be $B^*D = 0$. Thus 4.6.1 yields

$$0 = \langle u,\, (A^*D + C^*B)v\rangle = 2\langle u,\, C^*Bv\rangle.$$

The above equality shows that C^*Bv is parallel to v for each $v \in R^d$, that is $C^*B = \alpha\mathbb{1}$ for some $\alpha \in \mathbb{R}$. If $\alpha = 0$, then $A^*D = \mathbb{1}$ and thus $C = 0$ which is the wanted result. If $\alpha \neq 0$, then B is invertible and $C = \alpha B^{*_1}$. But this implies $A = 0$ and hence $-\mathbb{1} = C^*B = \alpha\mathbb{1}$, that is $\alpha = -1$. Accordingly the matrix would have the form

$$L = \begin{pmatrix} 0 & B \\ -B^{*-1} & 0 \end{pmatrix}$$

which sends $\mathcal{C}$ in its complement, contrary to our requirement.)

4.20 Prove Proposition 4.2.3.

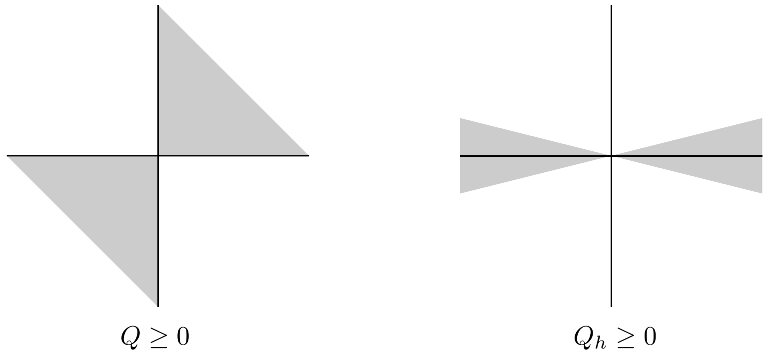


Figure 4.1: Cones

4.21 Show that a strictly monotone symplectic matrix can be put into the form

$$\begin{pmatrix} \mathbb{1} & \mathbb{1} \\ M & \mathbb{1} + M \end{pmatrix}$$

by multiplying it by Q-isometries on the left and on the right.

4.22 Prove Proposition 4.2.5.

4.23 Show that $V_U := \{(\xi, U\xi \in \mathbb{R}^{2d} \mid \xi \in \mathbb{R}^d\}$, $U = U^*$, belongs to the standard cone iff $U \geq 0$.

4.24 Prove Proposition 4.2.6.

4.25 Find a symplectic change of coordinates that transforms the standard form Q into the form Q_h defined by:

$$Q_h((x, y)) = \frac{1}{2}(h^2 \langle x, x \rangle - \langle y, y \rangle),$$

and draw the associate cone. (Hint: Consider

$$\begin{aligned} x = &\quad hx' - y' \\ y = &\quad hx' + y'. \end{aligned}$$

See the figure for the shape of the cones.)

4.26 Do the same as in Problem Problem 4.25 but with the cone $\{(v_1, v_2) \in \mathbb{R}^2 \mid \alpha \leq \frac{|v_1|}{|v_2|} \leq \alpha^{-1}\}$. (**make picture**

4.27 Given a cone $\mathcal{C}$ in a vector space $\mathbb{V}$ define the relation $v \succeq w$ if $v - w \in \mathcal{C}$. Prove that $v \succeq w$ and $w \succeq z$ implies $v \succeq z$ if and only if $\mathcal{C}$ is convex.

Show that $v \succeq w$ and $w \succeq v$ implies $v = w$ if and only if $\mathcal{C} \cup (-\mathcal{C}) = \emptyset$. If $\mathcal{C}$ has the two above properties, then $\succeq$ is an order realtion and $\mathbb{V}$ equipped with it is a Lattice

4.28 Let $C \in \mathbb{R}^n$ be a strictly convex compact set. For each two point $x, y \in C$ consider the line $\ell = \{\lambda x + (1 - \lambda y) \mid \lambda \in \mathbb{R}\}$ passing through x and y. Let $\{u, v\} = C \cap \ell$ and define

$$\Theta(x, y) = \left| \ln \frac{\|x - u\| \|y - v\|}{\|x - v\| \|y - v\|} \right|$$

(the logarithm of the cross ratio). Show that Θ is a metric in C (the Hilbert metric). Show that the distance of $x \in \text{int } C$ from ∂C is infinite. (Hint: The only non trivial task is to check the triangle inequality. Consider three points $x, y, z \in C$. If the points are collinear then the proof is easy. If they are not the consider the plane defined by them, we have now a two dimensional problem, thus it suffices to prove the result in $\mathbb{R}^2$. Consider the Figure 4.2 and remember that the cross ratio

$$R(x, y, u, v) = \frac{\|x - u\| \|y - v\|}{\|x - v\| \|y - v\|}$$

is a projective invariant. Then

$$\begin{aligned} R(x, z, u, v) &= R(x, w, x', y') \geq R(x, w, \alpha, \beta) \\ R(y, z, a, b) &= R(w, y, x', y') \geq R(w, y, \alpha, \beta) \end{aligned}$$

and the result follows since $R(x, w, \alpha, \beta) R(w, y, \alpha, \beta) = R(x, y, \alpha, \beta)$.)

4.29 Prove the same as the previous Problem for convex sets in a projective space.

4.30 Check that the metric defined in Problem 4.28 and Problem 4.29 applied to the projectivization of convex cones yields to the same metric defined in (4.3.1).(Hint: Let $\mathcal{C} \subset \mathbb{V}$ be a convex cone. The projectivization consists in considering the space $\mathbb{P}$ of the equivalence classes $[v]$ with respect to the equivalence relation $v \sim w$ iff $v = \lambda w$ for some $\lambda \in \mathbb{R}^+$. Let $\tilde{\mathcal{C}} = \{[v] \in \mathbb{P} \mid v \in \mathcal{C}\}$. Clearly $\tilde{C}$ is convex in the projective space $\mathbb{P}$.[24] So, if $[x], [y] \in \tilde{\mathcal{C}}$, the Hilbert metric is defined by the points $[u], [v] \in \partial\tilde{\mathcal{C}}$ on the line determined by $[x]$, $[y]$. By normalizing properly one obtains $[u] = [-\alpha x + y]$ and $[v] = [\beta x - y]$ form which the result follows.)

4.31 Compute the Hilbert metric for a disc and discuss its relation to the Poincarè metric (Hint: same?)

[24] The lines in $\mathbb{P}$ are given by $[\alpha v + \beta w]$ where $\alpha, \beta \in \mathbb{R}$ where $[v] \neq [w]$.

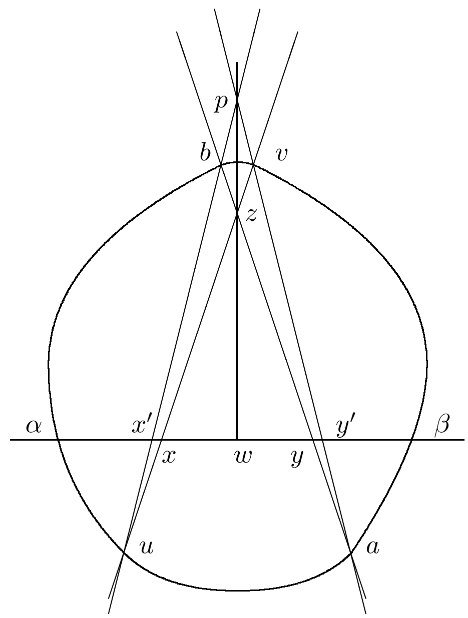


Figure 4.2: Hilbert metric

4.32 Kobayashi and Caratheodory metrics....

4.33 Show that the Hilbert metric for the cone $\mathcal{C}_\alpha$ defined in (4.3.8) is given by ???

4.34 Prove fromula (4.3.5).

4.35 Show that the Perron-Frobenius operator associated to a smooth expanding map of the circle has a spectral gap as an operator on $Lip(\mathbb{T}^2)$. (Hint: Check that there exists $b \in \mathbb{R}^+$ such that the norm

$$\|h\| := \|h\|_\infty + b\|h\|_{Lip}$$

is adapted to the cone. Define $\mathbb{V} = \{h \in Lip(\mathbb{T}^2) \mid \int h = 0\}$, notice that $\mathcal{L}\mathbb{V} = \mathbb{V}$. Then, for each $h \in \mathbb{V}$ there exists$\rho \in \mathbb{R}^+$ such that $h + \rho h_* \in \mathcal{C}_\alpha$, so

$$\|\mathcal{L}^n h\| = \|\mathcal{L}^n(h + \rho h_*) - \rho h_*\| \leq K\Lambda^n \rho.$$

Thus the spectral radius of $\mathcal{L}|_{\mathbb{V}}$ is less than Λ.)

4.36 Estimate the rate of mixing for Lipschitz functions for a smooth expanding map of the circle (Hint: use the spectral gap of the previous Problem.)

4.37 Prove that any continuous fraction of the form

$$\cfrac{1}{a_1 + \cfrac{1}{a_2 + \ddots}}$$

$a_i > 0$ is convergent provided the series $\sum_{n=1}^{\infty} a_n$ is divergent. (Hint: Let

$$\prod_{i=1}^{n} \begin{pmatrix} 1 & a_{2(n-i)} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ a_{2(n-1)+1} & 1 \end{pmatrix} \begin{pmatrix} 1 \\ u \end{pmatrix} = \begin{pmatrix} \beta_n \\ \alpha_n \end{pmatrix}$$

and verify, by induction, that $\frac{\alpha_n}{\beta_n}$ is exactly the $2n$ truncation of the continuous fraction. Thus the continuous fraction is a projective coordinate for the vector (α_n, β_n). Consider the cone $\mathcal{C}_+ = \{(x, y) \in \mathbb{R}^2 \mid x \geq 0;\ y \geq 0\}$. Then, for each $a, b \in \mathbb{R}^+$, holds

$$\begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ b & 1 \end{pmatrix} \mathcal{C}_+ \subset \mathcal{C}_+.$$

The result follows by computing the Hilbert metric contraction.

For a different approach see [91][Th14.1].)

4.38 Show that, Anosov systems form an open set in the $\mathcal{C}^{(1)}$ topology.

Notes

When cones first appeared?

The point of W. th. is that no estimate on the cone contraction is needed. To appreciate the advantage one can try to prove the existence of the LE via direct estimates for the Levovich map in example 4.1.1.c

Generalization of metrics on cones, see [90]

CHAPTER 5

Description of Billiard systems

Billiards are very widely studied model systems. In general they consists of a material point confined in some region of $\mathbb{R}^n$ or $\mathbb{T}^n$ with piecewise smooth boundaries;[1] in the simplest situation such a point moves with constant velocity until it reaches the boundary, and at the boundary it undergoes an elastic reflection. Such models include, e.g., a system of n hard spheres that interacts via elastic collisions (see section 5.6); the importance of such a systems as a basic model in statistical mechanics can be hardly overestimated.

These systems are conceptually extremely simple, yet they have an unpleasant feature: they lack smoothness. As we will see in the following there are three main type of non-smoothness: a) tangent collisions; b) collision with a corner; c) accumulation of infinitely many collisions in a finite time. Due to such pathologies these models, in spite of their simplicity, may present some incredibly annoying complications in their treatment.

Let $\mathcal{B}$ be the region in which the point is allowed to move and suppose that $\partial\mathcal{B}$ is a finite union of smooth manifolds with boundary. Clearly the motion can be seen as a flow ϕ_t on the unitary tangent bundle of $\mathcal{B}$ (in fact, given the initial position and the initial velocity the following motion is uniquely determined, moreover the modulus of the velocity will be constant through the motion, so it can be assumed equal to one without loss of generality).[2]

It can be checked directly that the flow is symplectic (Hamiltonian) in $\widetilde{X} := \mathcal{B} \times \mathbb{R}^n$ (see problem Problem 5.6). So, calling m the measure induced by Lebesgue on $X \equiv \mathcal{B} \times \{v \in \mathbb{R}^n \mid \|v\| = 1\}$, (X, ϕ_t, m) is a smooth flow with collisions (crf. Examples 3.7.1).

[1] Although one can easily consider billiards in a region of a Riemannian manifold with piecewise smooth boundaries, in this case the motion in the interior is just the geodesic flow; see [17] for such a general setting and section ?? for a similar generalization.

[2] A little thought will convince the reader that two motions with initial velocities that differ only in modulus will be exactly the same apart from the fact that they are run at different speeds.

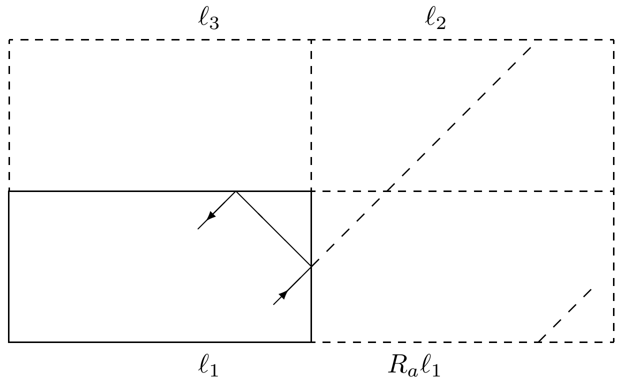


Figure 5.1: The trajectory beyond the mirror

5.0.1 Examples

5.0.1.a *Polygonal Billiards*

The name is self-explanatory: the domain $\mathcal{B}$ is a polygon. The simplest case is probably a rectangle: $\mathcal{B} = [0, a] \times [0, b] \subset \mathbb{R}^2$. Although the notion is fairly trivial, to study it we will employ a neat trick that has many other applications (e.g., see 5.4.1). Consider a trajectory $x + vt$ that reaches the wall $\ell_1 := \{(a, y)\}$. The law of reflection states that, if $v = (v_1, v_2)$, the reflected velocity is $(-v_1, v_2)$. Now define the map $R_a(x, y) = (2a - x, y)$. This is a reflection ($R_a^2 =$ identity) with respect to the wall $\{(a, y)\}$. Remark that $R_a\mathcal{B} = [a, 2a] \times [0, b]$, moreover $DR_a(-v_1, v_2) = v$. This means that, in the reflected box $R_a\mathcal{B}$, the reflected velocity is equal to the velocity before reflection.

The above algebraic discussion corresponds to a very intuitive geometrical fact: if the wall is a mirror, then the trajectory in the mirror is the continuation of the trajectory before collision (see figure 5.1).

After noticing this it is quite clear that one can understand better the trajectory in the “universal covering” of the box obtained be reflecting the box repeatedly with respect to its walls. In this covering the trajectory is simply a straight line and the trajectory in the original box is obtained by undoing the reflections (for the more mathematical inclined let us say that the plane is covered by equal boxes that are identified via reflections, see Problem 5.1). It is then obvious that, given the original velocity v only four velocities are possible: $(\pm v_1, \pm v_2)$. In fact, if we identify the opposite sides in figure 5.1 (that is, side ℓ_3 with side ℓ_1, side ℓ_2 with $R_a\ell_1$ and so forth) we obtain exactly a flat torus with sides twice as long as the ones of the original rectangle. In addition, the motion on such a torus corresponds precisely to the flow at unit speed in direction v. in other words the motion is equivalent to the rigid translations (geodesic flow) on the associated

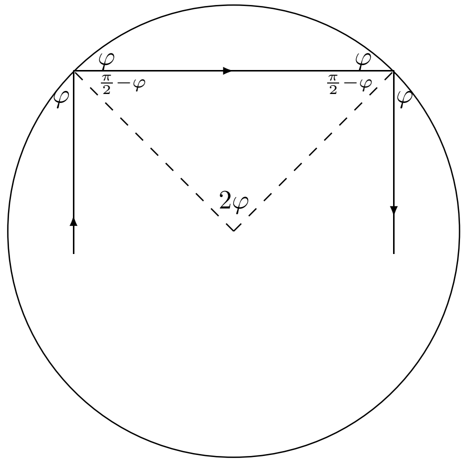


Figure 5.2: Trajectory in a Circular billiard

torus.

Accordingly, the motion is ergodic only if $\frac{v_1 b}{v_2 a}$ is irrational (see).

5.0.1.b *Circular Billiards*

In this case $\mathcal{B}$ is a disk of radius r. For convenience, let us center it at the origin of a Cartesian coordinate frame. Let us consider a point that has just collided with the boundary at the position $rn(\theta) := r(\cos\theta, \sin\theta)$, where θ is the angle with the x axis counted counterclockwise, and has velocity $v(\theta - \varphi) := (-\sin(\theta-\varphi), \cos(\theta-\varphi))$, which means that the velocity forms an angle φ with the tangent at the collision point. Accordingly, the trajectory will move along the cord of length $2r\sin\varphi$ and collide with the angle $\pi - \varphi$ which, after reflection, will be φ again.

This phenomena is nothing else than the conservation of the angular momentum (for the mechanical inclined) or of the Claroit integral (for the differential geometers).

All the above implies that, if $\frac{\varphi}{\pi} \in \mathbb{Q}$, then the motion will be periodic, otherwise the collision point will perform an irrational rotation on the boundary. In fact, let us choose as coordinates the distance τ form the last collision point computed along the trajectory; the distance s, computed along the circumference, of the last collision point from a fixed point on the circumference; and the angle φ. Then the phase space is

$$X = \{(\tau, s, \varphi) \in [0, r] \times S^1 \times [0, \pi] \mid 0 \le \tau \le 2r\sin\varphi\}$$

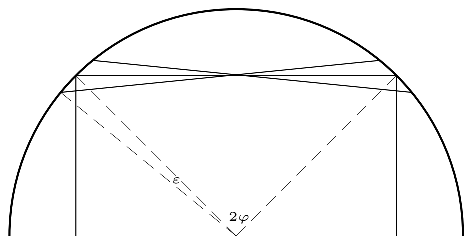


Figure 5.3: Focusing trajectories in a circular billiard

and the flow is noting else that a suspension flow (see section 1.3) with ceiling function $2r \sin\varphi$ constructed on the map T defined by

$$T(s, \varphi) = (s + 2\varphi, \varphi).$$

At the same time the middle point of the cords between two consecutive collisions will describe an irrational rotation on the circle of radius $r(1 - \cos\varphi)$. Note that this means that the trajectory under consideration (if $\varphi/\pi \notin \mathbb{Q}$) covers densely a two dimensional torus in the three dimensional space (see Problem 5.5) and it is ergodic restricted to it. The circle of radius $r(1 - \cos\varphi)$ circle is called *caustic*; the name derives from optic because if the trajectory is run by a beam of light that is the place with the highest luminosity.[3]

To understand better the concept of caustic consider a bunch of near by trajectories that all collide with the same angle φ, see Figure 5.3. If we consider two trajectories colling at s and $s + \varepsilon$, then it is an easy exercize in elementary geometry to compute the lenght of the segment on the first trajectory between the collision and the point in which the two trajectories intersect. The result is

$$2r \sin\frac{\varepsilon}{2} \cos(\varphi - \frac{\varepsilon}{2}) + \frac{2r \sin\frac{\varepsilon}{2} \sin(\varphi - \frac{\varepsilon}{2}) \cos\varepsilon}{\sin\varepsilon} = r \sin\varphi + \mathcal{O}(\varepsilon^2). \qquad (5.0.1)$$

That is the two trajtories meet (focus) at the middle point appart from an error of order ε^2. Thus a bunch of trajectories nearby to a given one will all focus approximatively on the middle point, the *burning* point indeed.

The above examples correspond to very regular motions ("integrable motion") that is exactly the opposite of what we mean to investigate. Unfortunately, to progress in the direction we are interested in many more technical

[3] In ancient Greek caustic ($\kappa\alpha\upsilon\sigma\tau\iota\kappa\acute{o}\zeta$) means "that burns". Of course, that would be an important concept if you want, e.g., burn a Roman ship. (check the Optic of Euclid)

tools are needed. Yet, before going on with general facts and definitions let us anticipate two concrete examples that will be particularly relevant in the following.

5.1 Sinai Billiard

The simplest example of Sinai billiards (introduced in [79] and studied in [80]) are given when $\mathcal{B} \subset \mathbb{T}^2$. More precisely, given a disk D, centered at the origin and with diameter $r < \frac{1}{2}$, let $\mathcal{B} = \mathbb{T}^2 \backslash D$. Calling $(x, v) \in \mathcal{B} \times \mathbb{R}^2$ the position and the velocity, respectively, the motion is described by a free flow

$$\phi_t(x, y) = (x + vt, v), \tag{5.1.1}$$

provided $\|x + vt\| \geq r$, that is provided the motion does not exist $\mathcal{B}$. When $x \in \partial\mathcal{B} = \partial D$ a collision takes place. Of course, at the collision it must be $\langle x, v\rangle \leq 0$, the velocity points toward D, otherwise the point would not have reached the obstacle D but rather would be flowing away from it. The collision law is, as already said, an elastic collision–namely the total energy and the momentum tangential to the collision plane must be preserved. Thus, calling v_- the velocity before collision and v_+ the velocity after collision, we require

$$\|v_+\| = \|v_-\| \,; \quad \langle Jx, v_-\rangle = \langle Jx, v_+\rangle,$$

where

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

so that $\langle Jx, x\rangle = 0$, that is $r^{-1}Jx$ is a unit vector tangent to the disk and oriented counterclockwise. This implies:

$$v_+ = v_- - \frac{2}{r^2}\langle x, v\rangle x. \tag{5.1.2}$$

5.1.1 Flow

From the above discussion it is clear that (X, ϕ^t, m) is a smooth flow with collisions, the only property that need to be checked is (3.7.1) (see Problem ???).

Let us call $V(x, v) = (v, 0)$ the vector field generating ϕ^t. A useful fact is the following.

Lemma 5.1.1 *If $w \in \mathcal{T}_\xi X$ and $\langle w, V(\xi)\rangle = 0$, then $\langle d\phi^t w, V(\phi^t(\xi))\rangle = 0$.*

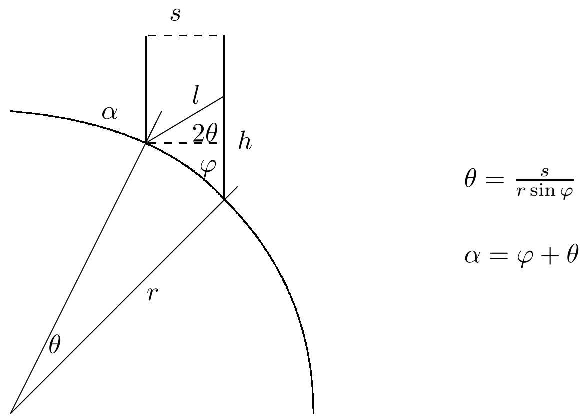


Figure 5.4: Collision

Proof. If no collision takes place, then the statement it is obvious by equation (5.1.1) and since for each $w = (w_1, w_2) \in \mathcal{T}X$ it must be $\langle w_1, v\rangle = 0$ (just differentiate $\|v\|^2 = 1$). Let us see what happens at collision.

Given the tangent vector $w = (w_1, w_2)$ at the point $\xi \in X$, we can consider the curve $\gamma(s) = \xi + ws$ that generates it ($\gamma'(0) = w$). Suppose that the next collision takes place with an angle φ. If we refer to the Figure 5.4 all we need to compute is the relation between h and l. A bit of geometry shows that

$$h = s \arctan\varphi + \mathcal{O}(s^2); \quad l = \frac{s}{\cos\theta}\arctan\varphi + \mathcal{O}(s^2) = s\arctan\varphi + \mathcal{O}(s^2).$$

Thus, if τ is the collision time of the trajectory starting at ξ and $\tilde{\gamma}(s) = \phi^{\tau_+}(\gamma(s))$, we have $\tilde{\gamma}'(0) = d_\xi\phi^{\tau_+}w := \tilde{w}$, and, calling v_+ the velocity after reflection, $\langle v_+, \tilde{w}\rangle = 0$, which proves the lemma. □

This means that in this case there is a particularly simple way to quotient out the flow direction (see ???): consider only vectors perpendicular to the flow.

5.1.2 Poincaré map

For many purposes it is useful to view the Sinai billiards as a symplectic map from a two dimensional domain to itself. Such a reduction is obtained via a general technique widely used in dynamical system: a Poincaré section (see 1.2). A Poincaré section consists in introducing some codimension one manifolds in the phase space X and then defining a map from such manifolds to themselves in such a way that to each point is associated its first return to the manifolds (if it exists). Let us be more concrete.

Historically the choice of the section to realize a Poincaré map as been based on $\partial\mathcal{B}$. In our case this consists of the boundary of the disk, that is a circle. Of course, it is also necessary to specify the velocity. Clearly there are two possibilities: one can consider velocities just before collision, which means $\langle x, v\rangle \leq 0$, (this is the Poincaré map from before collision to before collision) or one can consider the velocity just after collision, meaning $\langle x, v\rangle \geq 0$, (that is the Poincaré map from just after collision to just after the next collision). The two choices are equivalent, let us make the second.

If we identify the velocity by the angle φ between the tangent (directed clockwise) to the disk at the collision point and the velocity vector v, then the phase space is $\mathcal{M} = S^1 \times [0, \pi]$.

We can then define a map $T : \mathcal{M} \to \mathcal{M}$ in the following way: for each $\xi \in \mathcal{M}$, let $T\xi$ be the point just after the next reflection (if such a reflection exists). Note that, if no reflections would occur, almost all the trajectories would fill $\mathbb{T}^2$ densely,[4] hence T is defined almost everywhere.

It is natural to use as coordinate on the boundary the distance s, computed counterclockwise along the circle, from a given point. If we want to compute the induced invariant measure on the Poincaré section $\mathcal{M}$, according to section ?, we have to introduce the change of coordinates $\Xi : \mathcal{M} \times [0, \delta] \to X$ defined by

$$\Xi(s, \varphi, t) = (rn(sr^{-1}) + v(sr^{-1} + \varphi)t, sr^{-1} + \varphi - \frac{\pi}{2}),$$

where $n(\theta) = (\cos\theta, \sin\theta)$, $v(\theta) = (\sin\theta, -\cos\theta)$. In this coordinates a point is determined by its collision data (s, φ) and the time t past from the last collision.

A direct computation shows that

$$\begin{aligned}
\det \Xi &= \begin{vmatrix} -v(sr^{-1}) + r^{-1}n(sr^{-1} + \varphi)t & n(sr^{-1} + \varphi) & v(sr^{-1} + \varphi) \\ r^{-1} & 1 & 0 \end{vmatrix} \\
&= \begin{vmatrix} -v(sr^{-1}) & n(sr^{-1} + \varphi) & v(sr^{-1} + \varphi) \\ 0 & 1 & 0 \end{vmatrix} \\
&= \begin{vmatrix} -v(sr^{-1}) & v(sr^{-1} + \varphi) \end{vmatrix} = \sin\varphi.
\end{aligned}$$

So, given a set $A \subset \mathcal{M}$, calling $A_\varepsilon = \cup_{t=0}^{\varepsilon} \phi^t A$,

$$\mu(A) := \frac{1}{\varepsilon} m(A_\varepsilon) = \frac{1}{\varepsilon} \int_{A_\varepsilon} |det(\Xi)| ds d\varphi dt = \int_A \sin\varphi ds d\varphi.$$

Thus $d\mu = \sin\varphi ds d\varphi$ and $(\mathcal{M}, T, \mu)$ is a Dynamical Systems.

It is interesting to notice that μ becomes degenerate for $\varphi \in \{0, \pi\}$, which correspond to tangent collisions. Another annoying feature of the above choice

[4]Since for almost all velocities v we would have an irrational translation on $\mathbb{T}^2$, see Problem 5..

is that some trajectories never meet the boundary of the disk (for example consider the initial condition $x = (1, 0)$, $v = (0, 1)$) and other will travel an arbitrarily long time before the next collision.[5] These facts, although not catastrophic, may look unpleasant to someone. It is therefore relevant to notice that there are several other possible choices for the Poincaré section, each one with is own advantages and disadvantages. Let us see a couple of them.

Consider the fundamental domain $Q = [-\frac{1}{2}, \frac{1}{2}]^2$ of $\mathbb{T}^2$, choose ∂Q as the basis of the Poincaré section. Of course ∂Q it is not a smooth manifold (it consists of four lines). This problem is easily solved by extending the concept of Poincaré section to the case in which the section Σ is a finite (or even countable) union of smooth manifolds; a quick look at section ?? will convince the reader that this generalization is indeed immediate. This section has the the advantage of a simple structure, that there is a maximal time from Σ to itself, yet it does not solves the problem of the degeneration of the measure. Here as well we have problems with the trajectories that meet the section at very small angles.

To overcome such a problem one can choose the section $\Sigma \times [\delta, \pi - \delta]$. It is easy to see that if δ is chosen small enough then the only effect is to skip at most one crossing of the boundary Σ (see Problem 5..).

We will keep using the relation between the two dynamical systems (X, m, ϕ_t) and $(\mathcal{M}, \mu, T)$. In particular it is convenient to define the cone family on all $\mathcal{T}X$ instead that only on $\mathcal{T}\mathcal{M}$. We will see that an invariant cone family on $\mathcal{T}X$ induces an invariant cone family on $\mathcal{T}\mathcal{M}$.

5.1.3 Singularity manifolds

In this subsection we will study more precisely the singularities of the system and we will verify that they belong to the general setting developed in 3.6. We will consider two different Poincaré section to give the reader a more complete idea of the situation.

Let us start with the classical section *just after collision*. As already mentioned the phase space is $\mathcal{M} = S^1 \times [0, \pi]$. Clearly the only singularities of the map correspond to coordinates where the next collision is a tangent one. To analyze such a pathology it is more convenient to look at the billiard on the universal covering of the torus. In such a covering the obstacles will form a a lattice and the particles moves along a straight line between collisions (see figure 5.5).[6]

[5] This property is called *infinite horizon*. We will discuss it further in the sequel, see ???.

[6] This trick is very similar to the one used at the beginning of the chapter to discuss rectangular billiards, only now we take advantage of the periodicity of the torus rather than the invariant properties of the domain with respect to the reflections.

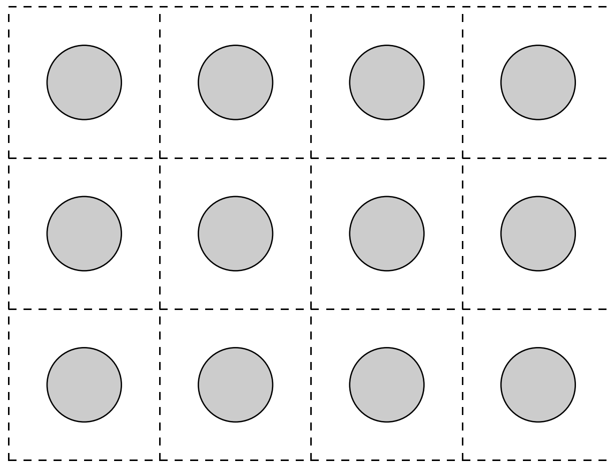


Figure 5.5: Universal covering of a Sinai billiard

The particle with coordinates (s, φ), just after collision, will move in the direction $v(r^{-1}s + \varphi)$ with unit speed.[7] Hence, if $C \in \mathbb{R}^2$ is the coordinate of the center of the obstacle with which the next collision will take place, the condition for a tangent collision reads

$$rn(r^{-1}s) + tv(r^{-1}s + \varphi) = C \pm rn(r^{-1}s + \varphi). \tag{5.1.3}$$

Where $t = t(s, \varphi)$ is the collision time. From (5.1.3 we can immediately extract two interesting informations multiplying it by $n(r^{-1}s + \varphi)$ and $v(r^{-1}s + \varphi)$ respectively

$$\langle C, v(r^{-1}s + \varphi)\rangle = t + r\sin\varphi > 0$$
$$F(s, \varphi) := \langle C, n(r^{-1}s + \varphi)\rangle - r\cos\varphi \pm r = 0$$

Taking the derivative of F with respect to φ we get

$$-r\sin\varphi + \langle C, v(r^{-1}s + \varphi)\rangle = t > 0,$$

thus we can apply the implicit function theorem ([1]) and conclude that the manifold corresponding to this discontinuity can be represented as the graph of a function $\varphi(s)$. In addition,

$$\frac{d\varphi}{ds} = -(\frac{1}{r} + \frac{\sin\varphi}{t}) < 0. \tag{5.1.4}$$

Since there are infinitely many obstacles with which the next collision can take place, there must be countably many discontinuity line (some of them are schematically represented in figure 5.6)

[7] Remember the convention $n(\theta) := (\cos\theta, \sin\theta)$ and $v(\theta) := (\sin\theta, -\cos\theta)$.

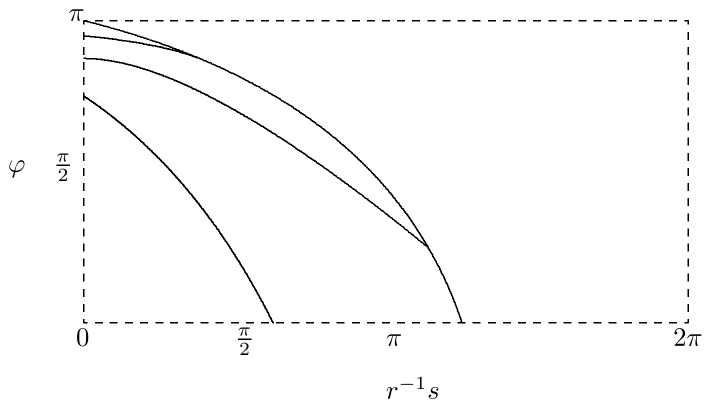


Figure 5.6: Few discontinuity lines in the Poicaré map

Next let us consider a different Poincaré section. In particular we would like to avoid to have a degenerate metric and symplectic form at the boundary of the section. Instead of using one of the proposal advanced in the previous section let us just consider the boundary of the obstacle together with all the angles in the interval $[\delta.\pi - \delta]$, for some sufficiently small δ. In this case we have two types of discontinuities: the tangencies and the preimmages of the boundary of the section (before they where the same thing). We will call this type of sections *regularized boundary section*.[8]

To analyze the preimages of the boundary of the section one can proceed in analogy with what we have done before, equation (5.1.3) in this case becomes

$$rn(r^{-1}s) + tv(r^{-1}s + \varphi) = C \pm rn(r^{-1}s + \varphi \pm \delta). \tag{5.1.5}$$

From (5.1.5) we obtain

$$\frac{d\varphi}{ds} = -(\frac{1}{r} + \frac{\sin\varphi}{t + r\sin\delta}) < 0. \tag{5.1.6}$$

Clearly the map is smooth up to, and including, this type of discontinuity, not so for the tangencies. In fact, it is easy to verify that the map is continuous crossing a tangency line (see Problem 7.8) but we will see immediately that it is not differentiable. By the discussion of section 5.3 (see in particular formulae (5.3.1) and (5.3.2)) it follows that if the next collision takes place with an angle $\varphi \notin [\delta, \pi - \delta]$, then calling τ_1 the time up to the tangent collision

[8]The meaning of the word "regularized" should be obvious but see also Problem 5.7.

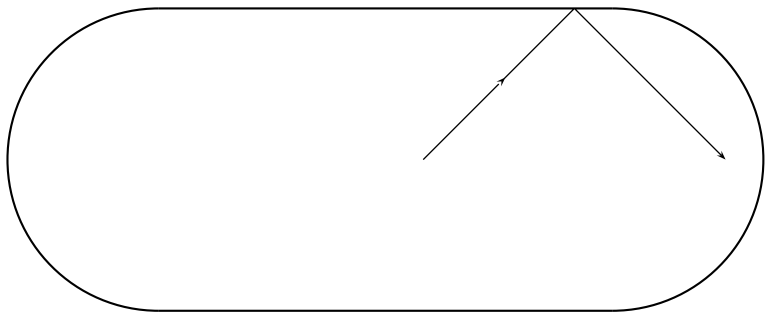


Figure 5.7: Bunimovich Stadium

and τ_2 the time from the tangent collision to the next, we have the formula

$$DT = \begin{pmatrix} 1 + \frac{2\tau_1}{r\sin\varphi} & \frac{2}{r\sin\varphi} \\ \tau_2(1 + \frac{2\tau_1}{r\sin\varphi}) & 1 + \frac{2\tau_2}{r\sin\varphi} \end{pmatrix}.$$

Clearly the norm of DT is bounded by a constant times $\frac{1}{\sin\varphi}$ (if in doubt do Problem 5.9). Now, if a point has distance ε from the singularity line, it will land at a distance $\sqrt{\varepsilon}$ from the tangency, which means that there exists a constant $c_t > 0$ such that, calling $\mathcal{S}$ the singularity line and ξ the point

$$|\sin\varphi| \geq c_t\sqrt{d(\xi,\mathcal{S})}.$$

Which means that the Derivative blows up only as a square root getting close to the singularity. By similar considerations it is possible to verify also that the second derivative blows up polinomially (see Problem 5.10).

5.2 Bunimovich Stadium

These billiards have been introduced [15] and first studied [16] by Bunimovich. In this case the table of the billiard is a convex subset of $\mathbb{R}^2$. The simplest, and original, one consists in two half circles joined by two straight lines (see Figure 5.7).

The name "stadium" is due to the shape of the domain $\mathcal{B}$ in which the motion takes place. The only difference is that now the curvature of the boundary is either zero (collisions against the straight segments) or negative (collision against the half circles).

5.2.1 Flow

We have seen that the flow in a square or in a circle is well defined and rather regular. Clearly the only relevant discontinuity in the Bunimovich Billiard arises when a trajectory hits the joining between the circumference and the straight lines.

5.2.2 Poincaré map

Use coordinates introduced for circle and (eventually) mention Lazutkin ....

....

5.3 Collision map and Jacobi fields

To compute, in general, the collision map it is helpful to introduce appropriate coordinates in $\mathcal{T}X$. A very interesting choice is constitute by the *Jacobi fields*.[9] Let X_- be the set of configurations just before collision. For each $(x,\, v) \in X \backslash X_-$ there exists $\delta > 0$ such that

$$\phi_t(x,\, v) = (x + vt,\, v) \quad 0 \le t \le \delta.$$

Let us consider the curve in $\mathcal{X}$

$$\xi(\varepsilon) = (x(\varepsilon),\, v(\varepsilon)),$$

with $\xi(0) = (x,\, v)$ and $\|v(\varepsilon)\| = 1$.

For each t such that $\phi_t(\xi(0)) \notin X_-$, let

$$\xi(\varepsilon,\, t) = (x(\varepsilon,\, t),\, v(\varepsilon,\, t)) = \phi_t(\xi(\varepsilon)).$$

The Jacobi field $J(t)$ is defined by

$$J(t) \equiv \left.\frac{\partial x}{\partial \varepsilon}\right|_{\varepsilon=0}.$$

[9] The Jacobi Fields are a widely used instrument in Riemannian geometry (see [36]) and have an important rôle in the study of Geodetic flows, although we will not insists on this aspect at present. Here they appear in a very simple form.

Note that, since $x(0,\, t) \notin X_-$, for $s < \delta$

$$\xi(\varepsilon,\, t+s) = \xi(\varepsilon,\, t) + (v(\varepsilon,\, t)s,\, 0),$$

so

$$J'(t) = \frac{dJ(t)}{dt} = \left.\frac{\partial v(\varepsilon,\, t)}{\partial \varepsilon}\right|_{\varepsilon=0}.$$

That is, $(J(t),\, J'(t)) = d\phi_t \xi'(0)$.

At each point $\xi = (x,\, v) \in X$ we choose the following base for $\mathcal{T}_\xi X$:[10]

$$\eta_0 = (v,\, 0);\;\; \eta_1 = (v^\perp,\, 0);\;\; \eta_2 = (0,\, v^\perp);$$

where $\|v^\perp\| = 1$, $\langle v,\, v^\perp\rangle = 0$.

The vector η_0 corresponds to a family of trajectories along the the flow direction and it is clearly invariant; η_1 to a family of parallel trajectories and η_2 to a family of trajectories just after focusing. It is very useful the following graphic representation. We represent a tangent vector by drawing a curve that it is tangent to it. A curve in $\mathcal{T}X$ is given by a base curve that describes the variation of the x coordinate equipped with a direction at each point (specified by an arrow) which show how varies the velocity (see Figure ??).

A direct check shows that each vector η perpendicular to the flow direction will stay so (see Lemma 5.1.1 and Problem 5.), i.e.

$$\langle d\phi_t\eta,\, (v_t,\, 0)\rangle = \langle d\phi_t\eta,\, d\phi_t(v,\, 0)\rangle = \langle \eta,\, (v,\, 0)\rangle = 0.$$

So the free flow is described by

$$d\phi^t\eta_0 = \eta_0; \quad d\phi^t\eta_1 = \eta_1; \quad d\phi^t\eta_2 = \eta_2 + t\eta_1,$$

that is, in the above coordinates

$$d\phi^t = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & t & 1 \end{pmatrix}. \tag{5.3.1}$$

Let us see now what happens at a collision.

Let $x_0 \in \partial\mathcal{B}$ be the collision point and let $\xi_c = (x_0, v)$ be the configuration at the collision. We want to compute $R_\varepsilon := d_{\phi^{-\varepsilon}\xi_c}\phi^{2\varepsilon}$, that is the tangent map from just before to just after the collision. Clearly $R_\varepsilon\eta_0 = \eta_0$. From the Figure 5.8 follows that, if $\gamma(s)$ is the curve associated to η_1 at the point $\phi^{-\varepsilon}\xi_c$,

$$d\phi^{2\varepsilon}\gamma(s) = \left(v_+^\perp\left[s + \varepsilon\frac{2s}{r\sin\varphi}\right], \frac{2s}{r\sin\varphi}\right) + \mathcal{O}(s^2)$$

[10]Here $v^\perp = Jv$ with

$$J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

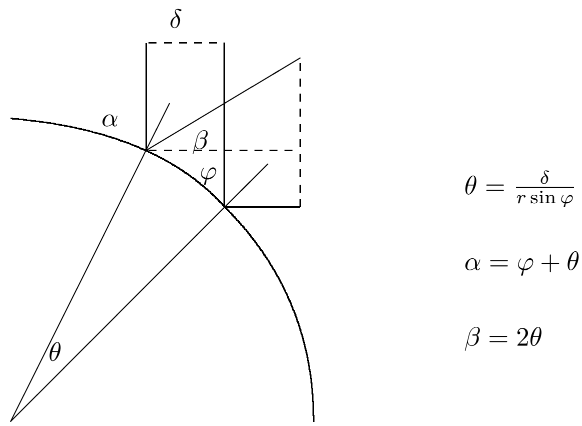


Figure 5.8: Collision

where r is the radius of the osculating circle (that is the circle tangent to the boundary up to second order) which is the inverse of the curvature $K(x_0)$ of the boundary at the collision point.

The above equation means that

$$J(\varepsilon) = (1 + \frac{2\varepsilon K(x_0)}{\sin\varphi})v_+^\perp.$$

Accordingly, calling $R = \lim_{\varepsilon\to 0} R_\varepsilon$ the collision map, we have

$$R\eta_1 = \eta_1 + \frac{2K}{\sin\varphi}\eta_2; \quad R\eta_2 = \eta_2.$$

Hence,

$$DR = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & \frac{2K}{\sin\varphi} \\ 0 & 0 & 1 \end{pmatrix}. \tag{5.3.2}$$

The above computations provide the following formula for the derivative of the Poincaré section from the boundary of the obstacle, just after collision, to the boundary of the obstacle just after the next collision

$$DT = \begin{pmatrix} 1 & \frac{2K}{\sin\varphi} \\ \tau & 1 + \frac{2\tau K}{\sin\varphi} \end{pmatrix}, \tag{5.3.3}$$

where τ is the flying time between the two collisions and φ the collision angle.

Formula (5.3.3) is sometimes called *Benettin formula* (e.g., [52], from ???).

5.4 Some general properties of billiards

A type of singularity that we have not yet discussed is the possibility of infinitely many collisions in a finite time. Let us see that, in two dimensions, such a pathology cannot occur for the above mentioned classes of billiards.

Lemma 5.4.1 *If the boundary of a convex billiard has bounded curvature, then there cannot be infinitely many reflections in a finite time.*

PROOF. .... □

For the case of unbounded curvature see Problem 7.4.

5.4.1 Finite collisions in an angle.

Let us now consider semi-dispersing billiards. Since the boundary is convex it follows that after a collision the particle cannot collide again with the same piece of boundary before colliding with some other piece or before traveling for some minimum time, in the case of motion on a torus. This means that infinitely many collision can, in principle, take place only where two smooth pieces of the boundary meet, that is at an angle. Here we will show the following fact.

Proposition 5.4.2 *In a two dimensional semi-dispersing billiard, if different pieces join non-tangentially, there a maximal number o collisions that can take place in a given time.*

To see this let us start to consider a particle in a flat angle. As we have already seen if we reflect the billiard at the collision point then the reflected trajectory is simply the continuation of the trajectory before collision. Applying this idea to the collision of a particle in a flat angle immediately yields a formula for the maximal number of collisions that can take place (see Figure 5.9). It is easy to convince oneself that when the boundaries are dispersing the number of collisions must be less than the number of collisions in a flat angle with the same angle.

discuss tangent angle+ convex + many dim

5.4.2 Applicability of Oseledec

....

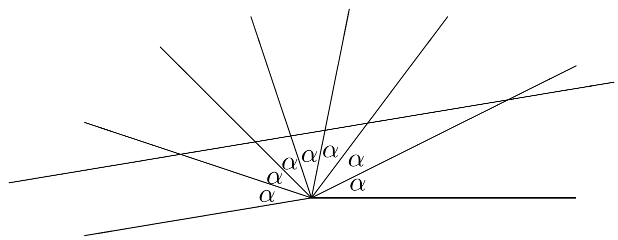


Figure 5.9: Reflecting a flat angle

Proposition 5.4.3 *The type of billiards described in Definition 5.5.1 are examples of smooth flows with singularities and their Poincaré section are dynamical systems systems with singularities.*

5.5 Different Tables, different games

Let us start with a bit of classification.

Definition 5.5.1 *Here are some standard classes of billiards:*

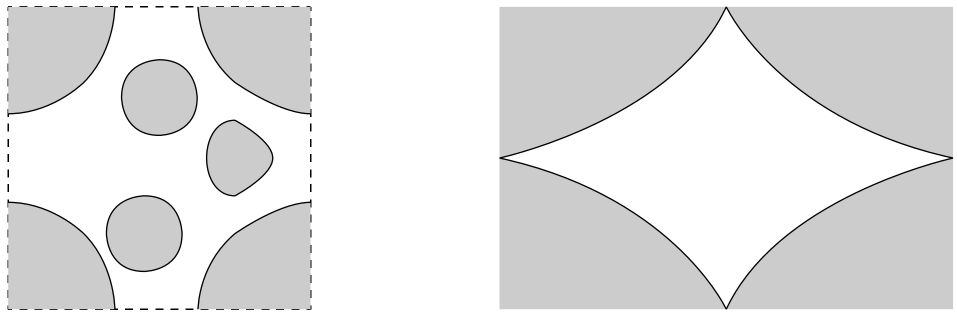
- *dispersing billiards are billiards with the boundary $\partial\mathcal{B}$ of the table is a finite union of strictly convex manifolds with boundary (this are often called Sinai billiards as well)*
- *semi-dispersing billiards are billiards with the boundary $\partial\mathcal{B}$ of the table is a finite union of strictly convex manifolds with boundary*
- *convex billiards are billiards where the tale $\mathcal{B}$ is a convex set.*

The remainder of the section is devoted to a more explicit description of several concrete examples of the above cases.

5.5.1 Dispersing

We have already seen the standard Sinai billiard in section 5.1. In general several convex obstacles may be present and they are not necessarily disjoint (see figure 5.10). One main issue in this class of billiards is the distinction between finite and infinite horizon. Finite Horizon means that there is a maximal time after which a collision must take place, infinite horizon means that there exists trajectories that never experience a collision.[11] The relevance

[11] Note that the other possibility (all the trajectories experience a collision in finite time but there does not exists an upper bound for such a time) cannot take place (see Problem 5.12).



Several Obstacles Diamond

Figure 5.10: Sinai Billiards

of such concept stems from the fact that orbit with no collision have zero Lyapunov exponents, hence the corresponding billiards cannot be uniformly hyperbolic.

5.5.1.a ***Infinite Horizon***

As already mentioned we have already seen the prototypical example in this class, yet it may be instructive to analize its properties a bit further. Consider the Sinai billiard described in section 5.1 and let r_1 the radius of the obstacle. Clearly it is necessary $r_1 < 1/2$ to have no self intersections of the obstacle. It is also obvious that if $r_1 < 1/2$ then there are trajectories that never collide. Let us study such trajectories a bit more in detail. First of all, since the system has a square symmetry, it is enough to consider trajectories with velocity in the first half of the first quadrant, i.e. velocities parallel to the directions $(1, \omega)$ with $\omega \in [0, 1]$.[12]

Let us consider the motion with no obstacle (a translation on the torus) and see if there are trajectories that never enter in the region $\|(x, y)\| \leq r_1$. Clearly such trajectories are trajectories for the billiard systems as well and precisely the trajectories that never experience a collision. For such trajectories it is particularly convenient to consider the Poincarè section determined by the line $S : \{x = -1/2\}$. If we look at the motion only when the particle intersects such a line we have that the corresponding map is given by $Ty = y + \omega \mod 1$, that is a rotation by ω of the circle $(-1/2, 1/2]$. If $\omega \notin \mathcal{Q}$ then the map T is ergodic (see) and this means that the trajectory will eventually collide. On the contrary, if $\omega = p/q$, $p, q \in \mathbb{N}$ and with no common

[12] If this it is not clear, read again the discussion of polygonal billiards in section 5.0.1.

divisors, then all the orbit will be periodic of period q (see) and it may be possible that some of them never collide.

Notice that a point in S with velocity parallel to $(1, \omega)$ will experience before a collision before meeting S again only if $y \in [-\omega/2 - r_1\sqrt{1+\omega^2},\ -\omega/2 + r_1\sqrt{1+\omega^2}]$. On the other hand, since the orbit of the point y has length q and because it is restricted to points of the type $y + n/q \mod 1$, which are exactly q, it follows that the orbit consists exactly of all such points. Accordingly, the orbit can avoid only intervals of size less than $1/q$. We can then conclude that there are orbits of the type p/q that never collide if and only if

$$2r_1\sqrt{1+\frac{p^2}{q^2}} < \frac{1}{q}. \tag{5.5.1}$$

If $q = 1$ then for $p = 0$, we have the already know result $r_1 < 1/2$; for $p = 1$ there can be no collisions only if $r_1 < \frac{1}{2\sqrt{2}}$. For $q \leq 2$ there are always collisions if $r_1 > [2\sqrt{5}]^{-1}$.

5.5.1.b *Finite Horizon*

The simplest case of Sinai billiard with finite horizon is obtained by employing two circular obstacles as in figure 5.11. We have thus a torus of size one together with a circular obstacle at the point $(0, 0)$ with radius r_1 and a circular obstacle at the point $(1/2, 1/2)$ with radius r_2.[13] Clearly

$$r_1 + r_2 < \frac{1}{\sqrt{2}}$$

in order for the obstacles not to intersect each other. By the discussion of the infinite horizon case it follows that we can choose $r_1 > 1/(2\sqrt{2})$ and $0 < r_2 < 1/\sqrt{2} - r_1$ to have a Sinai billiard with disjoint obstacles and finite horizon. For example one could choose $r_1 = 3/7$ and $r_2 = 1/4$ as in figure 5.11.

In the following we will need a more in depth undertanding of the above model. In particular we want to construct a nice Poincarè section in analogy with the ones introduced in section 5.1.3 and discuss the structure of the singularity lines for such a section.

The first step is to understand multiple consecutive tangencencies. Let us start with a double tangency of which the first is with the central obstacle. By symmetry one can limit the analysis to the case in which the second takes place with the upper right copy of the obstacle (see Figure 5.11. The position

[13]Remember that the coordinates are in the universal covering of the torus and that the points $(1/2, -1/2)$, $(1/2, 1/2)$, $(-1/2, 1/2)$, $(-1/2, -1/2)$ are identified.

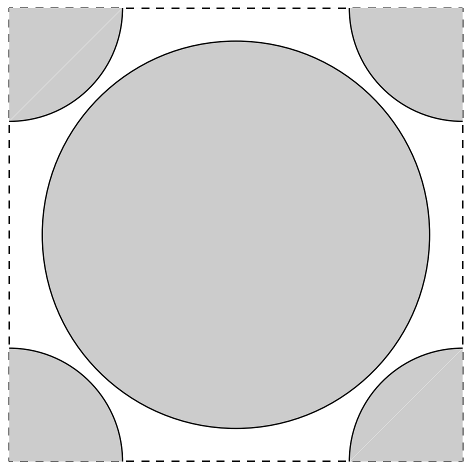


Figure 5.11: Simplest Sinai Billiard with finite horizon

of the particle at time t is given by $r_1 n(\theta) + v(\theta)t$, where $n(\theta) = (\cos\theta, \sin\theta)$ and $v(\theta) = (\sin\theta, -\cos\theta)$, so we have the next two equations

$$\|r_1 n(\theta) + v(\theta)t - p\| = r_2$$
$$\langle r_1 n(\theta) + v(\theta)t - p, v(\theta)\rangle = 0,$$

where $p = (1/2, 1/2)$ are the coordinates of the center of the second obstacle (of course we are working in the universal covering). The first equation determine the value of t for which the second collision takes place while the second impose that the collision is tangent. Solving the above equations yields

$$\frac{1}{\sqrt{2}} \cos\left(\frac{\pi}{4} - \theta\right) = \langle n(\theta), p\rangle = r_1 \pm r_2.$$

Accordingly we have four solutions: $\frac{\pi}{4} - \theta = \pm(\cos^{-1}\sqrt{2}(r_1 \pm r_2))$. In fact, only two are really relevant since the other two are obtained by symmetry around the line joining the two centers. It remains to check that the above trajectories do not intersect any other obstacle between the two tangencies. It turns out that the trajectories of the type $\frac{\pi}{4} - \theta = \pm(\cos^{-1}\sqrt{2}(r_1 - r_2))$ have a tangent collision with the central scatterer before colliding with the corner one. It is then easy to see that there can be at most four consecutive

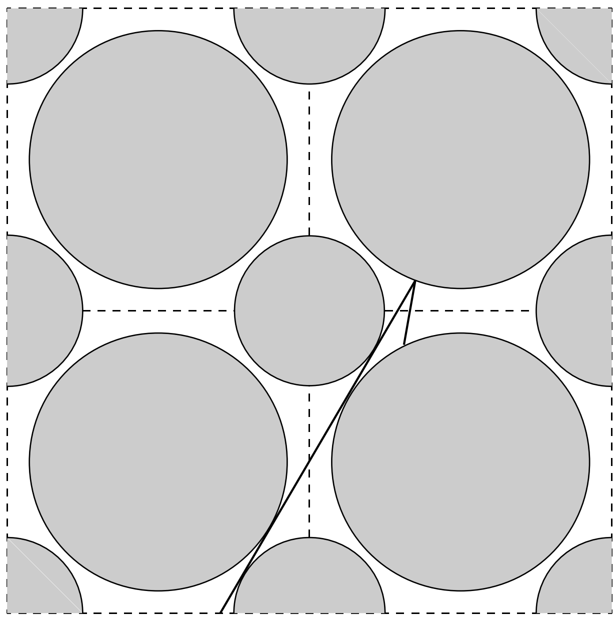


Figure 5.12: Multiple tangencies

tangencies after that the next collision will take place with an angle of more than 70 degree, see Figure 5.12.

At this points there are many different possibilities for the choice of the section, all more or less equivalent. Just to explore different possibility from the ones in section 5.1.3 we will make the choice depicted in figure 5.13. More precisely the Poincarè section consists of eight pieces (the configuration space part is the octagon in bold in the picture while the velocities are a subset of the inner ones–that is pointing toward the larger obstacle) $\{P_i\}_{i=1}^{8}$. For example, $P_1 = A_1 \times B$, where A_1 is the (horizontal) segment from the point $(3.75, 1)$ to the point $(.625, 1)$; while B is the set $[\pi/5, 4\pi/5]$.[14]. On the other hand $P_7 = A_7 \times B$ where A_7 is the segment, with an inclination of 45 degrees, from the point $(0, .625)$ to the point $(3.75, 1)$. All the other pieces of the section are defined by symmetry.

Describe the singularity lines and, maybe, make picture

[14]The choice of $\pi/5$ is completely arbitrary, any angle strictly larger than 0 and strictly smaller than $\pi/4$ will do as well.

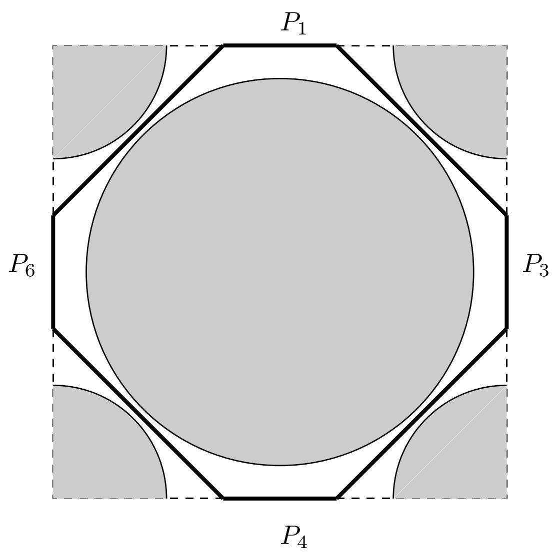


Figure 5.13: A Poincarè Section

5.5.1.c *Corners*

say something on the form of the discontinuity lines and the case of tangent corners.

5.5.2 Semi-dispersing

do def in two dim and comments in more

5.5.3 Convex

Cardiod

5.6 Hard spheres

We have already mentioned several times that the motions of several discs or balls that collide elastically among themselves is an example of billiard. It is finally time to look at this interesting fact in detail. We will treat explicitly the case of two disc in two dimensions and we will comment on the more general case.

5.6.1 Two balls, two dimensions

Let us start by considering the motion of two identical disks of mass one in $\mathbb{T}^2$. Again the motion is linear plus elastic collisions.

Clearly, the disks must have radius r sufficiently small in order to fit in the torus (see Problem 5.), for simplicity we assume $r < \frac{1}{4}$. The phase space is $X = \mathbb{T}^4 \times \mathbb{R}^4$.

If $x_1, x_2 \in \mathbb{T}^2$ are the coordinated of the center of the disks, the velocity changes at collision according to the law

$$\begin{cases} v_1^+ = v_1^- - \langle \eta, v_2^- - v_1^- \rangle (v_2^- - v_1^-) \\ v_2^+ = v_2^- + \langle \eta, v_2^- - v_1^- \rangle (v_2^- - v_1^-) \end{cases} \tag{5.6.1}$$

where η is a unit vector in the direction $x_2 - x_1$.[15]

Here there are more integral of motion than in the previous cases: the energy $E = \frac{1}{2}(\|v_1\|^2 + \|v_2\|^2)$ and the total momentum $P = v_1 + v_2$. Thus, if we want to obtain an ergodic systems, we have to reduce the system vie the integral of motion. we will then consider that phase spaces

$$X_{E,P} = \{(x_1, x_2, v_1, v_2) \in X \mid \frac{1}{2}(\|v_1\|^2 + \|v_2\|^2) = E;\ v_1 + v_2 = P\}.$$

Since, in the velocity space, the previous conditions correspond to the intersection between the surface of a four dimensional sphere (S^3) and a two dimensional linear space, the velocity vectors $(v_1 + v_2)$ is contained in a one dimensional circle. Thus, topologically, $X_{E,P} = \mathbb{T}^4 \times S^1$.[16] It is then natural to choose an angle θ as coordinate on S^1, moreover, since

$$2E = \|v_1\|^2 + \|v_2\|^2 = \frac{1}{2}\|v_1 - v_2\|^2 + \frac{1}{2}\|P\|^2,$$

it is hard to resist setting $v_2 - v_1 = v(\theta)$.[17] Hence,

$$\begin{cases} v_1 = \frac{1}{2}(P - v(\theta)) \\ v_2 = \frac{1}{2}(P + v(\theta)). \end{cases}$$

[15] To be precise $x_2 - x_1$ has no meaning since $\mathbb{T}^2$ it is not a linear space. Yet, at collision, the distance between the two disks is $2r$, so the global structure of $\mathbb{T}^2$ is irrelevant and we can safely confuse it with a piece of $\mathbb{R}^2$.

[16] Of course, we are considering only the cases $E \neq 0$, $P \neq 0$.

[17] As usual $v(\theta) = (\sin\theta, \cos\theta)$.

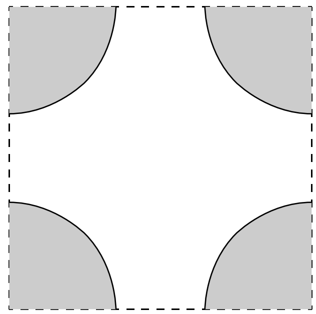


Figure 5.14: Reduced Systems

The free motion is then given by

$$\begin{cases} x_1(t) = x_1(0) + \frac{1}{2}(P - v(\theta))t \\ x_2(t) = x_2(0) + \frac{1}{2}(P + v(\theta))t. \end{cases}$$

Accordingly,

$$\begin{cases} x_1(t) + x_2(t) = x_1(0) + x_2(0) + Pt \mod 1 \\ x_2(t) - x_1(t) = x_2(0) - x_1(0) + v(\theta)t \mod 1. \end{cases}$$

It is then clear the need to introduce the two new variables $Q = x_1 + x_2$ and $\xi = x_2 - x_1$. The variable Q performs a translation on the torus, such a motions is completely understood and we can then disregard it. The only relevant motion is the one in the variables (ξ, θ). The reduced phase space is then $\tilde{=}\mathcal{B} \times S^1$ where $\mathcal{B} = \mathbb{T}^2 \backslash \{\|\xi\| \leq 2r\}$, that is the torus minus a disk of radius $2r$. The domain $\mathcal{B}$ is represented in Figure ?? and, apart the different choice of the fundamental domain, it corresponds exactly to the table of the simplest Sinai billiard.

The free motion corresponds to the free motion of a point as well, while at collision, from (5.6.1), we have

$$v(\theta^+) = v(\theta^-) - 2\langle \frac{\xi}{2r}, v(\theta^-)\rangle v(\theta^-)$$

that is exactly the elastic reflection from the disk! In conclusion the reduced system is exactly the simplest case of Sinai Billiard already studied in .

5.6.2 Many balls, many dimensions

It is now natural to wonder about n identical hard-balls moving on the d dimensional torus $\mathbb{T}^d$.

. . . .

5.7 Soft Billiards

For *soft billiard* it is meant billiard in which the particles moves subject to a potential field. They are called soft because the particle can penetrate the potential while the normal Billiards can be viewed as determined by an infinite (hard-core) potential that does not allow the particle to penetrate in certain regions. Since it is perfectly possible to consider cases in which the potential has an hard-core the name soft-billiards it is not always appropriate, *Billiards with potential* would be more accurate.

. . . .

5.7.1 Examples (More Billiards)

5.7.1.a *Kubo*

. . .

5.7.1.b *Knauf*

. . .

5.7.1.c *General results*

. . . [34] (donnay-liverani)

5.7.1.d *wedge with gravity*

. . .

Problems

5.1 Given a rectangular box $\mathcal{B}$, with its sides labeled by $\{1,2,3,4\}$ and let $R_i(\mathcal{B})$ be the reflection with respect to the side i of the box $\mathcal{B}$.[18] Let R_0 be the identity. Consider $G = \cup_{n=0}^{\infty}\{0,1,2,3,4\}^n$, if $g \in G$ then we define $R_g(\mathcal{B}) = R_{g_1}(\cdots(R_{g_n}(\mathcal{B}))\cdots)$ and, for each $g^i \in G$, $i \in \{1,2\}$, $g = g^2 \circ g^1 \in \{0,1,2,3,4\}^{n_1+n_2}$, is defined by $g_k = g^1_k$ for $k \le n_1$ and $g_k = g^2_{n-k}$, for $k > n_1$. Verify that $R_{g^2}(R_{g^1}(\mathcal{B})) = R_g(\mathcal{B})$. Introduce the equivalence relation $g_1 \sim g_2$ iff $R_{g^1}(\mathcal{B}) = R_{g^2}(\mathcal{B})$. Let $\tilde{G}$ be the collection of the equivalence classes. Verify the $\tilde{G}$ is a commutative group with respect to the operation "$\circ$″. (hint: Note that the geometrical meaning is simply that the final position of the box after a certain number of reflections does not depend on the order of the reflections. Clearly it suffices to check such a property for two reflections.)

5.2 Study the motion in a triangular billiard when the angle defining the triangle are all rational multiple of π. (Hint: use reflections again)

5.3 Study the motion in an elliptical billiard. (Hint: Verify that there exist an integral of motion.)

Lemma 5.7.1 *If the boundary of a convex billiard has bounded curvature, then there cannot be infinitely many reflections in a finite time.*

Proof. .... □

5.4 Show that for the Cardiod there cannot be infinitely many reflections in a finite time.

5.5 Verify that the caustics correspond to a two dimensional torus and the motion is ergodic restricted to it. (Hint: ...)

5.6 Check that the maps ϕ^t generated by a billiard flow are symplectic. (Hint: It is obvious for the free flow, it remains to check it for the reflections. This can be done by using formulae ???)

5.7 Find a change of variable that transforms the symplectic form in a regularized boundary section in the standard symplectic form.

5.8 Verify that, in a regularized boundary section, the map is continuous across a singularity line corresponding to a tangency.

[18] The label attached to the sides of the reflected box are the one obtained naturally from the old ones.

5.9 Prove that, given an $n \times n$ matrix A the norm $\|A\| := \sup_{v \in \mathbb{R}^n} \frac{\|Av\|}{\|v\|}$ where $\|v\| := \sqrt{\sum_n v_n^2}$ satisfies

$$\|A\| \leq \text{constant} \max_{ij} |A_i j|$$

and compute explicitly the optimal constant.

5.10 Determine the rate at which the second derivative explode as one gets near to a tangency singularity in the Sinai billiard with one circular obstacle in the torus.

5.11 Compute the number of collisions in a convex angle.

5.12 Show that in a Sinai billiard on $\mathbb{T}^2$ for which there exists trajectories that spend an arbitrary long time without colliding there must exists trajectories that never collide. (Hint: some continuity...)

5.13 Show that in a Sinai billiard on $\mathbb{T}^2$ almost all the trajectories must collide. (Hint: If a trajectory never collides then it performs a translation on the torus. Such translations are always ergodic apart from the case in which the ratio of the components of the velocity is a rational number. These trajectories are only countable.)

5.14 Let

$$L_i = \begin{pmatrix} 1 & 0 \\ t_i & 1 \end{pmatrix} \quad \text{and} \quad R_i = \begin{pmatrix} 1 & k_i \\ 0 & 1 \end{pmatrix},$$

for each $u \in \mathbb{R}^+$ write

$$\prod_{i=0}^{n} R_i L_i \begin{pmatrix} 1 \\ u \end{pmatrix} = \lambda_n \begin{pmatrix} 1 \\ u_n \end{pmatrix}.$$

Show that

$$u_n = k_n + \cfrac{1}{t_n + \cfrac{1}{k_{n-1} + \cfrac{1}{t_{n-1} + \ddots + \cfrac{1}{t_1 + \cfrac{1}{k_1 + \cfrac{1}{u}}}}}}$$

And find conditions for the convergence of the continuous fraction. (Hint: see Problem 4.37)

5.15 Study two disk with different masses.

5.16 Prove that the Poincarè map for the Sinai billiard is piecewise Hölder of Hölder exponent $\frac{1}{2}$.

Notes

Do a bit of history, maybe starting from Krylov on. About a bit of physical contest as well? or is it better to fit it in the chapter?

. . . .

CHAPTER 6

Examples of hyperbolic Billiards

In the previous chapter we have described some Billiards that, historically, have been very relevant in the development of the theory of hyperbolic systems. Yet, we have not investigated the hyperbolicity of such examples. To this task is devoted the present chapter.

6.1 Sinai Billiard

We define a cone family in the plane perpendicular to the flow direction $(v, 0)$, that is in the plane η_1, η_2,[1] this plane is naturally isomorphic to the tangent space of $\mathcal{M}$ (just project along the flow direction) in each non-tangent point. Let us sum up the result obtained in section 5.3. In the case in which no collision takes place we have seen that the parallel family η_1 stay parallel, while the most divergent family (the vector η_2) becomes less divergent (a linear combination of η_1 and η_2 with positive coefficients). This means that the first quadrant (in the η_i coordinates goes into itself but the η_1 side stays put–see Figure ??). Let us study what happens at a reflection. Looking at Figure 5.8 one can see that any divergent family of trajectories will be divergent after collision, and in particular the parallel family will be strictly divergent. To be more precise the η_2 family will go into itself after the collision while the parallel one will be strictly divergent. Again the cone goes strictly inside itself but one side (the η_2 one this time) stays put. Nevertheless, the combination of free motion and reflection clearly sends the cone strictly inside itself.

Since almost all trajectories experience a collision (see Problem 6.1) it follows that Wojtkowski theorem applies and that the Lyapunov exponents are different from zero almost everywhere for the dynamical system $(\mathcal{M}, T, m)$.

[1] See section 5.3 for the definition of η_1, η_2 and their properties.

6.2 Bunimovich stadium

The naïve understanding of the previous example is that the obstacle acts as a defocusing mirror and thus makes the trajectories diverge, whereby creating instability. This idea was already present in Krylov's work [58] and was considered the natural mechanism producing hyperbolicity. With this point of view in mind it seems that a table with convex boundaries (in which parallel trajectories are focused after reflections) is unlikely to yield hyperbolic behavior. This impression can be only confirmed by the presence of caustics in smooth convex billiards (see Problem 5.. and Notes of chapter 5). It came then as a surprise the discovery by Bunimovich that perturbations of the circle[2] could be hyperbolic. The main intuition is as follows: although the trajectories after reflection may be focusing, after some time they focus and then become again divergent. Hence, if there is enough time between consecutive collisions, we can still have divergent families going into divergent families (provided we look at the right place). Another, equivalent, point of view is that the instability is measured not just by the change in position but also by the change in velocity, from this point of view a very strong convergence it is not so different from a strong divergence.

To find a new invariant family of cones we have to remember a little the case of circular billiards (see Examples 5.0.1). We have seen that the collision angle is a conserved quantity of the motion. It is then natural to consider, at each point in phase space, the tangent vector η_3 associated to a family of trajectories that, at the next collision with one of the two half circles, will have the same collision angle.

We have defined η_3 in geometrical terms, clearly its expression in terms of η_1 and η_2 changes from point to point (see Problem 6.2). Yet, there are special points (the middle of the cord between two consecutive collision with the same half-circle) in which η_3 coincides with η_2 (this is seen immediately by geometric considerations, see Figure 5.3 and equation (5.0.1)).

Clearly, in a sequence of collisions with the same circumference the vector η_3 is invariant. Also, from the above considerations follows that before collisions η_3 corresponds to a diverging family, while immediately after a collision it corresponds to a convergent family.

What happens to the parallel family η_1? Since divergent families becomes convergent it is obvious that the parallel family, after reflection, becomes even more convergent. Hence, it will focus before the middle point to the next collision (the point where the family η_3 focuses).

The previous considerations suggest the cone $\mathcal{C}(x, v) = \{\xi \in \mathcal{TM} \mid \alpha\eta_1 + \beta\eta_3 \text{ with } \alpha\beta \geq 0\}$.

[2] Clearly non smooth perturbations such as the stadium, otherwise the KAM theorem would apply, see Problem 5..

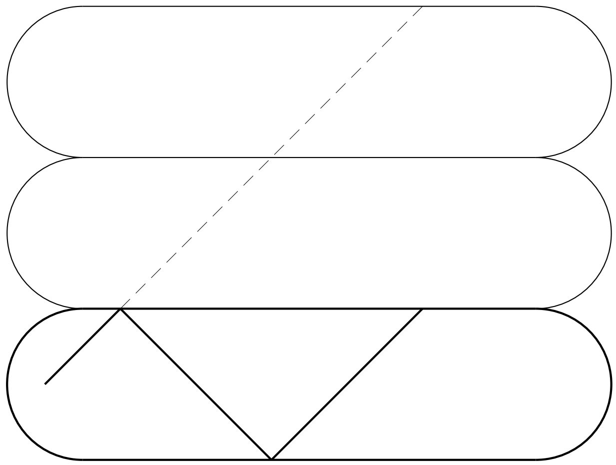


Figure 6.1: Reflecting trajectories in Bunimovich Stadium

Hence, for a trajectory that collides only with a half circle the cone just defined is invariant but not strictly invariant. Since this would be true also for a billiard inside a circle it is clearly not sufficient (the billiard inside a circle has zero Lyapunov exponents, since, as we have already remarked it has an integral of motion). Next, we need to see what happens if a trajectory goes from one circumference to the other (which will happen with probability one). In this case the infinitesimal motion is the same that would happen if the straight line would be not present. In fact, as we have already seen several times, if we reflect the billiard table along the straight lines we can imagine that the motion proceeds in a straight line (see Figure 6.1). Hence the family η_3 will first focus and then diverge for a longer time (and so get closer to the parallel family) than would happen if the collision would be in the same circle. This is exactly what we need to get strict invariance of the cone family.

In conclusion the cone family is strictly invariant each time that the trajectory goes from one half circle to the other. Since this happens almost surely, again we have proven hyperbolicity of the system.

It is interesting to notice that the cone family coincide with the one used in the Sinai Billiard (divergent trajectories) if one looks at it at the right point: the point laying in the intersection between the trajectory before collision and the circle of radius $r/2$ (if r is the radius of the half-circles forming the table) tangent to the the boundary at the next collision with a half-circle (but nowhere else).[3]

[3]If the last collision was with a flat wall, then the point is obtained by reflecting the

6.3 Cones

In the previous two sections we have analyzed two special examples using very geometrical arguments. This is very appealing and intuitive, yet, in general, it is useful to have a more algebraic approach. Given the formula for the derivative of the flow obtained in section 5.3 , it is very easy to implement the necessary formulae.

As we have already seen, provided we restrict ourself to looking in the right place, we can limit ourselves to considering the standard cone $\mathcal{C}_+ = \{\alpha\eta_1 + \beta\eta_2;\ \alpha\beta \geq 0\}$.

6.3.1 Examples

6.3.1.a *Dispersing (Sinai)*

As we have seen, in general dispersing billiards consist in a region $\mathcal{B} \in \mathbb{T}^2$ with strictly convex piecewise smooth boundaries. The cone family in all cases is always $\mathcal{C}_+$. By the results of section 5.3 the map from a collision to the next is given by

$$DT = \begin{pmatrix} 1 & 0 \\ \tau & 1 \end{pmatrix} \begin{pmatrix} 1 & \frac{2K}{\sin\varphi} \\ 0 & 1 \end{pmatrix}$$

where $K \geq K_- > 0$ is the curvature of the boundary at the collision point, τ is the time of fly between the two collisions, and φ the collision angle. Clearly $DT\mathcal{C}_+$ is strictly contained in $\mathcal{C}_+$, thus all the Sinai billiards are hyperbolic (since collisions take place almost surely, see Problem 5.13).

6.3.1.b *Bunimovich*

In this case, as we have already pointed out, the cone to consider is again $\mathcal{C}_+$, but only at special points. The cone family can then be obtained by transporting the cone forward ($d\phi^t\mathcal{C}_+$) until the next point where it is already defined.[4] The special point is the middle of the chord, that is the point at a distance $\tau = -\frac{\sin\varphi}{K}$ from the next collision; where $K = -r^{-1}$ is the curvature of the interior of the boundary of the half-circles. One can easily check algebraically all the geometric assertion of section 6.2 (but see Problems 6.4 Problem 6.5 to appreciate the advantage of this algebraic formulation).

---

billiard so the trajectory backward looks straight, determining the point and then reflecting back to find the real point on the trajectory.

[4]Note that this cone family is a bit different from the one defined in ???. Nevertheless, a bit of thought will shows that it is essentially equivalent. The point that I wish to make here is that several cone family can do the job and it is often convenient to consider different ones for different arguments (see ??? for an interesting use of such a principle).

6.3.1.c *Semi-dispersing*

Although the only possible cone family is constitute again by the divergent trajectories, as in the dispersing (Sinai) case, there is no general theory for the study of hyperbolicity of semi-dispersing billiards, thus their hyperbolicity must be explored case by case. Nevertheless, there are large classes of examples that are very well understood. Here we just mention a couple of cases.

6.3.1.d No straight lines

The easiest case is when $K = 0$ only on a set of measure zero of the boundary. Then, since the only trajectories that are not strictly monotone are the one either never colliding or colliding only with the parts of the boundary with zero curvature, and since such trajectories are of measure zero, the hyperbolicity follows by Wojtkowski theorem.

6.3.1.e Polygonal billiards with strictly convex obstacles

The easiest example is obtained by the Sinai billiard adding reflecting walls to delimit a fundamental domain. In this case we have a motion in a square containing an obstacle (the disk). For such an example the hyperbolicity follows from our previous discussion of the "ergodicity" of the motion in a rectangular box (see Examples 5.0.1–Polygonal Billiards) In particular, we have that for almost all the velocities $v = (v_1, v_2)$ (precisely the ones for which $v_1/v_2 \notin \mathbb{Q}$) the motion is ergodic on the set $\mathcal{B} \times \{(\pm v_1, \pm v_2)\}$. This means that almost all the trajectories will eventually collide with the obstacle whereby becoming strictly monotone, hence the hyperbolicity.

6.4 Geometric representation of cones in two-d

In the previous discussion we have seen that it is particularly important to keep track of the point at which an infinitesimal family focuses. This point of view can be brought to its extreme consequences by noticing that the distance of the focusing point along the trajectory can be used as a projective coordinate in the tangent space. It is in fact clear that all the language of cones is naturally translated in projective terms (the only thing that decides if a vector belongs or not to a cone is its direction and not its length). In formulas, if $\xi = \alpha\eta_1 + \beta\eta_2$ then the focusing point is at the distance $-\frac{\alpha}{\beta}$ (when $\alpha\eta_1 + \beta\eta_2 + \beta t\eta_1 = \beta\eta_2$). In this language the description of the systems becomes very geometric: a point in the tangent bundle becomes simply a point in Q (the point at which the moving point is situated) an oriented line (the direction in which the point is moving) and another point (the point at which the family of trajectories associated to the tangent vector would focus–note that this point can very well be out of Q). Even more drastically, we can forget about the first point, since different points correspond simply to different times, and the free evolution between collisions does not contain any relevant information. The relevant behavior of the Jacobi fields is then described by the so called "mirror equation:" calling f_0 the distance (measured in the direction of the motions) of the focusing point from the collision point before collision and f_1 the distance after, it holds

$$-\frac{1}{f_0} + \frac{1}{f_1} = \frac{2K}{\sin\varphi}.$$

This allows us to give to the cones a very geometric interpretation (see Figure ??)

. . . .

6.5 Convex billiard (two-D)

The representation discussed in the previous section is particularly suited to generalize Bunimovich stadium and obtain other convex billiards with non-zero Lyapunov exponents. Here we will give just an idea of what can be accomplished, for more details see [66], [93], [32].

Consider a sufficiently short $\mathcal{C}^{(2)}$ piece of a convex curve with the following property: each parallel family colliding with it will focus between each two consecutive reflections and focus again, at a distance less than some fixed constant, after leaving; see figure 10 (we will call pieces of curve "focusing arcs").

By the mirror equation we know that, if (φ, s) is the collision point, the parallel family ($f_0 = \infty$) focuses, after reflection, at the intersection between the trajectory and the circle of radius $\frac{1}{4K}$ tangent to the arc at the reflection point. From the figure it is clear that if such arcs are sufficiently far apart and are joined by straight lines, then there exists the wanted invariant cone family.

So far we have seen that it is possible to use some very intuitive geometric ideas to prove that a given system is hyperbolic. Yet, such systems are certainly not uniformly hyperbolic. To see this consider a trajectory that never experience a collision in a Sinai billiard or a periodic trajectory that never collides with the half circles in the Bunimovich stadium; both trajectories have zero Lyapunov exponents. In addition, we have already noticed that billiards systems are non-smooth.

The work of Sinai [79], [84] has shown that such obstacles can be overcome via the study of the "local ergodicity." In essence, the local ergodicity is a careful study of the points that cannot belong to the boundary of an ergodic component.

Such program has been carried out in great generality. For the smooth case a quite satisfactory result is the following [64] (but see also [50]) (here, for simplicity we consider only the two dimensional case).

6.5.1 Examples

6.5.1.a *Cardiod*

6.6 Billiards with Potentials

. . . .

6.7 More structure: increasing seminorms

This was the property originally used, together with the continuous fraction formalism, by Sinai to investigate billiards systems. **investigate applicability**

. . . .

6.8 Quadratic forms

The case of dimension larger than two is of obvious importance for the semi-dispersing case (since systems of hard-balls fall in this class see ???) but it has been investigated also in the convex case. In all cases it turns out to be convenient to employ the language of the quadratic forms introduced in section

???. We will discuss only the semi-dispersing case, for the (more complex) convex case references are provided in the Notes at the end of chapter.

If we have a domain $\mathcal{B} \subset \mathbb{T}^d$ or in $\mathbb{R}^d$ which has convex boundaries, calling $K(x)$ the curvature matrix of $\partial\mathcal{B}$ at the regular points $x \in \partial\mathcal{B}$, the convexity implies $K(x) \geq 0$.

Let $(p_0, v_0) \in Q \times \mathbb{R}^d$ to follow the trajectory starting from such a configuration it is convenient to consider the plane $\mathbb{V}(v_0) = \{\xi \in \mathbb{R}^d \mid \langle \xi, v_0 \rangle = 0\}$, let us call it orthogonal plane. It is clear that $\mathbb{V}(v_0) \times \mathbb{V}(v_0)$ is isomorphic to the tangent space of $Q \times \mathbb{R}^d$ at (p_0, v_0) quotiented with respect to the flow direction.[5] Let us use $b \in \mathbb{V}(v_0)$ and $\delta v \in \mathbb{V}(v_0)$ to designate the special and velocity part of a tangent vector and let us follow its evolution. If for a time t no collision takes place it is quite easy to see that, setting $(p_1, v_1) = \phi_t(p_0, v_0)$ and identifying in the obvious way $\mathbb{V}(v_0)$ with $\mathbb{V}(v_1)$, we have

$$d\phi_t(b, \delta v) = (b + \delta v t, \delta v).$$

In addition if (p_0, v_0) is a collision point, then we can describe what happens to the tangent vectors from immediately before to immediately after the collision. Let us call v_1 the velocity after collision, clearly

$$v_1 = v_0 - 2\langle v_0, \eta(p_0) \rangle \eta(p_0).$$

To write the formulas for the tangent vectors it is convenient to consider $\mathbb{V}(v_0)$ and $\mathbb{V}(v_1)$ as subspace of $\mathbb{R}^d \supset Q$. Let us call $(b', \delta v')$ the tangent vectors after the collision, then

$$\begin{aligned} b' &= \mathbb{I} b \\ \delta v' &= \mathbb{I}\left(\delta v - 2\langle v_0, \eta(p_0)\rangle \mathbb{P}^* K \mathbb{P} b\right)\ ; \end{aligned} \tag{6.8.1}$$

where $\mathbb{I}\xi = \xi - 2\langle \xi, \eta(p_0)\rangle \eta(p_0)$; $\mathbb{P}\xi = \xi - \frac{\langle \xi, \eta(p_0)\rangle}{\langle v_0, \eta(p_0)\rangle} v_0$ and K is the curvature of ∂Q at p_0.[6]

To write $DT : \mathcal{T}_{(p_0, v_0)}\mathcal{M} \to \mathcal{T}_{(p_1, v_1)}\mathcal{M}$ one can simply use the previous formulae and the operator $\mathbb{P} \times \mathbb{P}$ to project $\mathbb{V}(v_0) \times \mathbb{V}(v_0)$ onto $\mathcal{T}_{p_0} D \times \mathcal{T}_{p_0} D \cong \mathcal{T}_{(p_0, v_0)}$.

[5] Since nothing much happens in the flow direction, which in fact is not present after the reduction of the flow to its Poincaré map, it is helpful to eliminate it from the very beginning.

[6] Note that $\mathbb{P}$ is the operator that projects the space $\mathbb{V}(v_0)$ on the tangent space of ∂Q at p_0 along the direction v_0. Also, $\mathbb{I}(\mathbb{V}(v_0)) = \mathbb{V}(v_1)$; $\mathbb{I}^2 = \mathbb{1}$ and $\|\mathbb{I}\xi\| = \|\xi\|$ for each $\xi \in \mathbb{R}^d$, so $\mathbb{I}$ can be considered an identification of the spaces $\mathbb{V}(v_0)$ and $\mathbb{V}(v_1)$. Finally, by K we really mean an extension of $K : \mathcal{T}_{p_0} D \to \mathcal{T}_{p_0} D$ to an operator $K : \mathbb{R}^d \to \mathbb{R}^d$. Simply set $K\eta = 0$ and $\langle \eta, K\xi \rangle = 0$ for each $\xi \in \mathbb{R}^d$.

To compare the above formula with similar formulae present in the literature (see e.g., [30], [52]) it is helpful to notice that, setting $\mathbb{P}'\xi = \xi - \frac{\langle \xi, \eta(p_0)\rangle}{\langle v_1, \eta(p_0)\rangle} v_1$, holds $\mathbb{P}'\xi = \mathbb{P}\mathbb{I}^{-1}\xi$ for each $\xi \in \mathbb{V}(v_1)$ and $\mathbb{I}\mathbb{P}^*\xi = (\mathbb{P}')^*\xi$ for each $\xi \in \mathcal{T}_{p_0} D$.

To establish the hyperbolicity it suffice to verify that the quadratic form $\mathcal{Q}((b, \delta v)) = \langle b, \delta v\rangle$ is strictly increased from collision to collision [84], [63]. As already mentioned this implies immediately that all the Lyapunov exponents are different from zero for the dynamical system $(\mathcal{M}, T, \mu)$.

The quadratic form $\mathcal{Q}$ corresponds to the sector C defined by the two Lagrangian subspaces $\mathbb{V}_1 = \{(\delta q, \delta v) \in \mathbb{R}^{2d} \mid \delta v = 0\}$ and $\mathbb{V}_2 = \{(\delta q, \delta v) \in \mathbb{R}^{2d} \mid \delta q = 0\}$ (see [63]).[7]

The reader who finds the above formulae too abstract is invited to work out the special case of hard balls (see Problem 6.7).

6.9 Three Balls

We assume, for the time being, the following lemma

Lemma 6.9.1 *The trajectories for which one of the balls never collides with the other from a certain time t_* on, have zero measure.*

We can then discard such measure zero set of configurations.

If we have a collision at time t_1 between the spheres one and two then the only possibility for the $\mathcal{Q}$ not to increase is given by vectors of the form

$$\begin{cases} \delta x_1 = z + \lambda v_1 \\ \delta x_2 = z + \lambda v_2 \\ \delta v_1 = \delta v_2 = \delta v_3 = 0 \end{cases}$$

since this corresponds to the conservation of energy and total momentum of the two balls separately.

If the next collision takes places at time t_2 between the sphere one and three, then the $\mathcal{Q}$ from will increase unless, at time t_2,

$$\begin{cases} \delta x_1 = w + \mu v_1 \\ \delta x_3 = w + \mu v_3 \\ \delta v_i = 0 \end{cases}$$

The above system of linear relations implies

$$w - z = (\lambda - \mu) v_1(t_2^-).$$

Up to know we have considered a general situation, eventually by renumbering properly the spheres, now there can be be two different cases: either

[7] This is like to say $\mathcal{Q}(\xi) \geq 0$ is equivalent to $\xi \in C$.

the next different collision is between two and three or it is between one and two.

In the first case we have that at time t_3

$$\begin{cases} \delta x_2 = \alpha + \rho v_2 \\ \delta x_3 = \alpha + \rho v_3 \\ \delta v_i = 0 \end{cases}$$

thus

$$(\lambda - \rho)v_3(t_2^+) = (\mu - \rho)v_2(t_2^+) + (\lambda - \mu)v_1(t_2^-)$$

that is

$$(\lambda - \rho)(v_3(t_2^+) - v_2(t_2^+)) = (\lambda - \mu)(v_1(t_2^-) - v_2(t_2^+))$$

which means that the two velocities $v_3(t_2^+) - v_2(t_2^+)$ and $v_1(t_2^-) - v_2(t_2^+)$ must be parallel. Such a condition hold on a manifold of dimension$d - 1$, thus on a set of measure zero (see Problem 6..). Moreover, if $d \geq 3$, the set has codimension at least two and thus cannot separate the phase space in disjoint sets (see Problem 6..).

In the second case it must hold

$$\begin{cases} \delta x_1 = \alpha + \rho v_1 \\ \delta x_2 = \alpha + \rho v_2 \\ \delta v_i = 0 \end{cases}$$

which implies

$$\begin{cases} \alpha - z = (\mu - \rho)v_1(t_3) \\ \alpha - w = (\mu - \rho)v_2(t_3) \end{cases}$$

that is

$$(\mu - \rho)v_1(t_3) = \alpha - w + w - z = (\mu - \rho)v_2(t_3) + (\lambda - \mu)v_1(t_2),$$

hence **correct**

$$(\mu - \rho)(v_1(t_3) - v_2(t_3)) = ??(\lambda - \mu)v_1(t_2),$$

This means that $v_1(t_3)$, $v_2(t_3)$, $v_3(t_2)$ must be linearly independent. Again it is easy to check that, in $d \geq 3$, such a condition can be satisfied only a manifold of codimension one (see Prob 6.).

Let us go back to Lemma 6.9.1

Proof of Lemma 6.9.1. Suppose that the first ball is the one that never collides after the time t_*. We can assume that $v_1(t_*) \neq 0$ since this happens only on a set of zero measure. Consider the motion on the universal covering for a time t_l much larger than t_*. If we consider the initial conditions $\{(x_1(t_*), v) \in X \mid |v - v_1(t_*)| \leq \delta\}$ then in the time interval $[t_l, t_l + 2\|v_1(t_*)\|^{-1}]$ the particle can be anywhere in the set $\{x_1(t_0) + v(t_l + \tau\|v_1(t_*)\|^{-1}]) \mid |v - v_1(t_*)| \leq \delta,\ \tau \in [0,2]\}$ which, if t_l is large enough, contains completely a copy of the fundamental domain of the torus. This show that the initial conditions conditions for which no collision takes place cannot form an open set. Nevertheless, they could still be of positive measure. To investigate this possibility, and go from a topological to a measure theoretical result, we imitate what we have already done in the study of ergodicity (see Examples ??) and mixing (see ???).

Let E be the set of configurations for which the first ball never collides after some time, and suppose $\mu(E) > 0$. Then, by the Lebesgue density Theorem there are density points $\xi \in E$. Let $\xi = (x^*, v^*)$ be such a density point, then for each ε (to be chosen later) there exists δ such that, setting $B_\delta(\xi) = \{(x, v) \in X \mid \|x - x^*\| \leq \delta;\ \|v - v^*\| \leq \delta\}$,

$$\mu(B_\delta(\xi) \cap E) \geq (1 - \varepsilon)\mu(B_\delta(\xi)). \tag{6.9.1}$$

Now, by the same argument discussed before, $\Pi(\phi^t(B_\delta(\xi))$, where $\Pi(x, v) = x_1$, covers completely a fundamental domain if t is large enough. To be more precise, it suffices $t \geq 2\delta^{-1}$. Let us choose $t_* = 2\delta^{-1}$ and estimate the measure of $E \cap B_\delta(\xi)$. For each $(x, v) \in B_\delta(\xi)$, (x, v) can be in E only if the velocity v_1 is so that $x_1 + v_1 t_*$ is out of two disks of radius r (the other two balls). This means that there is a forbidden angle of size r/t_* and a forbidden interval of $\|v_1\|$ of the same order. By Fubini theorem theorem this implies that (x, v) cannot belong to a set of measure $c_0\delta^6$, for some constant c_0, while $\mu(B_\delta(\xi)) = c_1\delta^6$, for some other constant c_1.

Thus,

$$\mu(B_\delta(\xi) \cap E) \leq (1 - \frac{c_1}{c_0})\mu(B_\delta(\xi))$$

which contradicts (6.9.1), provided ε has been chosen sufficiently small. Hence, the assumption $\mu(E) > 0$ leads to a contradiction. □

6.10 N Balls (hard spheres)–few basic facts

If one wants to tackle the case of n balls on the $\mathbb{T}^d$ dimensional torus ($d > 1$, see Problem ?? for the case $d = 1$) it is clear that a simple minded analysis in the style of the previous chapter is doomed by the exponential explosion of the combinatoric of the various possibilities. To hope to handle in a reasonable

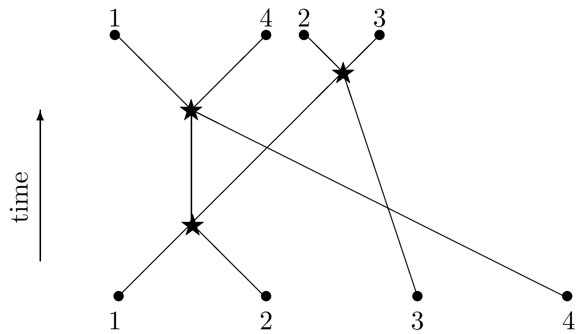


Figure 6.2: A simple collision graph (the stars are the collisions)

manner such a case some bookkeeping technology must be developed. In this section I will give a taste of what can be done.

First of all I will introduce a *collision graph* to describe pictorially the relevant features of a trajectory. The graph starts with n roots (each one representing one ball), from each root starts a line (representing the trajectory). A collision is represented by a starred node in the graph (with two entering lines–representing the incoming particles–and two exiting lines–representing the outgoing particles); see figure 6.2 for the case of four balls in which number one collides with two, then two with four, and finally two with three.[8] Next, let us call $\mathcal{G}$ a collision graph and let $V(\mathcal{G})$ be the collection of its vertexes, $\tilde{B}(\mathcal{G})$ the collection of its bonds and $B(\mathcal{G})$ the collection of bonds that connect starred vertices. In addition for each bond $b \in B(\mathcal{G})$ let $\nu(b), \nu_+(b)$ be the two vertices joined by the bond.[9]

So far the collision graph describes the combinatorics of the collision but nothing else, to make it useful it is necessary to add informations concerning the evolution of tangent vectors.

Form our analysis in section 6.9 it follows that when two balls i and j collide the only possibility for a tangent vector of the form $(\delta q, 0)$ not to enter strictly in the cone is if there exist λ and z such that

$$\begin{aligned} \delta q_i &= z + \lambda v_i \\ \delta q_j &= z + \lambda v_j . \end{aligned} \tag{6.10.1}$$

If we want to follow the history of a vector of type $(\delta q, 0)$ that stubbornly

[8] The rule for tracing the graph is that the order of the balls is not changed at collision, so the line on the left represents the particles entering the collision node from the left. Remark that the collision graph is only a symbolic device and does not respect the geometry of the actual collisions, so the ordering of the balls is only a device to tell them apart and has no relation with the actual geometry of the associated configuration. Keeping this in mind, in figure 6.2 the final disposition of the balls is: one, four, two, three.

[9] By convention $\nu(b)$ corresponds to the lower collision and $\nu_+(b)$ to the upper.

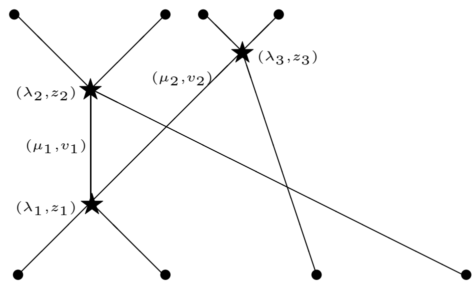


Figure 6.3: A decorated collision graph

refuses to enter strictly in the cone it is then convenient to specify at each vertex the values (λ_ν, z_ν) appearing in the associated equation (6.10.1). Of course, to recover the tangent vectors from the $\{(\lambda_\nu, z_\nu)\}_{\nu \in V(\mathcal{G})}$, it is necessary to specify the velocities. To this end we specify for each bond the velocity $v(b)$ of the particle associated to such a line. We can then decorate a graph with the above informations and we obtain a full description of the history of a tangent vector that keeps being not increased by the dynamics in the trajectory piece described by the graph (of course provided such a vector exists at all).

Now consider a bond $b \in B(\mathcal{G})$, if it represents the trajectory of the particle j between the collision corresponding to the the vertex $\nu(b)$ and the one corresponding to the vertex $\nu_+(b)$, then the corresponding component of the tangent vector at such times can be written both as $\delta q_j = z(\nu(b)) + \lambda(\nu(b))v(b)$ and $\delta q_j = z(\nu_+(b)) + \lambda(\nu_+(b))v(b)$. Accordingly, the following compatibility condition must be satisfied:

$$z(\nu(b)) - z(\nu_+(b)) = [\lambda(\nu_+(b)) - \lambda(\nu(b))]v(b). \tag{6.10.2}$$

It is then natural to define another decoration, this time associated to bonds that connect two collision vertexes,

$$\mu(b) := \lambda(\nu_+(b)) - \lambda(\nu(b)). \tag{6.10.3}$$

By decorated collision graph, we will mean a graph with $(\lambda(\nu), z(\nu))$ attached to each vertex and $\mu(b), v(b)$ to each bond connecting two collisions, with a mild abuse of notations we will call such decorated graph $\mathcal{G}$ as well, see figure 6.3.[10]

As time progress the graph will grow more complex, in particular it may develop *cycles*. By a cycle I mean a connected path of bonds that leave a vertex and go back to it, e.g. the thick bonds in the graph of figure 6.4.

[10]Note that the above description is quite redundant due to (6.10.2), yet we will see in the following that such a description is quite convenient.

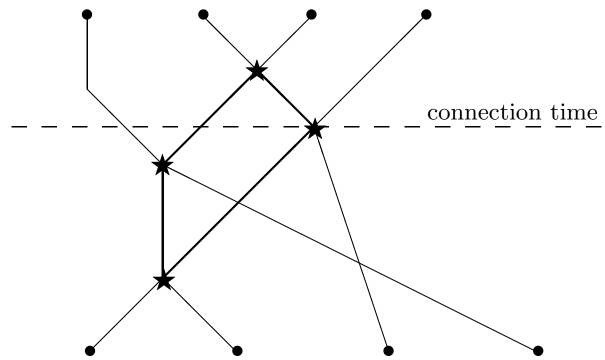


Figure 6.4: A cycle

Once a cycle is formed a remarkable compatibility condition can be derived. In fact, let $C \subset \mathcal{G}$ be a cycle, let us run it counterclockwise and define, for each bound $b \in C$, $\varepsilon_C(b) = 1$ if the bond is run from bottom to top and $\varepsilon_C(b) = -1$ if it is run from top to bottom. We have, by definition (6.10.3), $\sum_{b\in B(C)} \varepsilon_C(b)\mu(b) = 0$. In addition, we can sum equation (6.10.2) for each bond in the cycle and obtain

$$\begin{aligned}\sum_{b\in B(C)} \mu(b)\varepsilon_C(b)v(b) = 0\\ \sum_{b\in B(C)} \varepsilon_C(b)\mu(b) = 0.\end{aligned} \tag{6.10.4}$$

The above formula is essentially the *closed path formula* introduced by Simanyi in [1]. Such a formula expresses a compatibility condition that puts a clear restriction on the possible existence of the decorated collision graph, and hence of the corresponding non increasing vector.

With a moment thought the reader will realize that the analysis in section 6.9 boiled down to showing that equation (6.10.4), corresponding to the first cycle involving all three balls, can have a non trivial solution only on a codimension one manifold. It is then clear that, to treat the general case, one must follow the graph in time, collect the various equations (6.10.4) following the formation of new cycles and then try to analyze such a set of equations to show that, at a certain point, they do not admit any non trivial solution. To do that, it is first necessary to understand a bit better the process of cycle formation.

First of all one must notice that it suffices to consider connected graphs.[11] Indeed, if there is no chain of collision whatsoever connecting two given balls,

[11] That is graphs in which there are paths that join any two given roots.

then there exists an obvious non increasing vector. Given a graph there will then exists a well defined *connecting time*, that is the first time for which the graph is connected, (see figure 6.4). Before such a time the graph can be decomposed into connected components. Let us call *collision clusters* the set of balls associated to a connected components of the graph.

Second notice that, given a graph, the set of cycle contains many more objects than we are interested in. In fact, given two cycles, we can always get a new cycle by joining them or, given a cycle, we can get a new one by running back and forward on a bond that touches, but does not belong to, the cycle; yet the equations (6.10.4) associated to such "new" cycles do not carry any new information. For such a reason, and for deeper ones that will be apparent later, it is necessary to define uniquely a convenient subset of "independent" cycles.

Definition 6.10.1 *By* minimal cycle *we mean a cycle such that no subset of bonds still forms a cycle.*

Clearly we will be interested only in minimal cycles, yet when new cycles forms it may be possible to choose among several minimal cycles.

Definition 6.10.2 *Given a cycle we call* top bonds *the bonds that enter in the highest vertex and* bottom bonds *the ones exiting from the lowest one.*

For each minimal cycle we will chose one bond and we will call it *forbidden bond*. In the following we will choose as the forbidden bond the top bond exiting from the lowest vertex.

We will use the following rule for choosing minimal cycles: *when a collision creates new cycles*[12] *then we choose a new minimal cycle that does not contain any forbidden bond of previously chooses cycles.* Note that, if such a rule can be applied, then a forbidden bond occurs only in one minimal cycle since, being a top bond, was not present before being chosen as forbidden and after will not belong to the minimal cycles by construction.

Lemma 6.10.3 *The above rule fixes a unique choice of a minimal cycle each time new cycles are born. In addition, all the minimal cycles present at any given time can be obtained by composing the chosen ones*[13] *and if one deletes from the graph all the forbidden bonds, then no cycle is left.*

Proof. The proof is by induction on the number of cycles. Clearly the first time cycles are created there is only one minimal cycle and no cycle if the forbidden bond is erased.[14] Now suppose we have already gathered m

[12] That is cycles that where not present before such a collision.

[13] This means that each other minimal cycle is a subset of the union of the chosen ones.

[14] Given two minimal cycles, it would readily imply the presence of a cycle at a previous time, contrary to the assumption.

minimal cycles, and suppose a new cycle is created. First of all notice that if there are two new minimal cycles, then they must include the new vertex, hence they must have the same top bonds. Thus, putting them together and erasing the top bonds they must still contain a cycle. By hypothesis such a cycle can be generated by the minimal cycles already chosen. Hence, it suffices to chose only a new minimal cycle to generate all the new cycles present in the graph. In addition, suppose that the two cycles do not contain any forbidden bond. Then the cycle constructed above cannot contain any forbidden bond but this is impossible by the induction hypothesis. Hence the uniqueness. Next suppose, that there is a cycle that does not contain any forbidden bond, hence it cannot visit the new vertex but then again it cannot exist by the induction hypotheses. We are left with the existence.

Choose any a new minimal cycle and suppose that it contains forbidden bonds $\{\underline{b}_i\}$. Let $\{C(\underline{b}_i)\} =: \{C_i\}$ be the corresponding cycles temporally ordered. Clearly instead of using the incriminated forbidden bond we can run along the other bonds of C_1, we have thus a new cycle that does not contain the forbidden bond of C_1, moreover C_1 does not contains forbidden bonds of earlier cycles by induction. If the cycle is not minimal reduce it to a minimal cycle. We have now a new set of forbidden bonds contained in the cycle, but they all belong to cycles appearing after C_1. Iterating the above argument we can remove all the forbidden bonds. □

We will call such cycles the *canonical cycles* associated to the graph. Next let us see how cycles are created.

Lemma 6.10.4 *At each new collision either the number of collision clusters decreases by one or a new cycle is formed.*

PROOF. The two colliding balls can either belong to different cluster or to the same. In the first case clearly the number of cluster decreases by one. In the second case notice that the two bonds that join to create the new vertex cannot belong to any existing cycle since they do not lead anywhere before the last collision. On the other hand, such bonds either come from the same vertex, and then we have a new two cycle which clearly is minimal, or come from different vertices. In the latter case each of such vertices will be connected to some root of the graph. On the other hand, since the graphs is connected, there will be a path connecting two such roots. We have just described a cycle containing the new vertex and the two incoming bonds. □

In view of the above Lemma is then natural to divide the vertices in cluster *merging* and in *cycle* creating vertexes. The Lemma 6.10.4 means that, calling $N_{cy}(\mathcal{G})$ the number of cycles, $N_{cl}(\mathcal{G})$ the number of clusters and $N_v(\mathcal{G})$ the number of collision vertexes the quantity $N_v(\mathcal{G}) - N_{cy}(\mathcal{G}) + N_{cl}(\mathcal{G})$ remains constant as the graph grows. Since at time zero it has the value n it follows

$$N_v(\mathcal{G}) - N_{cy}(\mathcal{G}) + N_{cl}(\mathcal{G}) - n = 0. \tag{6.10.5}$$

On the other hand, in a connected graph, from each vertex exits two upward bonds each one of such bonds either enters in another vertex or gets at the top of the graph. Thus, calling $N_b(\mathcal{G})$ the number of bonds connecting two vertexes it holds true

$$N_b(\mathcal{G}) = 2N_v(\mathcal{G}) - n. \tag{6.10.6}$$

Equations (6.10.5) and (6.10.6) implies that, after the connection time, holds true

$$N_b(\mathcal{G}) = 2N_{cy}(\mathcal{G}) + n - 2. \tag{6.10.7}$$

Since each cycle yields a close path formula ($d+1$ equations) and the total number of variables $\mu(b)$ in such equations is exactly the number of bounds involved in the cycles, in order to have an hope for the equations to have only the trivial solution it must be $N_b(\mathcal{G}) \leq (d+1)N_{cy}(\mathcal{G})$ which is implied by $N_b(\mathcal{G}) \leq 3N_{cy}(\mathcal{G})$. That is it is necessary to have at least $n-2$ cycles.[15]

Now that one see how to generate a (hopefully sufficient) number of equations (6.10.4) it remains the much harder task of analyzing them.

As a simple example let us analyze the situation when the number of balls n equals the number of dimensions of the space d plus one which generalizes the case of three balls in two dimensions already treated (Simanyi-inventiones papers).

....

Problems

6.1 Show that in a Sinai Billiard almost all the trajectories eventually collide.

6.2 Write η_3 as a linear combination of η_1 and η_2.

6.3 on focusing arcs.

6.4 Consider the Bunimovich stadium in which the half circles are substituted by half ellipses. Find conditions for which there exists a stable periodic orbit.

6.5 Consider the Bunimovich stadium in which the half circles are substituted by half ellipses. Find conditions that insure hyperbolicity.

[15]Note that for three balls this means one cycle in agreement with the analysis in section 6.9.

6.6 baldwin non ergodic

6.7 Compute the tangent map relative to the collision of two identical balls of mass 1 and radius R. (Hint: Let (q_1, p_1), (q_2, p_2) the coordinates of the two balls. Clearly they collide when $\|q_1 - q_2\| = 2R$. If $(\delta q_i, \delta p_i)$ are the tangent vectors at the collision time, then the first task is to flow the infinitesimal trajectories so that they all collide at the same time. This corresponds to finding λ so that $\langle \delta q_1 - \delta q_2 + \lambda(p_1 - p_2), q_1 - q_2\rangle = 0$. That is

$$\lambda = -\frac{\langle \delta q_1 - \delta q_2, n\rangle}{\langle V, n\rangle}$$

where $n := (2R)^{-1}(q_1 - q_2)$ and $V := p_1 - p_2$. The reflection law is given by

$$\begin{aligned} q_i^+ &= q_i^- \\ p_1^+ &= p_1^- - \langle V, n\rangle\, n \\ p_2^+ &= p_2^- + \langle V, n\rangle\, n. \end{aligned} \tag{6.10.8}$$

For a curve of trajectories colliding at the same time we can simply differentiate (6.10.8) and, remembering that we must to flow back (along the vector field) the tangent vector after collision, we have

$$\begin{aligned} \delta q_1^+ &= \delta q_1^- + \lambda(p_1^- - p_1^+) = \delta q_1^- - \langle \delta q_1^- - \delta q_2^-,\, n\rangle\, n \\ \delta q_2^+ &= \delta q_2^- + \lambda(p_2^- - p_2^+) = \delta q_1^- + \langle \delta q_1^- - \delta q_2^-,\, n\rangle\, n \\ \delta p_1^+ &= \delta p_1^- - \langle \delta p_1^- - \delta p_2^-, n\rangle\, n - (2R)^{-1}\langle V, \widetilde{\xi}\rangle\, n - (2R)^{-1}\langle V, n\rangle \widetilde{\xi} \\ \delta p_2^+ &= \delta p_2^- + \langle \delta p_1^- - \delta p_2^-, n\rangle\, n + (2R)^{-1}\langle V, \widetilde{\xi}\rangle\, n + (2R)^{-1}\langle V, n\rangle \widetilde{\xi}, \end{aligned}$$

where $\widetilde{\xi} := \widetilde{\delta q_1} - \widetilde{\delta q_2}$ and $\widetilde{\delta q_i} := \delta q_i + \lambda p_i$.)

6.8 Use the formulae obtained in Problem 6.7 to compute the increase of the standard quadratic form when two balls collide. Verify that such a formula agrees with the one obtainable from equation (6.8.1). Use it to characterize the vectors for which the form does not increases. (Hint: By calling $\delta q := (\delta q_1, \delta q_2)$ and $\delta p := (\delta p_1, \delta p_2)$ and using the same notation of Problem 6.7, a straightforward computation yields

$$\langle \delta q^+,\, \delta p^+\rangle = \langle \delta q^-,\, \delta p^-\rangle - \frac{\langle V, n\rangle}{2R}\|\widetilde{\xi}\|^2.$$

Thus we have no increase only if $\widetilde{\delta q_1} = \widetilde{\delta q_2}$. Hence, setting $z := \widetilde{\delta q_1}$, we have $\delta q_1 = -\lambda p_1 + z$, $\delta q_2 = -\lambda p_2 + z$.)

6.9 Use the formulae obtained in Problem 6.7 to show that, if $\delta p_1 = \delta p_2 = 0$, then $\delta p_1^+ = -\delta p_2^+$ and δp_1^+ is orthogonal to $V^+ := p_1^+ - p_2^+$. (Hint: The first statement is obvious the second follows by computing and noticing that $\langle \widetilde{\xi}, n \rangle = 0$.)

6.10 Prove that for each system of n-hard balls there exists an invariant set Λ of positive measure such that the Lyapunov Exponents for points in Λ are all $\neq 0$. (Hint: find a set U of positive measure and $t > 0$ such that $\sigma(d_x \phi^t) > 1$ for each $x \in U$. Then set $\Lambda = \cup_{n \in \mathbb{N}} T^{-n} U$. Use the pairing of the exponents (Problem ??) to conclude.)

Notes

Discuss convex billiards in higher dimensions.

CHAPTER 7

Foliations–the minimum

In this chapter we will investigate some particular cases of the general results put forward in chapter 3; in particular, we will restrict our study to symplectic systems. More precisely, we will discuss theorem....

7.1 Existence of the foliations–smooth 2D

In this section we will prove Theorem 3.4.8 in a special case that suffices for our purposes. For simplicity, we will consider only the case in which there exists a **continuous** eventually strictly invariant cone family. This is certainly not the most general case but it presents all the basic ideas of the proof apart from the introduction of the so called *Pesin set*, [71].[1]

The existence of a continuous (eventually strictly invariant) cone family implies immediately that the distributions are continuous a.e. (see Lemma 4.4.1)[2] while if the distribution are continuous then there always exists a continuous cone family (see Problem 7.5).

To be more precise, here we treat the case when $X = \mathbb{T}^2$, with the standard symplectic form,[3] the map T is symplectic (i.e., $\det(DT) = 1$ since we are in two dimensions) and eventually strictly monotone with respect to a continuous cone family. It must be stressed that we allow non-uniformly hyperbolic maps. For uniformly hyperbolic maps the proof of existence would be much simpler, in fact almost the same than the fixed point case (see 3.3.2), see Problem 7.4 for more details.

[1] In some sense the Pesin sets are sets on which there exists a continuous cone family

[2] Contrary to the geneal case in which they can be only measurable, this, accordingly to Lusin Theorem, means that there exist sets of arbitrarily small measure outside of which the distribution is continuous, but this is clearly much less.

[3] This it is not a restriction since the following are all local and locally any symplectic form can be put in the standard form thanks to the Darboux Theorem [5] (see Problem 8.1 and Problem 8.2).

The analysis will be local. Let $\bar{x} \in \mathbb{T}^2$ be a strictly monotone point in the past, that is a point for which it exists $\bar{n} \in \mathbb{N}$ such that

$$D_{\bar{x}} T^{-\bar{n}} \mathcal{C}(\bar{x})^c \subset \text{Int } \mathcal{C}(T^{-\bar{n}}\bar{x})^c \cup \{0\},$$

where the " ^{c} " means the complement. Clearly the above relation means that

$$D_{T^{-\bar{n}}\bar{x}} T^{\bar{n}} \mathcal{C}(T^{-\bar{n}}\bar{x}) \subset \text{Int } \mathcal{C}(\bar{x}) \cup \{0\}. \tag{7.1.1}$$

Remark 7.1.1 *In the following we will deal only with unstable manifolds. The discussion of stable manifolds is completely identical since if a cone family is eventually stricly invariant for a map T, then its complement is stricly eventually invarinat for the map T^{-1} (crf. Problem 7.1).*

7.1.1 Coordinate system (Linear algebra)

Consider the space $\mathbb{R}^2$ with the standard symplectic form and two cones $\mathcal{C}$, $\bar{\mathcal{C}}$ such that $\bar{\mathcal{C}} \subset \text{Int } \mathcal{C} \cup \{0\}$.[4] We introduce here a convenient system of coordinates to view quantitatively such a situation.

First of all it is convenient to use the edges of the cone $\mathcal{C}$ as coordinate axix. Clearly we can assume that one edge is the x axix without loss of generality.[5] Suppose that in the new coordinates the second edge has direction $(1, u)$, $u > 0$.[6] We can then use the change of coordinates determined by the matrix

$$\begin{pmatrix} 1 & -u^{-1} \\ 0 & 1 \end{pmatrix}.$$

It is immediate to check that in the new coordinates the come $\mathcal{C}$ is given exactly by the first and the third quadrant (the standard sector $\mathcal{C}_+$), as wanted.

Let us consider, in the above coordinates, the cone $\mathcal{C}_\alpha := \{v \in \mathbb{R}^2 \mid \alpha \leq \frac{v_2}{v_1} \leq \alpha^{-1}\}$, $\alpha \in (0, 1)$; clearly $\mathcal{C}_\alpha \subset \mathcal{C}_+$. Note that all the matrices of the form, $a \neq 0$,

$$L_a = \begin{pmatrix} a & \frac{1}{a(\alpha^{-1}-\alpha)} \\ \alpha a & \frac{1}{a\alpha(\alpha^{-1}-\alpha)} \end{pmatrix}$$

have the property $L_a \mathcal{C}_+ = \mathcal{C}_\alpha$ (in fact, these are the only symplectic matrices with such a property, see Problem 4.26.

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[4] The impatient reader wanting to know immediately why we are worrying about such a situation can think about the tangent space at $\bar{x}$ and (7.1.1).

[5] Remember that, in two dimensions, the symplectic group consists just of the matrices with determinant one, hence it contains the rotations, see Problem 4..

[6] Also this can be achieved without loss of generality, since $u \neq 0$ follows form the fact that the cones are not degenerate, while if $u < 0$, then we can choose this edge first as x-axix and then the other will have the wanted representation.

By (4.1.2) we have

$$\sigma(L_a) = \inf_{v\in\mathcal{C}_+}\sqrt{\frac{Q(L_a v)}{Q(v)}} = \frac{1+\alpha}{\sqrt{1-\alpha^2}} := \sigma_\alpha, \tag{7.1.2}$$

and, by (4.3.6),

$$\operatorname{diam}(\mathcal{C}_\alpha) = \ln\alpha^{-2}.$$

On the other hand, by a Q-isometry (see problems Problem 4.16, Problem 4.19 for general properties of Q-isometries) we can transform the cone $\bar{C}$ determined by the two edges $(1,u)$, $(1,v)$, $v > u > 0$, in a cone of the type C_α. Indeed, consider the change of coordinates determined by the matrix

$$\begin{pmatrix} [uv]^{\frac{1}{4}} & 0 \\ 0 & [uv]^{-\frac{1}{4}} \end{pmatrix},$$

then the cone $\bar{\mathcal{C}}$ becomes exactly the cone $\mathcal{C}_\alpha$ with $\alpha = \sqrt{\frac{u}{v}}$.

Collecting the above considerations yields the following fact.

Lemma 7.1.2 *If $L\mathcal{C} \subset \mathcal{C}$, for some cone $\mathcal{C}$ and symplectic matrix L, with $\sigma(L) \geq \sigma_\alpha$ where $\alpha \in (0,1)$, then there exists a symplectic change of coordinates such that, in the new coordinates, the cone is the standard sector $\mathcal{C}_+$ and $L\mathcal{C}_+ \subset \mathcal{C}_\alpha$.*

Remark 7.1.3 *The above discussion is quite transparent if one thinks that cones are a projective notion. We have just seen that, as usual in projective spaces, three points (edges) can be put in an arbitrary position by a transformation that preserves the structure at hand, not so for four points (edges) since an invariant must be respected (in our case the diameter of the cone which, as we have seen in chapter 4, is related to the cross ratio, a projective invariant).*

7.1.2 Coordinate system (A neighborhood)

Here we introduce convenient coordinates in a neighborhood of $\bar{x} \in \mathbb{T}^2$. By uniform continuity of the cone family it follows that there exists and open set U and a cone $\bar{\mathcal{C}} \supset \mathcal{C}(\xi)$ for each $\xi \in U$ such that[7]

$$D_{T^{-\bar{n}}\xi}T^{\bar{n}}\mathcal{C}(T^{-\bar{n}}) \subset \operatorname{Int}\bar{\mathcal{C}} \cup \{0\} \quad \forall \xi \in U. \tag{7.1.3}$$

[7] Here we are using the local Cartesian structure of $\mathbb{T}^2$ to identify tangent spaces. This simplify matters. To treat general manifolds one needs to use explicit coordinates or a connection.

Let $\alpha \in (0,1)$ such that[8]

$$\inf_{\xi \in U} \sigma(D_{T^{-\bar{n}}\xi} T^{\bar{n}}) = \sigma_\alpha. \tag{7.1.4}$$

Using the Cartesian structure of $\mathbb{T}^2$ and the results of the previous subsection we can then introduce Cartesian coordinates in U such that $\bar{\mathcal{C}}$ becomes the standard sector $\mathcal{C}_+$ and

$$D_{T^{-\bar{n}}\xi} T^{\bar{n}} \mathcal{C}(T^{-\bar{n}}) \subset \text{Int } \mathcal{C}_\alpha \cup \{0\} \quad \forall \xi \in U. \tag{7.1.5}$$

Note that in this coordinate $Q_\xi(v) \geq Q(v)$ for each $\xi \in U$ and each tangent vector v (here Q stands for the standard quadratic form).

7.1.3 Growth of vectors

If the cone family were strictly invariant then the quadratic form would increase exponentially in time. As we saw in chapter 4, in the more general case only almost every point will show an exponential increase. To better exploit this phenomenon it is convenient to have a better understanding of the set of points for which such an increase takes place.

Lemma 7.1.4 *For each $\xi \in \mathbb{T}^2$ for which $T^n \xi \in U$, $n \geq \bar{n}$ holds*

$$Q_\xi(D_\xi T^n v) \geq \sigma_\alpha Q(v) \quad \forall v \in \mathcal{T}_\xi \mathbb{T}^2.$$

Proof. As already remarked, $Q_\xi(v) \geq Q(v)$ for each $\xi \in U$, thus the result follows immediately by the definition (7.1.4) and the monotonicity of the map:

$$Q_\xi(D_\xi T^n v) \geq Q_{T^{n-\bar{n}}\xi}(D_{T^{n-\bar{n}}\xi} T^{\bar{n}} v) \geq \sigma_\alpha Q(v).$$

□

Even more interesting is the following fact.

Lemma 7.1.5 *If $\xi \in U$ and $n \geq \bar{n}$ is such that $T^n \xi \in U$, then*

$$Q(D_\xi T^n v) \geq \sigma_\alpha Q(v)$$

for all tangent vectors v.

[8]In this case

$$\sigma(D_{T^{-\bar{n}}\xi} T^{\bar{n}}) = \inf_{v \in \mathcal{C}(T^{-\bar{n}}\xi)} \sqrt{\frac{\bar{Q}(D_{T^{-\bar{n}}\xi} T^{\bar{n}} v)}{Q_{T^{-\bar{n}}\xi}(v)}},$$

where $\bar{Q}$ is the quadratic form associated to the cone $\bar{\mathcal{C}}$.

PROOF. By construction $\mathcal{C}(\xi) \subset C_+$, thus

$$D_\xi T^n \mathcal{C}_+ \subset D_\xi T^n C(\xi) \subset C_\alpha.$$

The estimate follows then by (7.1.2). □

From the above Lemma it is clear the interest of the points that come back to U. It is then natural to introduce the set

$$\Omega(U) = \{\xi \in U \mid \#(\{T^{-n}\xi\}_{n\in\mathbb{N}} \cap U) = \infty\}, \tag{7.1.6}$$

that is the set of point in U that visit U infinitely many times in the past. Remark that, by Poincaré Return Theorem 1.6.5, $m(\Omega(U)) = m(U)$.

Lemma 7.1.6 *For each $\xi \in \Omega(U)$ holds*

$$\lim_{n\to\infty} \inf_{v\in\mathcal{C}(T^n\xi)} Q(D_{T^{-n}\xi}T^n v) = \infty.$$

PROOF. Let $\{n_i\}_{i\in\mathbb{N}}$ be times such that $n_{i+1} - n_i \geq \bar{n}$ and $T^{n_i}\xi \in U$, then

$$\begin{aligned} Q(D_{T^{n_i}\xi}T^{-n_i}v) &\geq \mu Q_{T^{n_i-\bar{n}}\xi}(D_{T^{n_i}\xi}T^{-n_i-\bar{n}}v) \geq \mu Q(D_{T^{n_i}\xi}T^{-n_{i-1}}v) \\ &\geq \mu^i Q(v) \end{aligned}$$

□

7.1.4 Unstable manifolds!

The above result is the key to prove the wanted Theorem. It shows that each time the point goes back to the neighborhood U a fix expansion takes place. To conclude we must take care only of two minor points: 1) the expansion is measured by the form rather then by the norm as we would like; 2) we have established expansion in the tangent space but we clearly need a local, not an infinitesimal, statement.

Let us take care of this two problems.

Lemma 7.1.7 *If $v \in \mathcal{C}_\alpha$ then*

$$\sqrt{2}\sqrt{Q(v)} \leq \|v\| \leq \sqrt{\alpha + \alpha^{-1}}\sqrt{Q(v)}.$$

PROOF. It clearly suffices to consider vectors of the type $v = (1, u)$, for such vectors

$$\frac{\|v\|^2}{Q(v)} = u^{-1} + u,$$

the result follows since $u \in [\alpha, \alpha^{-1}]$. □

Remark 7.1.8 *Note that Lemma 7.1.7 tells us that, for vectors in $\mathcal{C}_\alpha$, we can use the Q-form, instead of the norm, to measure the length of a vector. This is clearly helpful in our context since the form is increasing.*

The second point requires few definitions. As we have seen in chapter 1 and 3 the strategy to construct an unstable manifold is to start in the past with some curve and iterate it forward hoping that it converges to the unstable manifold. To do that we need first to specify the class of curves we intend to iterate, then to make precise what we mean by *converging*.

Let us choose $\delta \in \mathbb{R}^+$ so small that there exists a non-empty open set U_δ such that its $\sqrt{\alpha + \alpha^{-1}}\delta$-neighborhood is completely contained in U.

Definition 7.1.9 *For each $\xi \in U_\delta$ and $\delta \in \mathbb{R}^+$ we define the set of $C^{(1)}$ curves $M_\xi = \{\gamma : [-\delta, \delta] \mid \gamma(0) = \xi;\ \gamma'(s) \in \mathcal{C}_\alpha;\ Q(\gamma'(s)) = 1\}$, we will call them unstable-like curves at ξ.*

Note that all the unstable-like curves are completely contained in U since, by Lemma 7.1.7, their maximal length is $\delta\sqrt{\alpha + \alpha^{-1}}$. Let us define the distance[9] between two curves in M_ξ

$$d(\gamma_1, \gamma_2) := \sup_{s \in [-\delta,\, \delta]} \Theta(\gamma_1'(s), \gamma_2'(s)). \tag{7.1.7}$$

Lemma 7.1.10 *The function on M_ξ^2 defined in (7.1.7) is indeed a distance, moreover M_ξ is complete with respect to such a distance.*

Proof. Since Θ is a metric on $\{v \in \mathcal{C}_\alpha \mid Q(v) = 1\}$, see Lemma 4.3.1, then d is clearly a distance. Moreover, by Lemma 4.3.7, it follows

$$\|\gamma_1'(s) - \gamma_2'(s)\| \leq \text{cost.}\, \Theta(\gamma_1'(s), \gamma_2'(s)).$$

Thus if $\{\gamma_n\} \subset M_\xi$ is a Chauchy sequence with respect to d it is so also in the $C^{(1)}$ topology. Accordingly, there exists $\gamma_* \in C^{(1)}$ such that $\gamma_n \to \gamma_*$. Obviously, $\gamma_*(0) = \xi$, $Q(\gamma_*'(s)) = 1$ and $\gamma_*'(s) \in \mathcal{C}_\alpha$, that is $\gamma_* \in \mathcal{C}_\alpha$ which proves the claim. □

Now the setting is sufficiently precise to prove the wanted result.

Theorem 7.1.11 *Almost all the points in U_δ have an unstable manifold.*

Proof. Let $\Omega_\delta := \Omega(U_\delta)$.[10] If $n \geq \bar{n}$ and $\xi \in U_\delta$, $T^{-n}\xi \in U_\delta$, then we can study $T^n M_{T^{-n}\xi}$. If $\gamma \in M_{T^{-n}\xi}$ then the length of $T^n\gamma$ can be estimated

[9] See section 4.3, and in particular Example 4.3.1 Cones and Q-forms, for the definition of the metric Θ and its properties.

[10] See 7.1.6 for a defintion of Ω.

as follows. First notice that is $T^n\gamma$ it is not completely contained in U then its Q-length must exceed δ, at least in one direction.[11] Thus we need only to compute the length in one direction assuming that all the curve is contained in U:

$$\int_0^\delta \sqrt{Q(D_{\gamma(s)}T^n\gamma'(s))}ds \geq \sigma_\alpha\delta \geq \delta.$$

For each $\xi \in \Omega(U_\delta)$ let $\{n_j\}_{j\in\mathbb{N}}$ be the times such that $T^{-n_j}\xi \in U_\delta$. It follows that, for each $\gamma \in M_{T^{-n_j}\xi}$, calling $\tilde{\gamma}$ the curve $T^{n_j}\gamma$ parametrized so that $Q(\tilde{\gamma}') = 1$, we can define $\tilde{T}_j\gamma(s) = \tilde{\gamma}(s)$ for each $s \in [-\delta, \delta]$.

Accordingly, the sets $M_\xi^j := \tilde{T}_j M_{T^{-n_j}\xi} \subset M_\xi$ must all contain the unstable manifold at ξ (if it exists and is smooth). On the other hand, equality 4.3.7 together with Lemma 7.1.5, implies that

$$\operatorname{diam}(M_\xi^j) \leq \operatorname{const}\sigma_\alpha^{-2j}.$$

Thus $\cap_j M_\xi^j$ consists of just one curve $\gamma_\xi \in M_\xi$. The curves $\{\gamma_\xi\}_{\xi\in\Omega(U_\delta)}$ are exactly the unstable manifolds we are looking for. To see this we must show that, for each $x, y \in \gamma_\xi$, (see definition 3.4.1)

$$\lim_{n\to\infty} d(T^{-n}x,\, T^{-n}y) = 0.$$

What we certainly know up to now is that, if $\{n_j(\xi)\}$ are the backward return times of ξ in U_δ, then

$$d(T^{-n_j}x,\, T^{-n_j}y) \leq \sqrt{\alpha+\alpha^{-1}}\sigma_\alpha^j\delta.$$

Yet, it is still possible that the backward trajectory of ξ spends more and more time outside the neighborhood U_δ and, during such long trips, the length of $T^{-n}\gamma_\xi$ grows back to order one.

Let us estimate the measure of such possibly troublesome points. Let $E_n = \{\xi \in U_\delta \mid T^{-k}\xi \notin U_\delta \,\forall\, k \in \{1,\dots,n\}\}$. Looking at the proof of the Poincaré Return Theorem 1.6.5 it is imediately clearly that $m(E_n) \leq \frac{1}{n}$. It is then natural to introduce the set

$$\bar{\Omega}(U_\delta) := \left\{\xi \in U_\delta \;\Big|\; \exists\,\{n_j\} \,:\, T^{-n_j}\xi \in U_\delta \text{ and } \limsup_{j\to\infty}\frac{n_{j+1}-n_j}{j} = 0\right\}.$$

The following is very similar to Lemma 3.4.7 with the only difference that we do not have the strong control outside the return set that is ensured there by (3.4.1). This problem is taken care of by the next Lemma.

[11] Obviously, by Q-length of a curve γ we mean

$$\int \sqrt{Q(\gamma'(s))}ds.$$

Lemma 7.1.12 *If $\xi \notin \bar{\Omega}(U_\delta)$, then the series*

$$S_n(\xi) := \frac{1}{n}\sum_{k=1}^{n} \chi_{U_\delta}(T^{-k}\xi)$$

either is not convergent or converges to zero.

PROOF. The argument is by contradiction. Suppose that $\lim_{n\to\infty} S_n(\xi) = \ell > 0$. Since $\xi \notin \bar{\Omega}(U_\delta)$, calling $\{n_j(\xi)\}$ the backward returns to U_δ, there exists $c > 0$ and $\{j_m\}$ such that $n_{j_m+1} - n_{j_m} > cj_m$. Next, choose $\varepsilon < \min\{c^{-1}, \frac{c\ell^2}{3}\}$ and let $\bar{n} > \varepsilon^{-1}$ such that $|S_n(\xi) - \ell| < \varepsilon$ for all $n \geq \bar{n}$. At last choose $\bar{m}$ such that $n_{j_{\bar{m}}} > \bar{n}$. Note that, by definition,

$$S_{n_{j_m}} = \frac{j_m}{n_{j_m}}.$$

Then

$$\begin{aligned}
|\ell - S_{n_{j_{\bar{m}}+1}}| &= |\ell - \frac{j_{\bar{m}}+1}{n_{j_{\bar{m}}+1}}| \geq \ell - \frac{j_{\bar{m}}+1}{n_{j_{\bar{m}}} + cj_{\bar{m}}} \\
&\geq \frac{\frac{j_{\bar{m}}}{n_{j_{\bar{m}}}} + \frac{1}{n_{j_{\bar{m}}}}}{1 + c\frac{j_{\bar{m}}}{n_{j_{\bar{m}}}}} \geq \ell - \frac{\ell + 2\varepsilon}{1 + c(\ell + \varepsilon)} \\
&\geq \frac{c\ell^2 + (c\ell - 2)\varepsilon}{1 + c\ell} > \varepsilon
\end{aligned}$$

contrary to the assumption. □

On the other hand Birkhoff Theorem implies that $\chi_{U_\delta}^{-}(\xi) := \lim_{n\to\infty} S_n(\xi)$ exists almost everywhere and it is an invariant function. Thus setting $A = \{\xi \in \mathbb{T}^2 \mid \chi_{U_\delta}^{-}(\xi) = 0\}$ we have $T^{-1}A \subset A$ and

$$0 = \int_A \chi_{U_\delta}^{-}(\xi) m(d\xi) = \int_A \chi_{U_\delta}(\xi) m(d\xi) = m(A \cap U_\delta).$$

Collecting the above considerations it follows that $m(\bar{\Omega}(U_\delta)) = m(U_\delta)$. □

Remark 7.1.13 *The previous proof is very similar in spirit to the strategy used in chapter 3 to prove Hadamard-Perron Theorem 3.3.2. Only it takes advantage of the lesson, learned there, of the importance of a cone field to establish regularity. In fact, the proof yields immediately that the manifold are $\mathcal{C}^{(1)}$. The disadvantage is that some preliminary work is needed to construct appropriate metrics over cones so, in same sense, the proof is less elementary although it is certainly more natural.*[12]

[12] Actually, one could have the doubt that the above theorem is weaker since, by starting immediately by considering smooth curves, does not rule out the possibility of the existence of Lipschitz curves, which 3.3.2 does. This turns out to be a superficial impression due to my choice to present the result in its simplest form, see Problem 7.2 for details.

Clearly the stable manifold can be constructed exactly in the same way arguing about T^{-1} rather than T.[13] Actually teh regularity of the invariant manifolds is more that $\mathcal{C}^{(1)}$; we will discuss this in more detail in section 7.2.

7.1.5 The non-smooth case

In this section we will extend some of the previous results to the case of smooth dynamical systems with singulartity. For simplicity we will discuss only the uniformly hyperbolic case.

In the case of uniform hyperbolicity, as we have already pointed out, the proof of existence for the unstable manifolds, in the smooth case, is much simpler (crf. Problem 7.4). The simplification comes from the fact that there is no need to struggle in order to gain a fixed rate of expansion, each iteration of the map provides the wanted expansion.

Nevertheless, if syngularities are prensent an extra difficulty arises: the curves that we are iterating to approximate the manifolds may cross a syngularity and be cut. This means that they could be much shorter than expected and, in principle, even of zero legth. If such a situation could take place, we would not have any unstable manifold at all. Yet, we have already seen in Example 3.6.1 (Discontinuous Arnold Cat) that, typically, the invariant manifolds exists and do have some length; here we will argue in the same direction but in greater generality and by using a more flexile strategy. The basic idea in the following is to consider sets of the type

$$\begin{aligned} X_{\delta,n} &:= \{x \in X \mid d(T^{-n}x, \mathcal{S}) \geq \tfrac{\delta}{(n+1)^2}\}; \\ X_\delta &:= \cup_{n=0}^{\infty} X_{n,\delta} \end{aligned} \tag{7.1.8}$$

Remark 7.1.14 *In the dfiniton of X_δ, and in the following, we use the distance, rather than the Q-form used in the previou chapters. This simplifies the argument in the uniformly hiperbolic case. Yet, to argue using only the cone family would be very instructive for the following, see problem Problem 7.7.*

Lemma 7.1.15 *If $(\mathbb{T}^2, T, m)$ is a uniformly hyperbolic symplectic system with syngularity, then there exists $c_* > 0$ such that any $x \in X_\delta$ has an unstable mabifold of size $c_*\delta$.*

Proof. First of all notice that one can assume, withou loss of generality, that there exists a striclty invariant cone field $\mathcal{C}(x)$, (crf. Problem 7.6). Let us call $\lambda_- > 1$ the minimal expansion for vectors in the cone family. Then, in analogy with Definition 7.1.9, we consider the sets

$$M(x, \delta) = \{\gamma : [-\delta, \delta] \mid \gamma(0) = x;\ \gamma'(s) \in \mathcal{C}(\gamma(s));\ \|\gamma'(s)\| = 1\}.$$

[13] Obviously the associate cone family consists of the complement of the cones for the forward map (see Problem 7.1 for details).

Note that if $x \in X_\delta$, then all the curves in $M(T^{-n}x, c_-\lambda_-^{-n}\delta)$ do not intersect $\mathcal{S}$, provided c_- is choosen small enough.[14] Now, if $\gamma \in M(T^{-n}x, c_-\lambda_-^{-n}\delta)$, then T^γ, reparametrized by arc-lenght and restricted to the lenght $2c_-\lambda_-^{-n+1}\delta$ is a curve in $M(T^{-n+1}x, c_-\lambda_-^{-n+1}\delta)$, for the same arguments explained in Theorem 7.1.11. Accordingly, we can construct a sequence of curves $\gamma_n \in M(x,\delta)$ by choosing a curve in $M(T^{-n}x, c_-\lambda_-^{-n}\delta)$, iterating it in the future n times, considering the connected componet contianing x, reparametrizing it by arc-length and, finally, taking a pice of length δ on each size of x. Since the cone family is strictly invariant, the exact same argument used in Theorem 7.1.11 proves that the curves form a Chauchy sequence and the limiting curve is the manifold we are looking for. $\square$

Corollary 7.1.16 *If $(\mathbb{T}^2, T, m)$ is a uniformly hyperbolic symplectic system with syngularity, then almost every point has unstable manifold.*

PROOF. Just notice that, calling Y_δ the complement of X_δ, holds

$$\begin{aligned}
m(Y_\delta) &\leq \sum_{n=0}^{\infty} m(\{x \in \mathbb{T}^2 \mid d(T^{-n}x, \mathcal{S}) \leq \frac{\delta}{(n+1)^2}\}) \\
&\leq \sum_{n=0}^{\infty} m(T^n\{x \in \mathbb{T}^2 \mid d(x, \mathcal{S}) \leq \frac{\delta}{(n+1)^2}\}) \\
&= \sum_{n=0}^{\infty} m(\{x \in \mathbb{T}^2 \mid d(x, \mathcal{S}) \leq \frac{\delta}{(n+1)^2}\}) \\
&\leq \operatorname{cost}\delta \sum_{n=0}^{\infty} \frac{1}{(n+1)^2} \leq \operatorname{cost}\delta
\end{aligned}$$

where we have used the definition of systems with singularities (crf. 3.6) and the fact that a δ neighborhood of a singularity manifold has measure $\operatorname{cotst}\delta$ (see Problem 7.9 for details). Accordingly, $\cup_{\delta>0} X_\delta$ has full measure and this proves the claim.[15] $\square$

7.2 Regularity of the invariant distributions

In Lemma 4.4.1 we have seen that if the system is smooth and the eventually monotone cone family is continuos then the distributions is continuos. In fact,

[14] To be precise, if $\ln\lambda_- \geq 2$ it sufficies $c \leq 1$, while if $\ln\lambda_- < 2$ one needs $c \leq \frac{e^{(\ln\lambda_-)^2}\lambda_-^{-2}}{3-\ln\lambda_-}$.

[15] The probability aware reader has certainly noticed an alternative approach: to use Borel-Cantelli. See chapter 9 for such an approach.

more it is true. Since the theory of the regularity of the invariant distributions is a rather wide subject here we will address only few basic facts that should give a precise idea of which type of results are available. Let us start with a special istanc of a general result for Anosov maps.

Lemma 7.2.1 (A naïve argument) *For a two dimensional smooth Symplectic Dynamical System with a strictly monotone continuous cone family the invariant distributions are uniformly Hölder continuos.*

PROOF. For each $x \in \mathcal{M}$ let $v^u(x)$ be the unitary vector in the unstable direction and, for each $h \geq 0$ let $\mathtt{osc}_h v^u(x) := \sup_{\|y-x\|\leq h} \|v^u(x) - v^u(y)\|$. By Lemma 4.4.1 $\mathtt{osc}_h v^u(x)$ is a continuos function. Let us fix $x \in \mathcal{M}$, for each $n \in \mathbb{N}$ there existes a neighborhood $U_n \ni T^{-n}x$ such that $T^n U_u$ is contained in a coordinate neighborhood U of the type described in the previous section. Then

$$\begin{aligned}
\mathtt{osc}_h v^u(x) &\leq c_1(x) \sup_{\|y-x\|\leq h} \Theta(v^u(x),\, v^u(y)) \\
&= c_1(x) \sup_{\|y-x\|\leq h} \Theta(D_{T^{-n}x}T^n v^u(T^{-n}x),\, D_{T^{-n}y}T^n v^u(T^{-n}y)) \\
&\leq c_1(x) \sup_{\|y-x\|\leq h} \Theta(D_{T^{-n}x}T^n v^u(T^{-n}x),\, D_{T^{-n}x}T^n v^u(T^{-n}y)) \\
&\quad + c_1(x) \sup_{\|y-x\|\leq h} \Theta(D_{T^{-n}x}T^n v^u(T^{-n}y),\, D_{T^{-n}y}T^n v^u(T^{-n}y)).
\end{aligned}$$

Where $c_1(x)$ is a continuous function of x. Next, since the cone family is stricly monotone it follows that there exists $\sigma < 1$ such that $Q(D_z T^n v) \geq \sigma^{-n}\Theta(v)$ for each $v \in \bar{\mathcal{C}}$ and $z \in U_n$. By equation (4.3.7) it follows that

$$\Theta(D_{T^{-n}x}T^n v^u(T^{-n}x),\, D_{T^{-n}x}T^n v^u(T^{-n}y)) \leq \sigma^{2n}\Theta(v^u(T^{-n}x),\, v^u(T^{-n}y)).$$

The last needed information is an estimate on the distance between $T^{-n}x$ and $T^{-n}y$. If $\mu = \|DT^{-1}\|_\infty$, then

$$d(T^{-n}x,\, T^{-n}y) \leq \mu^n d(x,y).$$

Accordingly,

$$\begin{aligned}
\mathtt{osc}_h v^u(x) &\leq \sigma^{2n} c_1(x) \sup_{\|z-w\|\leq \mu^n h} \Theta(v^u(z),\, v^u(z)) + c_2(x,n)h \\
&\leq c_3(x)\sigma^{2n}\|\mathtt{osc}_{\mu^n} h v^u\|_\infty + c_2(x,n)h.
\end{aligned}$$

Note that all the constants c_i are continuous in x. We can then take the sup of the previous inequality and obtain

$$\|\mathtt{osc}_h v^u\|_\infty \leq C_1\sigma^{2n}\|\mathtt{osc}_{\mu^n h} v^u\|_\infty + C_2(n)h.$$

Next, choose $\alpha > 0$ such that $\sigma^2\mu^\alpha < 1$ (this type of conditions are often called *bunching conditions*) and n_* such that $C_1\sigma^{2n_*}\mu^{\alpha n_*} = \rho < 1$ and $C_3 := C_2(n_*)$ yields

$$\frac{1}{h^\alpha}\|\mathtt{osc}_h v^u\|_\infty \leq \frac{1}{(\mu^{n_*}h)^\alpha}\|\mathtt{osc}_{\mu^{n_*}h} v^u\|_\infty + C_3.$$

Iterating the above expression immediately yield the wanted Hölder continuity, which will be automaitically uniformly since the estimates do not depend on the point. □

The previous result is far from optimal but we will not push it for two reason: on the one hand we will not need a finer result, on the other hand a similar refinement will be discussed explicitely in the subsection 7.3.1.

Nevertheless, for completeness, we conclude with a brief discussion on the smootheness of the invariant manifolds. This amounts to stating the regularity of the invariant distributions in their own direction.

Lemma 7.2.2 *The stable and unstable manifolds are $\mathcal{C}^{(1)}$.*

PROOF. .... □

7.3 The Holonomy (regularity properties)

As we have already seen in subsection 2.2.2, the absolute continuity of the stable and unstable foliations is a key ingredient in the proof of ergodicity for hyperbolic systems. In essence this means that, as far as measure theoretical considerations are concerned, we can use a coordinate system in which the above foliations are straight lines. It turns out that the most convenient formulation of this property is based on the holonomy map.

Definition 7.3.1 *Given two $C^{(1)}$ manifolds V_1, V_2 uniformly transversal to the unstable foliation and such that, for a given $\delta \in \mathbb{R}^+$, $W^u_\delta(z) \cup V_2$ consists of exactly one point for each $z \in V_1$, we can define the function $\phi_{V_1,V_2} : V_1 \to V_2$ defined by*

$$\{\phi_{V_1,V_2}(z)\} := W^u_\delta(z) \cup V_2.$$

The function ϕ_{V_1,V_2} is called unstable Holonomy map.

Clearly, the above definition can be extended to the stable holonomy and, more generally, to any foliation of the space.

Definition 7.3.2 *A foliation is called* absolutely continuous *if the holonomy map associated to any two uniformly transversal manifolds is absolutely continuous.*

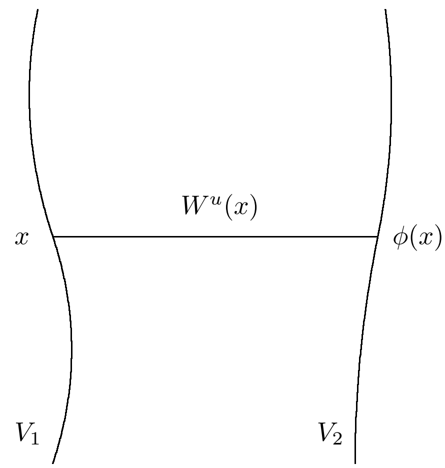

Figure 7.1: Unstable Holonomy

In the following we will prove that the unstable foliations for two-dimensional uniformly hyperbolic symplectic system are indeed absolutely continuous. In fact, here we will consider only holonomies between stable manifolds. Yet, the general case can be treated similarly (see section 7.5 where a more general point of view is needed or solve Problem Problem 7.).

7.3.1 The smooth case

The first step in the study of the property of the holonomies is to investigate their regularity. This investigation can be done at various levels of sophistication and uses some of the most typical ideas in hyperbolic dynamical theory. To give an idea of such techniques we will present here two approaches, similar in the spirit but exhibiting different levels of technical complexity. The first, more rough, is certainly non optimal but it is simple and robust (the same argument works in a much more general setting) and shows the basic idea in its purest form. The second, sharper, will use some special feature of our situation (two dimensions) and it will give an idea of which techniques can be used to carry out more delicate estimates (in particular *distortion estimates*). The basic idea is to try to get a quantitative estimate of the distance between $\phi(x)$ and $\phi(y)$ in terms of the distance between x and y. As usual in the hyperbolic business, the strategy is to go back and forward in time taking advantage of the contracting and expanding properties of the system.

Lemma 7.3.3 (A naïve argument) *The unstable holonomy between two,*

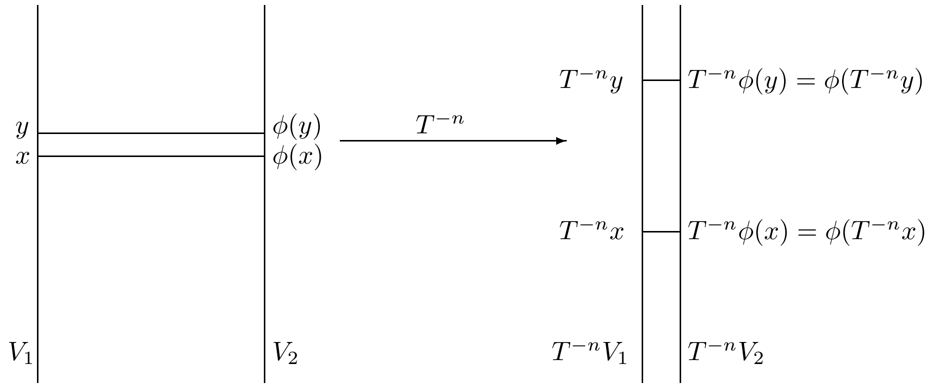

Figure 7.2: Unstable Holonomy backward in time

sufficiently close, stable manifolds V_1, V_2 is Hölder continuous.[16]

Proof. Let us remember that, by Lemma 4.4.2 the systems under consideration are Anosov. Thus there are $c > 0$, $\lambda_+ \geq \lambda_- > 1$ and $\mu_+ \geq \mu_- > 1$ such that

$$
\begin{aligned}
&c\lambda_-^n \leq \|D_x T^n v\| \leq c^{-1}\lambda_+^n \quad \forall x \in \mathbb{T}^2, v \in E^u(x),\ \|v\| = 1 \\
&c\mu_-^n \leq \|D_x T^{-n} v\| \leq c^{-1}\mu_+^n \quad \forall x \in \mathbb{T}^2, v \in E^s(x),\ \|v\| = 1
\end{aligned}
\tag{7.3.1}
$$

This means that going back in time, if $x, y \in V_1$, we have

$$
\begin{aligned}
&c^{-1}\lambda_-^{-n} d(x,y) \geq d(T^{-n}x, T^{-n}\phi(x)) \geq c\lambda_+^{-n} d(x,y) \\
&c^{-1}\mu_+^n d(x,y) \geq d(T^{-n}x, T^{-n}y) \geq c\mu_-^n d(x,y).
\end{aligned}
\tag{7.3.2}
$$

Here, and in the following, if z, w belong to the sme manifold (either stable or unstable) by $d(z,w)$ we mean the distance measure along the manifold (which, in our case, is just a $\mathcal{C}^{(1)}$ curve). The all idea is to go backward in time, the effect of which is schematically represented in figure 7.2

The key fact is that if $d(T^{-n}x, T^{-n}y) \geq 3d(T^{-n}x, T^{-n}\phi(x))$ then, due to the fact that there exists a maximal and minimal angle between the stable and the unstable direction, then $d(T^{-n}\phi(x), T^{-n}\phi(y))$ cannot be much bigger than $d(T^{-n}x, T^{-n}y)$. Although it is a quite intuitive fact we will spend few words in making it precise, since it is the cornerstone of our argument. Notice that $d(T^{-n}x, T^{-n}y) \leq 3d(V_1, V_2)$, thus the relevant points in figure 7.2 lie in a relatively small neighborhood. By Lemma 7.2.1 in such a neighborhood

[16] The exponent of Hölder continuity is given by (7.3.8) and is related to the so called *buncing conditions*, see [51] for more details.

all the manifolds are essentially straight lines. Since such lines are uniformly transversal then

....

In conclusion, there exists $a > 0$ such that, provided $d(T^{-n}x, T^{-n}y) \geq d(T^{-n}x, T^{-n}\phi(x))$,

$$d(T^{-n}\phi(x), T^{-n}\phi(y)) \leq a d(T^{-n}x, T^{-n}y). \tag{7.3.3}$$

Call $\bar{n} = \inf\{n \in \mathbb{N} \mid d(T^{-n}x, T^{-n}y) \geq d(T^{-n}x, T^{-n}\phi(x))\}$ and collect the previous inequalities:

$$\begin{aligned} d(\phi(x), \phi(y)) &\leq \tfrac{d(\phi(x),\phi(y))}{d(T^{-n}\phi(x),\, T^{-n}\phi(x))} d(T^{-n}\phi(x),\, T^{-n}\phi(x)) \\ &\leq a \tfrac{d(\phi(x),\phi(y))}{d(T^{-n}\phi(x),\, T^{-n}\phi(x))} d(T^{-n}x,\, T^{-n}x) \\ &\leq a \tfrac{d(\phi(x),\phi(y))}{d(T^{-n}\phi(x),\, T^{-n}\phi(x))} \tfrac{d(T^{-n}x,\, T^{-n}x)}{d(x,y)} d(x,y) \end{aligned} \tag{7.3.4}$$

It turns out to be convenient not to estimate the quantity

$$\rho_n^n := \frac{d(T^{-n}x,\, T^{-n}x)}{d(x,y)} \leq \mu_+^n \tag{7.3.5}$$

since it will produce a cancellation later on. For the rest we use the estimates (7.3.2) on the expansion and obtain

$$d(\phi(x), \phi(y)) \leq ac^{-1}\rho_{\bar{n}}^{\bar{n}}\mu_-^{-\bar{n}} d(x, y). \tag{7.3.6}$$

Next we need an upper bound for $\bar{n}$, this is readily done noticing that $d(T^{-n}x, T^{-n}y) \geq d(T^{-n}x, T^{-n}\phi(x))$ is implied by

$$\rho_n^n d(x, y) \geq c^{-1}\lambda_-^{-n} d(x, \phi(x))$$

which means

$$\bar{n} \leq \frac{\ln\left[c^{-1}\frac{d(x,\phi(x))}{d(x,y)}\right]}{\ln \rho_{\bar{n}}\lambda_-}. \tag{7.3.7}$$

To conclude it suffices to substitute our estimate (7.3.7) into inequality (7.3.6).

$$\begin{aligned} d(\phi(x), \phi(y)) &\leq ac^{-1}\left[\rho_{\bar{n}}\mu_-^{-1}\right]^{\frac{\ln\left[c^{1}\frac{d(x,\phi(x))}{d(x,y)}\right]}{\ln \rho_{\bar{n}}\lambda_-}} d(x, y) \\ &= ac^{-1}\left[c^{1}\frac{d(x,\phi(x))}{d(x,y)}\right]^{\frac{\rho_{\bar{n}}\mu_-^{-1}}{\rho_{\bar{n}}\lambda_-}} d(x, y), \end{aligned}$$

The above inequality means that there exists a constant $c_1 > 0$, depending only V_1, V_2 and $a, c, \lambda_\pm, \mu_\pm$ such that

$$d(\phi(x), \phi(y)) \leq c_1 d(x, y)^{\frac{\ln \lambda_-\mu_-}{\ln \lambda_-\mu_+}}. \tag{7.3.8}$$

$\square$

Remark 7.3.4 *The above Lemma shows once more the power of the idea of transporting back (or forward) in time the quantities of interest before arguing about them. This was its didactic purpose. Its limit should be quite clear to the reader who has digested the proof: it compares the maximal expansion with the minimal expansion on all the space, but in the proof we always considered points that are, at each time, very close to each other, it is hard to believe that we cannot take advantage of this.*

In fact, the above remark is competely valid in two dimensions as we will see shortly, in higher dimensions different direction can expand at different rates so different expansion rates are, up to a point, inevitable.[17]

Lemma 7.3.5 (Distortion) *Given δ sufficiently small, two points $x, y \in \mathbb{T}^2$ and vectors $v, w \in D_{T^{-1}x}T\mathcal{C}(T^{-1}x) \subset \mathbb{R}^2$, let us define $x_i := T^i x$, $y_i := T^i y$ and $v_i := \frac{D_xT^iv}{\|D_xT^iv\|}$, $w_i := \frac{D_xT^iw}{\|D_xT^iw\|}$ for each $i \in \mathbb{Z}$. If*

$$\|x_i - y_i\| + \Theta(v, w) \leq \delta$$

for each $i \in \{0, \dots, \bar{n}\}$, then there exists a constant $D > 0$ (depending only on the map T) such that

$$e^{-D\delta} \leq \frac{\|D_xT^iv\|}{\|D_yT^iw\|} \leq e^{D\delta} \quad \forall i \in \{0, \dots, \bar{n}\}.$$

PROOF. First of all, since the cone family is continuous (and thus uniformly continuous), it follows that for each $\xi, \eta \in \mathbb{T}^2$ and $\bar{v} \in D_{T^{-1}\xi}T\mathcal{C}(T^{-1}\xi) \subset \mathbb{R}^2$ if $\|\xi - \eta\| + \Theta(\bar{v}, \bar{w}) \leq \delta_1$, then $\bar{w} \in \mathcal{C}(\eta)$, provided δ_1 is small enough. From this follows that there exists $K > 0$ such that $\Theta(v_i, w_i) \leq K\delta$ for each $i \leq \bar{n}$. Accordgly, by choosing δ so that $K\delta < \delta_1$, it follows that $w_i \in \mathcal{C}(x_i)$. Let us prove the claimed inequality by induction. It is true by hypothesis for $i = 0$, suppose it to be true for i. Then $w_i \in \mathcal{C}(x_i)$ since, obviously, $v_i \in D_{T^{-1}x_i}T\mathcal{C}(T^{-1}x_i)$. Notice that, since the cone family is strictly invariant, there exists $\sigma < 1$ such that the cone metric is contracted at least by σ. Hence,

$$\begin{aligned}\Theta(v_{i+1}, w_{i+1}) = \Theta(D_{x_i}Tv_i, D_{y_i}Tw_i) &\leq \Theta(D_{x_i}Tv_i, D_{x_i}Tw_i) + \Theta(D_{x_i}Tw_i, D_{y_i}Tw_i)\\ &\leq \sigma\Theta(v_i, w_i) + C\|x_i - y_i\| \leq (\sigma K + C)\delta\end{aligned}$$

and the result follows provided $K = (1 - \sigma)^{-1}C$. In fact, iterating the above formula on can get the more precise estimate that will be needed in the following

$$\Theta(v_{i+1}, w_{i+1}) \leq \sigma^i\Theta(v, w) + C\sum_{j=0}^{i}\sigma^{i-j}\|x_j - y_j\|. \tag{7.3.9}$$

[17] See *buncing conditions* in [51] for a more detailed discussion.

To continue, note that $\|D_xT^iv\| = \|D_{x_{i-1}}Tv_{i-1}\|\|D_xT^{i-1}v\|$, thus, by induction

$$\|D_xT^iv\| = \|v\|\prod_{j=1}^{i-1}\|D_{x_j}Tv_j\|.$$

Consequently,

$$\begin{aligned}|\ln\|D_xT^iv\| - \ln\|D_yT^iw\|\,| &\leq \sum_{j=1}^{i-1}|\ln\|D_{x_j}Tv_j\| - \ln\|D_{y_j}Tw_j\|\,| \\ &\leq \sum_{j=1}^{i-1}|\ln\|D_{x_j}Tv_j\| - \ln\|D_{x_j}Tw_j\|\,| \\ &\quad + \sum_{j=0}^{i-1}|\ln\|D_{x_j}Tw_j\| - \ln\|D_{y_j}Tw_j\|\,| \\ &\leq C_1\sum_{j=1}^{i-1}(\Theta(v_i,w_i) + \|x_i - y_j\|)\end{aligned}$$

To conclude, define $z = W^u_{2\delta}(x)\cap W^s_{2\delta}(y)$,[18] by the uniform transversality of the manifolds follows that there exists $c_0\in\mathbb{R}^+$ such that $\|z-x\| + \|z-y\| + \|T^{-i}z - x_i\| + \|T^{-i}z - y_i\| \leq c_0\delta$. This means that, for some $c_1 > 1$,

$$\begin{aligned}\|x_j - y_j\| &\leq \|T^jz - x_j\| + \|T^jz - y_j\| \leq c_1\lambda^{-i+j}\|T^iz - x_i\| + c_1\lambda^{-j}\|z-y\| \\ &\leq c_0c_1\{\lambda^{-i+j} + \lambda^{-j}\}\delta.\end{aligned} \tag{7.3.10}$$

Accordingly, by (7.3.10) and (7.3.9),

$$\begin{aligned}|\ln\|D_xT^iv\| - \ln\|D_yT^iw\|\,| \leq c_0c_1C_1&\sum_{j=1}^{i-1}\left[\sigma^j\Theta(v,w) + C\sum_{k=0}^{i}\sigma^{j-k}(\lambda^{-i+k} + \lambda^{-k})\delta\right] \\ &+ c_0c_1C_1\sum_{j=1}^{i-1}(\lambda^{-i+k} + \lambda^{-k})\delta := D\delta.\end{aligned}$$

□

Remark 7.3.6 *Note that the above Lemma is essentially two dimensional. In higher dimension there can be different rate of expansions in different directions and one cannot hope to obtain such a strong result for the derivative. Nevertheless, a similar arguments works in general for the Jacobian (e.g.,* $\det(DT|_{E^u})$*).*

[18]The point z is uniquely defined, see Problem 7.3.

Lemma 7.3.7 (A two dimensional argument) *The unstable holonomy between two stable manifolds V_1, V_2 is Liptshics.*

PROOF. The proof goes exactly as the one of Lemma 7.3.3 untill equation (7.3.4) that is convenient to rewrite as follows

$$\begin{aligned}
d(\phi(x),\phi(y)) &\leq a\frac{d(\phi(x),\phi(y))}{d(T^{-n}\phi(x),d(T^{-n}\phi(x)))}\frac{d(T^{-n}x,d(T^{-n}x))}{d(x,y)}d(x,y)\\
&= a\frac{\int_{\gamma_1}|\gamma_1'|}{\int_{\gamma_1'}|DT^{-n}\gamma_1'|}\frac{\int_{\gamma_2'}|DT^{-n}\gamma_2'|}{\int_{\gamma_2}|\gamma_2'|}d(x,y)\\
&= a\frac{d(x,y)}{d(\phi(x),\phi(y))}\int_{\gamma_2'}ds\frac{1}{\int_{\gamma_1'}d\tau\frac{D_{\gamma_1(\tau)}T^{-n}\gamma_1'(\tau)}{D_{\gamma_2(s)}T^{-n}\gamma_2'(s)}}d(x,y)
\end{aligned}$$

where γ_1 is a parametrization, by arclength, of the curve from $\phi(x)$ to $\phi(y)$ and γ_2 of the curve fron x to y. Next, we use the estimate obtained in Lemma 7.3.5, applied to T^{-1}, to conclude

$$d(\phi(x),\phi(y)) \leq ae^{D\delta}d(x,y).$$

□

Remark 7.3.8 *The above Lemma starts to give an idea of what can be achieved with this type of arguments. In fact, it is possible to obtain more explicit formulae but this transcends the scopes of the present notes. We present sharper results only in a special case suited to our subsequent needs.*

7.4 Absolute continuity

Consider the measures m_{V_i} induced by the the Lebesgue measure on the manifolds V_i. For we know that the Holonomy is continuous, this means that we can define the measure $\bar{m}$ on V_1

$$\bar{m}(f) = m_{V_2}(f\circ\phi).$$

In the two dimensional setting the absolutely continuity of the Holonomy is equivalent to the absolute continuity of $\bar{m}$ with respect to m_{V_1} (see [74] or solve Problem 7.10 if in doubt).

Since, by Lemma 7.3.7, the Holonomy is Liptshics, then it is absolutely continuous (again see [74] or solve Problem 7.11).

It is interesting that a slight modification of the previous arguments yield a much deeper knowledge of the regularity properties of the Holonomy.

Lemma 7.4.1 *The unstable holonomy ϕ between two stable manifolds V_1 and V_2 has Jacobian*

$$J\phi(x) = \lim_{n\to\infty} \frac{D_x T^{-n} v^s(x)}{D_{\phi(x)} T^{-n} v^s(\phi(x))}$$

PROOF. First of all let us check that the limit in the statment is indeed convergent. ....

To prove the statment we will slightly refine the estimates in Lemma 7.3.3 with the improvements described in Lemma 7.3.7. Consider $x \in V_1$ and, for each each $h \in \mathbb{R}^+$, let $x_h \in V_1$ such that the distance between x and x_h (measured along V_1) is exactly h.

Due to the continuity, and hence the uniform continuity, of the stable and unsatble distributions (see ???), for each $\varepsilon \in \mathbb{R}^+$ there exists δ such that in each ball of size δ the distributions vary by less than ε.

In strict analogy with Lemma 7.3.3 we choose the first n such that $d(T^{-n}x, T^{-n}x_h) \geq d(T^{-n}x, T^{-n}\phi(x))$. Note that this will happen for $n \leq \frac{\ln dh^{-1}}{\ln \lambda_-\mu_+}$, where $d := \text{dist}(v_1.V_2)$. This means that that all picture in Figure 7.2, after applying T^{-n} is contained in a neighborhood of size $d^{\frac{\ln\lambda_-}{\lambda_-\mu_+}} h^{\frac{\ln\mu_+}{\lambda_-\mu_+}} < \delta$, provided we have choosen h small enough. Accordingly, the same arguments used in Lemma 7.3.3 really prove that

$$1 - \varepsilon \leq \frac{d(T^{-n}\phi(x),\, T^{-n}\phi(y))}{d(T^{-n}x, T^{-n}x_h)} \leq 1 + \varepsilon.$$

After that, the estimates in Lemma 7.3.7 prove the result.
.... □

By using the explict formula provided by Lemma 7.4.1 it is possible to obtain more stringet information on the holonomy.

Lemma 7.4.2 *The Jacobian of the unstable holonomy is Lipshics and, in addition*

$$\|1 - J\phi\| \leq c\, dist(V_1,\, V_2)$$

for some cosntant c depending only on T.

PROOF. .... □

7.4.1 Non-unifom hyperbolicity

only results and ideas

7.5 The non-smooth case

In this section we discuss the question of the regularity of the foliations for smooth dynamical systems with singularities. This is a rather vast subject and it is therefore necessary to limit the scope of the presentation. As before the discussion will be restricted to the uniformly hyperbolic case. The aim of the presentation is twofold: on the one hand I want to develop only the strictly needed part of a rather rich and complex theory, on the other hand I want to provide the reader with the basic ideas needed to tackle the more general results in the literature.

Let us start with a generalization of the distortion bound given in Lemma 7.3.5.

Lemma 7.5.1 (Distortion) *Consider an open subset $U \in \mathbb{T}^2$, such that $U \subset X_{\delta,n}$.*[19] *Then, given $x, y \in U$, $x_i = T^{-i}x$, $y_i = T^{-i}y$, such that x_n and y_n are joined by an unstable curve* [20] *contained in $T^{-n}U$, there exists a constant $D > 0$ (depending only on the map T) such that, for each $v, w \in D_{Tx}T^{-1}\mathcal{C}(Tx)$, holds that*

$$e^{-D\{\Theta(v,w)+\delta^{-2c_2}\ diam(U)\}} \leq \frac{\|D_xT^{-i}v\|}{\|D_yT^{-i}w\|} \leq e^{D\{\Theta(v,w)+\delta^{-2c_2}\ diam(U)\}} \quad \forall i \in \{0,\dots,n\},$$

provided $diam(U) \leq \delta$. [21]

Proof. We start by noticing that there exists a constant $d_0 > 0$ such that, if two points are joined by a unstable curve, then the lenght of the curve is bounded by d_0 times the distance between the points (see Problem 7.). Calling γ_n the unstable curve joining x_n and y_n, and seting $\gamma_i := T^{n-i}\gamma$, it follows that $T^i\gamma \cap \mathcal{S} = \emptyset$ for all $i \in \{0,\dots,n\}$. Accordingly,

$$\|x_i - y_i\| \leq \ell(\gamma_i) \leq c_1^{-1}\lambda^{-i}\ell(\gamma_0) \leq c_1^{-1}\lambda^{-i}d_0 \operatorname{diam}(U)$$

which replaces, in this simpler situation, (7.3.10) of Lemma 7.3.5.

As in Lemma 7.3.5 we define $v_i = \frac{D_xT^{-i}v}{D_xT^{-i}w}$ and $w_i = \frac{D_yT^{-i}v}{D_yT^{-i}w}$ Thus, the same argument as in Lemma 7.3.5 yields

$$\begin{aligned}\Theta(v_{i+1}, w_{i+1}) &\leq \sigma^i\Theta(v,w) + C\delta^{-c_2}\sum_{j=0}^{i}\sigma^{i-j}\|x_j - y_j\| \\ &\leq \sigma^i\{\Theta(v,w) + C_2\delta^{-c_2} \operatorname{diam}(U)\}.\end{aligned} \tag{7.5.1}$$

[19] see (7.1.8) for the definition of $X_{\delta,n}$.

[20] By "unstable curve" I mean any $\mathcal{C}^{(1)}$ curve $\{\gamma(t)\}$ such that $\gamma'(t) \in \mathcal{C}(\gamma(t))$ for each t.

[21] The constant c_2 is the one from Definiton 3.6 of smooth systems with singularities.

Thus,

$$
\begin{aligned}
|\ln\|D_xT^iv\|-\ln\|D_yT^iw\|\,| &\leq C_3\sum_{j=1}^{i-1}\sigma^i\{\Theta(v,w)+\delta^{-c_2}\ \mathrm{diam}(U)\}+\sum_{j=1}^{i-1}\int_{T^{n-j}\gamma}\left\|\frac{D_\xi^2T}{D_\xi T}\right\|d\xi\\
&\leq C_3(1-\sigma)^{-1}\{\Theta(v,w)+\delta^{-c_2}\mathrm{diam}(U)\}+D_*\sum_{j=1}^{i-1}\int_{\gamma_j}d(\xi,\mathcal{S})^{-2c_2}d\xi\\
&\leq C_3(1-\sigma)^{-1}\{\Theta(v,w)+\delta^{-c_2}\mathrm{diam}(U)\}+D_*\sum_{j=1}^{i-1}\lambda^{-j}j^{4c_2}\delta^{-2c_2}\ \mathrm{diam}(U)\\
&:= D\{\Theta(v,w)+\delta^{-2c_2}\ \mathrm{diam}(U)\}.
\end{aligned}
$$

□

The above Lemma allows to control the distortion on manifolds whose immage lies completely in $X_{\delta,n}$, unfortunately this set is too small for our purposes. The problem comes from the fact that a manifold of a point in X_δ is certainly long at least δ but often it is much longer. Thus if we consider the holonomy between manifolds at a given fixed distance a good deal of manifolds that connect such manifolds may belong to ponints in the intersections of the first manifold and X_δ only for a *delta* much smaller than the distance between the two manifold. Accordingly we would need to control points that may have an immage much closer to the singularity lines than is allowed to points in X_δ. This will become more clear in Capter 9, for the time being it is simply a justification for the following sharper results.

As we have seen in the previous sections the distortion plays an import rôle in the study of absolute continuity. Since we are interested in specific examples and not in the most general theory we will assume a simplifying condition on the distortion.

Definition 7.5.2 *We will say that a smooth map with singularities has mild distortion if there exists $D_*>0$ and $\alpha\in[0,1)$ such that, for each $x\in\mathcal{M}\backslash\mathcal{S}$, holds*

$$
\|D_xT\|+\|D_xT^{-1}\|\leq D_*d(x,\mathcal{S})^{-\alpha}\,;\quad \frac{\|D_x^2T\|}{\|D_xT\|}\leq D_*d(x,\mathcal{S})^{-1}.
$$

7.5.1 Examples

7.5.1.a *Billiards*

The billiard in 5.5.1 (finite Horizon) has mild distortion . . . .

The relevance of the above concept is illustrated by the next result.

Let $X_\delta^{(k,n)}:=\cup_{i=k}^nT^iX_{i,\delta}$.

Lemma 7.5.3 *Let an open set $U \subset \mathbb{T}^2$ be such that $U \subset X_\delta^{(k,n)}$ and $T^{-j}U \cap \mathcal{S} = \emptyset$ for all $j \in \{0,\dots,k\}$. Consider two stable curves $\gamma_i \subset U$ and assume that there exists an unstable curve $\gamma \subset T^{-n}U$ with $T^n\gamma \cap \gamma_i \neq \emptyset$. In addition, require that there exists $\bar c, \underline{c} > 0$ such that $d(T^{-j}\gamma_i, \mathcal{S}) \geq \bar c \ell(T^{n-j}\gamma)$, for all $j \in \{0,\dots,k\}$, and $\max\{\ell(T^{-j}\gamma_1), \ell(T^{-j}\gamma_2)\} \leq \underline{c}\ell(T^{n-j}\gamma)$.*[22] *Then there exists $C_* > 0$, depending only on T, $\bar c$ and $\underline{c}$, such that*

$$C_*^{-k} \leq \frac{\ell(\gamma_1)\ell(T^{-n}\gamma_2)}{\ell(\gamma_2)\ell(T^{-n}\gamma_1)} \leq C_*^k.$$

Proof. Since, for each $x \in U$,

$$d(T^{-n}(T^{-k}x), \mathcal{S}) \geq \frac{\delta}{(n+k)^2} \geq \frac{\delta k^{-2}}{n^2},$$

it follows that $T^{-k}U$ satisfies the hypotheses of Lemma 7.5.1. Accordingly,

$$\begin{aligned}&\frac{\ell(T^{-k}\gamma_1)\ell(T^{-n}\gamma_2)}{\ell(T^{-k}\gamma_2)\ell(T^{-n}\gamma_1)} = \frac{\int dt_1 \|DT^{-k}\gamma_1'(t_1)\| \int dt_2 \|\gamma_2'(t_1)}{\int dt_1 \|DT^{-k}\gamma_2'(t_1)\| \int dt_2 \|\gamma_1'(t_1)}\\ &\leq e^{2D\delta^{-1}k^2 \operatorname{diam}(T^{-k}U)}.\end{aligned}$$

Since $T^{-k}U$ contains, by hypothesis, an unstable curve, it follows that its diameter is larger than some constant (depending only on the cone field) times the length of the curve. Accordingly $\operatorname{diam}(U) \geq \text{cost}.\lambda^k \operatorname{diam}(T^{-k}U)$, hence

$$\frac{\ell(T^{-k}\gamma_1)\ell(T^{-n}\gamma_2)}{\ell(T^{-k}\gamma_2)\ell(T^{-n}\gamma_1)} \leq e^{2D_1\delta^{-1}\operatorname{diam}(U)}.$$

Clearly the lower bound follows by interchanging the rôle of γ_1 and γ_2.

The problem remains to handle the first k iterations. Now the curves can get very close to the singlurities. Hence, for $j \leq k$, let $\gamma^i := T^{-j+1}\gamma_i$, parametrized by arc-length, then

$$\frac{\ell(T^{-j+}\gamma_1)}{\ell(T^{-j+1}\gamma_2)} = \frac{\ell(T\gamma^1)}{\ell(T\gamma^2)} = \frac{\int dt \|DT(\gamma^1)'(t)\|}{\int ds \|DT(\gamma^2)'(s)\|}. \tag{7.5.2}$$

By the hypotheses follows that, for each $t, s > 0$,

$$d(\gamma^1(t), \gamma^2(s)) \leq (1+2\underline{c})\ell(T^{n-j}\gamma) \leq \bar c^{-1}(1+2\underline{c}) \min\{d(\gamma^i, \mathcal{S})\}.$$

This means that

$$\frac{\|DT(\gamma^1)'(t)\|}{\|DT(\gamma^2)'(s)\|} \leq C\sigma^j + e^{D_*(\min\{d(\gamma^i,\mathcal{S})\})^{-1}d(\gamma^1(t),\gamma^2(s))} \leq C + e^{D_*\bar c^{-1}(1+2\underline{c})} := C_*. \tag{7.5.3}$$

[22]For any curve $\tilde\gamma$, by $\ell(\tilde\gamma)$ we designate the length of the curve.

Equations (7.5.2) and (7.5.3) together imply

$$\frac{\ell(T^{-j+1}\gamma_1)}{\ell(T^{-j+1}\gamma_2)} \leq C_* \frac{\ell(T^{-j}\gamma_1)}{\ell(T^{-j}\gamma_2)}$$

which, applied inductively, yields the Lemma. □

We are now ready investigate the property of the Holonomy for discontinuous systems. The first remark is that, due to Lemma 7.1.15, each $x \in X_\delta$ has an unstable manifold of size at least $c_*\delta$. This means that is we have two stable manifolds at a maximal distance $c_*\delta$ then the Holonomy is well defined on $V_1 \cap X_\delta$; yet the domain $D(V_1, V_2)$ of the holonomy is typically much larger. It is possible to investigate the regularity of the holonomy directly on its full domine, this yelds poorer regularity. Not to enter in too technical details we will content ourselves with the minimal result needed in the sequel.

We are now ready to investigate the regularity of the Holonomies. Unfortunately, for further needs, we need to study the holonomy not just between stable manifold but, more generally, between any two manifolds transversal to the unstable distribution. To do this requires a bit of preliminary work.

Since the argumet, as usual, is local let us consider $\bar{x}$ such that $\bar{x} \notin \cap_{i=0}^{\bar{n}} T^{-i}\mathcal{S}^+$ for some $\bar{n}$ to be defined later. Clearly, in a neighorhood of such a point the cone field $D_{T^{\bar{n}}z}T^{-\bar{n}}\mathcal{C}(T^{\bar{n}}z)$ is well defined and continuous, moreover such cones will have diameter $\sigma^{\bar{n}}$, for some $\sigma \in (0, 1)$ depending only only on T, in $\mathcal{C}(z)$. ....

Lemma 7.5.4 *For each $\delta > 0$ the unstable holonomy between two manifolds V_1, V_2 in... such that $d(V_1, V_1) \leq \delta$, restricted to the set $V_1 \cap X_\delta$, is Liptshics.*

PROOF. ....*just like Lemma 7.3.7 by applying the previous lemmata*.................... □

Lemma 7.5.5 *For each V_1, V_2 and $\varepsilon > 0$ the unstabel holonomy ϕ_{V_1,V_2}, restricted to $D(\phi_{V_1,V_2}) \cap X_\varepsilon$, is Liptshics.*

PROOF. Consider a sequence of manifolds $\{\tilde{V}_i\}_{i=1}^n$ such that $V_1 = \tilde{V}_1$, $\tilde{V}_n = V_2$ and $d(\tilde{V}_i, \tilde{V}_{i+1}) \leq \varepsilon$. Then

$$\phi_{V_1,V_2}\big|_{D(\phi_{V_1,V_2})} = \phi_{\tilde{V}_n,\tilde{V}_{n-1}} \circ \cdots \circ \phi_{\tilde{V}_2,\tilde{V}_1}\big|_{D(\phi_{V_1,V_2})}.$$

By Lemma 7.5.4 follows that, for each $i \in \{1, \ldots, n\}$, $\phi_{\tilde{V}_i,\tilde{V}_{i-1}}$ is Lipshics on $D(\phi_{\tilde{V}_i,\tilde{V}_{i-1}}) \cap X_\varepsilon$. The result then follows since the compositon of Lipshics maps is Lipshics. □

7.6 More comments on regularity

put general results–take from the appendix of [BKL]

Problems

7.1 Show that if the cone family $\mathcal{C}(x)$ is eventually stricly monotone for the map T, then the family $\mathcal{C}_-(x) := \overline{\mathcal{C}(x)^c}$ is eventually stricly monotone for the map T^{-1}. (Hint: Let $X_n := \{x \in X \mid D_xT^n\mathcal{C}(x) \subset \text{int}\,(\mathcal{C}(T^n(x))\}$, then, by definition, $\cup_n X_n$ is a set of full measure. On the other hand, by Problem 4.6 it follows that for $x \in T^nX_n$ holds $D_xT^{-n}\mathcal{C}_-(x) \subset \text{int}\,(\mathcal{C}_-(T^{-n}x))$.)

7.2 Show that the invariant manifolds constructed in Theorem 7.1.11 are the unique invariant set. (Hint: Take the two curves on the boundary of M_ξ and show that the set contained between them shrinks to the curve already obtained.)

7.3 Show that for an Anosov Systems there exists δ such that, in each neighborhood U of diameter less than δ, for each $x, y \in U$ $W^u(x) \cap W^s(y)$ consists of exactly one point. (Hint: transversality...)

7.4 Prove the existence of the stable and unstable manifolds for uniform hyperbolic systems without using the cone metric. (Hint: View the manifolds as a graph ...)

7.5 Prove that if a symplectic dynamical system with syngularities admits a continuous cone family, then the systems is uniformly hyperbolic. (Hint: prove contraction (using symplecticity) and uniform transversality)

7.6 Prove that a uniformly hyperbolic two dimensional symplectic dynamical system wtih singularities admits a strictly invariant cone family. (Hint: By definition 3.1.1 it follows that the stable and unstable distributions are uniformly transversal. We can therefore consider the cone family $\mathcal{C}_a(x) = \{v \in \mathbb{R}^2 \mid |\langle v, v^u(x)\rangle| \geq (1-a)\|v\|\}$. Such a cone family it clearly non degenerate and it is easy to check that it is stricly invariant provided ....

7.7 Redo the argument in section 7.1.5 using the Q-form rather than the sidtance in the definition of X_δ.

7.8 Generalize the content of section 7.1 to an arbitrary two diemnsional symplectic manifold. (Hint: same stuff ...)

7.9 Given a codimension one manifold $\mathcal{M} \subset \mathbb{R}^d$ and a subset $K \subset \mathcal{M}$ such that ∂K is the finite union of submanifolds with boundaries, show that a δ neighborhood of K has measure bounded by a constat times δ.

7.10 put problems on absolute continuity, relation between the notion for functions and the notions for measures, see garyepy-evans

7.11 Prove that the measures $\bar{m}$ and m_{V_1} (defined in section 7.4) are equivalent. (Hint: Let us start by proving that $\bar{m}$ is absolutely continuous with respect to m_L. This is equivalent to saying that if, for some measurable set A, $\bar{m}(A) = 0$ then $m_L(A) = 0$. Since, by definition, $\bar{m}(A) = m(\Psi(A))$ this means that, for each $\varepsilon > 0$, it is possible to cover the set $\Psi(A)$ with a collections of pseudo-rectangles S_n such that

$$\sum_n m(S_n) \leq \varepsilon.$$

But, by ??, $m(S_n) \geq c^{-1} m_L(C_n)$, where $S_n = \Psi(C_n)$. Hence,

$$m_L(A) \leq \sum_n m_L(C_n) \leq c \sum_n m(S_n) \leq c\varepsilon$$

which shows that $m_L(A) = 0$.

Next, let us prove that m_L is absolutely continuous with respect to $\bar{m}$. To prove this we will show that if for a measurable set A, $\bar{m}(A) > 0$, then $m_L(A) > 0$. We will argue by contradiction.

Suppose that there exists a measurable set A such that $\bar{m}(A) > 0$ and $m_L(A) = 0$. Since both $\bar{m}$ and m_L are Borel measures they are regular. Accordingly, for each $\varepsilon > 0$ there exists an open set $U \supset A$ such that

$$m_L(U) \leq \varepsilon,$$

and there exists a compact set $K \subset A$ such that

$$\bar{m}(K) \geq \frac{1}{2}\bar{m}(A).$$

Since Ψ is continuous together with its inverse, $\Psi(K)$ is compact and $\Psi(U)$ is open. It is then easy to construct a finite disjoint collection of cubes Γ_n such that $\Gamma_n \subset U$ and

$$m(\cup_n \Gamma_n \cap \Psi(K)) \geq \frac{1}{2}\bar{m}(K).$$

In each such cube we can fit a pseudo-rectangle S_n such that $m(S_n) \geq c_q m(\Gamma_n)$. Collecting the above considerations and since $m(S) \leq c m_L(C)$ holds that

$$\begin{aligned} m_L(U) &\geq m_L(\cup_n \Psi^{-1}(S_n)) \geq c^{-1} \sum_n m(S_n) \geq c^{-1} c_q \sum_n m(\Gamma_n) \\ &\geq \tfrac{c_q}{2c} \bar{m}(K) \geq \tfrac{c_q}{4c} \bar{m}(A) \end{aligned}$$

which leads to the announced contradiction provided ε has been chosen small enough.)

7.12 Prove that the invariant distributions of a simplectic smooth two dimensional dynamical system are Lipsitcs. (Hint: use in Lemma 7.2.1 the distortion estimate of Lemma 7.3.5 in anology with the logic employed in Lemma 7.3.7.)

Notes

Put a brief summary of the regularity properties of the foliations in general.

CHAPTER 8

Local ergodicity for smooth systems

8.1 Some general results

The prototype of all the results we will discuss in this and in the next chapters is the following seminal Anosov's Theorem.

Theorem 8.1.1 (Anosov [4]) *Let X be a connected compact smooth Riemannian manifold and call m the Riemannian measure. Every smooth uniformly hyperbolic dynamical system (X, μ, T), such that $\frac{d\mu}{dm} \in \mathcal{C}^{(2)}(X)$, is ergodic.*

In fact, it is possible to show much more, for example that such systems are Bernoulli [11] or to weaken the regularity properties ($\mathcal{C}^{(1+\alpha)}$ suffices [51]). In the non uniformly hyperbolic or non-smooth case (or both) new difficulties arises: in principle the stable and unstable manifolds can be very short and can wiggle in an unbounded way. Yet much can be said even in this cases by using few natural extra assumptions, we will discuss non-uniform hyperbolicity in this chapter and uniformly hyperbolic discontinuous systems in the next.

8.2 Non-uniform hyperbolicity

In chapter 6 we have seen several examples of hyperbolic systems, it is now natural to ask if such systems are indeed ergodic. This chapter is a first step toward answering such a question. Here we start to see how the Hopf argument explained chapter 2 can be generalized to non-uniform smooth hyperbolic systems.

Theorem 8.2.1 (Local Ergodicity) *Let (X, μ, T) be a dynamical system where X is a connected compact symplectic manifold, μ is the symplectic measure, and T is smooth. Suppose that there exists a* **continuous** *eventually strictly invariant cone family. Then for each $x \in X$ such that, for some $n \in \mathbb{N}$, $D_x T^{-n}\mathcal{C}(x) \subset int\mathcal{C}(T^{-n}x) \cup \{0\}$, x belongs to an open ergodic components. More precisely, there exists an open set $U \subset X$, $x \in U$, and an invariant set $\Lambda \subset X$, such that $(\Lambda, \mu, T\big|_\Lambda)$ is ergodic and*

$$\mu(U \backslash (U \cap \Lambda)) = 0.$$

The above theorem shows that the only points candidate to live on the boundary of different ergodic components are the one for which not only the Lyapunov exponents are zero but for which no expansion whatsoever takes place. A careful study of such points allows, eventually, to claim that a given system is indeed ergodic. This is the strategy usually employed to prove ergodicity and we will illustrate it in details in the following.

We will start by proving the above theorem in the two dimensional case.

8.3 Two Dimensions–the tool

Given a point $\bar{x} \in X$, stricly monotone in the past, we introduce a neighborhood U and a system of coordinate analogous to the one introduced in section 7.1.2. The only difference being that, for graphic convinience, we rotate it of $45°$ degrees. Let us recall the construction.

The fact that the point is stricly monotone in the past means that there exists $\bar{n} \in \mathbb{N}$ such that $D_{T^{-\bar{n}}\bar{x}} T^n \mathcal{C}(T^{-\bar{n}}\bar{x}) \subset \operatorname{Int}\mathcal{C}(\bar{x}) \cup \{0\}$.[1] Now let us consider a Darboux chart (see ?? or problem ??) of a neighborhood U_0 of $\bar{x}$ small enough so that $D_{T^{-\bar{n}}\xi} T^n \mathcal{C}(T^{-\bar{n}}\xi) \subset \operatorname{Int}\mathcal{C}(\xi) \cup \{0\}$ for each $\xi \in U_0$. We will use the Cartesian structure of the chart to identify the tangent spaces. By the continuity of the cone field, and the fact that T is $\mathcal{C}^{(1)}$, it follows that there exists a neighborood $U \subset U_0$ and a cone $\bar{\mathcal{C}}$ such that $D_{T^{-\bar{n}}\xi} T^n \mathcal{C}(T^{-\bar{n}}\xi) \subset \operatorname{Int}\bar{\mathcal{C}} \cup \{0\} \subset \mathcal{C}(\xi)$, for each $\xi \in U$. By eventually further restring the neighborhood U, we can then perform a change of coordinates such that the volume is the standard cartesian volume (see Probelm 1). We are now in the same situation we where in section 7.1.1, only instead to trasfrom the cone $\bar{\mathcal{C}}$ into the standard cone $\mathcal{C}_+$ we find it more convenient to make a linear symplectic change of coordinates[2] that transforms it in the cone $\bar{\mathcal{C}} := \{(u, v) \in \mathbb{R}^2 \mid |v| \le |u|\}$. The tranformation that transforms the cone $\mathcal{C}_+$ in the cone $\bar{\mathcal{C}}$ is the rotation

$$U := \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}.$$

[1]If in doubt, look at problem Problem 4.6.

[2]That is a change of coordinates with determinanat one.

The same map transfroms the cones $\mathcal{C}_\alpha$, of section 7.1.1, into the cones $\mathcal{C}^a := \{(u,v) \in \mathbb{R}^2 \mid |v| \le a|u|\}$ with $a = \frac{1-\alpha}{1+\alpha} \in (0,1)$.

Note that from (7.1.2) it follows that for each $L : \mathbb{R}^2 \to \mathbb{R}^2$ such that $L(\bar{\mathcal{C}}) \subset \mathcal{C}_\alpha$ holds that

$$\sigma(L) \le \frac{1}{\sqrt{a}} \tag{8.3.1}$$

Summing it up, we have introduced a neighborhood U of $\bar{x}$ and coordinates such that all the cones contain the cone $\mathcal{C}^+$, moreover there exists $a \in (0,1)$ such that if $x \in U$ and $T^n x \in U$, for some $n \ge \bar{n}$, then

$$D_x T^n \mathcal{C}^+ \subset D_x T^n \mathcal{C}(x) \subset D_{T^{n-\bar{n}}x} T^{\bar{n}} \mathcal{C}(T^{n-\bar{n}}x) \subset \mathcal{C}^a.$$

Next we consider a square $[-\ell_1, \ell_1] \times [-\ell_2, \ell_2] := Q \subset U$, such that $a\ell_1 < \ell_2 < \ell_1$, and we define its stable and unstable cores.

Definition 8.3.1 *Given a rectangle $Q = [-\ell_1, \ell_1] \times [-\ell_2, \ell_2]$ we define the following subsets:*

- *by unstable core of Q, called $C_u(Q)$, we mean the set $\{(u,v) \in Q \mid \ell_2 - a\ell_1 - a|u| \ge |v|\}$;*
- *by stable boundary of Q, called $C_s(Q)$, we mean the set $\{(u,v) \in Q \mid \ell_1 - \ell_2 - |u| \ge |v|\}$;*
- *by core of Q, called $C(Q)$, we mean the set $C_u(Q) \cap C_s(Q)$;*
- *by unstable boundary of Q we meand $\partial_u Q := \{(u,v) \in Q \mid v \in \{-\ell_2, \ell_2\}\}$;*
- *by stable boundary of Q we meand $\partial_s Q := \{(u,v) \in Q \mid u \in \{-\ell_1, \ell_1\}\}$;*

Note that $C_u(Q) \ne \emptyset$ iff $a\ell_1 \le \ell_2$, $C_s(Q) \ne \emptyset$ iff $\ell_2 \le \ell_1$; moreover $\partial Q = \partial_u Q \cup \partial_s Q$. See Figure 8.1 for a pictorial representation of the core of a box, from the picture it is also obvious the next Lemma. Lemma which is the motivation of the definition of core.

Lemma 8.3.2 *The stable and unstable fibers intersecting the core of Q interserct its unstable and stable boundary (respectively). In addition, any two (stable and unstable) manifolds intersecting the core of Q must have a unique intersection in Q.*

Proof. Obvious (Do like in chap. 2) .... □

By the above Lemma we can then define a change of coordinates that trivializes the foliation. To do so, define $\Psi : C(Q) \to Q$ by requiring that $\Psi(x,y)$ be the unique intersection in Q of the stable manifold of the point

$(x, 0)$ and the unstable manifold of the point $(0, y)$.[3] Note that Ψ is defined only almost everywhere, since a zero set of points may not have a stable or unstable manifold, yet the absolute continuity of the foliation implies that the set of points on the horizontal axis that have stable manifold has full measure and the same for the point on the vertical axis having unstbale manifold (see Problem 8.).

The main point of the above change of coordinates is illustraded by the next proposition.

Proposition 8.3.3 *If we perform the change of coordinates $(x, y) := \Psi(u, v)$, then in the coordinates (u, v) the foliations consists of straight lines. In addition, the map Ψ is absolutely continuous.*

PROOF. .... □

We are now able to prove the wanted theorem. The reader has certainly noticed that we have reduced the general setting of smooth non-uniformly hyperbolic maps to a situation completely analogous to the toy example discussed in chaper 2. It will then not be a surprise that the proof of ergodicity is almost identical to the one in section 2.2.1, let us quickly remember how it goes.[4]

We start by considering an arbitrary contionuos function f and the backward and forward ergodic verages f^- and f^+.[5] We will show that such functions are almost everywhere constant on $C(Q)$. Since the space of continuous functions is separable and it is dense in L^1 it follows that we can construct a dense set of functions in L^1 such that their ergodic averages are all constant, in $C(Q)$, ouside of the same zero measure set. Clearly, by a standard approximation argument, this implies that all the ergodic averages of the L^1 functions are almost everywhere constant in $C(Q)$. In particular this means that there cannot be invariant sets A such that the measure of $A \cap C(Q)$ is different both from zero and from the full measure of $C(Q)$. This means exactly that $C(Q)$ is contained in one ergodic component appart for a zero measure set. Of course the ergodic component "containing" $\bar{x}$ can be obtained by choosing any open set $O \subset C(Q)$, containing $\bar{x}$, and constructing the set $E = \cup_{n\in\mathbb{Z}} T^n O$. Such a set is obviously invariant; moreover, E is a union of open sets hence it is itself open. We have thus constructed an invariant open set containing the point $\bar{x}$ such that the dynamics, restricted to it, is ergodic. This is exactly what we wanted to prove.

[3] Some times for such a point it is used the notation $[(x, 0), (0, y)]$.

[4] The only real difference is that now we have a local argument while in the simple case of linear automorphism it was possible to do global consideration on all the phase space.

[5] The reader not familiar with these concepts is warmly invited to read section 2.2 before proceeding.

To conclude the proof we need only to show that for any continuous function the ergodic averages are almost everywhere constant on $C(Q)$. To see it we remember that the forward and backward ergodic averages are almost everywhere equal (see Lemma 2.2.1). In addition, since the function is continuous, the forward average is constant on stable manifolds while the backward average is constant on unstable manifolds. By restricting the ergodic averages on $C(Q)$ and performing the change of coordinates discussed above we obtain that f^+ is costant along all vertical lines while f^- is constant on almost all horizontal lines, hence they must be almost everywhere equal to a given constant with respect to the Lebesgue measure in the new coordiantes. By Proposition 8.3.3 it follows that the set on which they are not constant is of zero measure also in the original coordinates and this concludes the proof.

8.4 Two Dimensions–the application

The preceeding results can be used to prove ergodicity of many interesting systems.

First of all notice that Theorem 8.1.1, in two dimensions, readily follows since in that case there always exists a continuous cone family (see Problem .) and every point is stricly monotone. Accordingly Theorem 8.2.1 can be applied at each point. Clearly when two neighborhoods belonging to one ergodic component overlap they have a positive measure intersection, hence they must both belong to the same ergodic component.[6] This imediately shows there there exists only one ergodic component, that is the map is ergodic.

The basic consequence of the local ergodicity point of view is illustrated by the following trivial Corollary

Corollary 8.4.1 *In the hypotheses of Theorem 8.2.1 the boundaries of the ergodic components consists of non monotone points.*

This allows to get a good idea of the structure of the ergodic components. For example a first result is the following particular case of a Theorem by Pesin [2]

Lemma 8.4.2 *If almost all the points of a smooth dinamical system, satisfying the hypotheses of Theorem 8.2.1, are strictly monotone, then the system has, at most, countably ergodic components.*

PROOF. Since we can apply Theorem 8.2.1 to a full measure set, we get an open covering of such a set. In fact, since if two ergodic components intersect then they are the same, we get an open disjoint covering. But each element of the covering has strictly positive measure hence there can be only countably many elements. □

[6] It should be obvious but the non convinced reader can do Problem 8..

To get a better idea of what can be done let us see few concrete examples.

8.4.1 Examples

8.4.1.a *Levowich-like maps*

This is a class of maps $T : \mathbb{T}^2 \to \mathbb{T}^2$ defined as follows: for each $h \in \mathcal{C}^3(\mathbb{T}^1, \mathbb{T}^1)$ we define the map $T : \mathbb{T}^2 \to \mathbb{T}^2$ by

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x + h(x) + y \\ h(x) + y \end{pmatrix} \mod 1 \tag{8.4.1}$$

We moreover require the following properties

1. $h(0) = 0$ (zero is a fixed point);
2. $h'(0) = 0$ (zero is a neutral fixed point)
3. $h'(x) > 0$ for each $x \neq 0$ (hyperbolicity)

Note that condition (3) implies in particular that $h'(x)$ be positive in a neighborhood of zero, which forces

$$h''(0) = 0; \quad h'''(0) \geq 0.$$

We will restrict to the generic case

4. $h'''(0) > 0$.

Since the derivative of the map is given by

$$D_{(x,y)}T = \begin{pmatrix} 1 + h'(x) & 1 \\ h'(x) & 1 \end{pmatrix}. \tag{8.4.2}$$

$\det(DT) = 1$, thus the Lebesgue measure m is an invariant measure (the maps are symplectic). We consider then the dynamical systems $(\mathbb{T}^2, T, m)$.

In the special case $h(x) = x - \frac{1}{2\pi}\sin(2\pi x)$ this is exactly the already discussed Levowich map (see (3.3.10)).

As already noted for the Levowich map in Examples 4.1.1.c, the matrix has only positive entries thus the cone $\mathcal{C}_+ = \{(u, v) \in \mathbb{R}^2 \mid uv \geq 0\}$ is invariant. In addition, note that $(0, 0)$ is the only fixed point of the map.[7] To check the strict invariance of the cone field one notices that the cone is not stricly invariant only if one of the matrix elements is zero and this can happen only if $h'(x) = 0$. Let

[7] Since $T(x, y) = (x, y)$ imediately implies $y = 0$ and $x = h(x)$ which, by hypothesis, has the only solution $x = 0$.

$\Lambda = \{x \in \mathbb{T} \mid h'(x) = 0\}$, then Λ is a countable set.[8] On the other hand if $x_* \in \Lambda$ then $T(x_*, y) = (x_* + h(x_*) + y, h(x_*) + y)$ hence

$$\begin{aligned} D_{(0,y)}T^2 &= \begin{pmatrix} 1 + h'(h(x_*) + y) & 1 \\ h'(h(x_*) + y) & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 + h'(h(x_*) + y) & 2 + h'(h(x_*) + y) \\ h'(h(x_*) + y) & 1 + h'(h(x_*) + y) \end{pmatrix} \end{aligned}$$

which is always stricly monotone unless $h'(h(x_*) + y) = 0$. Again h' can have at most countably many zeroes.

The above discussion implies that we can apply Theorem 8.2.1 to all but, at most, countably many points on the set of lines $\{(x, y) \mid x \in \Lambda\}$. By Lemma 8.4.2 follows that there can be, at most, countably many open ergodic components. In addition, Corollary 8.4.1 tells us that the only candidate for the boundary of an ergodic component are the non strictly monotone points, but we simply do not have enough points to make up a boundary, hence there must be an unique ergodic component. In conclusion, all the maps under consideration are ergodic.

8.4.1.b *Burns-Gerber*

....

....

8.4.1.c *Donnay sphere*

....

8.4.1.d *D-L Potential*

....

8.5 Higher Dimension-comments

. . . .

[8]This, for continuos function, depends on Sard's ? theorem but here can be proven easily since the funcion is $C^{(1)}$, see Problem 8..

8.5.1 Examples

8.5.1.a *Geodesic flow on manifolds with non-positive curvature*

8.5.1.b *?????*

?????

The last obstacle to overcome along these lines is the non-smoothness. Many progress have been made in this direction. Katok and Strelcyn [52] have extended the above mentioned results of Pesin on stable manifolds to a large class of maps with discontinuities; while in [64] there exists an analogous of Theorem 3.5 that applies to such a general class of discontinuous maps.

The exposition of the general theory goes well beyond the scopes of the present course, yet in the next section we will discuss in detail a very simple case. The simplicity of the model should not deceive: the majority of the keys ideas are already present at such a simplified level.

Problems

8.1 Let $\Delta dx\wedge dy$ be a non degenerate volume form in a $\mathbb{R}^2$.[9] Show that each point $x\in\mathbb{R}^2$ has a neighborhood U where one can perform a change of coordinates that transforms the volume into the standard Cartesian volume. (Hint: Suppose, without loss of generality, that the point under consideration is $(0,0)$. Consider the functions $\Psi:\mathbb{R}^2\to\mathbb{R}^2$ of the form

$$\Psi(x,y)=(x+\varphi(x,y),y).$$

If we perform the change of coordiantes $(u,v)=\Psi(x,y)$ whe have

$$\mathrm{vol}(\Psi(A)):=\int_{\Psi(A)}dudv=\int_A|\det(D_{(x,y)}\Psi)|dxdy.$$

The problem is then solved if we have $|\det(D_{(x,y)}\Psi)|=\Delta(x,y)$. Now, $|\det(D_{(x,y)}\Psi)|=|1+\partial_x\varphi|$ accodingly we need

$$\partial_x\varphi(x,y)=\Delta(x,y)-1 \tag{8.5.1}$$

since in this case $1+\partial_x\varphi=\Delta>0$. The solution of (8.5.1) is given by

$$\varphi(x,y)=\int_0^x\Delta(\xi,\,y)d\xi-x.$$

Since $|\det(D_{(0,0)}\Psi)|=\Delta(0,0)\neq 0$, by the inverse function Theorem it follows that there exists a neighboord U of $(0,0)$ where Ψ is invertible. We have thus the wanted coordinate change.)

[9]This means that $\Delta\in C^{(1)}$, $\Delta>0$ and, for each measurable set $A\subset\mathbb{R}^2$, holds $\mathrm{vol}(A):=\int_A\Delta(x,y)dxdy$.

8.2 higher dim in symplectic spaces

Notes

try to do Hopf for SRB and put in a problem

. . . .

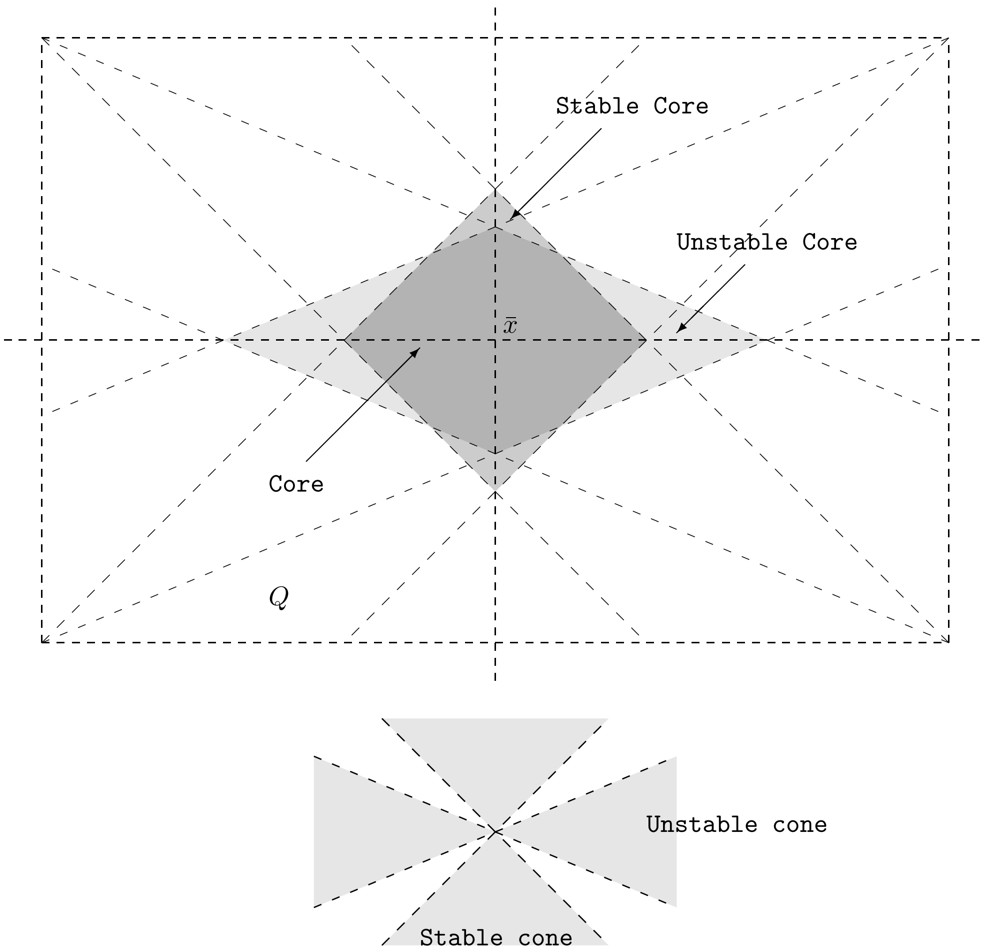

Figure 8.1: The coordinate neighborhood

CHAPTER 9

Ergodicity for non smooth maps– An introduction

Chapter 2 was devoted to the investigation of the ergodicity of a linear automorphism of $\mathbb{T}^2$; it cannot be a surprise that piecewise linear automorphism can be used as a paradigm to study systems with discontinuities. Such examples can be obtained by using the same model discussed in Chapter 2 but with $a \notin \mathbb{Z}$.

Accordingly, we will start discussing the ergodic properties of a discontinous linear automorphism of the two dimensional torus first, then we will see how the ideas illustrated by such a toy model can be applied to a much more general setting. Among other things we will prove the ergodicity of a Sinai Billiard wioth finite horizon.

9.1 The easiest case: a piecewise linear map

Here we will discuss the piecewise linear map defined in Example 3.6.1 (Discontinuos Arnold Cat). As we have seen there in particular, and more generally in chapter 7, for these type of maps there still exist stable and unstable manifold, although they may be very short. Accordingly, serious problems arise if one wants to try to apply the Hopf method. Let us analyze the situation more carefully.

9.1.1 What is left of the Hopf method

Next we will try to apply the Hopf method naïvely and see how far it leads us.

For any continuous function $f : \mathbb{T}^2 \to \mathbb{R}$ the forward ergodic average f^+ is constant on the stable leaves and the backward ergodic average f^- is constant on the unstable leaves. Let us call a point $x \in \mathbb{T}^2$ f-typical, if $f^+(x)$, $f^-(x)$, $W^u(x)$ and $W^s(x)$ are well defined and $f^+(x) = f^-(x)$. The set of f-typical points has full measure by Proposition 3.6.4, Birkhoff Theorem 1.6.1 and Corollary 1.6.3. Thus a stable (or an unstable) leaf contains a set of f-typical points of full arc-length, except for a family of leaves of total measure zero. To see this consider a coordinate system with the axis parallel to the unstable and stable directions respectively. Let B be the set of non-typical points, tehn

$$0 = m(B) = \int d\xi \int d\eta \mathbb{1}_B(\xi, \eta),$$

thus, by Fubini, for almost all ξ the one dimensional Lebesgue measure of $B \cap \{(x, y) \in \mathbb{T}^2 \mid x = \xi\}$ must be zero. But $\{(\xi, y)\}$ is, by construction, a segment parallel to the stable direction and Proposition 3.6.4 tells us that it is the union of, at most, countably many stable leaves of positive lenght. Accordingly, teh intersection of B whith any of these leaves has zero one dimensional Lebesgue measure. hence all the leaves whose intersection with B has positive measure must belong to the set of exceptional lines, therefore to a set of zero measure.

If $W^s(x)$ is not one of those exceptional leaves, then the set

$$C_1 = \bigcup_{y \in W^s(x),\, y \text{ is } f-\text{typical}} W^u(y)$$

has positive measure and $f^- = f^+ = \mathit{const}$ on C_1. In fact, calling $\ell(y$ teh lenght of $W^u(y)$ we have that $\ell(y) > 0$ for almost all $y \in W^s(x)$, since $W^s(s)$ is typical. Thus, by Fubibi Theorem, if $\bar{\ell}$ is the lenght of $W^s(x)$, tehn

$$m(C_1) = \int_0^{\bar{\ell}} dy \ell(y) > 0.$$

Moreover, f^- is constant on each $W^u(y)$ and f^+ is constant on $W^s(x)$. Hence, for almost all $y_1, y_2 \in W^s(x)$

$$f^-(y_1) = f^+(y_1) = f^+(y_2) = f^-(y_2).$$

Accordingly, f^- is a.e. constant on C_1 and the statment follows since $f^- = f^+$ a.e..

We can proceed by adding all the stable leaves through f-typical points in C_1 to obtain C_2, etc., but a priori there is no reason to expect that we will be able to cover all of the torus in this way. (Indeed one can imagine that there is a dividing line between two ergodic components of our system and that all the stable and unstable leaves stop short of crossing this line.) That is where the Hopf method breaks down. It can only tell us that the ergodic components have positive measure and, therefore, that there are at most countably many of them.[1]

9.2 The Sinai Method

We have seen, in the previous section, that the Hopf method is not sufficient to prove the ergodicity of a discontinuous map because the stable and unstable leaves may be short. The Sinai method amounts to establishing that most of the stable and unstable leaves are, in a certain sense, sufficiently long. The first (highly nontrivial) step in this method is to formulate precisely what is meant by "sufficiently long". As before, we do it only for the unstable leaves; the changes necessary in the case of stable leaves are obvious.

Let $\mathcal{U} \subset \mathbb{T}^2$ be a (small) square with the sides parallel to unstable and stable directions respectively (since all the following arguments take place in $\mathcal{U}$, we choose convenient coordinates to make the geometry simpler; the unstable direction is horizontal and the stable direction vertical). For any $0 < c < 1$ we construct a sequence $\mathcal{G}_n(c), n = 1, 2, \ldots,$ of coverings of $\mathcal{U}$ in the following way. Without loss of generality we can let

$$\mathcal{U} = \{(u, v) \mid -b < u < b,\ -b < v < b\}.$$

We consider the net $\mathcal{N}(n, c)$ defined by

$$\mathcal{N}(n, c) = \{\frac{c}{n}(m,\ k) \in \mathcal{U} \mid m, k \in \mathbb{Z}\}.$$

Now the covering $\mathcal{G}_n(c)$ is the collection of squares having centers at points from $\mathcal{N}(n, c)$ and sides, of length $\frac{1}{n}$, parallel to the sides of $\mathcal{U}$. If $c < \frac{1}{2}$ then $\mathcal{G}_n(c)$ is a covering of $\mathcal{U}$ (otherwise $\mathcal{G}_n(c)$ may cover only a smaller square). The parameter c will be chosen later to be very small, so that many squares in $\mathcal{G}_n(c)$ overlap. However, once c is fixed, a point in $\mathcal{U}$ may belong, at most, to a fixed number, independent of $n = 1, 2, \ldots,$ of squares in $\mathcal{G}_n(c)$; we denote this number by $k(c)$ (one can easily establish that $k(c) \leq (\frac{1}{2c} + 1)^2$, but we will not use any explicit estimate).

We call two squares, in $\mathcal{G}_n(c)$, immediate neighbors if the distance between their centers is $\frac{c}{n}$.Two immediate neighbors overlap on $1-c$ part of their areas.

[1]To be more precise, the previous argumetn does not suffice to claim that C_1 belongs to one ergodic component. To argue this we have to modify our argument by taking a sequence of continuous functions dense in L^1 and considering the set of points which are f-typical for all the functions f in the sequence. This set, as the intersection of countably many sets of full measure, has full measure. We can then use it in the definition of C_1 and claim that $f^- = f^+ = const$ on C_1 for all the functions in our dense sequence. This implies that such C_1 does belong to one ergodic component. It follows easily that every invariant subset of positive measure contains an ergodic component of positive measure. Hence all ergodic components have positive measure.

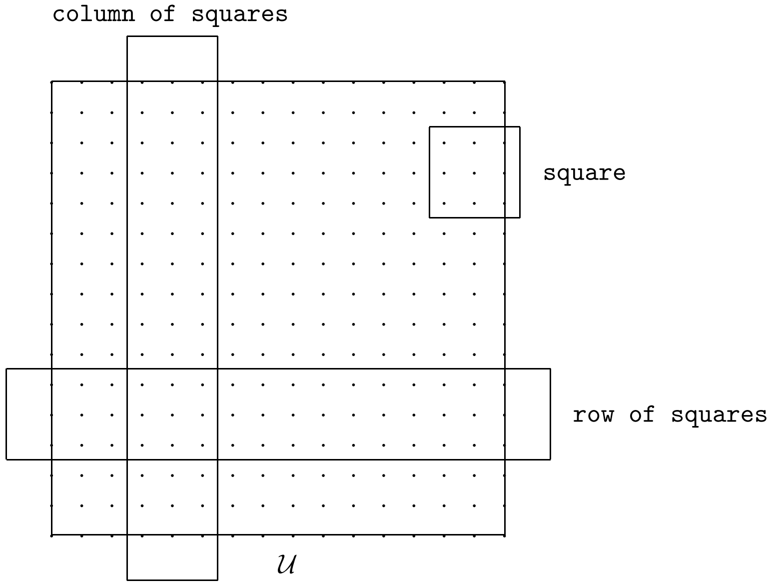

Figure 9.1: The neighborood $\mathcal{U}$

One can naturally define a column of squares and a row of squares as special collections of squares in $\mathcal{G}_n(c)$ (see Figure 9.1). For example, a sequence $\{R_i\}_{i=1}^l$ of squares from $\mathcal{G}_n(c)$ is called a column of squares if, for every $i = 1, \ldots, l-1$, R_i and R_{i+1} are immediate neighbors, R_{i+1} is above R_i, and there is no square in $\mathcal{G}_n(c)$ below R_1 or above R_l.

For each square $R \in \mathcal{G}_n$ we introduce the stable, $\partial_s R$, and unstable, $\partial_u R$, boundaries of R; $\partial_s R$ is the union of the two boundary segments of R which have the stable (vertical) direction and $\partial_u R$ is the union of the two boundary segments of R which have the unstable (horizontal) direction. Given a point $x \in R$, the unstable leaf $W^u(x)$ may intersect both segments in $\partial_s R$ or it may be too short to reach one of them (or both). In the first case we say that $W^u(x)$ is long in R, or that it is connecting in R , in the second that it is short in R or that it is not connecting in R.

Definition 9.2.1 *Given α, $0 < \alpha < 1$, we call a square $R \in \mathcal{G}_n(c)$ α-connecting if the measure of the set of points $x \in R$ whose unstable leaf $W^u(x)$ is long in R is at least α part of the total area of R.*

Sinai formulates the property that most of unstable leaves are sufficiently long in the following way.

Theorem 9.2.2 (Sinai Theorem) *There is $\alpha_0 < 1$ such that for any $\alpha \in (0, \alpha_0]$ and any $c \in (0,1)$,*

$$\lim_{n\to+\infty} n\ \mu\left(\bigcup\{R \in \mathcal{G}_n(c) \mid\ R \text{ is not } \alpha\text{-connecting }\}\right) = 0.$$

In other words, the theorem says that if α is sufficiently small, then the union of the squares in $\mathcal{G}_n(c)$ which are not α-connecting has measure $o(\frac{1}{n})$.

The proof of Sinai Theorem can be found in section 9.3. Before going into it let us show how Sinai Theorem 9.2.2 can be used to get information about ergodic components. Notice that Definition 9.2.1 and the Sinai Theorem 9.2.2 can be repeated for stable leaves.

Proposition 9.2.3 *If the Sinai Theorem holds for both unstable leaves and stable leaves in a square $\mathcal{U} \subset \mathbb{T}^2$, then $\mathcal{U}$ belongs to one ergodic component of T.*

In view of the arbitrariness of the square $\mathcal{U}$ to which we can apply this Theorem we obtain immediately

Corollary 9.2.4 *The map T is ergodic.*

Proof of Proposition 9.2.3. Let us fix α sufficiently small so that the Sinai Theorem holds for α-connecting squares both in the unstable and stable versions. Next we fix c smaller than α. As a consequence two α-connecting squares in $\mathcal{G}_n(c)$, which are immediate neighbors, contain in their intersection a set of connecting leaves of positive measure. The reason is that immediate neighbors intersect over $1-c$ part of their areas and hence the guaranteed α part of the square covered by connecting leaves cannot fit into the remaining c part of the square. In the following we will not change the values of α or c and, for simplicity, we will call connecting any square which is α-connecting both with respect to stable and unstable leaves.

Consider any continuous function f on the torus. We call a point $y \in \mathbb{T}^2$ f-typical if the forward time average f^+ and the backward time average f^- are well defined at y and $f^+(y) = f^-(y)$. The set of f-typical points has full measure. We call a stable (unstable) leaf f-typical if its points, except for a subset of zero arc-length, are f-typical. The union of leaves which are not f-typical is a set of measure zero.

For any connecting square R let us define

$$W^{u(s)}(R) = \{x \in R \mid W^{u(s)}(x) \text{ is } f\text{-typical and long in } R\}.$$

Although we cannot apply the Hopf argument to the whole torus we can use it in a connecting square R to claim that f^+ is constant on all of $W^s(R)$ and f^- is constant on all of $W^u(R)$ with the two constants coinciding. Note that we say here (and we mean it) "all of $W^{s(u)}$" and not almost all. Indeed, first of all f^+ is constant on each of the stable leaves in $W^s(R)$. Further let us fix an unstable leaf in $W^u(R)$. The stable leaves from $W^s(R)$ intersect this unstable leaf in f-typical points, except for a set of stable leaves of total measure zero. Hence excluding these exceptional stable leaves the value of f^+ on the stable leaves has to coincide with the constant value of f^- on the distinguished unstable leaf. We conclude that f^+ is constant almost everywhere on $W^s(R)$ and the constant is equal to the constant value of f^- on the unstable leaf. Since we could have used any other unstable leaf in $W^u(R)$ it follows that f^- is constant on all of $W^u(R)$. By symmetry f^+ is constant on all of $W^s(R)$. (The reader must have noticed the implicit use of the Fubini Theorem in the arguments above. It is only natural since the stable and unstable leaves are parallel segments. In the nonlinear case one has to use the "absolute continuity" of the foliations into stable and unstable manifolds. This property is all that we need, to make the present argument work.)

Further for two connecting squares R_1 and R_2 which are immediate neighbors f^+ is constant on $W^s(R_1) \cup W^s(R_2)$ and f^- is constant on $W^u(R_1) \cup W^u(R_2)$ with the two constants coinciding. Indeed at least one of the intersections $W^u(R_1) \cap W^u(R_2)$ (if one square is above the other) or $W^s(R_1) \cap W^s(R_2)$ (if one square is next to the other) must have positive measure and hence is nonempty, forcing the constant value of f^+ or f^- to be the same for both squares.

After this observation we proceed to prove that the time average of f is almost everywhere constant in $\mathcal{U}$. To that end let $y, z \in \mathcal{U}$ be two f-typical points with f-typical leaves, $W^u(y)$ and $W^s(z)$ respectively. Our goal is to prove that $f^-(y) = f^+(z)$.

We say that $W^u(y)$ ($W^s(z)$) intersects completely a column (row) of squares in $\mathcal{G}_n(c)$ if it is connecting in one of the squares of the column (row). The Sinai Theorem allows us to claim that, for sufficiently large n, $W^u(y)$ intersects completely at least one column of connecting squares in $\mathcal{G}_n(c)$, i.e. a column in which all the squares are connecting, and $W^s(z)$ intersects completely at least one row of connecting squares. Indeed, suppose to the contrary that every column of squares in $\mathcal{G}_n(c)$ intersected completely by $W^u(y)$ contains at least one non-connecting square. Since the number of columns intersected completely by $W^u(y)$ grows linearly with n and the measure of one square in $\mathcal{G}_n(c)$ is $\frac{1}{n^2}$, we obtain that the measure of the union of non-connecting squares would be $O(\frac{1}{n})$ which contradicts the Sinai Theorem. (Here we have used the fact that the squares in $\mathcal{G}_n(c)$ cannot overlap more than $k(c)$ times.)

Let us fix a column and a row of connecting squares which are intersected completely by $W^u(y)$ and $W^s(z)$ respectively. Let R be the (unique) square which belongs both to the column and the row. Let further R_1 denote a square in which $W^u(y)$ is connecting and R_2 denote a square in which $W^s(z)$ is connecting. By the construction $y \in W^u(R_1)$ and f^- is constant on the, possibly disjoint, set $W^u(R_1) \cup W^u(R)$. Similarly $z \in W^u(R_2)$ and f^+ is constant on $W^s(R_2) \cup W^s(R)$. It follows that $f^-(y) = f^+(z)$. In view of the arbitrariness in the choice of the f-typical leaves $W^u(y)$ and $W^s(z)$ we obtain that the time average of f must be constant in $\mathcal{U}$.

The proof is then concluded thanks to the approximation argument already discussed in the continuous case. That is, for each invariant measurable set $A \subset \mathcal{U}$ it follows that either A or $\mathcal{U} \backslash A$ must have zero measure, hence $\mathcal{U}$ must belong to one ergodic component. $\square$

9.3 Proof of the Sinai Theorem

The proof of the Sinai Theorem does not require a rigid geometric structure of the coverings $\mathcal{G}_n(c)$; it holds for any sequence of coverings by squares with side $\frac{1}{n}$ as long as there is a uniform bound on the number of squares covering one point. However, the lattice structure of the centers of the squares in $\mathcal{G}_n(c)$ allows to work with columns and rows of squares, as we did in the above application of the Sinai Theorem.

The first step in the proof is the choice of α_0. To that end we consider the smallest sector $\mathcal{C}$ in $\mathbb{R}^2$ symmetric about the horizontal (unstable) line which contains the lines with the two directions of the sides of $\mathcal{M}^-$, i.e., the directions of the segments in $\mathcal{S}^-$. Let

$$\mathcal{C} = \{(\xi, \eta) \mid \ |\eta| \le \kappa(a)|\xi|\}.$$

It can be checked that $\kappa(a) < 1$ for any $a \neq 0$. We put $\alpha_0 = \frac{1}{2}(1 - \kappa(a))$. The reason for this choice is that, for any square with vertical and horizontal sides crossed by a line with the direction contained in $\mathcal{C}$, the shaded area in Figure 9.2 does not exceed $1 - 2\alpha$ part of the area of the square.

Let us observe that all of the segments in $\bigcup_{i=0}^{+\infty} T^i \mathcal{S}^-$ have directions contained in the sector $\mathcal{C}$. Indeed a linear hyperbolic map pushes lines towards the unstable direction except for the stable line, which stays put.

It follows from the construction of the unstable leaves (Proposition 4.1) that an unstable leaf has endpoints on forward images of $\mathcal{S}^-$ under T. Hence if an unstable leaf is short in a square then the square must be intersected by

$$\bigcup_{i=0}^{+\infty} T^i \mathcal{S}^- \ .$$

Although this does not look like a severe restriction, since we can expect that the last set is dense, it has far reaching consequences. The reason being,

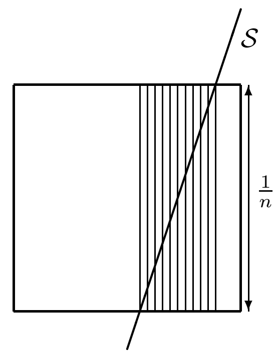

Figure 9.2: Stable manifolds cut by only one singularity in a box

heuristically, that the singularity lines $T^i\mathcal{S}^-$ become more and more horizontal as $i \to +\infty$ and they cannot cut effectively unstable leaves which are themselves horizontal.

We claim that, for any fixed $M \geq 1$, the singularity lines

$$\mathcal{S}_M^- = \bigcup_{i=0}^{M} T^i\mathcal{S}^-$$

by themselves can produce only few squares which are not α-connecting so that their total measure is $O(\frac{1}{n^2})$. To make this precise (and clear) we introduce an auxiliary notion of an M-bad square in a covering $\mathcal{G}_n(c)$. We say that a square $R \in \mathcal{G}_n(c)$ is M-bad if the measure of the set of points $y \in R$ such that the unstable leaf $W^u(y)$ has an endpoint in $R \cap \mathcal{S}_M^-$ (so that it is short in R) is greater than $1 - 2\alpha$ part of the measure of the square. (Loosely speaking a square is M-bad if it is not connecting because of the singularity lines in $\mathcal{S}_M^-$.)

If a square R intersects only one segment in $\mathcal{S}_M^-$ then the measure of points in R whose unstable leaves have endpoints on the intersection of this segment with R does not exceed $1 - 2\alpha_0 = \kappa(a)$ part of the measure of the square since the direction of the segment is in the sector $\mathcal{C}$, see Figure 9.2. Hence an M-bad square has to intersect at least two segments in $\mathcal{S}_M^-$.

But the singularity set $\mathcal{S}_M^-$ is a fixed finite collection of closed segments with only fixed finite number of intersection points (i.e., belonging to several segments). Away from the intersection points the segments are fairly wide apart and a small square cannot extend from one to another, see Figure 9.3.

Hence, for sufficiently large n, an M-bad square in $\mathcal{G}_n(c)$ cannot be farther from one of the intersection points than $\frac{const}{n}$. It follows that the total measure of M-bad squares does not exceed $\frac{const}{n^2}$, where the constant depends only on a, c, α and M.

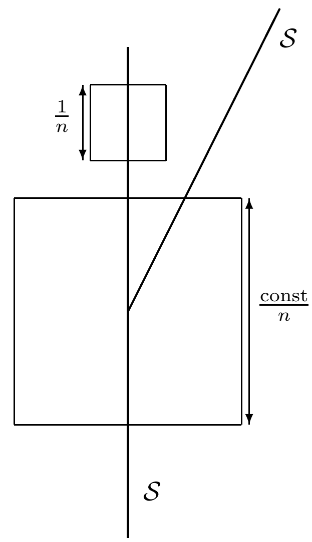

Figure 9.3: Away from double singularities

In this way we took care (in some sense) of the finite number of singularity lines in $\mathcal{S}_M^-$; we now face the problem of controlling the effects of the 'tail' $\bigcup_{i=M+1}^{+\infty} T^i\mathcal{S}^-$.

Let us suppose that a square $R \in \mathcal{G}_n(c)$ is not α-connecting and it is not M-bad. Hence at least α part of its area is covered by short leaves with endpoints in

$$R \cap \bigcup_{i=M+1}^{+\infty} T^i\mathcal{S}^-.$$

Let $W^u(y)$ be such a leaf short in R with an endpoint on $T^i\mathcal{S}^-$. Then

$$T^{-i}\left(W^u(y) \cap R\right) \subset X_{t_i}$$

where $t_i = n^{-1}\lambda^{-i}$ and, as before, $X_t = \{x \in \mathcal{M}^- \mid d(x, \mathcal{S}^-) \leq t\}$. Indeed, under the action of T^{-1}, an unstable leaf contracts by a factor of λ and the length of the part of $W^u(y)$ in R does not exceed $\frac{1}{n}$.

In view of this observation we can claim that each square which is not α-connecting and which is not M-bad has at least α part of its area covered by

$$\bigcup_{i=M+1}^{+\infty} T^i X_{t_i}.$$

Since each point in $\mathcal{U}$ is covered by, at most, $k(c)$ squares from $\mathcal{G}_n(c)$, then the measure of the union of squares in $\mathcal{G}_n(c)$ which are not α-connecting and which are not M-bad does not exceed

$$k(c) \times \frac{1}{\alpha} \sum_{i=M+1}^{+\infty} \frac{const}{n\lambda^i} = \frac{1}{n} \left(\frac{k(c)}{\alpha} \sum_{i=M+1}^{+\infty} \frac{const}{\lambda^i} \right),$$

(here the constant is equal to the total length of $\mathcal{S}^-$). We have thus estimated the measure of the union of squares in $\mathcal{G}_n(c)$, which are not α-connecting and which are not M-bad, by the size of an individual square times the M-tail of a fixed convergent series. Some of the readers may have noticed that this completes the proof. For clarity, let us do it explicitly.

Let us take an arbitrary $\epsilon > 0$. We choose and fix $M = M(\epsilon)$ so large that the last series does not exceed $\frac{\epsilon}{2n}$, i.e.,

$$\frac{k(c)}{\alpha} \sum_{i=M+1}^{+\infty} \frac{const}{\lambda^i} < \frac{\epsilon}{2}.$$

Given M we can still choose $n_0 = n_0(\epsilon, M)$ so large that, for any $n \geq n_0$, the measure of the union of M-bad squares in $\mathcal{G}_n(c)$ is less than $\frac{\epsilon}{2n}$. To estimate the measure of the union of squares in $\mathcal{G}_n(c)$, for $n \geq n_0$, which are not α-connecting we split them into those which are M-bad and those which are not. For both families of squares the measure of their union is less than $\frac{\epsilon}{2n}$. This proves our claim. $\square$

Remark 9.3.1 *Notice that we have proven more that what it is stated in the theorem, namely that the long fibers we have constructed enjoy the following propertirs: for each* $n \in \mathbb{N}$ *there exists* $M(n) \in \mathbb{N}$ *such that* $T^k W^s \cap X_{\frac{1}{n}\lambda^{-k}} = \emptyset$ *for all* $k > M(n)$. *This does not play any rôle at the moment but will be relevant in the following when we will discuss more general systems.*

9.4 Sinai Billiard with finite horizon

In this section we will discuss the local ergodicity of the billiard described in section 5.5 (subsection Finite horizon).

Since the strategy will be to make the setting as similar as possible to the linear case, the reader who has digested the previous sections of this chapter should find the following quite natural.

We begin by recalling some needed results already obtianed in the previous chapters.

In section 6.3.1 we have proven that such a billiard is hyperbolic.

To proceed is convenient to reduce the billiards flow to a map. This can be done in various ways, for example let us use the section depicted in Figure 5.13. Clearly almost every trajectory intersects the section, thus we can apply Theorem 1.5.2 and the ergodicity of the billiard flow is implied by the ergodicity of the section. It suffices then to investigate the Poincarè map. The resulting Poincaré map is a Smooth Dynamical System with Singularities (see Proposition 5.4.3). In section 7.1.5 is proven the existence of the invariant manifolds and 7.5 is devoted to the proof of the abosolute continuty of the stable and unstbale foliations.

As we have already done in section 8.3 we will introduce a local system of coordinates in order to reduce the case at hand to a situation similar to the one treated in sections 9.2 and 9.3.

First of all we will use the coordiantes (s, ξ), where $\xi = -\cos\varphi$. It is easy to check that, in this coordinates, $\mathcal{M}$ is the union of pieces of $\mathbb{R}^2$ with the standard simplectic form (see ??). Form now on we will use the cartesian structure of $\mathbb{R}^2$ to identify all the tangent spaces, in addition we will identify the tangent space with the phase space.

Second consider a point $\bar{x} \in \mathcal{M} \setminus \mathcal{S}$ such that there exist $n_0 \in \mathbb{N}$ for which $\sigma(DT^{-n_0}) > 3$, then there exists a neighborhood $\mathcal{U}_0$ of $\bar{x}$ such that $\mathcal{U}_0 \cap \mathcal{S}_{n_0} \neq \emptyset$ and $\sigma(DT^{-n_0}) > 3$ on $\mathcal{U}_0$.

Remark 9.4.1 *Note that the only points for which n_0 does not exists are the points which never experince a collision (see ???).*

Next, by the continuity of the cone family and of DT^{n_0} we can find a neighborhood $\mathcal{U} \ni \bar{x}$ and a cone $\bar{\mathcal{C}}$ such that $\bar{\mathcal{C}} \subset \mathcal{C}(x)$ for each $x \in \mathcal{U}$, $DT^{-n_0}\bar{\mathcal{C}} \supset \mathcal{C}(T^{n_0}(x))$ and $\sigma(DT^{-n_0}) =: a > 3$, where this time σ is computed by using the quadratic form associated to $\bar{\mathcal{C}}$. We can then construct a coordinate systems centered at $\bar{x}$ similar (but a bit more simmetric with respect to time) to the one already used in section 8.3.

As usual we can start by making a linear symplect change of coordinates so that $\bar{\mathcal{C}}$ is mapped onto $\mathcal{C}_+$ and $DT^{n_0}\mathcal{C}(T^{-n_0}x) \subset \mathcal{C}_\rho$, $\rho := \frac{a^2-1}{a^2+1} \leq 1$ (see (7.1.2)). To abide to our previous conventions to have unstbale direction more or less horizontal we can then apply the symplectic change of coordiante

$$L = \frac{1}{\sqrt{2}} \begin{pmatrix} a^{-\frac{1}{2}} & a^{-\frac{1}{2}} \\ -a^{\frac{1}{2}} & a^{\frac{1}{2}} \end{pmatrix} \quad ; \quad a = \sqrt{\frac{1+\rho}{1-\rho}}.$$

It is imediate to verify that $L\mathcal{C}_+ = \mathcal{C}^a$ and $L\mathcal{C}_\rho = \mathcal{C}^{a^{-1}}$ so that the complement of $\mathcal{C}^a$ (stable cone) and $\mathcal{C}^{a^{-1}}$ (unstbale cone) are simmetric with respect to $\mathcal{C}^+$, see Figure 9.4.

At this point we can construct a covering exactly as we did at the beginning of section 9.2, and our aim it to prove again Sinai Theorem in this new case:

Theorem 9.4.2 (Sinai Theorem) *There is $\alpha_0 < 1$ such that for any $\alpha \in (0, \alpha_0]$ and any $c \in (0, 1)$,*

$$\lim_{n\to+\infty} n\ \mu\left(\bigcup\{R \in \mathcal{G}_n(c) \mid R \text{ is not } \alpha\text{-connecting }\}\right) = 0.$$

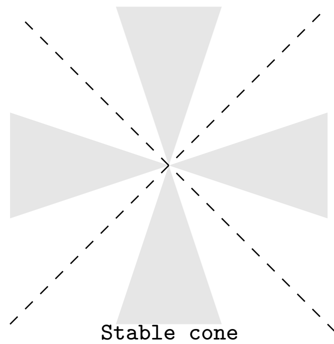

Figure 9.4: The stable and unstbale cones ($a = 3$)

The proof of the Sinai Theorem holds essentially unchanged with respect to 9.2.2, only a little more care must be taken since the stable and unstable manifolds are no longer straight line with a precise orientation but only curves with tangent vector in the respective cone. To handel this extra little problem it is helpful to use the concept of *core* already introduced in Definition 8.3.1.

The same holds for the proof of the analogous of Proposition 9.2.3, only now we must use the results on absolute continuity of the foliations from section 7.5.

This completes the proof of the local ergodicity.

9.5 The general case

discuss the general theorem in [LW]–that is discuss the conditions and omit the proofs. Discuss also general situation for billiards (including last correction).

Problems

Notes

CHAPTER 10

Examples of ergodic billiards

At last we have all the technology needed to investigate the ergodicity of several highly non trivial examples. This chapter is dedicated precisely to such a task.

10.1 Ergodicity of the map

Remark 10.1.1 *Let us point out that the property that the sector $\mathcal{C}$, defined by the directions of the segments in $\mathcal{S}^-$, is sufficiently narrow ($\kappa(a) < 1$) can be relaxed. For a general hyperbolic piecewise linear map it is sufficient that the segments in $\mathcal{S}^-$ are not parallel to the stable direction. In such a case we can find a natural N such that all the segments in $\bigcup_{i=N+1}^{+\infty} T^i \mathcal{S}^-$ have directions contained in a chosen narrow sector $\mathcal{C}$ (N is the number of iterates of T which do not put the singularity lines $\mathcal{S}^-$ into the chosen sector $\mathcal{C}$). Then the argument above applies to any square neighborhood $\mathcal{U}$ which does not intersect*

$$\mathcal{S}_N^- = \bigcup_{i=0}^{N} T^i \mathcal{S}^- \ .$$

Similarly in the version of the Sinai Theorem for the stable leaves we would have arrived at a natural N' such that the claim holds for any square $\mathcal{U}$ which does not intersect

$$\mathcal{S}_{N'}^+ = \bigcup_{i=0}^{N'} T^{-i} \mathcal{S}^+ \ .$$

Hence, it follows from Proposition 2.3 that any open square, with horizontal and vertical sides, which does not intersect $\mathcal{S}_N^- \cup \mathcal{S}_{N'}^+$ belongs to one ergodic component. This implies that the partition of $\mathbb{T}^2$ into ergodic components is coarser than the partition into (open) connected components of

$$\mathbb{T}^2 \setminus \left(\mathcal{S}_N^- \cup \mathcal{S}_{N'}^+\right) \ .$$

Since $\mathcal{S}_N^- \cup \mathcal{S}_{N'}^+$ is a finite collection of segments we obtain that there are at most finitely many ergodic components. To argue that there is only one component let us note that $\mathcal{S}_{N-1}^- \cup \mathcal{S}_{N'}^+$ and $T^N\mathcal{S}^-$ intersect in at most finitely many points which split the segments in $T^N\mathcal{S}^-$ into finitely many segments $\{I_k\}_{k=1}^{K_N}$ so that the interior of every I_k lies in the boundary of at most two connected components of $\mathbb{T}^2 \setminus \left(\mathcal{S}_N^- \cup \mathcal{S}_{N'}^+\right)$, i.e., it has only one connected component on each side. Suppose that for such a segment I_k is in the boundary of two different ergodic components. Then TI_k is also in the boundary of two different ergodic components. But TI_k and $\mathcal{S}_N^- \cup \mathcal{S}_{N'}^+$ have only finitely many points of intersection, so that whole open sub-intervals of TI_k must end up inside one connected component of $\mathbb{T}^2 \setminus \left(\mathcal{S}_N^- \cup \mathcal{S}_{N'}^+\right)$ and thus it must have the same ergodic component on both sides. This contradiction implies that I_k does not take part in the splitting of $\mathbb{T}^2$ into ergodic components so we can drop it. In this way we can drop all of $T^N\mathcal{S}^-$ and claim that the partition into ergodic components is coarser than the partition into connected components of

$$\mathbb{T}^2 \setminus \left(\mathcal{S}_{N-1}^- \cup \mathcal{S}_{N'}^+\right) \ .$$

It is now clear that we can proceed by dropping $T^{N-1}\mathcal{S}^-$ and $T^{-N'}\mathcal{S}^+$ as possible boundaries for the ergodic components and arriving eventually at $\mathcal{S}^+ \cup \mathcal{S}^-$ as the only possible boundaries we see that even these can be dropped. Hence there is only one ergodic component.

Let us spell out the property of T which is basic in this argument:

Although some points of $\mathcal{S}^-$ return to $\mathcal{S}^-$ under iterates of T, no interval in $\mathcal{S}^-$ can do it.

10.2 Sinai Billiard–finite horizon

10.3 Sinai Billiard–infinite horizon

10.4 Two balls

10.5 Convex Billiards

10.5.1 Bunimovich Stadium

10.5.2 Wojtkowski Cardioid

10.5.3 Focusing Billiards

10.6 Generalized Sinai Billiards

particles in an external potential, particles in a wedge (introduce before) and general W-models.

10.7 Three Balls

10.8 Hyperbolicity for a gas of hard balls–few considerations

We have already seen that one of the sources of lack of hyperbolicity in hard balls systems is the possibility that the balls divide into two or more groups that never interact among each other. In the following we show that such a phenomena can occur only for a set of trajectories of zero measure. To establish hyperbolicity it then remain to prove that if a trajectory has sufficiently many collisions (a "rich" collision structure in the hungarian language) then it is sufficent–i.e. the cone goes strictly inside. This last fact it is quite complex and we do not address it here.

Theorem 10.8.1 (Simany-Szasz [2]) *Given n hard balls in $\mathbb{T}^d$, the set of initial conditions for which the balls can be divided into two groups such that in the future or in the past no balls for different groups collide has measure zero.*[1]

PROOF. Let us consider $G_1, G_2 \subset \{1, \ldots, n\}$ with $G_1 \cup G_2 = \{1, \ldots, n\}$ and $G_1 \cup G_2 = \emptyset$. Let $A \subset \mathbb{T}^d$ be the set of initial conditions for which no

[1] In Lemma 6.9.1 the above result was obtained for the special case of three balls.

collision takes place among the balls of G_1 and G_2 in the future.[2] We can then consider the following change of coordinates: let (Q_i, P_i) be the position of the center of mass and the total momentum of the ball belonging to G_i, respectively; for each ball in G_i let $(x_k^i, \eta_k^i) := (q_k - Q_i, p_k - P_i)$. In other words we have separated the motion of the center of mass from the morion relative to the center of mass in both groups of particles. Once we have done this is it more convenient to look at the motion on the cover of the torus rather than on the torus itself.

Notice that $P_1 + P_1 = 0$ and $Q_1 + Q_2 = 0$.

Let us choose a particle k_i form G_i, we will prove that the set $B \subset A$ for which k_1 never collides with k_2 in the future has zero measure. Given the arbitrariness of our choice of the particles k_i, this clearly implies that A has zero measure, whereby proving the theorem.

Let $P = P_2 - P_1$, and $\xi = \xi_2 - \xi_1 + Q_2 - Q_1$, then, if the initial condition belong to B, it must be

$$\|2tP - \xi\| \geq 2R \quad \forall r \geq 0 \tag{10.8.1}$$

....

□

Problems

10.1 ......

Notes

[2]The past is treated in exactly the same way.

CHAPTER 11

The state of the art

This chapter is intended to give a brief review of current research related to the arguments discussed in this book. Given the pace at which the field developes the present discussion will be probably obsolete already at pubblication time, yet I hope it will provide the reader at least an idea of the current developments and lines of research.

Problems

11.1 ......

Notes

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[1] **find it**

[2] **find it**

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